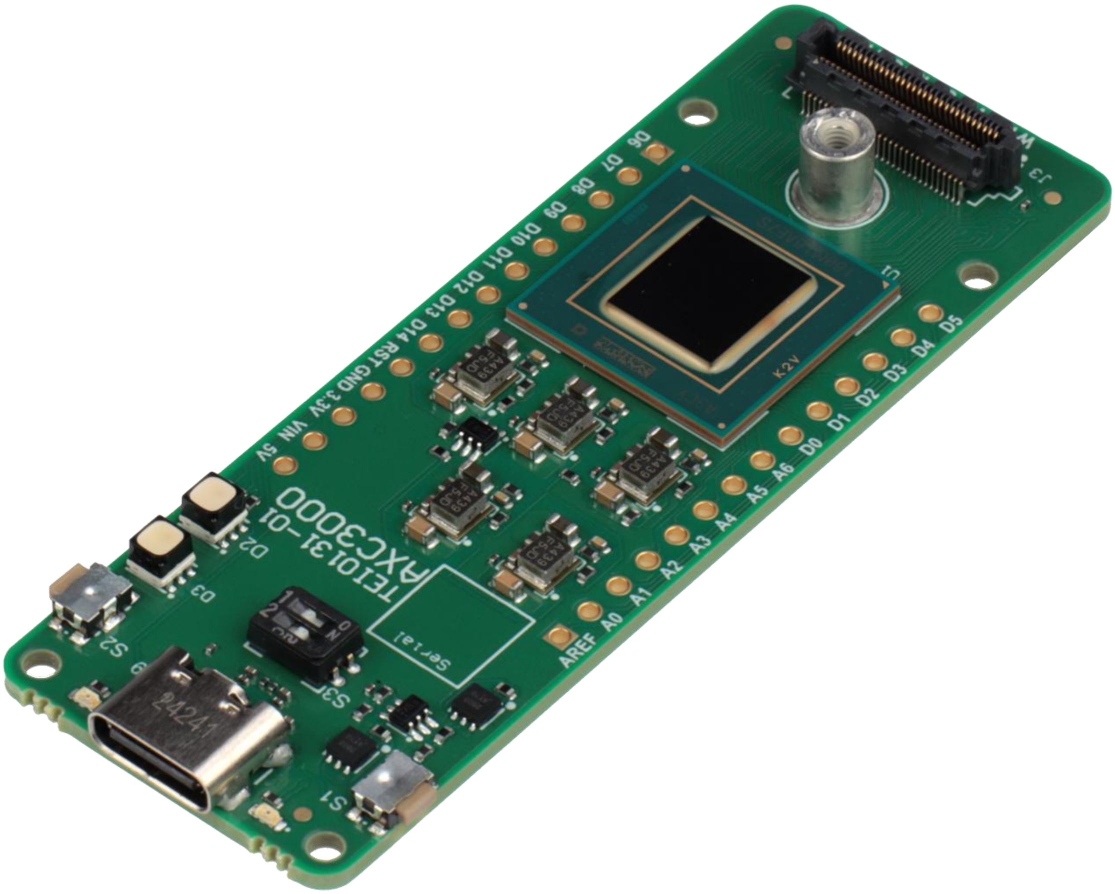
AI Solutions with Agilex® 3 and ONE WARE



Software and hardware requirements to complete all exercises

Software Requirements:	Quartus® Prime Pro version 25.1.1
Hardware Requirements:	AXC3000 Agilex-3 Series Board

Document Control

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1 OneWare AI Workshop for Altera FPGA

This OneWare AI workshop, which is created for participants with no prior knowledge of AI, covers the following topics:

- Introduction to OneWare: What it is and comparison to other AI implementations
- Step-by-step workshop flow for AI machine learning (ML) including:
    - Design requirements.
    - Acquiring and preparing a small single digit dataset, along with exploring dataset options.
    - Model Settings.
    - Training the model.
    - Exporting the results.
    - Testing the model with your handwriting
    - Visualizing the results.
    - Importing the trained model into an FPGA.
    - Programming and verifying the FPGA results.

2 Introduction to OneWare

The German based company, OneWare, has developed ONE WARE Studio with the ONE AI extension. This software enables rapid development and generation of custom AI models that can be used for different applications, hardware, and industries. "The ONE AI extension enables every developer to create, train, and integrate custom and efficient AI into the desired hardware and allow AI researchers to gain insights many times faster."

Within this software, no manual model coding or even knowledge of which specific neural network model would be best for your dataset is needed, rather a custom AI model can be generated with a single click.

This workshop targets Agilex 3 devices but the OneAI software can target other Altera device families such as Max 10, Cyclone and others.

3 One AI software vs other AI solutions

There are different ways to implement AI in an Altera device.

3.1 TinyML

TinyML has the following features:

- Very low-level hardware implementation of machine learning.
- Method for implementing machine learning (ML) on extremely low power, small size, edge-AI applications devices.
- Capable of interfacing with Nios V (soft-core processor) or ARM (hard-core processor).
- Free to use (no cost).
- Python -TensorFlow software (or other open-source ML software) to create a model, prepare the dataset, and run the training.
- Specific, models i.e. LeNet-5, Rosetta-Net, etc. need to be chosen before implementation.
- Uses Windows, WSL or Linux for operating systems

3.2 One Ware Studio

OneWare Studio and ONE AI extension have the following features:

- Software studio that offers a higher level of abstraction as compared to Tiny ML.
- Creates custom AI models based on the GUI settings. The resulting model is tailored to your specifications, hardware and dataset rather than using pre-designed model like LeNet-5 or Resnet-50.
- No prior knowledge of AI models is required.
- Often creates AI models that are more efficient than custom hand-created AI models
- Compatible with Nios V (FPGA)
- Generates RTL (VHDL) code
- Exported code can be used in the Quartus Platform Designer for integration with other FPGA IPs (this is the approach in this workshop)
- Operates under Windows OS

3.3 FPGA AI Suite

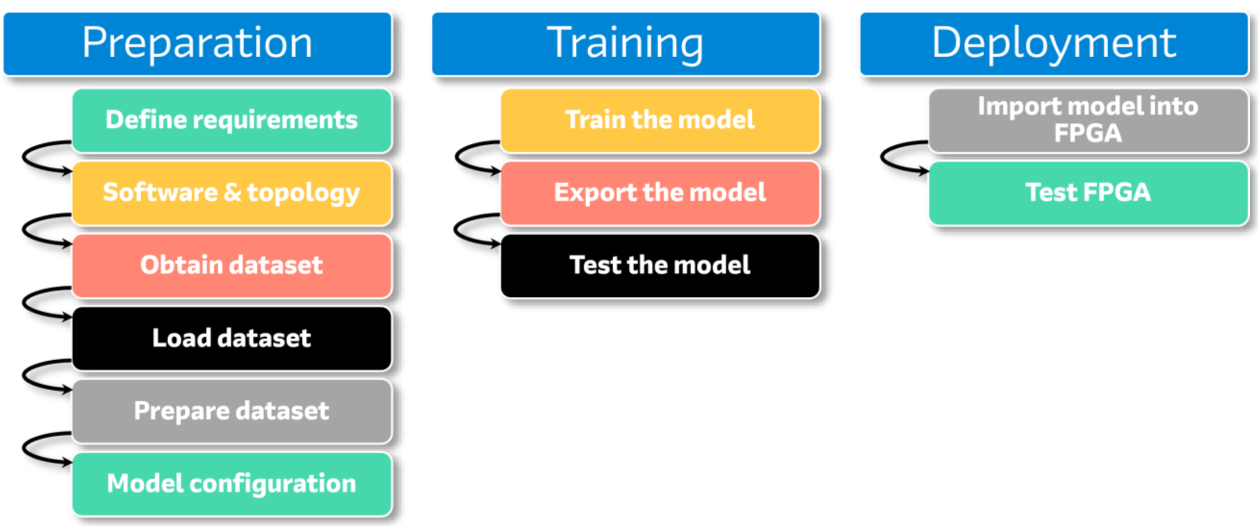
FPGA AI Suite is:

- AI inference development platform
- A way to convert a pre-trained AI model and implement it in Quartus
- Knowledge of specific AI models is needed.
- Provides as output resources and performance
- Commonly utilized on larger FPGAs, such as Agilex 5 and higher, which typically feature large external memory.
- Capable of delivering high-performance solutions, achieving high frames per second (fps)
- Command line software
- Operates on Linux OS or WSL.

4 Workshop steps

- The following steps describe the flow of the AI workshop.
- Design requirement.
- Software and network topology.
- Obtain a dataset.
- Load Dataset.
- Prepare data set, i.e. filters and augmentations.
- Model Configuration.
- Train the model.
- Export the model.
- Test the model.
- Import model into a FPGA.
- Test the FPGA implementation.

The overall workflow of this workshop can be visualized in three stages, as illustrated in the following diagram. You will be guided through each of these stages step by step.

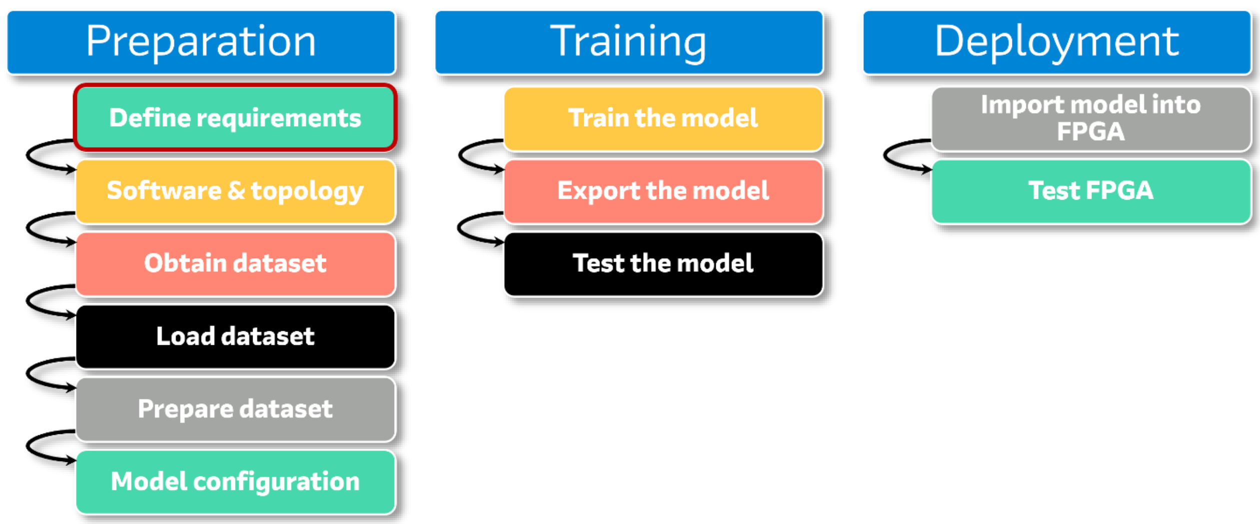


5 Design Requirements

The first step for any design process is to clearly define the requirements. This step is very important because it affects

- Design
- Dataset selection.
- Choice of models to be implemented.
- Target device.
- Results.

This foundational step is visually represented in the red box of the workflow diagram.



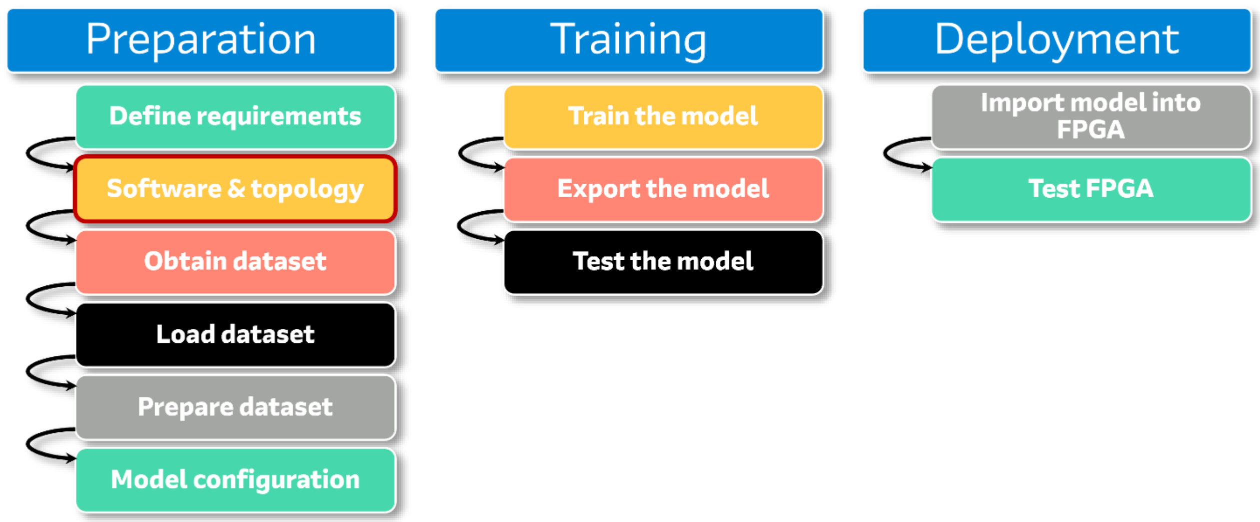
The design requirements for this workshop are as follows:

- Utilize ML to classify a single digit from the number 0 to 9, resulting in 10 distinct output categories.
- Employ OneWare Studio and ONE AI extension software to create a small, custom AI model. There will be no hand-coding of the model.
- Use a sub-set of the NIST (National Institute of Standards and Technology) database, which contains a large collection of handwritten digits and letters (letters will not be used in this case).

For this workshop, only a small sub-set of the database will be used: 3,000 images (300 per digit) for training and 500 images for testing. Using a subset of a large dataset with the One Ware software means that training can be faster using a Cloud server, and it can be faster to test a prototype. (If needed, larger datasets can be used).

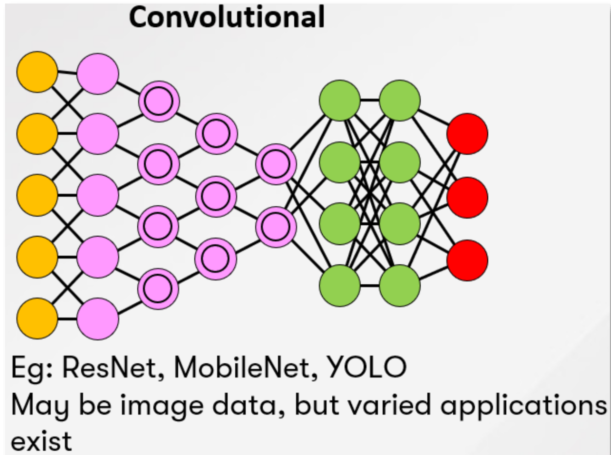
6 Software and Network Topology

The next step in the workflow is to determine the network topology and select the appropriate framework (software). This step is visually represented in the red box with the workflow diagram.



In this workshop, the software and network topology are defined. OneWare Studio with the OneAI extension will be the software platform for model creation. The network topology selected for this task is image detection.

Convolutional Neural Networks (CNN) being the preferred choice due to their effectiveness in processing visual data. The topology of CNN (for a refresher) looks like:



From the previous figure, the input layers are yellow “neurons” (where the data is input), hidden layers (i.e. the “calculation layers and classification layers”) would be the purple and green “neurons”. Purple neurons represent the convolutional and pooling layers which are the computational layers. The green neurons, which are the fully connected layers are the classification layers. The output layer is the red “neurons”.

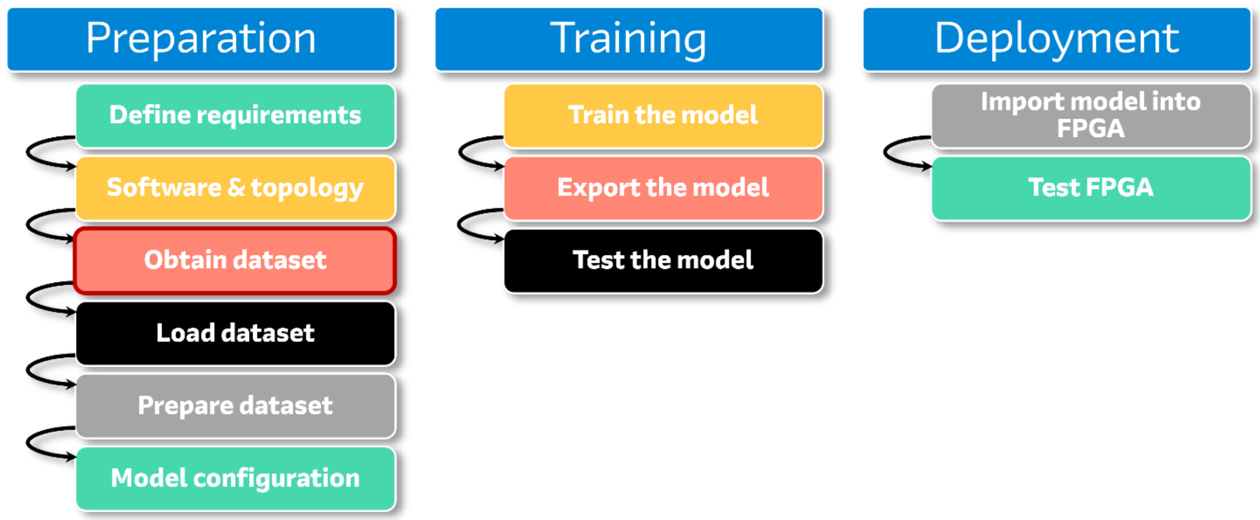
One Ware Studio and One AI extension will generate a custom model. This model will also be reviewed later in this workshop.

In this step, the software would be downloaded and installed. The version used for the workshop is OneWare Studio 0.21.10.0 and the OneAI Extension is version 0.6.3. (If you use other versions of the software, the screenshots and functionality might be different).

However, when using cloud-based computers for the workshop, the OneWare Studio is pre-installed and set up for you.

7 Obtain a Dataset

This step in the flow is highlighted in the red box in the following diagram.



There are different ways to obtain a dataset for machine learning:

- Download datasets from the internet. There are many datasets available.
- With software ML packages such as TensorFlow.
- Create from scratch.

A less time-consuming and less expensive method, especially when starting with AI, is to use a pre-created dataset rather than creating a custom dataset from scratch.

When looking at hand-writing datasets, the NIST (National Institute of Standards and Technology, which is an agency of the US government) created a free download database. The latest version is called Special Database 19 (SD-19), where this database is at Special Database Catalog | NIST.

Within SD-19, this dataset contains handwritten upper- and lower-case letters and hand-written numbers. 3,600 writers handwrote details in forms, before over 800,000 characters were extracted from these forms to create the dataset. As an example, just one file, alone, which is called by_class.zip, is about 984 MB even in .zip form. This is a large file.

There are variations of this MNIST SD-19 dataset such as:

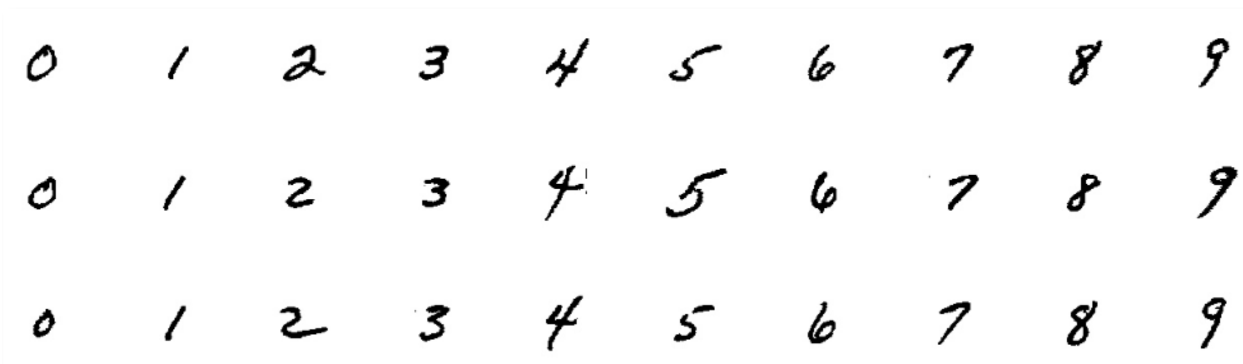
- MNIST (Modified National Institute of Standards and Technology) which includes only digits from 0 to 9. In this set, there are 60,000 training images and 10,000 testing images. The data in this dataset is set to $28x28$ pixels.

- **ENMIST** (extended MNIST). This dataset, which is also derived from the NIST Special database 19, is a newer dataset than the MNIST dataset. This dataset includes digits, and upper- and lower-case letters. The data in this set is also set to $28x28$ pixels.

For this workshop, a small subset of the SD-19 dataset will be used. The SD-19 dataset was chosen over alternatives like MNIST or EMNIST because it offers scalability in image resolution. Unlike the fixed 28×28 pixel size of MNIST and EMNIST, the SD-19 dataset allows for 128×128 pixel resolution. This larger size is particularly advantageous as it provides more space for your handwriting input, and it enables clearer number representation.

The subset dataset **nist_sd19_subset_sorted.zip**, can be downloaded from the OneAI_demo_datasets GitHub repository: (**don't do this**, it has been provided in the lab for convenience)

This file contains 3,.000 images, 300 images per digit allocated for training and 500 images for the testing. Despite being a subset of the original SD-19 dataset, it can deliver accurate results in handwriting recognition tasks. Below are some examples of the dataset:



The dataset is pre-installed as part of the project files at

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\nist_sd19_subset_sorted

7.1 Review the Dataset.

7.1.1 Extract the Dataset.

7.1.1.1 Open Windows Explorer and **navigate** to c::\altera_workshops\axc3000\OneWare_AI_lab\OneWare

7.1.1.2 Right click on nist_sd19_subset_sorted.zip. Select **Extract All**

Note there are two sub-directories called Test and Train. Within the Test and Train directories, there are further sub directories of 0_1 to 9_10. This is used for auto labelling the images. The first digit is translated to a unique ID. The second represents the actual contents. The folder 0_1 contains all the images of the digit '1'. The folder 9_10 contains all the images of the digit '0'.

***NOTE*:** It is crucial not only to train models using the training data but also to verify the accuracy using a separate data set. The train set is separate from the test set. It is very important to check the accuracy of the model with the training set. The goal is to achieve model accuracy as close to 100% as possible, which means the output data matches what is expected.

7.1.2 Open various .png files (within the 0 to 9 sub directories) by clicking on the files

A .png, which is a portable network graphic file, is an image file commonly used on websites.

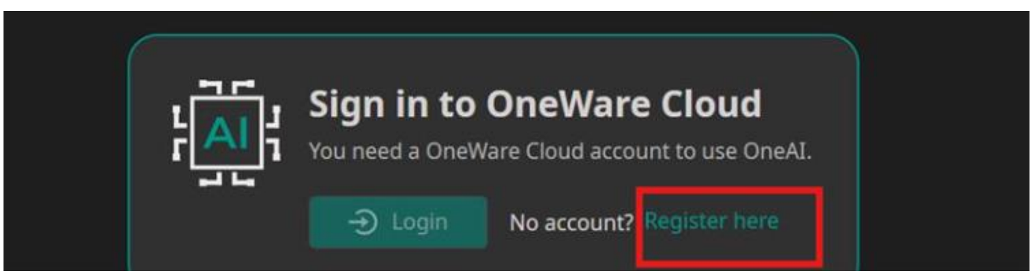
You should see various handwriting images where the images are somewhat different from each other. This difference is expected because people's handwriting is different from each other.

Before the dataset can be used, a New Project needs to be created. This is the location where generated files will be placed.

7.2 Launch OneWare Studio

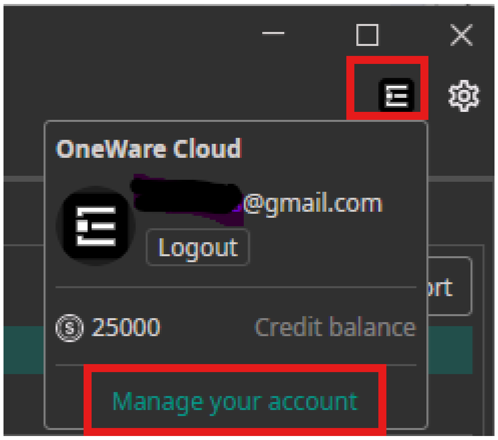
7.2.1 From the Windows start menu -> Select and Open the OneWare Studio

7.2.2 **Ignore** the **Update Available** pop-up window. It will disappear after a few seconds**.** Register for a OneWare account. Enter your email address and password. If you have previously registered, login using those credentials.

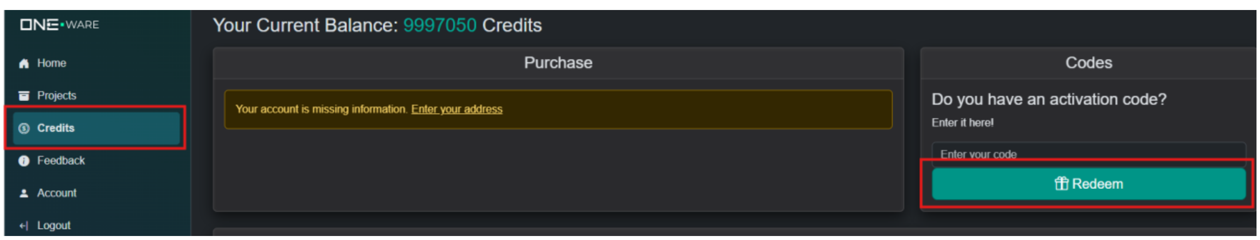


7.2.3 Redeem OneWare Cloud **training credits**.

7.2.3.1 Click on the account icon and then click **Manage your account**. This will open the OneWare portal in a browser.



7.2.3.2 Select the **Credits** tab. Press the **Redeem** button. The coupon code is **arrow25**.

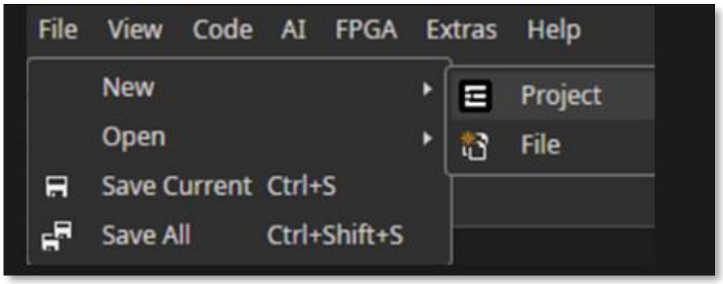


The count will be credited with 25,000 training credits. The training used by this lab typically requires 500 credits. Currently, OneWare is refunding half of the credit charged back to the account.

After you have completed the workshop day, you may use the remaining credits for many training sessions.

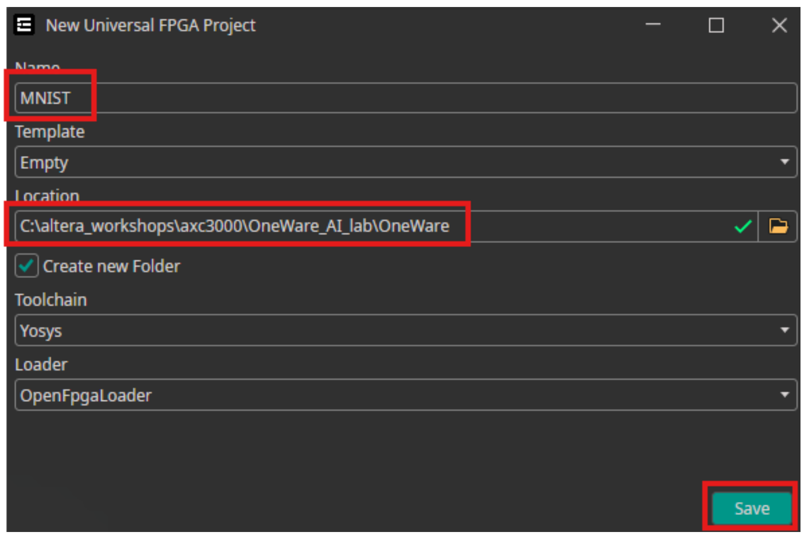
7.2.4 Open a new project

7.2.4.1 Select File → New → Project



A project defines where the setup files and the results will be placed.

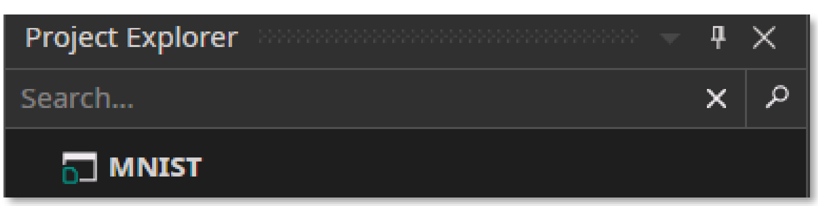
7.2.4.2 Make the changes shown below.



For reference, **Yosys** toolchain, which is the open-source Open Synthesis Suite, can synthesis the output (HDL) files for FPGAs. The **OpenFPGALoader** is a universal utility used for FPGA programming. In this case, Quartus will be used for synthesis and programming.

7.2.4.3 Select Save

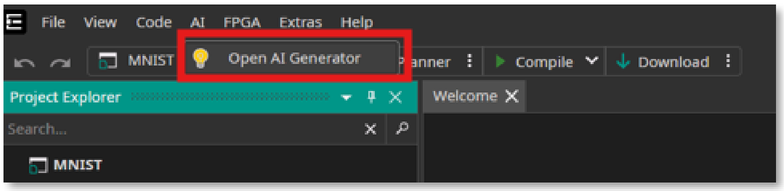
The Project Explorer should now look like



7.2.5 Initialize the AI project.

This is where the settings for the AI will be added before the model can be generated.

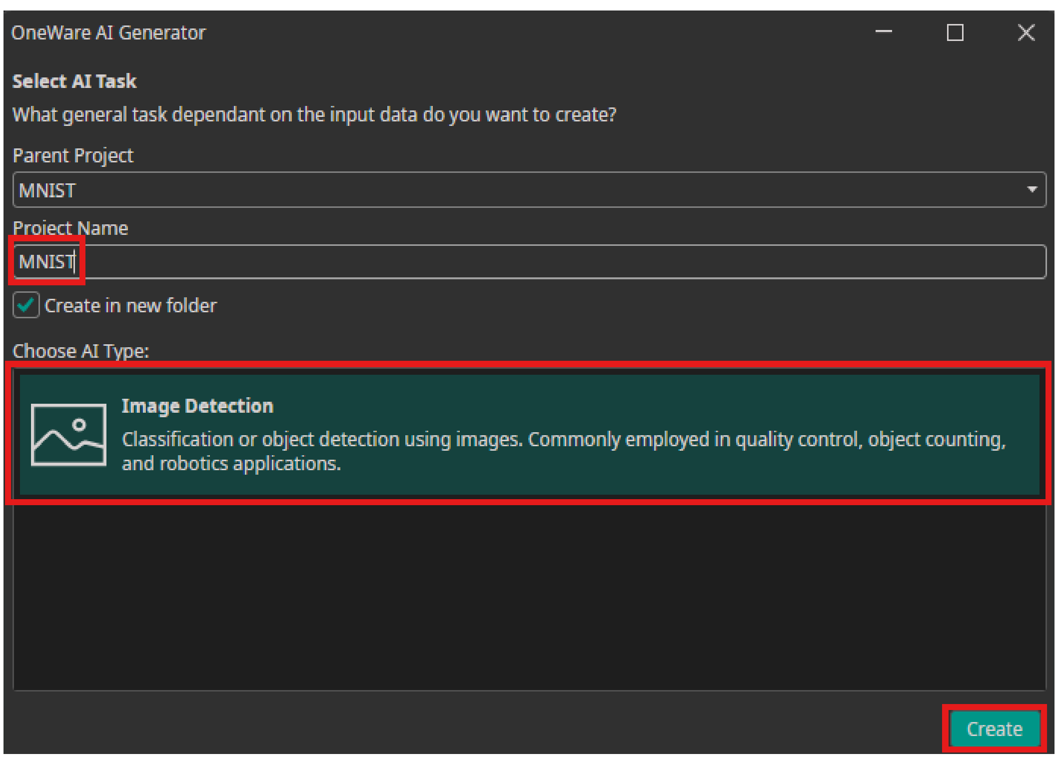
7.2.5.1 Select AI → OpenAI Generator



NOTE: If you do not see the AI in the top line of the Menu part, this means that the setup is not correct, thus refer to the installation manual.

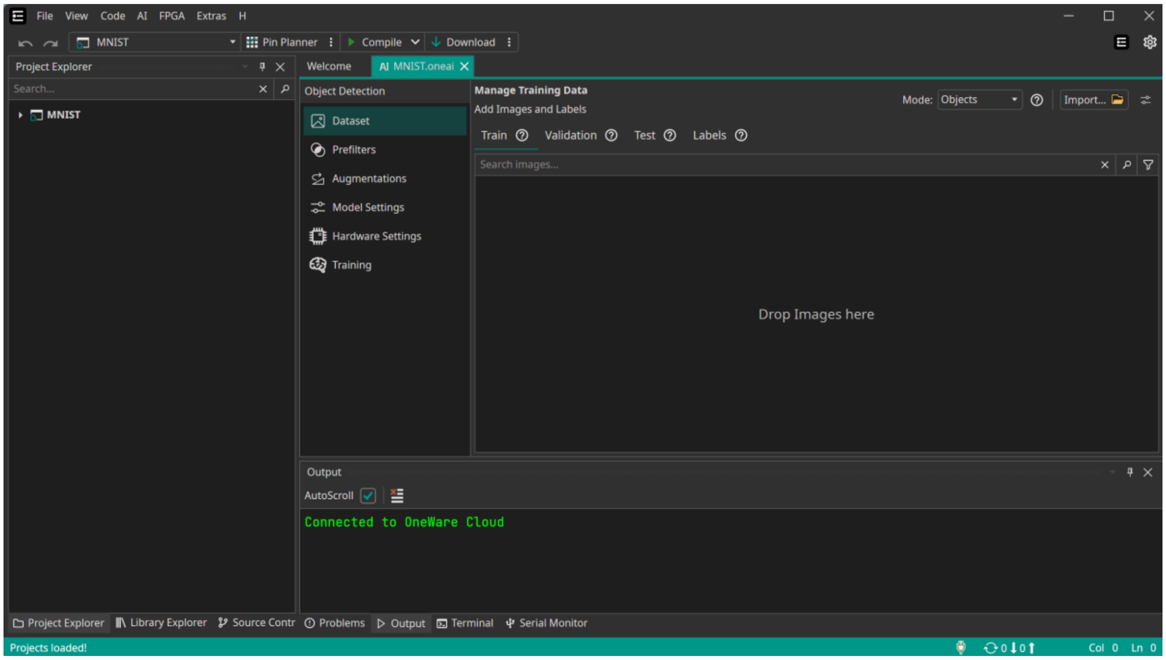
7.2.5.2 Enter the Project name (MNIST) and select the AI type.

There is only one type of AI in this context, which is image detection. This category encompasses image classification and object detection. If you do not select the Image Detection, the Create button will be "dimmed" until you select it.



7.2.5.3 Select Create

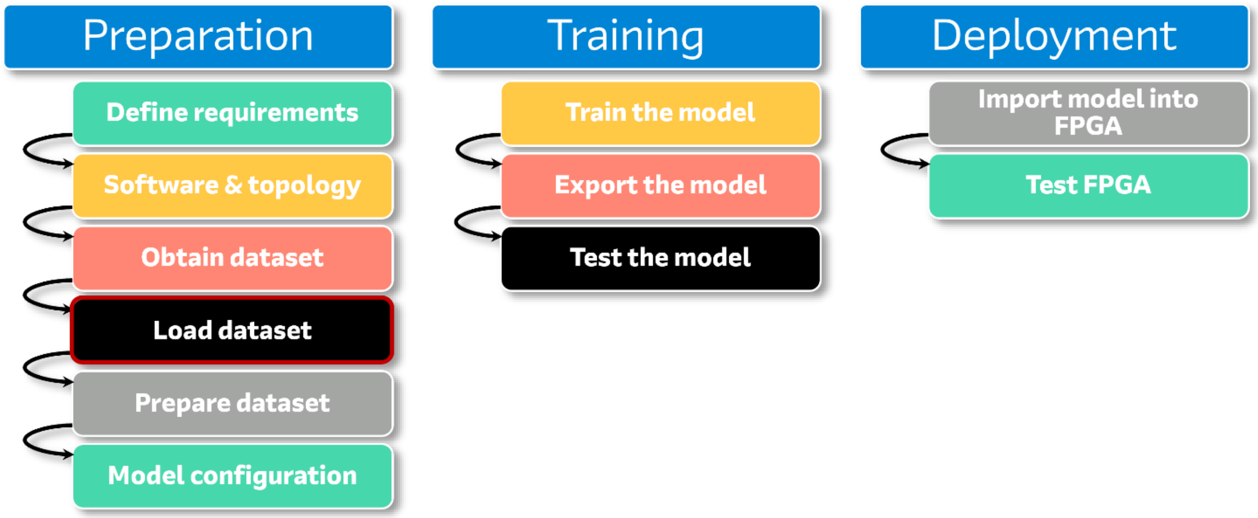
This window will now appear as follows:



Please note the section on the right side labeled “Drop Images here”. Images will not be added manually to this area; instead, all the files will be uploaded simultaneously. Since there are thousands of images, the drag and drop method is not suitable in this case.

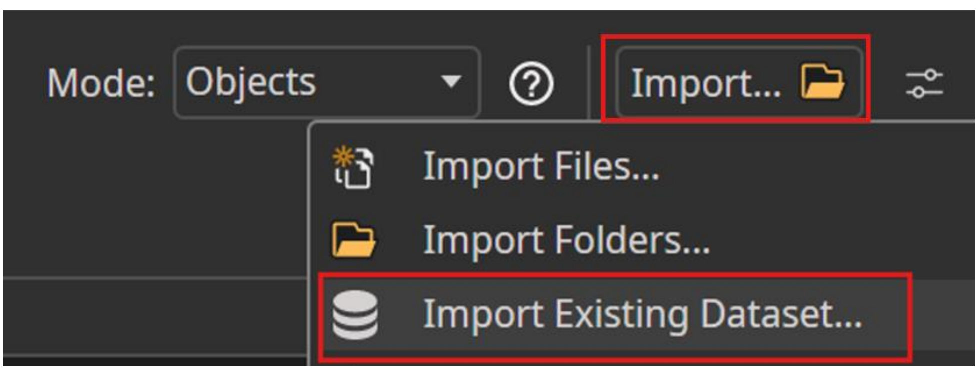
8 Load the Dataset

The next step is to load the dataset, as indicated by the red box, into OneAI Extension.

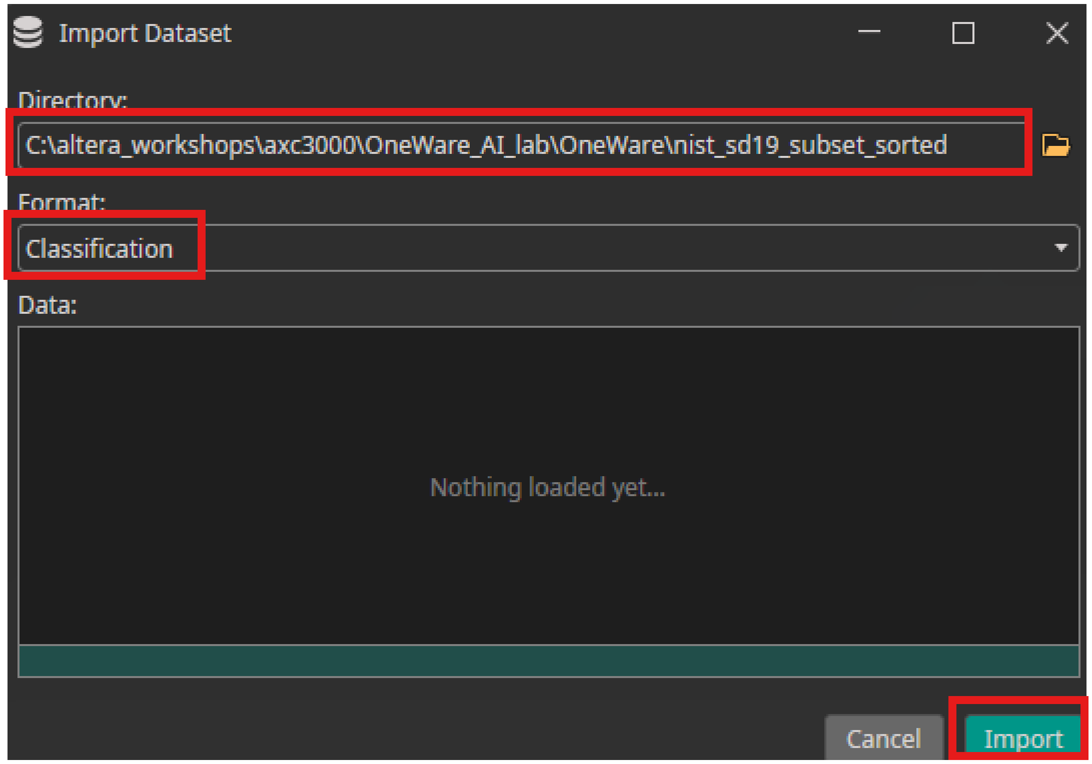


8.1 Import a Dataset

8.1.1 Select Import -> Import Existing Dataset...



8.1.2 In the Pop-UP window, change the following settings



Note for the directory option, you are not pointing to the specific test or train directories but one level above these directories. The software will be able to import to the correct directory and put the files in the train/test directories.

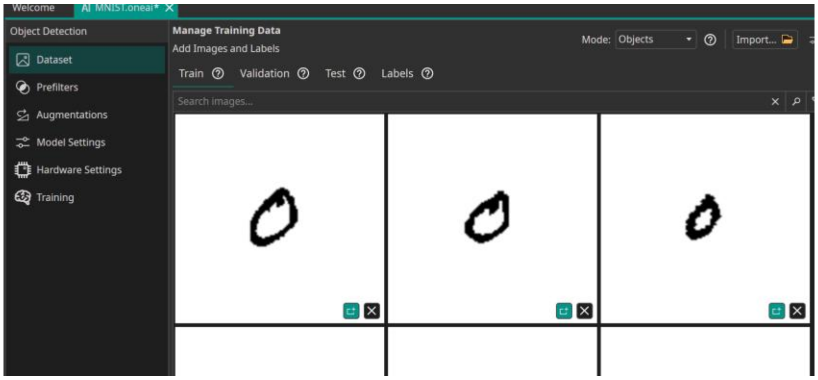
The format of classification means that the output data will be sorted by "output buckets" ie is the number 0, 1, 2, etc.

8.1.3 Select the Import button

The software will then import all the dataset files and place them in the Train and Test section.

Importing can take a short while to complete.

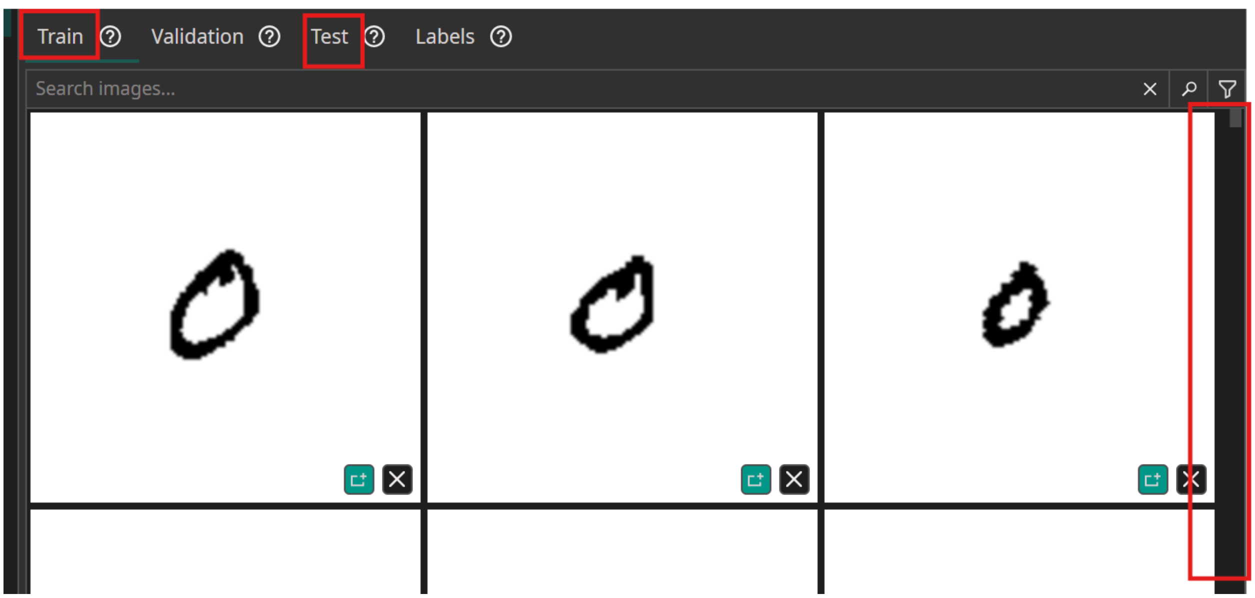
You should now see the following data in the window



8.2 Review the Dataset contents

8.2.1 Select Train and Test to see the various data that was uploaded.

Use the slider on the right-hand side to view the various data including other numbers as 1, 2, 3 and so on.

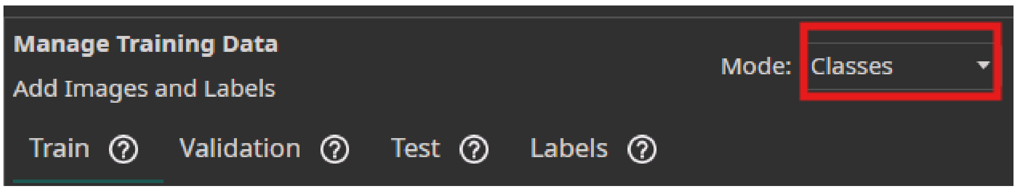


Every AI model requires a training and testing dataset. There are more images allocated for training than for testing, allowing the model's accuracy to be evaluated using "clean" data that was not seen during training.

There are no files under Validation, which is expected because the dataset .zip file had only the Train and Test directories. There is an option in the OneAI software to use some files for validation.

8.2.2 Change the Data Mode from Objects

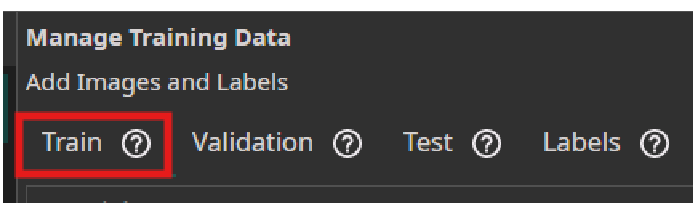
To



The reason for this change is that Object (Detection) means that it involves classification and defining the coordinates (location) of the object shown in a bounding box.

Classification means that the object is assigned to specific categories and thus assigned a label. In this case, the numbers will be placed in specific categories (“output buckets”) of 0 to 9.

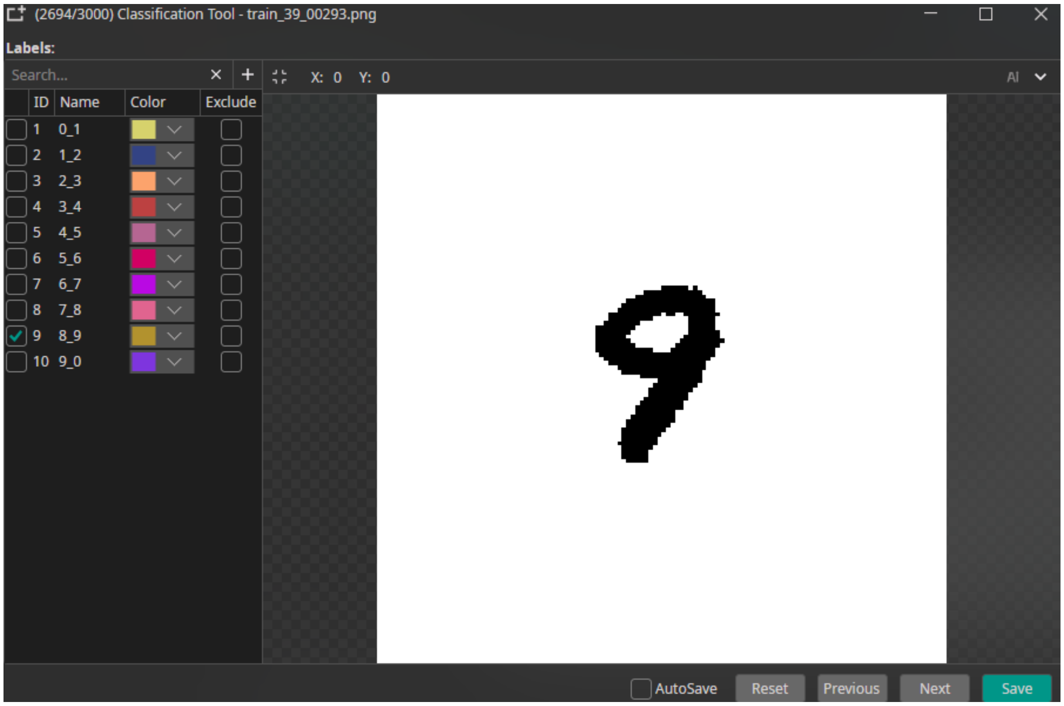
8.2.3 Reselect the Train dataset



If you do not go back to the Train dataset, you will be unable to proceed with the following steps.

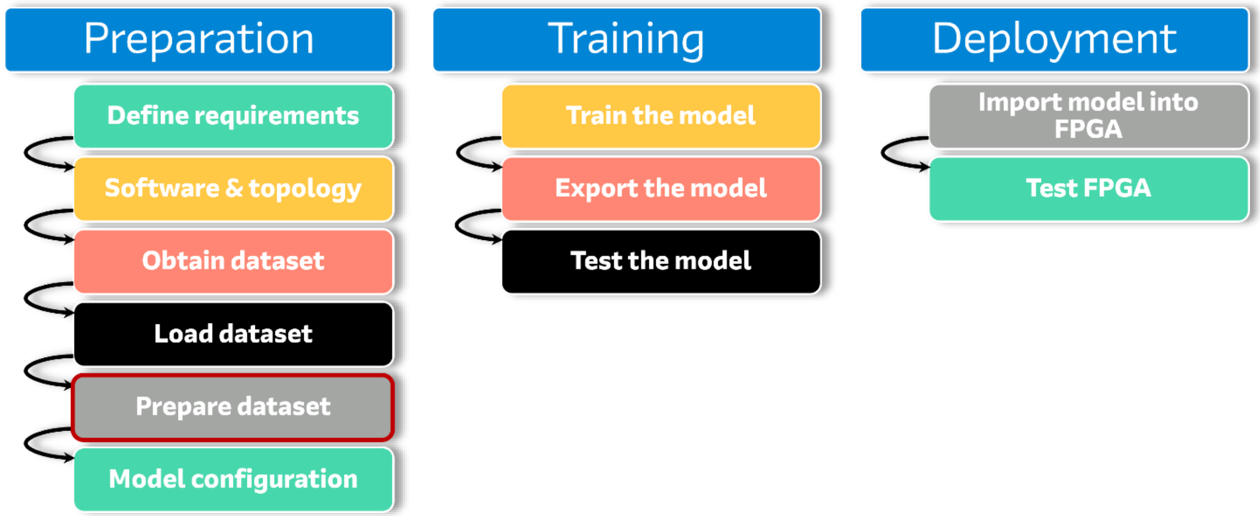
8.2.4 Review the Dataset classification

Double click on a digit in the Train dataset. The images are automatically classified based on the naming convention of the folder they were located in. The first letter represents an ID and the second represents the label of the image. This image is allocated the name ‘9’ with an ID of 9. OneWare studio reassigned it an internal ID of 10. ID 0 is reserved.


(2694/3000) Classification Tool - train_39_00293.png
Labels:
Search...
X: 0 Y: 0
AI
ID Name Color Exclude
1 0_1
2 1_2
3 2_3
4 3_4
5 4_5
6 5_6
7 6_7
8 7_8
9 8_9
10 9_0
AutoSave
Reset
Previous
Next
Save

9 Prepare the Dataset

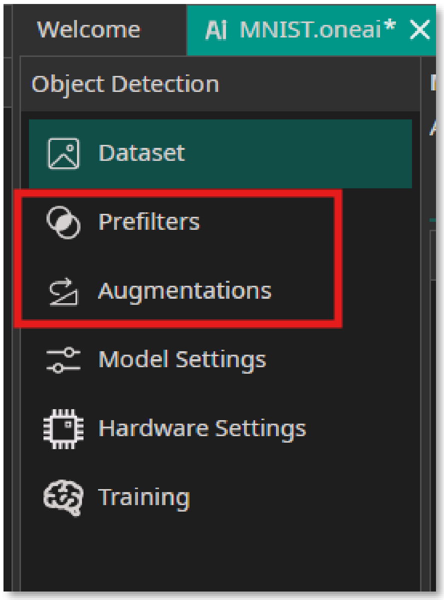
This step in the process is as follows:



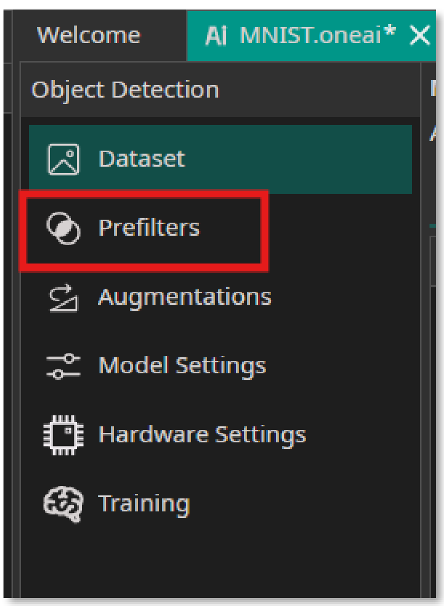
Before generating the model, it is necessary to adjust certain settings in the software. These configurations will directly impact the resulting model.

9.1 Specify Pre-filters and Augmentations

The steps for configuring pre-filters and augmentations are as follows:

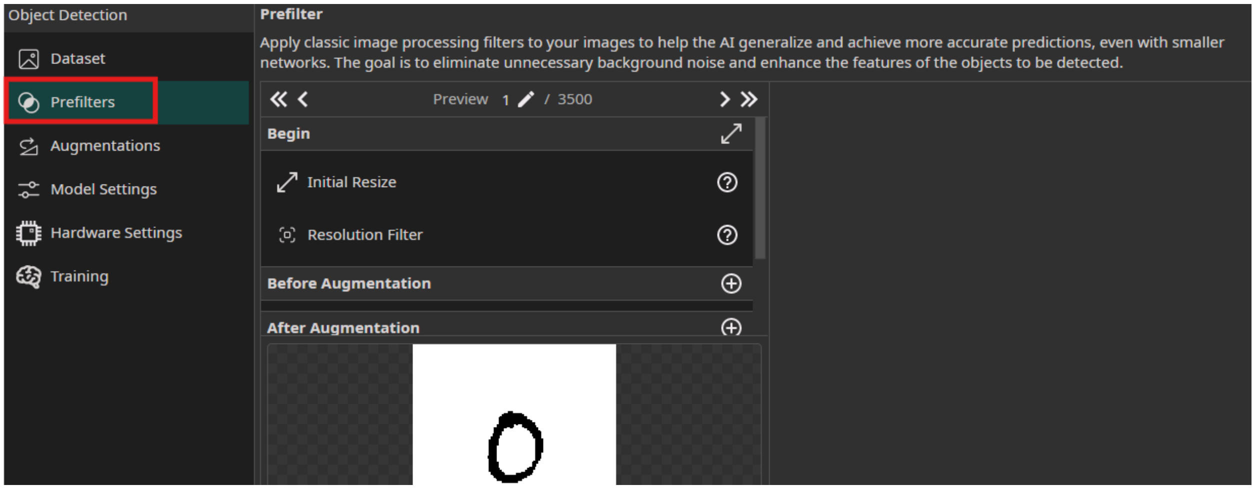


9.1.1 Select the Prefilters section



Prefilters, which are traditional image processing filters, are valuable in ML because they can help the AI make more accurate predictions. Prefilters enhance relevant features within the images and help eliminate potential noise.

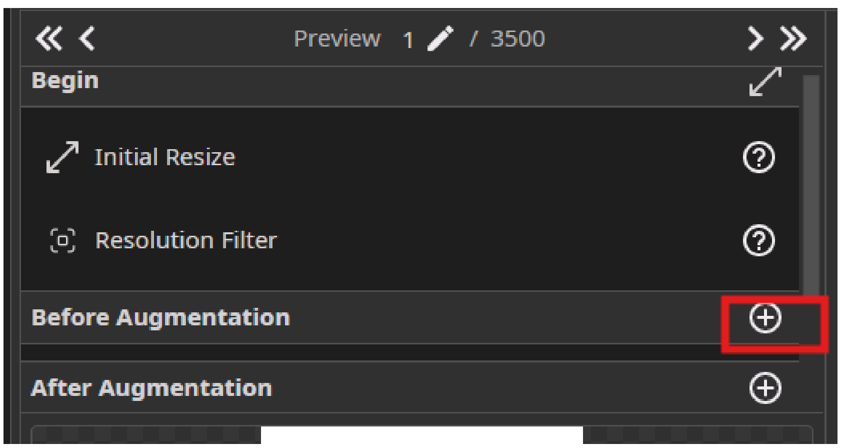
These filters can be seen in the following section:



If you have another number in the window, this will not affect the results.

NOTE: There are two sections for pre-filters. There are pre-filters applied before augmentation and pre-filters after augmentation. Augmentation refers to applying filters that increase the amount of training data, such as rotation and translation, but without having to manually add more training data.

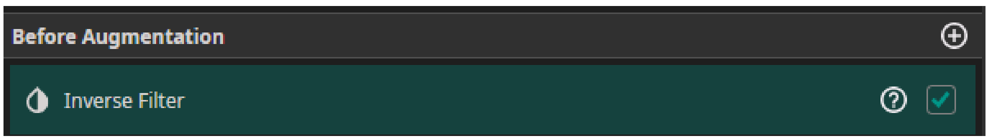
9.1.2 Expand the Before Augmentation



Note the number of options that are possible. There will be one change to the setting while the other one will be set to default.

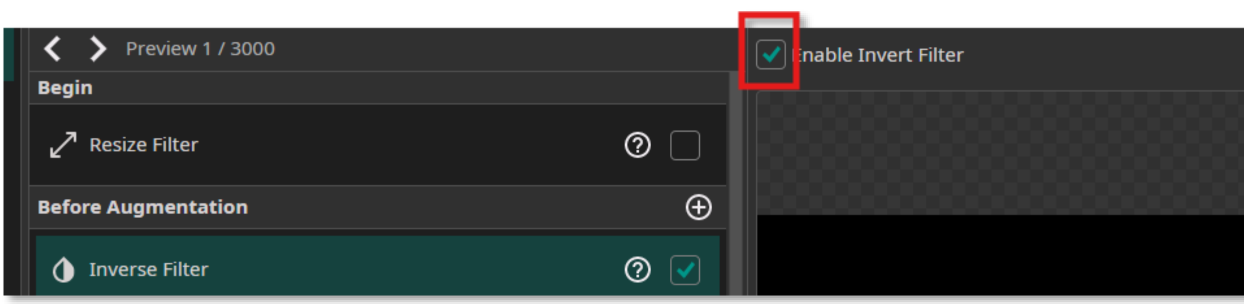
9.1.3 Select the Inverse Filter

9.1.4 Click on Inverse Filter. The check box has a check mark indicating the filter is enabled.

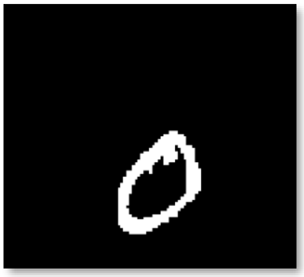


Note the original number in the window is black on white

9.1.5 Select Enable Invert Filter as

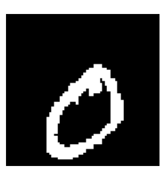


Note that the Image is now



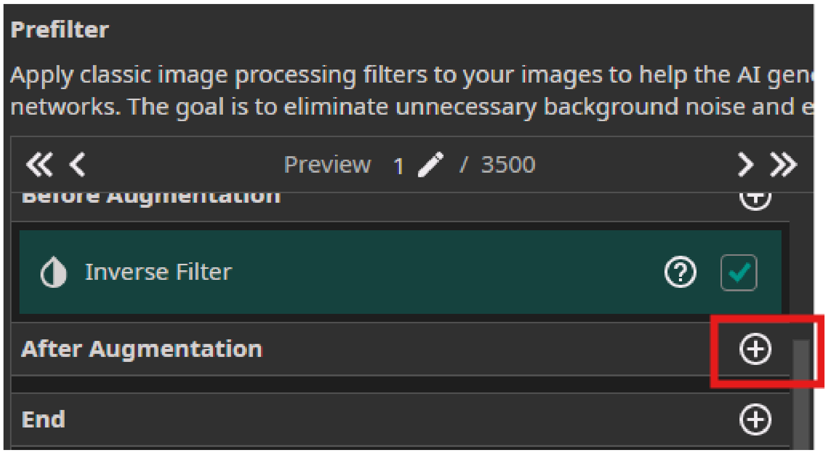
The inverse filter converts the black numbers on a white background to white numbers on a black background. This adjustment is necessary since, during subsequent augmentation steps, certain areas may be filled with black pixels; as a result, the background will match these padded (black) regions.

One might ask why the other filters are not enabled. The details regarding the remaining filters are provided below.

Filters	Original setup	Changed to	Why used/not used?
Crop Filter	N/A	N/A.	Not used because it is used to remove irrelevant parts of the image.
Frequency filters Transform the image to the frequency domain and use to smooth (Low Pass filter) and/or sharp (High Pass filter)	Original image. 	New image. 	The images can be used as the default since the objective is to simplify the objects; however, in this case the images are already sufficiently simplified.
Sharpen Filter Enhance the clarity and definition of object edges to make them more distinct and recognizable.	Original image. 	No changes with the filter on. 	No visible difference between original and new image. This filter should remain disabled, as it may reduce performance without offering any advantage.
Color Filter Apply various color filters to make features more distinguishable; for	N/A	N/A	This filter is not used, as the dataset images are not in color.

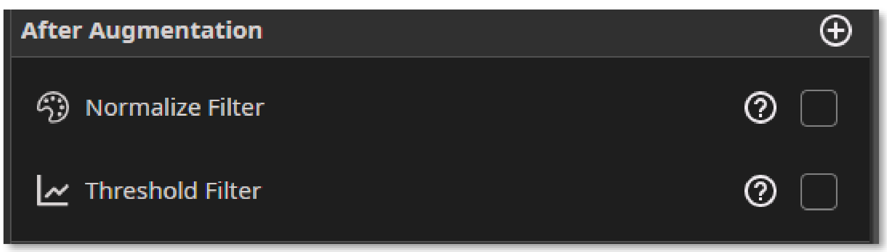
Filters	Original setup	Changed to	Why used/not used?
example, adjust brightness, contrast, saturation and hue as needed.			
Normalize Filter Rescales the image's brightness values – dark areas become black and lighter areas become white – to enhance contrast.	Original image 	No changes with the filter on. 	No change is needed because the image is already black and white.
RGB to HSV Filter Convert RGB (Red, Green, Blue) to HSV (Hue, Saturation, Value).	N/A	N/A	This option is not needed, as the images are not in color.
HSV to RGB filter Convert HSV (Hue, Saturation Value - brightness) to RGB (Red, Green Blue). HSV is primarily used for color, while RGB is the standard format for displaying images on a monitor.	N/A	N/A	This option is not needed, as the images are not in color.
Threshold Filter Remove unnecessary background in an image by usings sharpen.	N/A	N/A	There are no background images. It is only black.

9.1.6 Expand After Augmentation

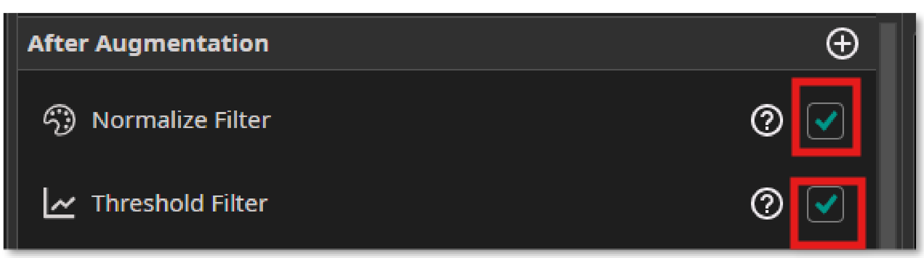


9.1.7 After Augmentation, select the **+option** to use the **Normalize** and **Threshold** Filter.

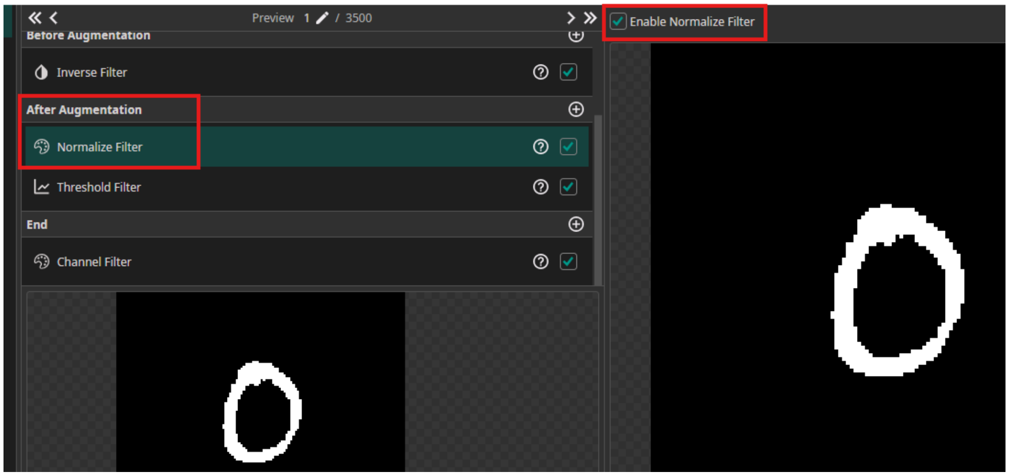
The filter choices available for **After Augmentation** are the same as Before Augmentation. The stage where the filters are implemented are different. Make the OneWare application window **large enough** so the selections below are **visible**. Use the scroll bar on the right to see the new additions.



9.1.8 Ensure that there is a checkmark associated with both filters to enable them.



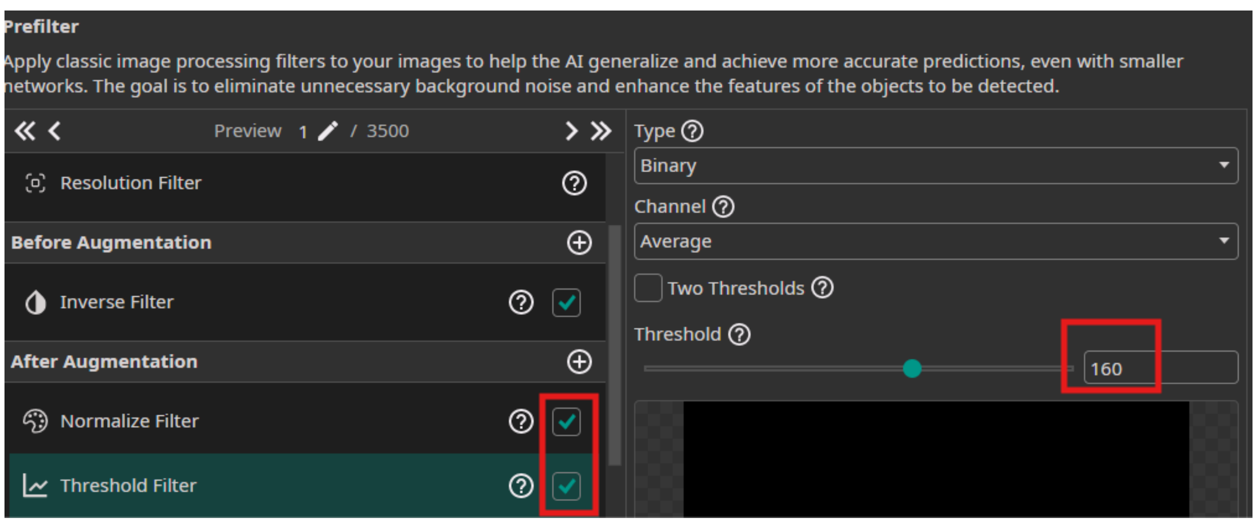
9.1.9 Enabling Normalize Filter means that "this will rescale the brightness values of the image and thus increase the robustness against lighting variations".



9.1.10 Change the settings for the threshold filter, if it is not set by default.

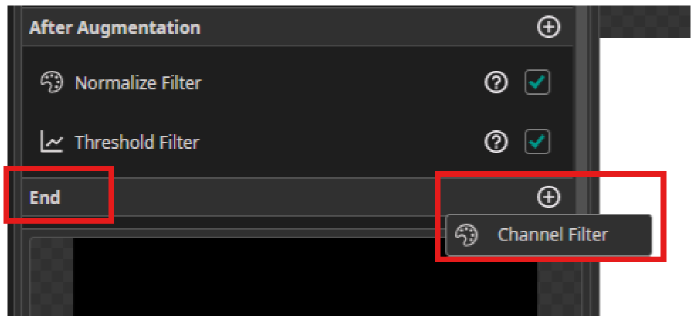
Threshold Filter removes "unnecessary background information in the image."

For the threshold filter, increase the value to 160, if it is not set by default. Threshold filter works by comparing each pixel's brightness to this value. If a pixel's value is greater than 160, it will be set to black; if it is less than 160, it will be set to white. This results in a binary image (black or white, i.e., 0 to 255), which simplifies processing and improves computational efficiency. Although 160 is higher than half of 255, it has been chosen to yield more effective results.

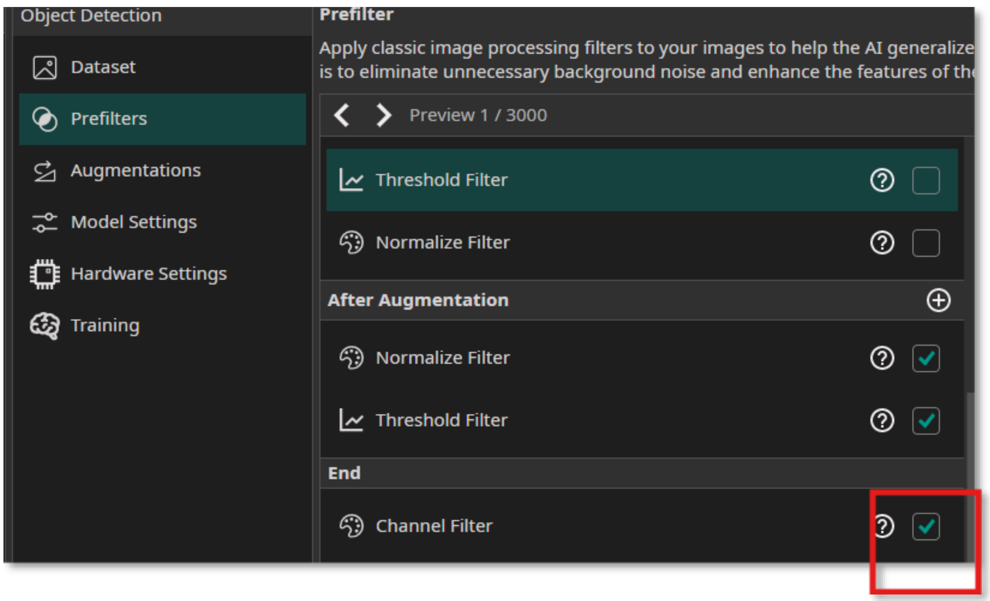


The final step for pre-filters is to add a channel filter.

9.1.11 Press + in the End section. Select Channel filter.

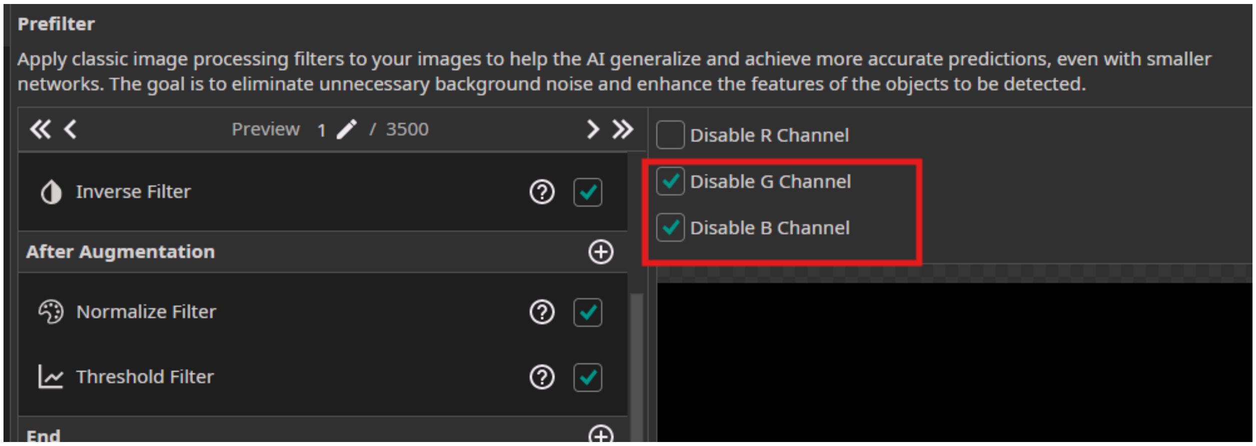


9.1.12 Enable the Channel Filter by selecting the check mark (if it is not enabled).

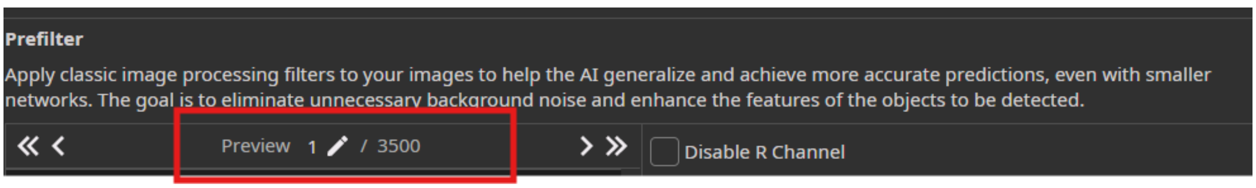


The Channel Filter is used to isolate specific RGB color channels. One might ask: Why enable the Channel Filter if the image has no color? The reason is that, after applying the threshold filter, the image is reduced to grayscale values, making all three colors channels unnecessary. To optimize performance, two channels should be disabled. In this case, the red channel is retained, but it would function the same if either of the other two channels were kept instead.

9.1.13 Disable the G and B Channels

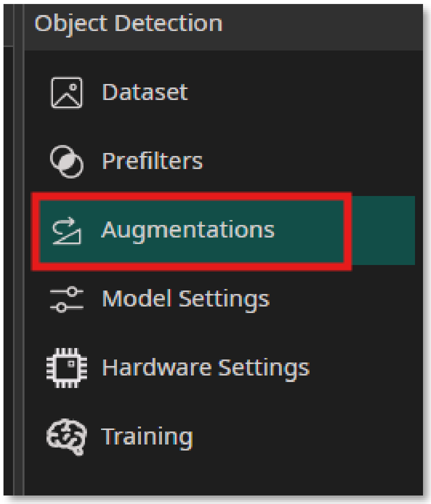


You could, if you wish, preview the settings on a specific dataset value. Preview 1/3500 shows that the 1st dataset value is being previewed. Previewing different values could be used with complex images to ensure that the images are filtered correctly.

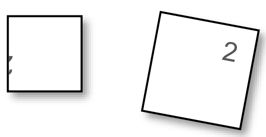


The Next step is to make changes to the Augmentations.

9.1.14 Select Augmentations



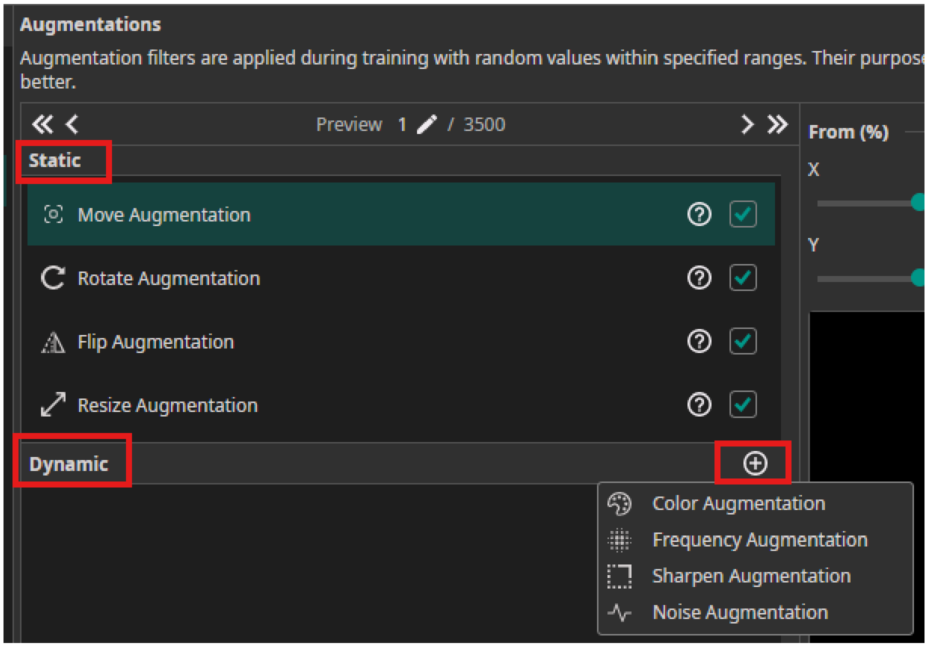
"Augmentation filters are applied during training with random values." Augmentation will increase the number of images in the training set, apply more diversity, without collecting more images thus allowing the AI mode to learn and generalize better. For example, if there is an upside down or rotated number, the number should still be recognized but without having to collect more data.



In this case, many after augmentation settings will be used to increase the training set without importing more training data.

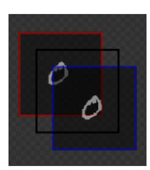
There are different types of augmentation-static and dynamic augmentation. Static augmentation, which refers to fixed changes to the entire training dataset, is simple to implement. Static implementation examples include flip, move and rotate. Dynamic augmentation, which refers to real time adjustments during the training, is more complex to implement. Dynamic augmentation examples include noise, frequency and sharpening.

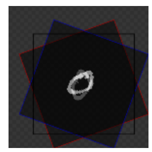
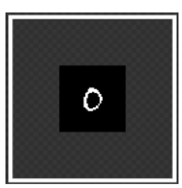
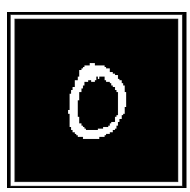
9.1.15 Press the the + button to see the **dynamic** augmentation settings.

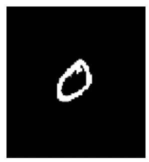


9.1.16 Make the setting changes in the software as directed the last column in the table.

The table describes the desired settings and the reason for their selection.

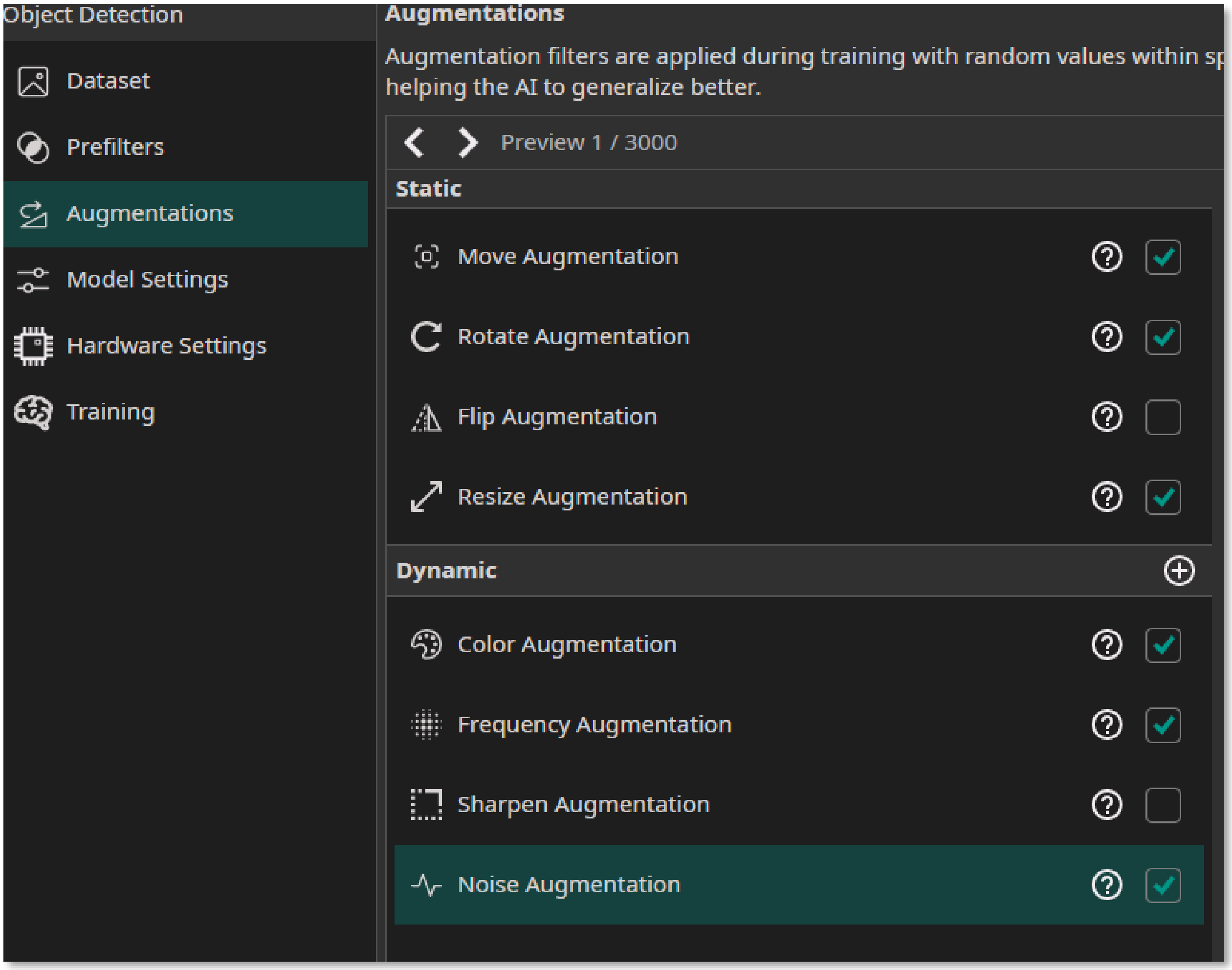
Augmentation Filters	Image before the filter	Image after the filter	Setting to be set and explanation
Static Augmentation			
Move Enabling the image to be moved up to +/- % in X and Y dimensions.			Set **both** the X and Y to be **+/- 20** to improve robustness in number placement. The number that you

Augmentation Filters	Image before the filter	Image after the filter	Setting to be set and explanation
			are testing may not always appear in the exact same position.
Rotate Enabling the image to be rotated into a certain position.			Set the rotation range to be **-20 to +20** to enhance robustness against variations in the image orientation.
Flip Enabling the image to be flipped.			This option is not enabled, (**remove the enable checkmark**) as it is assumed that the numbers will not be flipped in this case. This will also not make a difference for some numbers like 0 or 1.
Resize Change the image size.			This setting could be used with webcams. Entering the values as **40 and 125** for the X and Y directions means that if you would like to test the software with a Webcam (after the

Augmentation Filters	Image before the filter	Imag e after the filter	Setting to be set and explanation
			workshop only), this can be done.
Dynamic Augmentation			
Color Compensation for different lighting conditions.			This setting is used with webcams. (In this workshop, a webcam is *not* used). Adjust the **brightness range to (-30, 30**) and the **contrast range to (80, 120).** Leave the saturation and hue at their default values. Entering the values means that OneWare software could be used with a Webcam (after the workshop only)
Frequency Use this to clean up images out of focus.			**Enable the low pass filter** (to smooth out images) and set the values to be **0 and 1.5**.
Sharpen Make the images more concise.			This does not need to be enabled, as even when set to high values (e.g. 9 or 50), it produces

Augmentation Filters	Image before the filter	Image after the filter	Setting to be set and explanation
			no noticeable change in the result.
Noise Add robustness against camera noise.			Add variation to the background. Set the values to **0, 10** which means that 10% noise is acceptable. Note the noise affect in the image

The settings should now look like

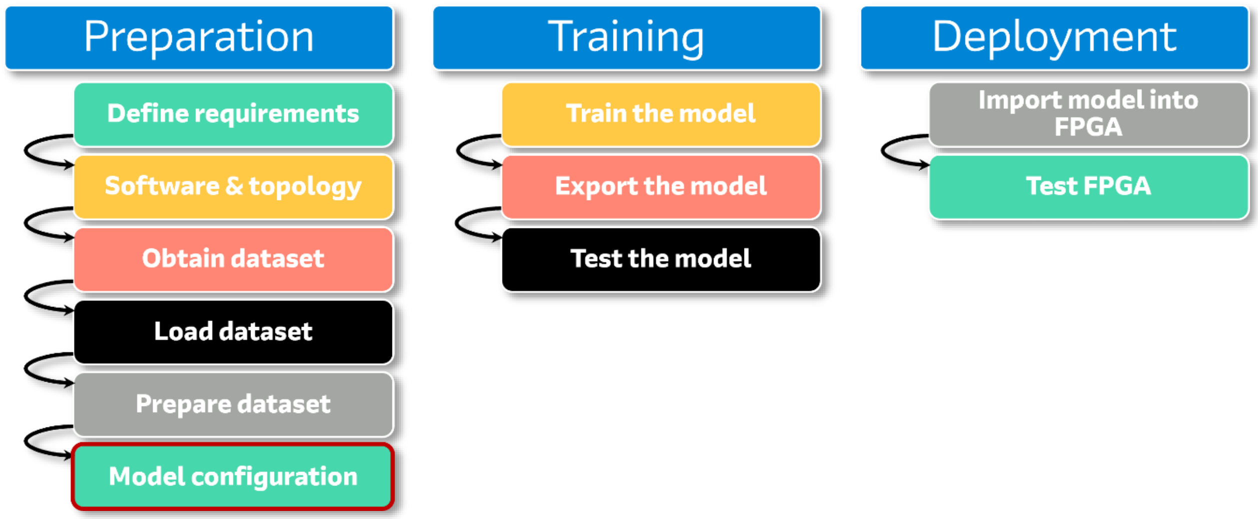


There may be a question: Will the prefilter settings affect the augmentation settings? The answer is yes; they do influence/affect each other.

For instance, in the pre-filter section, the Normalize fitter (which adjusts brightness by converting darker pixels to black and the lighter pixels to white) and Threshold filter (which increase the threshold for camera) will have a direct impact on the Color Augmentation (which compensates for lighting variations) and Noise Augmentation (which add robustness). The reason for using these settings is that the model will be more robust, have better accuracy, and be able to generalize better under different conditions (i.e. different lighting conditions).

10 Model Settings

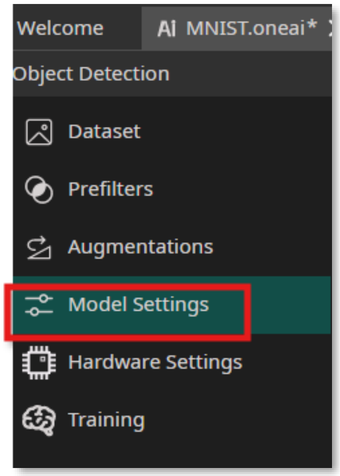
The next setting is to configure the model. This step in the flow is:



In OneWare Studio with the AI extension, the software will automatically generate a custom AI model after doing model generation. This automatic model is different than other AI model approaches. When implementing a TinyML or even the FPGA AI Suite, the model needs to be created ahead of time.

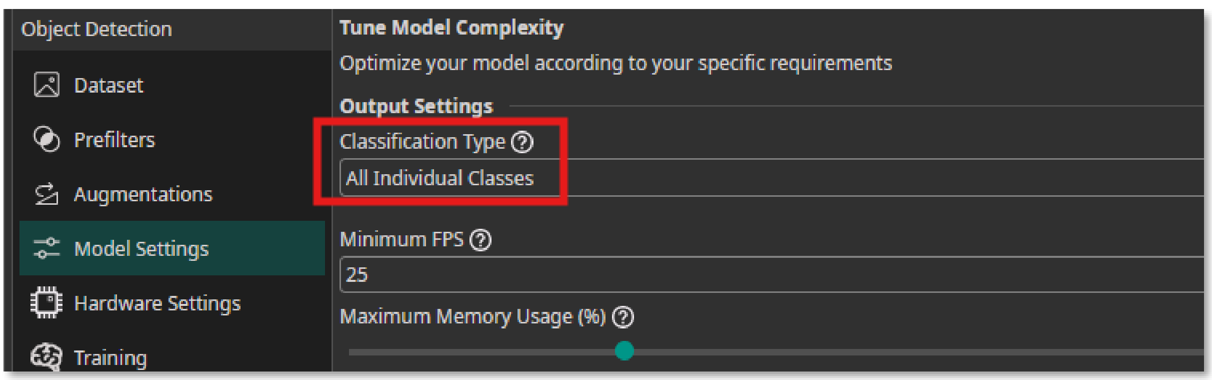
10.1 Model Settings selections

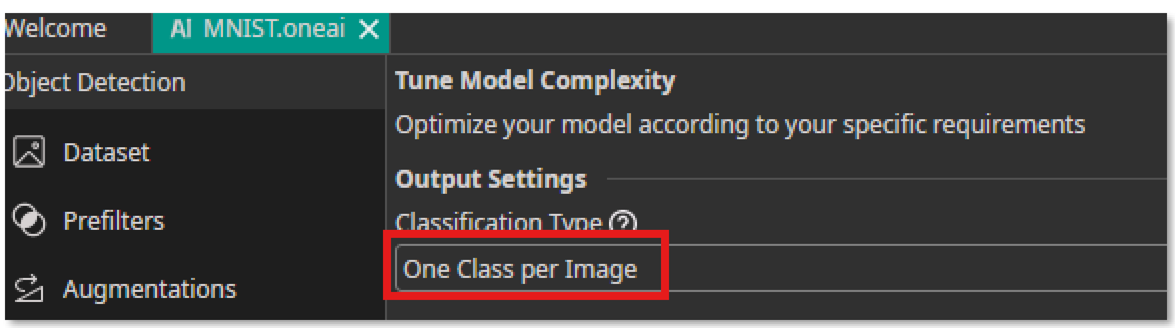
10.1.1 Select the Model Settings



These are the settings that will be used for the model.

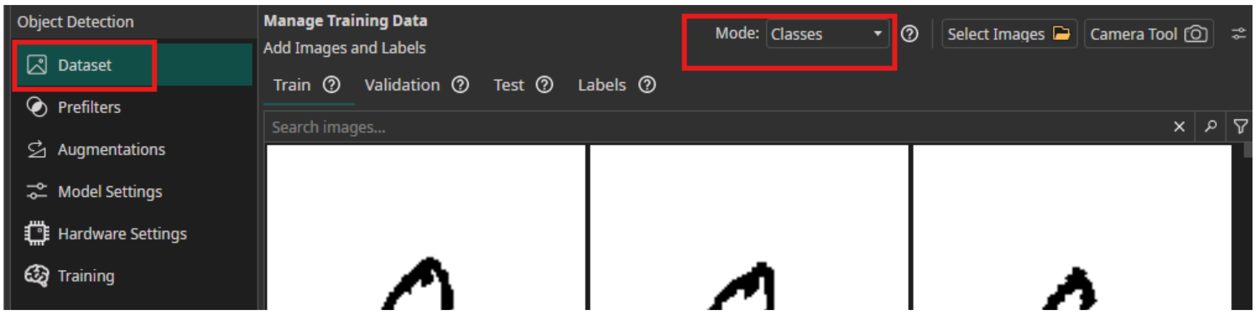
10.1.2 Change the Classification Type from All Individual Classes to One Class per Image





One Class per Image means that each number output will only belong to one "output bucket". For example, the number 4 will only belong to the number 4 output bucket not to the number 6 output bucket.

> ***NOTE:*** If the classification type of One Class per image is not seen, go to the Dataset, and ensure that Mode Class is set, and then go to the Model Settings again.



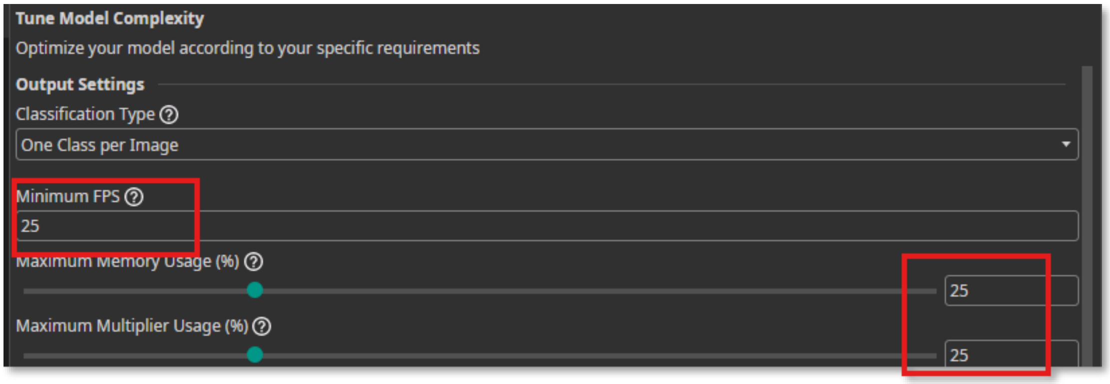
As a reminder, Classification refers to assigning an object to a specific category and labeling it accordingly. In this case, the numbers range from 0 to 9, so any prediction must be one of these digits.

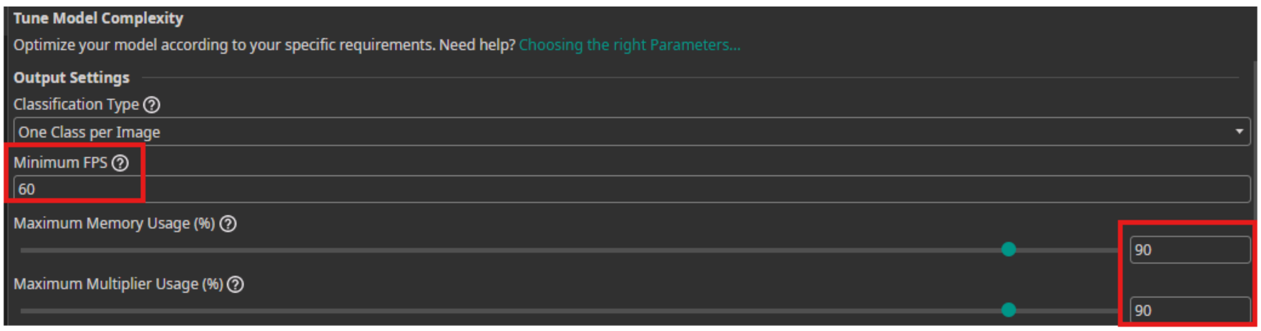
A question may be: If the training and test datasets contain only single digit numbers, what will happen if you test the model with two or more digits? The answer is that, since the model was not trained for multi-digit inputs, it will not be able to predict them accurately. It may return no result or just make a random guess.

When working with an AI model for prototype or production, it is important to have large training and testing datasets (including corner cases) to try to reduce possible random/incorrect guess situations.

The next step is to optimize the model.

10.1.3 Change the settings for Minimum FPS from 25 to 60, Maximum Memory Usage from 25% to 90% and, Maximum Multiplier Usage from 25% to 90 %.





The reasons for changing the settings are detailed below.

- Minimum FPS

  FPS stands for frames per second. A rate of 60 fps (compared to 25 fps or 30 fps) is generally the threshold at which video appears smooth. At slower rates, such as 25 fps, motion can appear choppy. Since the target device is a smaller device, the fps should not be excessively high (e.g., not 4K nor 8K rates). Instead, 60 fps is an appropriate starting point.

- Maximum Memory Usage

  This setting instructs the model to utilize the device's memory as possible for calculations. AI processing is resource intensive, and setting the value as 50% means the model can use up to that amount of memory if needed. At this stage, the target device has not yet been selected, but it will be specified in the Hardware Settings. If a high value, such as 90%, is set but the model requires less resources, it will only use the actual number of resources needed. However, if you set the percentage too low, OneWare Studio will notify you if the model cannot fit into the device.

- Maximum Multiplier Usage

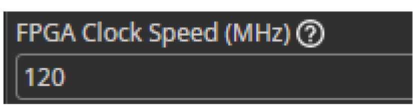
  This setting instructs the model to use as many multipliers as possible within the device for its calculations. The exact number of available multipliers will depend on the specific device, which will be defined later in the Hardware Settings.

  Setting this value to 50% allows the model to access half the number of multipliers possible. A higher value, such as 90%, provides the model with greater computational capacity. Depending on the device family, these multipliers will typically be implemented using the DSP blocks

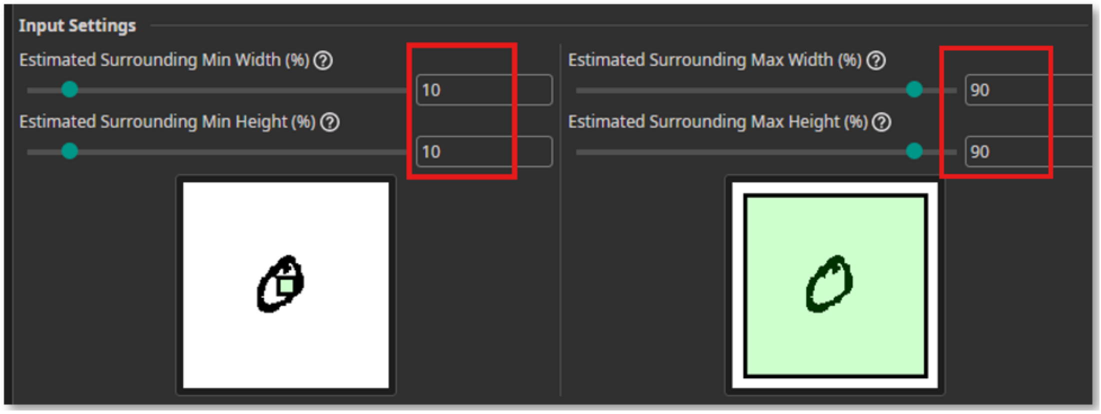
- FPGA Clock Speed (MHz)

  This is the internal clock speed of the FPGA. Setting this to a higher frequency than the default (50MHz) means that the VHDL code generated will use less FPGA resources. This happens because a higher performance AI hardware in the FPGA can share more functions than a slower one. The hardware can be used, thus ensuring that less FPGA resources are used. This is also affected by the Minimum FPS setting. There a higher number might require more FPGA resources.

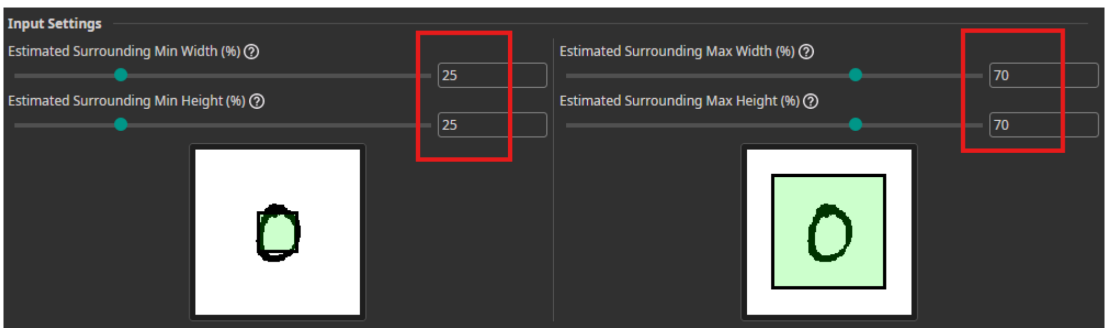
  Set the FPGA clock speed to 120MHz. The design uses an Agilex 3 FPGA which has higher performance than some of the previous families of devices.



10.1.4 Change the Input Settings from

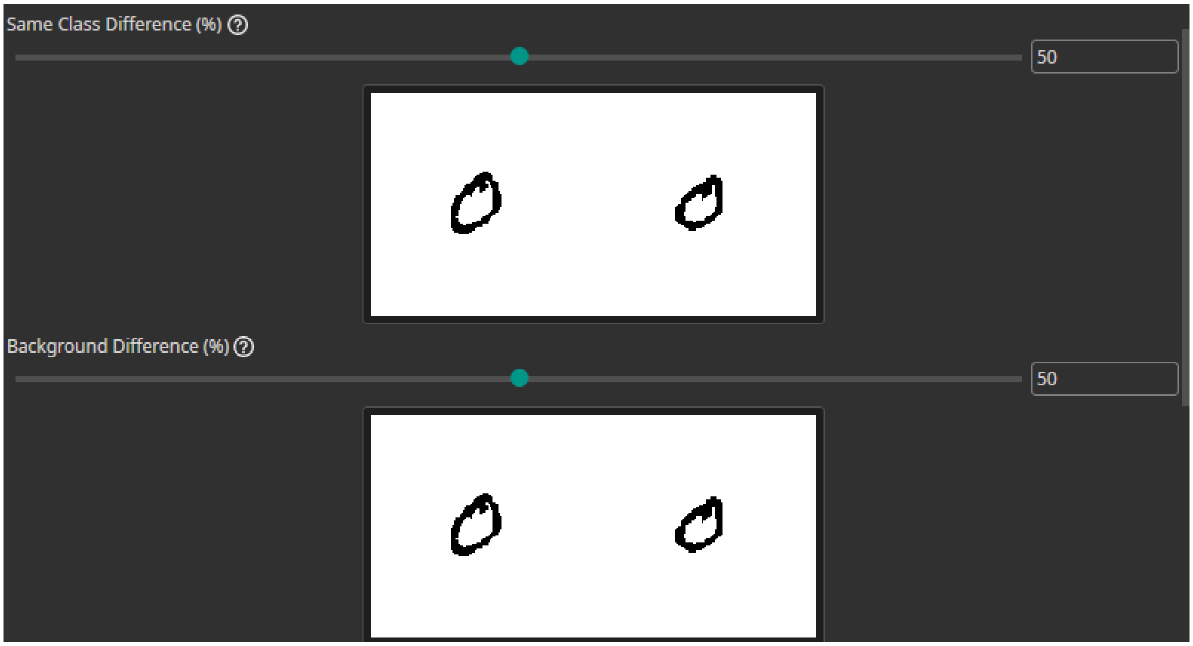


to



The Input Settings are used to increase the minimum and maximum width and height of the image processed by the model. As shown in the image size (red box), enlarging the input dimensions allows the model to detect even the smaller handwriting.

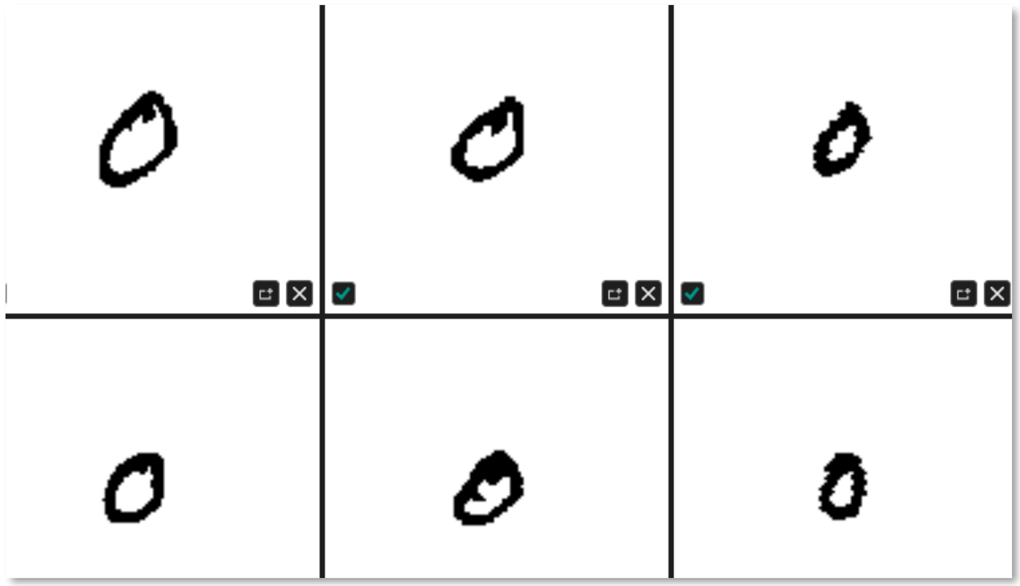
10.1.5 Change the Same Class Difference and Background Difference from



to

- Same Class Difference

  This refers to the variation of the images in the same class. For example, looking at the training dataset for the number 0, there are differences in the size, angle, thickness/jerkiness of the lines for the number 0.



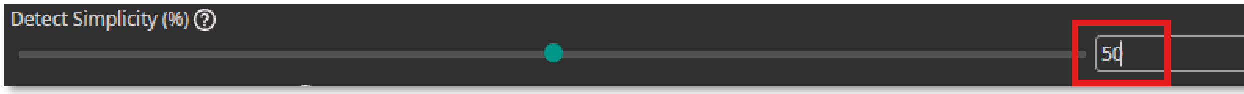
  The range of the Same Class Difference spans from 10% (minimal variation, such as examining a bottle label under different lighting) to 85% (maximum variation, involving differences in shapes and other features). Setting this value to 40% places below the midpoint between 10% and 85%, indicating a moderate level of variation.

- Background Difference

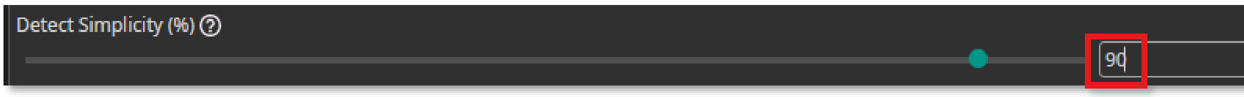
  This setting represents the color variation of the background image. In this case, there is no variation, as the background is uniform white with black numerical text. The value can range from 0% to 100%, where:

  - 15% - slight variation, such as items moving on a conveyor belt.
  - 30% - moderate variation.
  - 100% - highly varied background, such as objects in different locations.

10.1.6 Change the Detect Simplicity from



to



- Detect Simplicity

  It is used to determine how easy an object class can be detected. The higher the value, the easier it is for the system to identify the object.

  - 5% - complex tasks, such as determining different cat breeds.
  - 40% - moderate tasks, such as detecting surface scratches.
  - 90% - simple task, such as detecting black circles. In this case, it detects black images on a white surface.

10.1.7 Change the estimated size for Objects from

to

Estimated Size defines the percentage of the image's which width and height for minimum, average and maximum object dimensions.

If the minimum object width is set to 5%, the smallest object will appear very small, whereas at 50% the smallest object would be half the image's width.

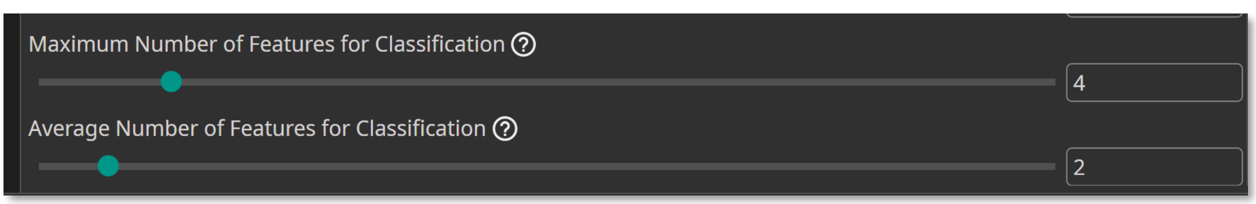
By changing the minimum width and height from 10% to 25%, the smallest object will be one-quarter of the image size.

The average object width and height, set at about 50%, indicate that the average object will occupy roughly half of the image.

The maximum object width and height, set at about 70%, mean that the largest will cover most of the image.

10.1.8 Change the Number of Features from

to



Features are attributes of the input data that help distinguish one data sample from another. Models use these attributes without the need for manual coding, to identify the differences between objects and determine what each item represents.

As an example, when examining the numbers from 0 to 9, certain attributes reveal differences between them: a downstroke (as in the number is 1, 7 and part of 9), a circle (as in 0 and part of 9), a horizontal line (as in 4 or the top of 5), curve (as in parts of 2, 3 and 6). The red boxes show some of the attributes.

The key point to note is that models automatically identify these attributes as the data is processed. The model will typically first perform edge detection, then search for features such as horizontal and vertical lines, curves, and then further break down the object into further basic building blocks. If these features were coded manually instead of using a model, the process would be time consuming, as every possible variation would need to be accounted for.

The maximum number of features and the average number of features settings guide the software in determining whether the model should be complex or simple. A lower number of features results in a less complex model.

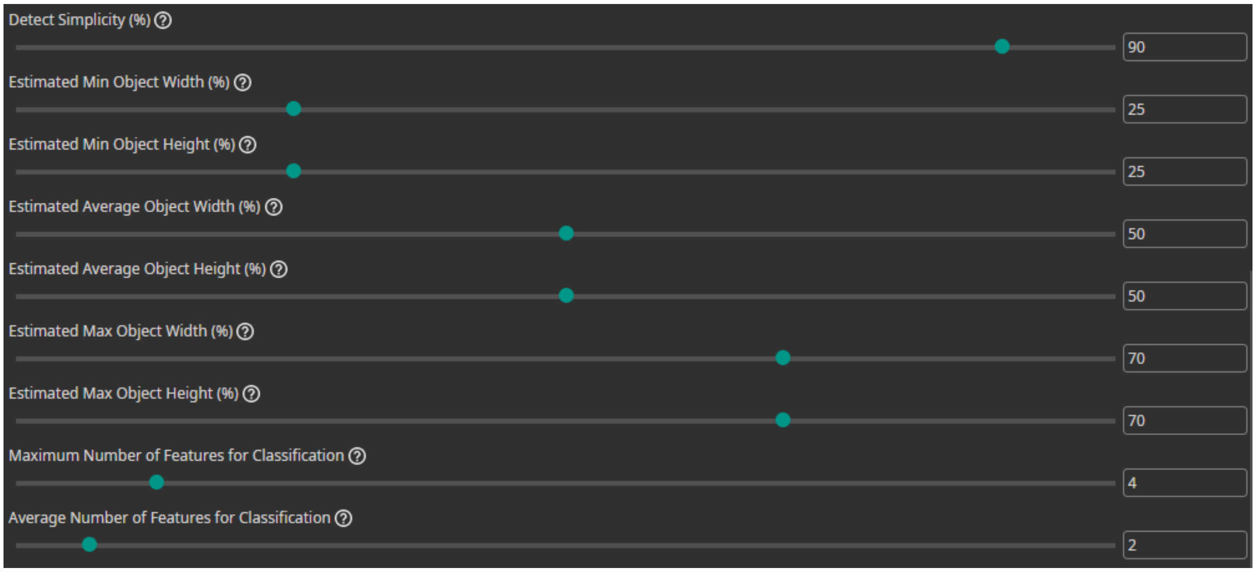
The range for the maximum feature setting is as follows:

- **2-5:** Simple objects, such as numbers (the reason for selecting this range in this case).
- **10-15:** Moderate complexity, such as varying patterns or textiles.
- **20-32:** Highly complex, detailed analysis.

Average Number of Features for Classification setting guides the model's complexity.

- **2-4:** Uniform objects (e.g., in this case, the numbers in black on a white surface).
- **5-10:** Moderate capacity.
- **15-25:** High capacity.

The average number of features should always be lower than the maximum number of features. In summary, the settings **should now look like**:

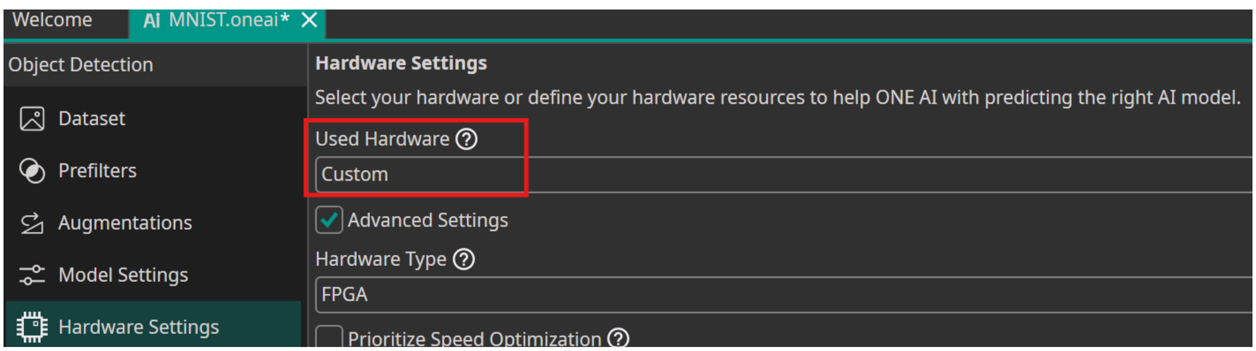


The Group setting does not need to be changed.

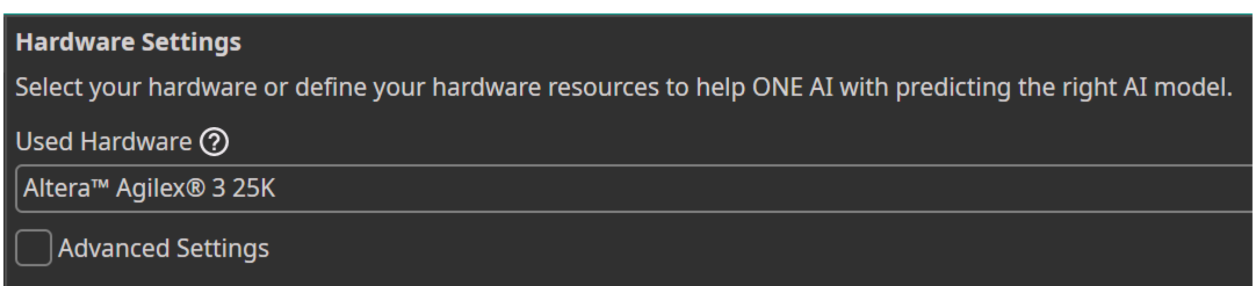
10.2 Hardware Settings

The next step is to select the Hardware device that this is being targeted.

10.2.1 In the Hardware Settings, select the pull down menu for Used Hardware

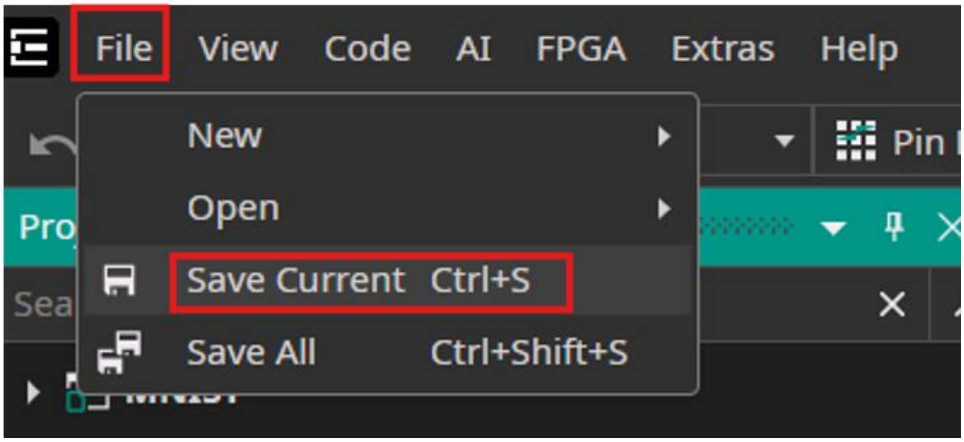


10.2.2 Select Agilex 3

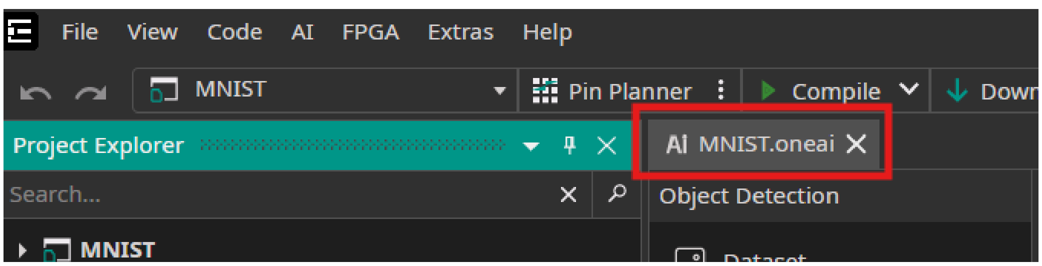


For the OneWare version of the 0.21.8.0 and on, the Agilex 3 devices are now included in the software.

10.2.3 Save your settings as File -> Save Current



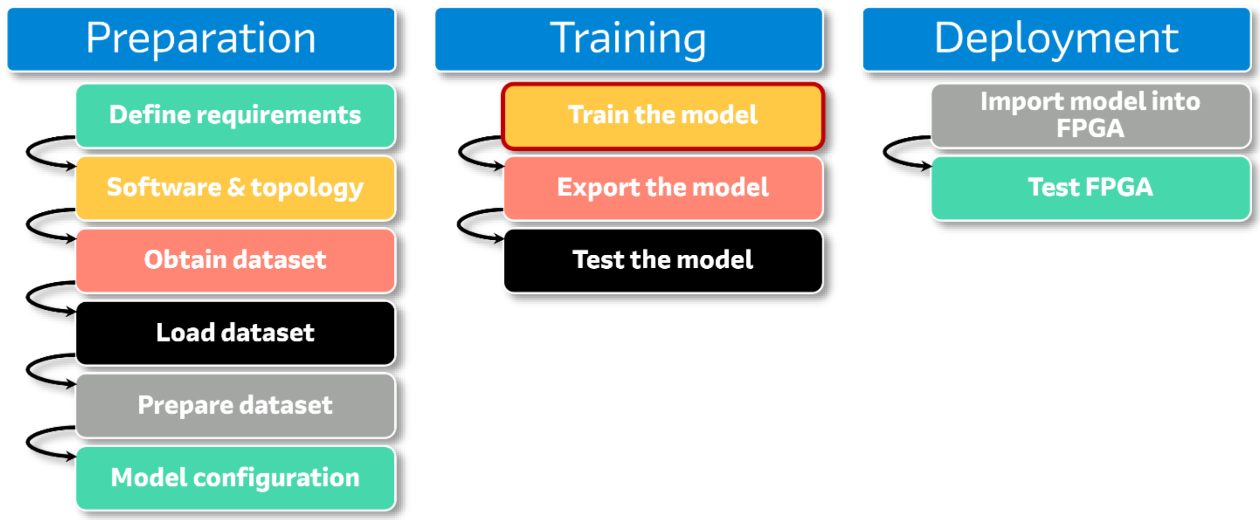
So that it looks like



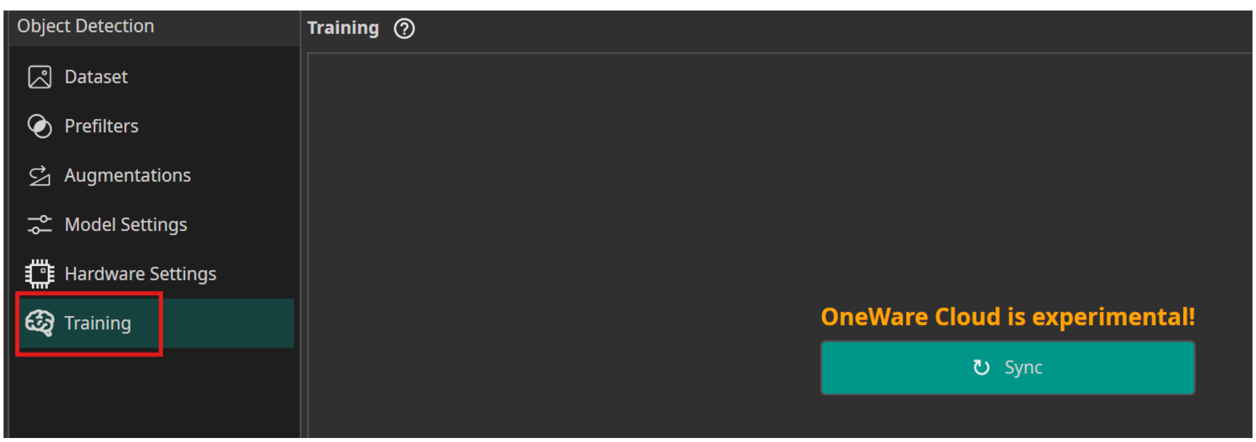
Where there is no longer a * by the name. Your configurations are now saved.

11 Train the Model

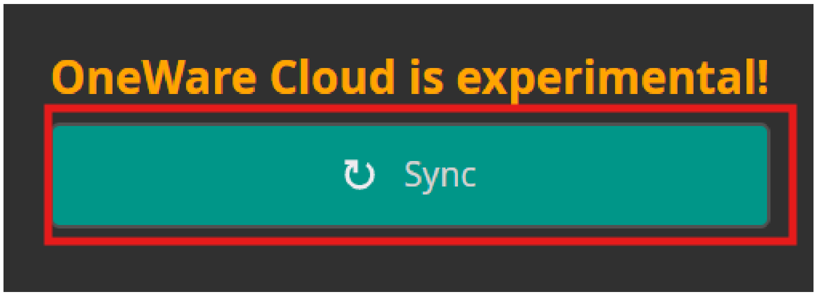
In previous sections, the dataset was prepared. From this point onward, the model will be trained.



11.1 Select the Training option



11.1.1 Click Sync to synchronize the OneWare Studio project setting with the OneWare cloud.



This will now give you access to the OneWare Cloud.

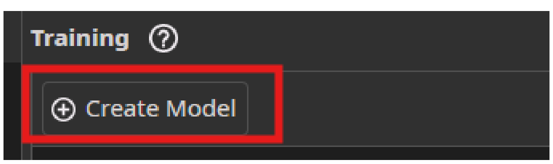
On the off chance that you are unable to log into the Cloud Services, you can still download a trained model from the internet for use in the later export step. The download link is available at:

OneAI_demo_datasets/models/nist_sd19_sorted/onnx/nist_sd19_sorted.onnx at main · one-ware/OneAI_demo_datasets · GitHub

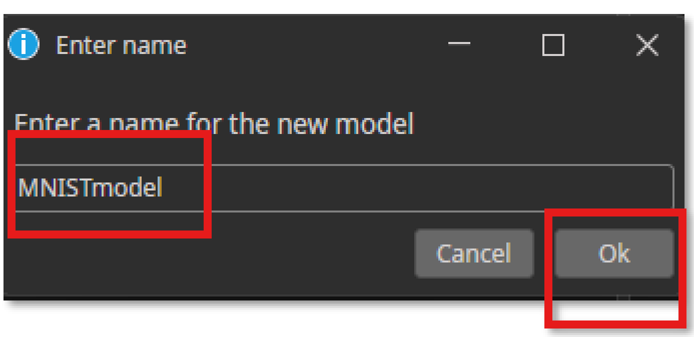
Where the file is called nist_sd19_sorted.onx. The output file is an ONNX file, and thus you can skip the following training steps and go straight to visualization section.

11.2 Create the Model.

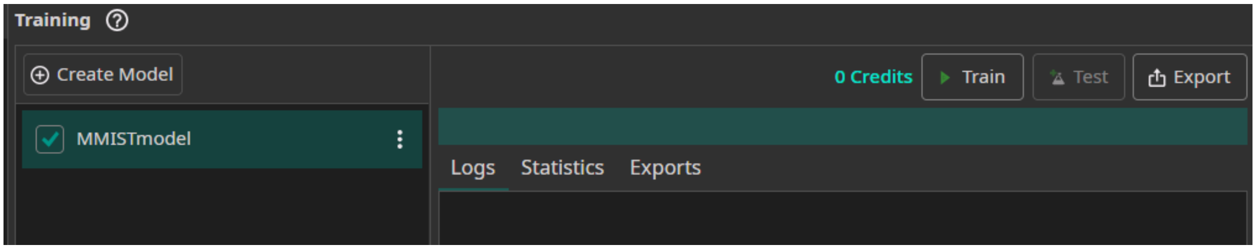
11.2.1 Click Create Model



11.2.2 Enter a name for a new model, and then click on OK button. This name can be any name such as

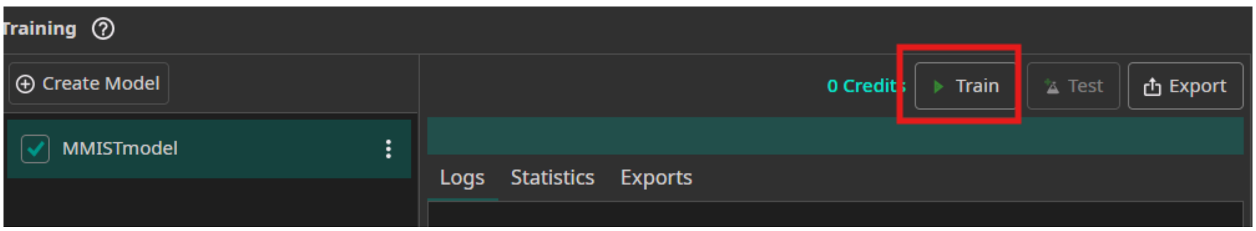


It will then look like this

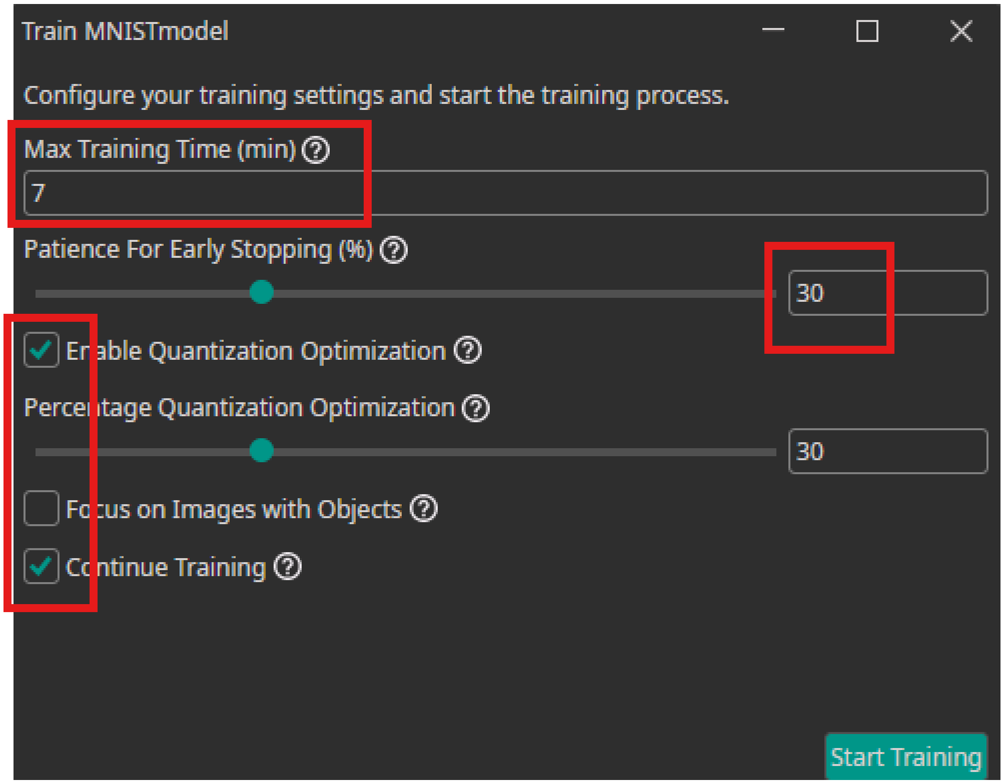


11.3 Start Training

11.3.1 Select Train



11.3.2 Modify the configuration by adjusting the items highlighted in the red boxes as follows:



- Max Timing time (min)

  The training time specifies how long the training process will run before stopping. A duration of 10 minutes is sufficient to train this model. During training, the results (such as errors) can be monitored in real time. Patience for early stopping is a step to stop the training earlier. This setting can remain at 30%, which is the software's default value, which means that the accuracy is close enough, the training process may stop earlier in about 7 minutes.

- Enable Quantization Optimization

  Enable Quantization Optimization, as the model will be imported into a FPGA. Quantization Optimization allows the model to be compressed, resulting in a smaller implementation. This compression is achieved by converting higher-precision values (e.g., floating point) into lower-precision values (e.g., 8-bit). Although this greatly reduces the model's size, it does not significantly affect accuracy. Quantization is widely used in machine learning because it enables even small devices to run models.

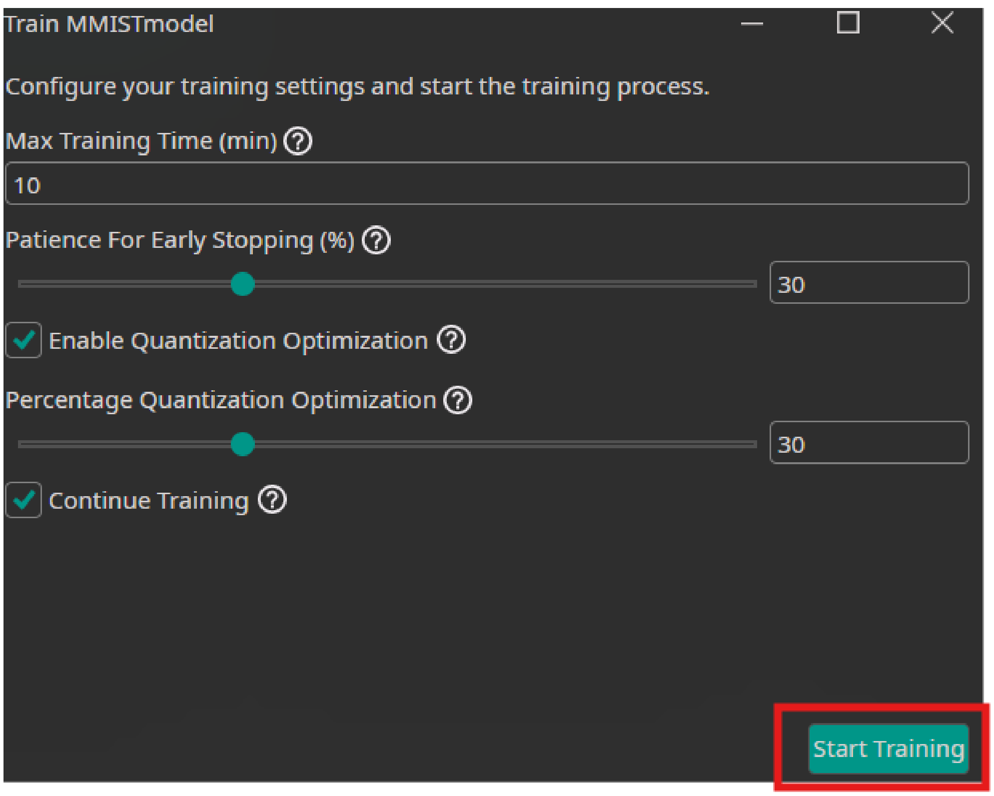
NOTE: If the Enable Quantization Optimization is not activated, and you attempt to train the model and export the RTL files, an error message will appear indicating that Quantization must be enabled. Quantization ensures that the RTL code can fit into the FPGA.

The Percentage Quantization Optimization setting defines the portion of training performed in the normal state before applying quantization. This value can be set to 30%.

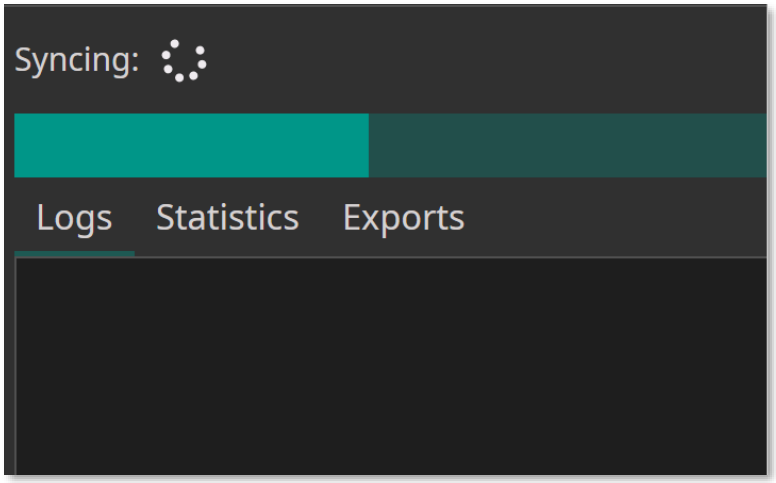
- Continue Training

  If you have trained a model before, you can continue to train the model again. If you have not trained the model before, this setting will make no difference.

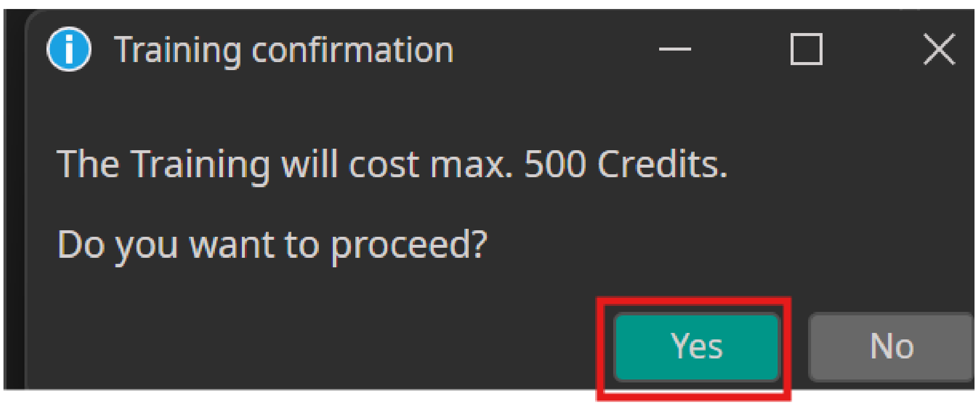
11.3.3 Click on Start Training



The data will be transferred to the OneWare Cloud.

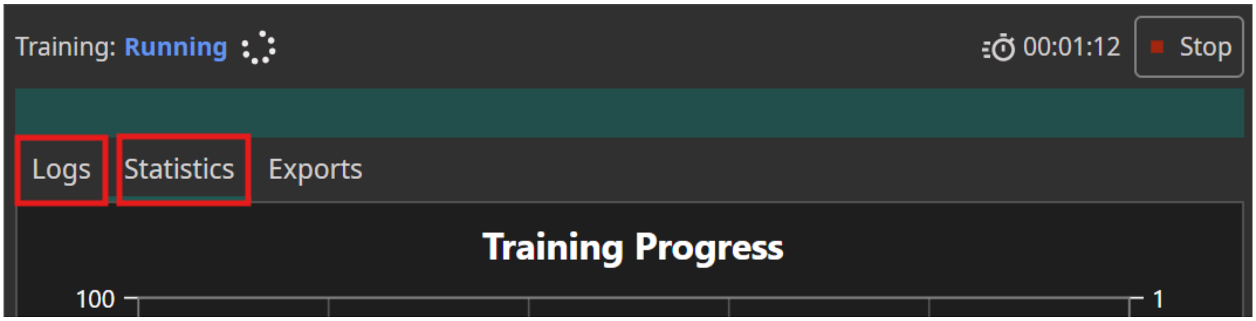


11.3.4 If you see a window like the following window regarding Training Credit, select Yes.



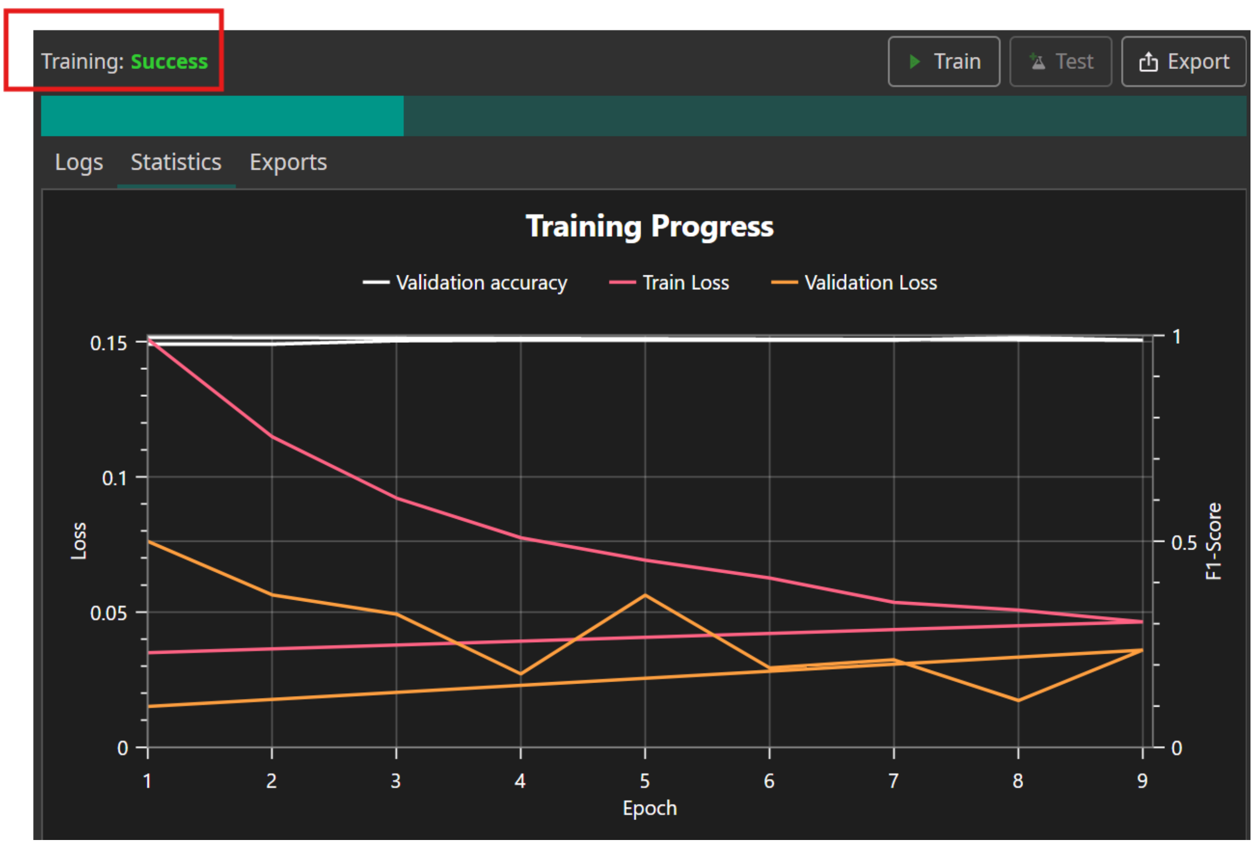
The transfer speed depends on the internet connection and the cloud server. In practice, the upload has been observed to take about 10 to 15 minutes over a wireless connection, although this time may vary.

Once the upload is complete, the process will enter the running stage. It will run for a maximum of 10 minutes, as specified in the settings but it will finish faster. Under the Statistic tab, the training progress is displayed as follows:



As the training process runs between 5 to 10 minutes, the values on the Statistics tab will update and the values for the loss will decrease and the accuracy will increase. The lines will change as it runs.

The training will run until it states the word Success as seen below.

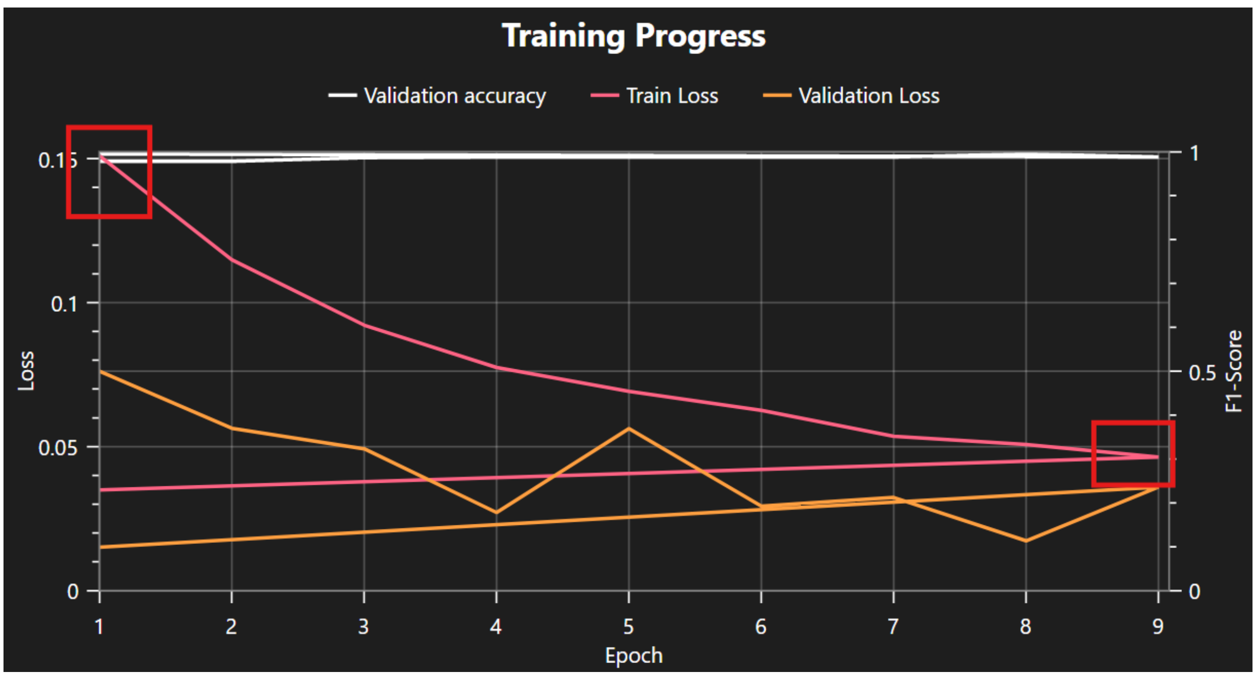


Your results might be somewhat different than those shown above. The trend should be the same. During training the validation accuracy increases and thus the inverse, the training loss, is lowered. Results can vary from one training session to the next.

Here are some items to note

Epoch refers to the number of times that the model processes the entire dataset. In this case, it runs approximately 10 times. As a general guideline, setting up to 100 epochs can be a good starting point to observe how the loss trends. The train and validation loss should be trending towards zero.

A common question is: When do you know it's time to stop running epochs? Training can be stopped when the loss value becomes sufficiently low. As shown in the screenshot below, during the first few epochs, the loss is very high. However, after more than 9 epochs, the training loss decreases, indicating improvement. Training loss can be seen as approximately the inverse of accuracy, thus as training loss goes to 0%, the accuracy increases to 100%.



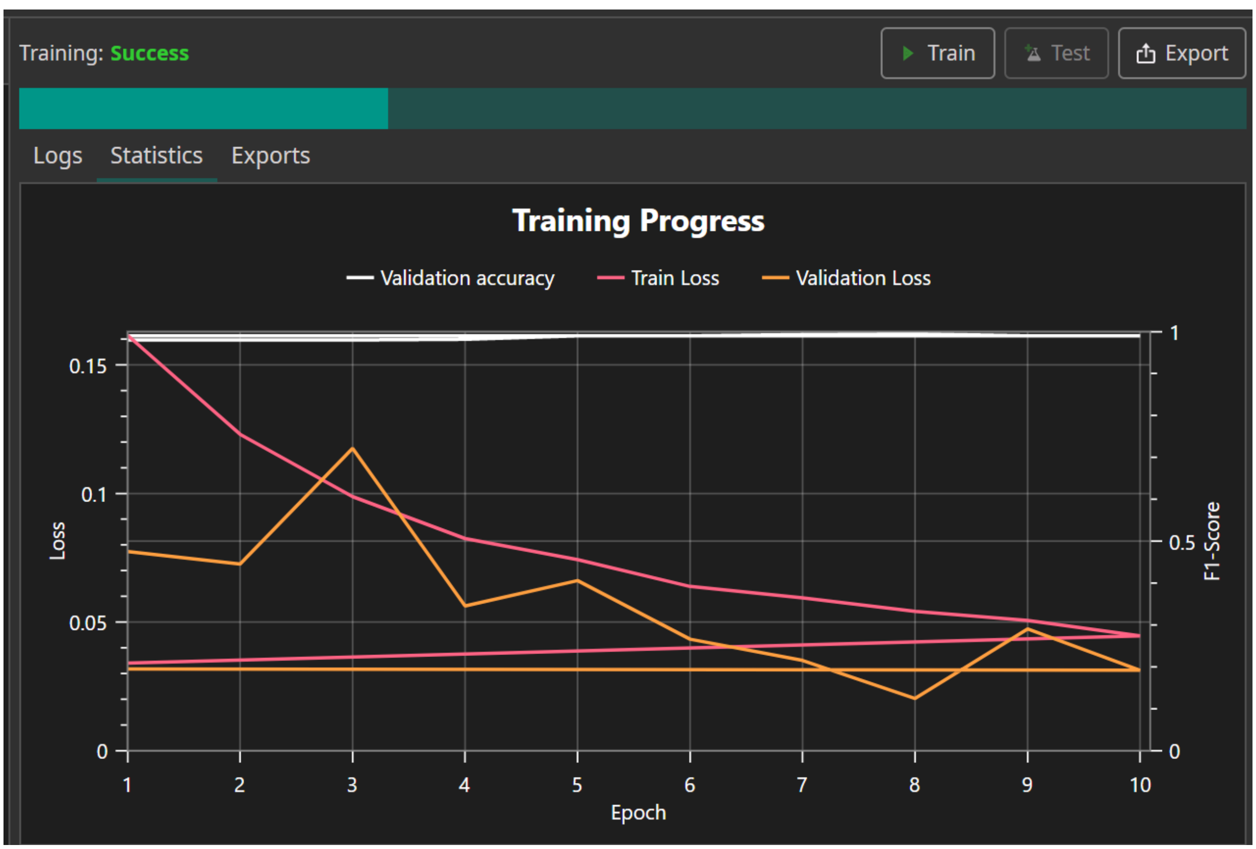
Training loss measures how well the model fits the data from the training dataset. As this value decreases, the model is "learning" the patterns in the data. Loss is calculated by comparing the predicted label to the actual label. For example, if the model predicts an 8 but the correct (label) value is 6, it will then adjust the model parameters accordingly.

Mathematically, there are various loss functions. In this case, categorical cross-entropy loss commonly used for classification tasks with multiple classes, such as the 9 classes here is applied. In this software, the value is computed automatically, but in other platforms such as TensorFlow, built-in functions can also perform this calculation, for example:

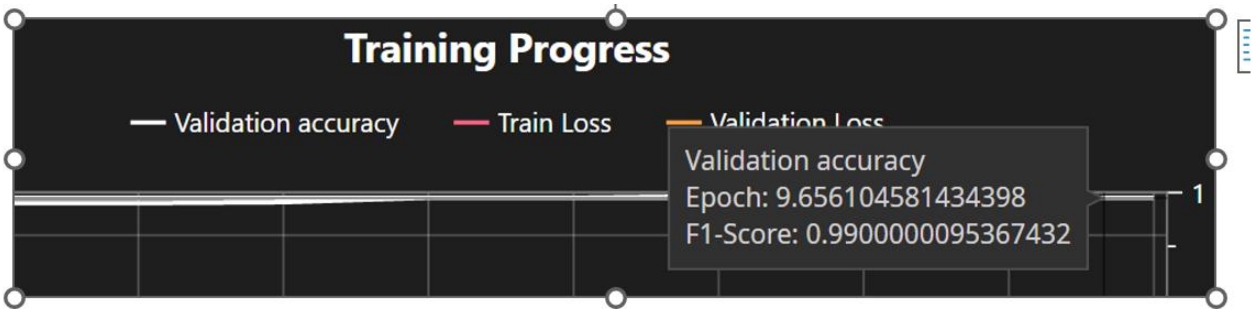
```
model.compile(loss = tf.keras.metrics.categorical_crossentropy,
optimizer = "adam", metrics = ['accuracy'])
```

There will also be a result for validation loss. Validation loss measures how accurately the model fits the data from the test dataset. As this number decreases, the model becomes better at predicting data it has not seen before.

There may be a question of what is an F1-Score? The F1-Score is on right hand side of the chart. A F1-Score, which can be used for classification models, is defined by Wikipedia as the "harmonic mean of precision and recall." The range for F1- scores is between 0 and 1, where 1 is a perfect score.

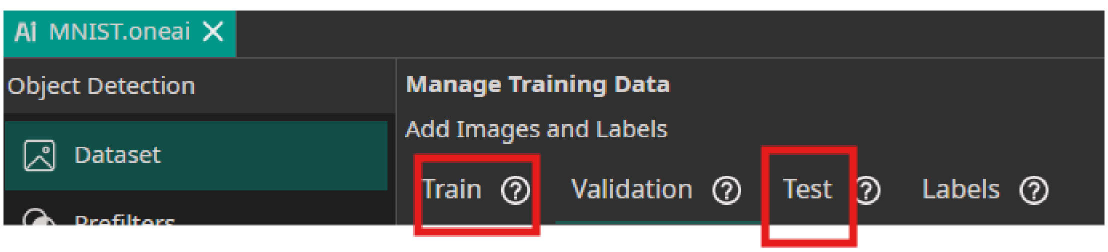


If you select on the validation accuracy line near the epoch 10, note that the F1-score is 0.99 which is almost perfect

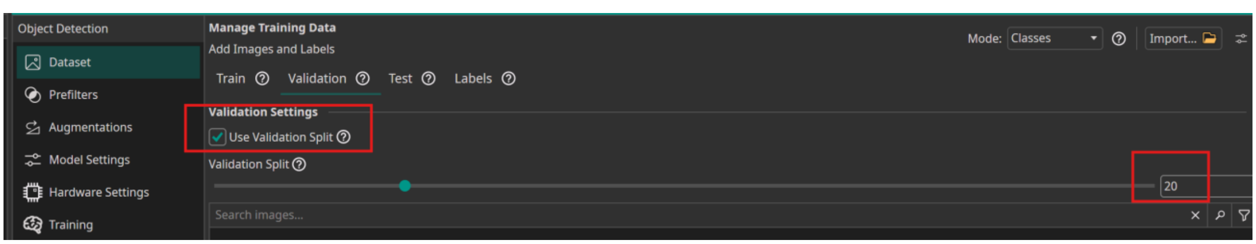


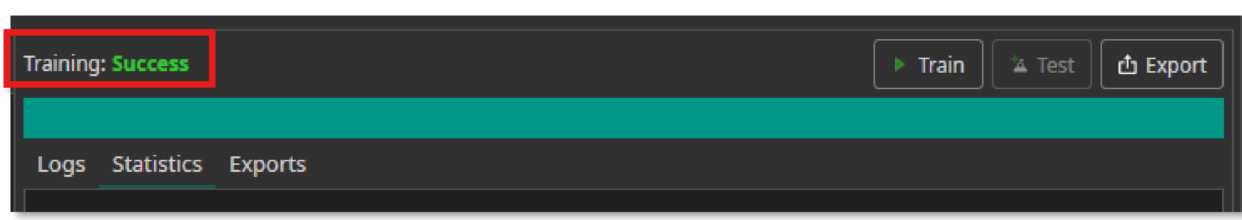
If you select on the loss values, the F1-score will be less.

Why is validation loss and validation accuracy reported if only the train and test datasets were uploaded into the dataset?

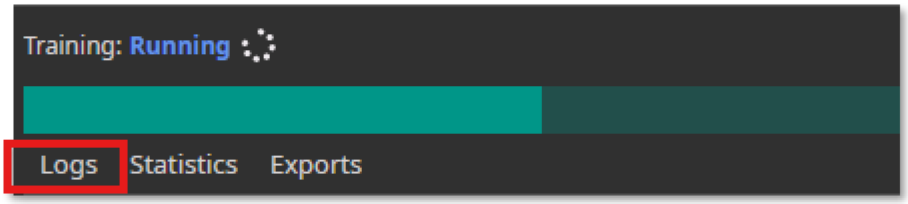


The reason for this is that by default in the Validation dataset, the validation image percentage was set to 20% by default.





You can also monitor the *Logs* while the process is running. Here are some key points to note regarding the values displayed under the *Logs* tab.



Results show items such as:

```
Starting Task: bbf9885e-8dfd-4f82-ab30-960dbc1ee4ab
Starting OneAi task with command: -m oneai.worker --gpu 0 --ram 0.3 --vram 0.45 -
Starting ONE AI...
Loaded dataset with 11 label types.
Found training dataset with 3000 images, validation dataset with 0 images, test d
Class types: ['No Object', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10']
Minimum label count for removal: 1
Labels to remove based on count: [0]
Removing label ID: [0] from dataset.
Analyzed dataset
Model search successful
Selected model: 44.0 KB RAM usage, 33850 parameters, 2.0% DSP usage
Starting Training
```

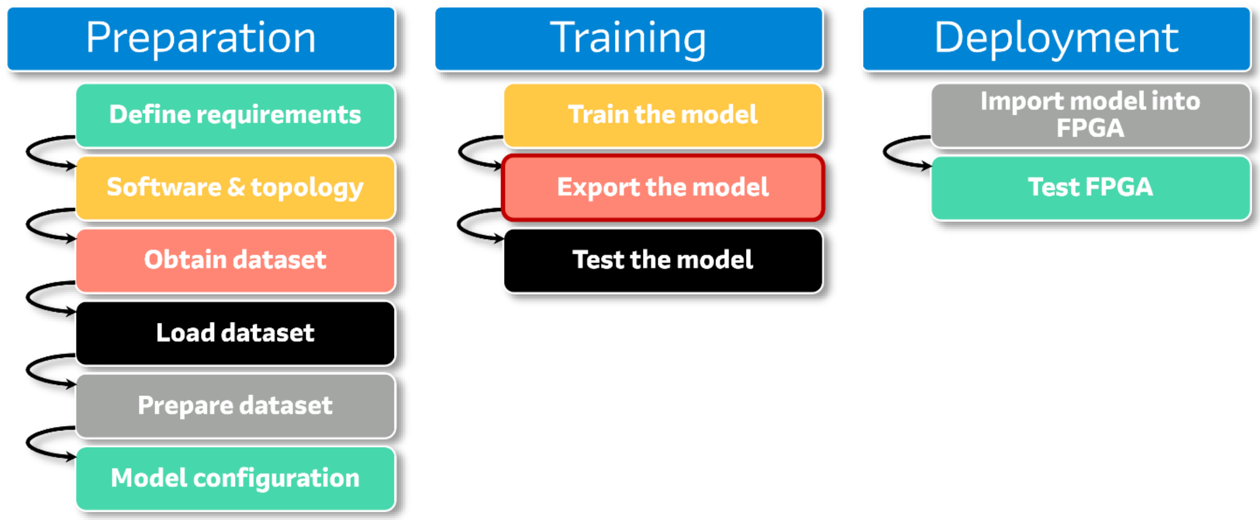
-Training dataset of 3000 images and test dataset of 500 images which you uploaded.

-There may be a Question of why it does it state the lines of “labels to remove based on count[0], minimum label count for removal: 1 and removing label ID [0] from dataset”? The reason for these lines is that since you can select what class that each image is, there is a reserved class where there is no class selected for an image. For example, you could have an image with no class, and thus this could be detected. In this case, where it states removing label ID [0] from dataset, this means that the software detected no image without a number in the dataset, thus the case with no number in the image was ignored.”

-Amount of memory and DSP usage that you will need in an Altera device. It does not show the amount of logic. If the resource usage exceeds the capacity of the selected device, it will report a “no-fit.”

12 Model Export

This step in the flow is:

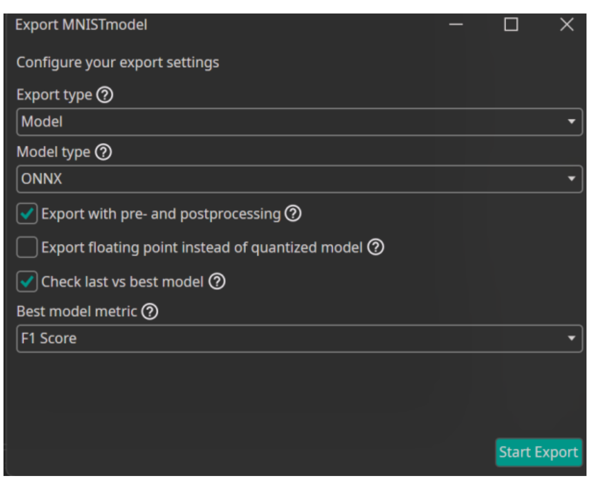


The model will now be exported so that it can be tested and deployed. There will be different files for export.

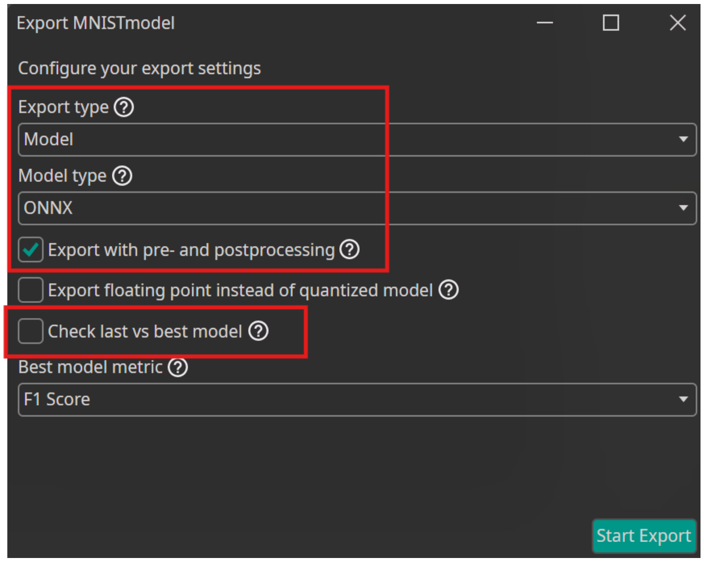
12.1 Export the ONXX model

12.1.1 Select Export as

The following window will appear:



12.1.2 Disable the last vs best model check box.



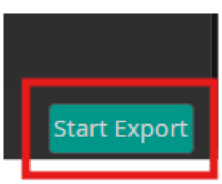
This step will download the ONNX file, which will be used for model visualization.

Export pre- and post-processing is enabled so that the filters are incorporated into the model to have correct results.

Export floating Point option is the only format available for the ONNX model; therefore, selecting or not selecting it will make no difference in the results.

Check last vs best model is disabled because this setting is applicable only for object detection, whereas in this case the task is object classification.

12.1.3 Select Start Export



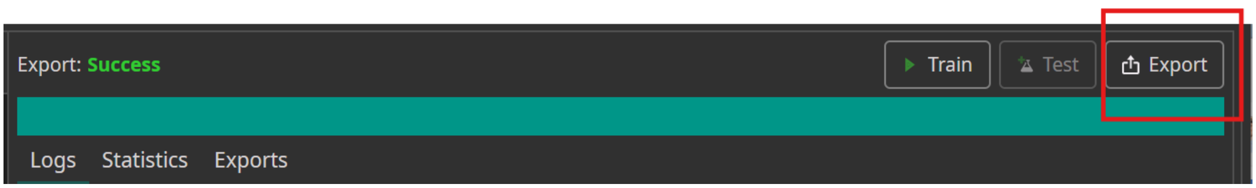
It will show Export Success once the file is created as follows

Notice that it exported the floating point model.

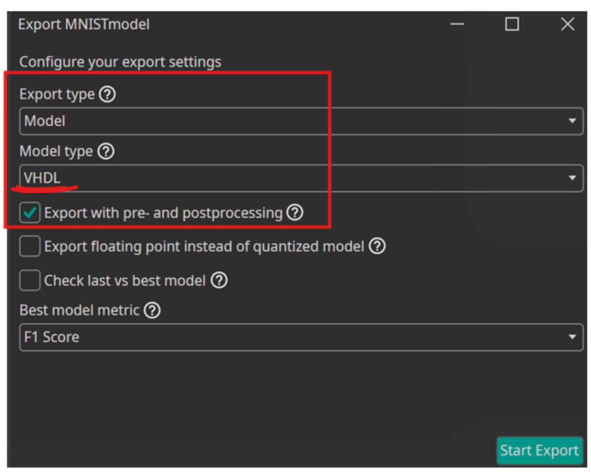
For reference, this file is not downloaded until a later step, so you will not see it in your directory. It has been prepared for export.

12.2 Export the VHDL Model

12.2.1 Select Export again as follows:

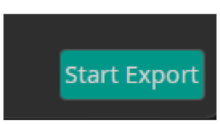


12.2.2 Ensure that the settings are set as:



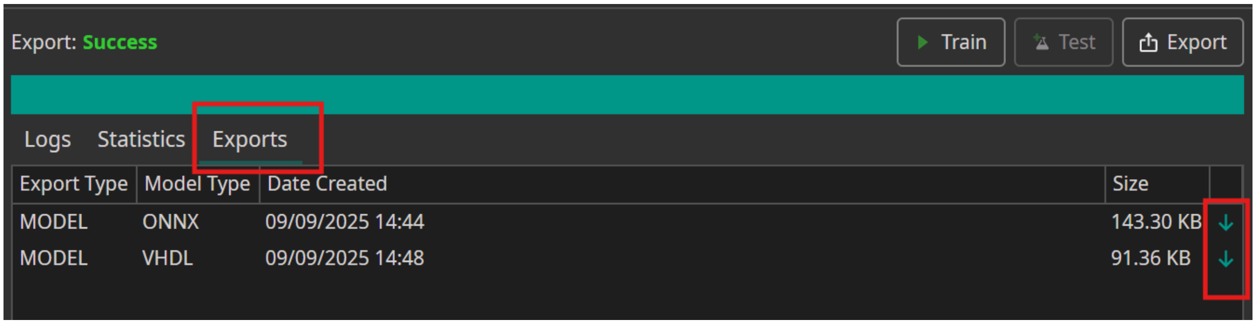
This time the VHDL files will be downloaded.

12.2.3 Select Start Export

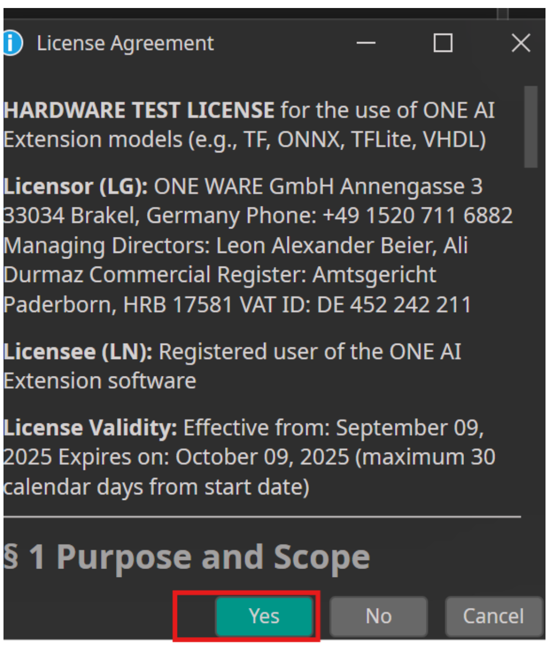


12.3 Download the ONNX and VHDL Models

12.3.1 Select the Export tab as follow and then select the down arrows one by one.

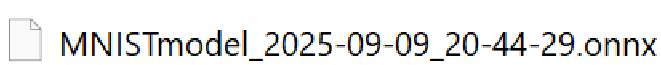


12.3.2 When you try to download the ONNX model, a License Agreement pop up window will appear. Read it, and then select Yes



The ONNX model will be downloaded in the following directory

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\MNIST\MNIST\Models



When you try to download the VHDL file, you will see the license agreement window again.

12.3.3 Select Yes to download the VHDL files

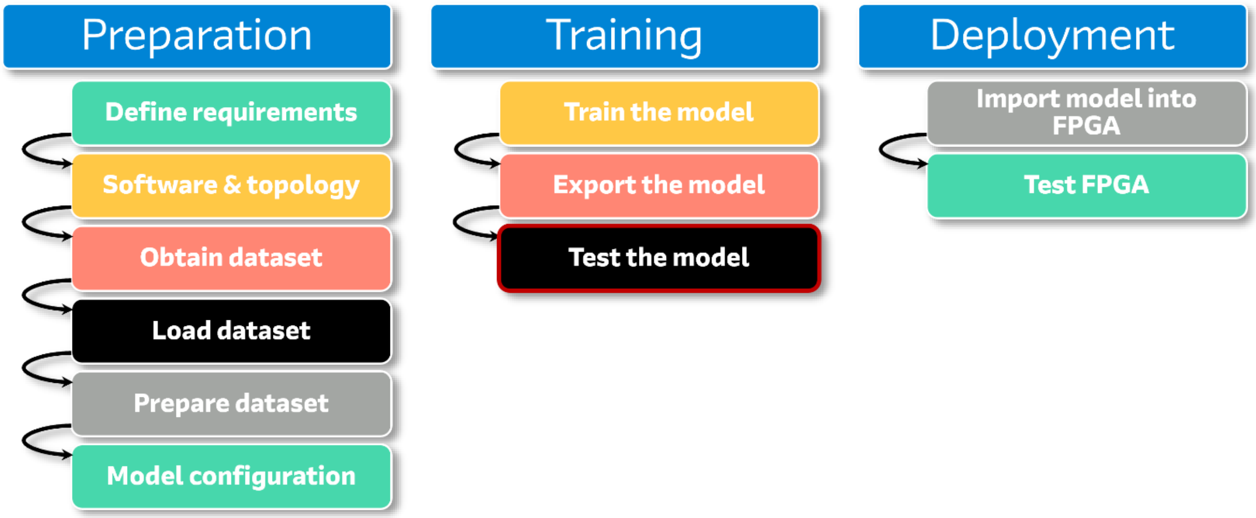
The VHDL files will be downloaded in the following directory

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\MNIST\MNIST\Export\MNISTmodel\export_files

- Core
- Filters
- Quartus_IP
- ONEAI_Config_Package
- ONEAI_Data_Package
- ONEAI_Simulation
- ONEAI_Top
- Test_Data_Package

13 Test the Model

This step is dedicated to testing the model before deploying it.

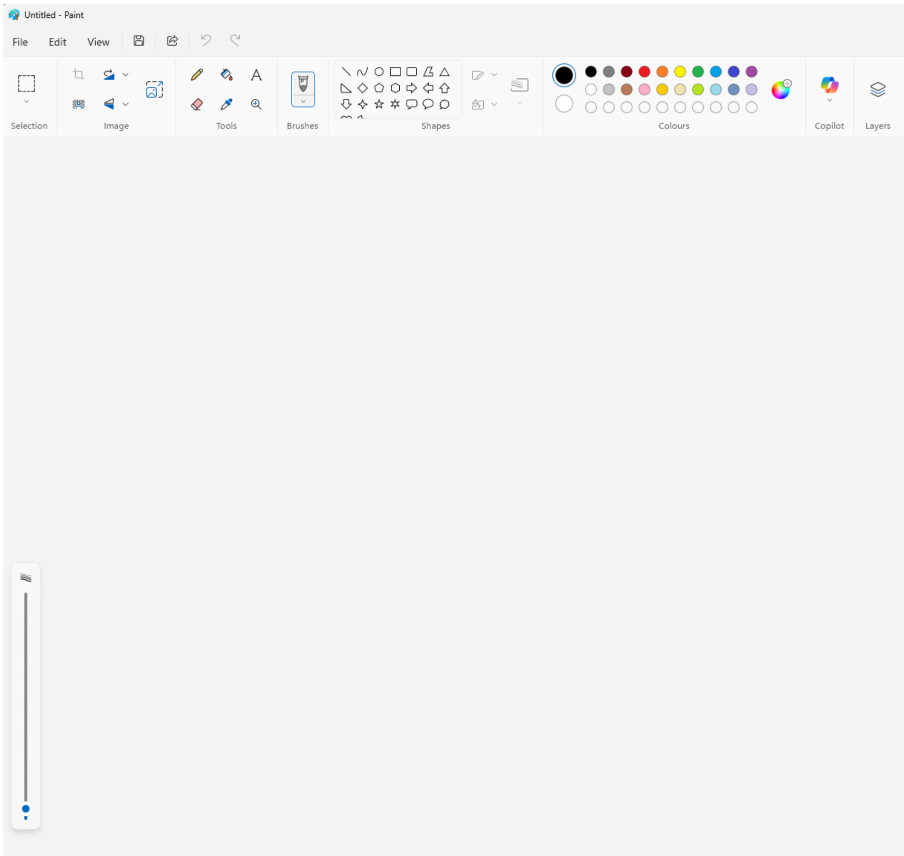


Now, let's see the robustness of the model, by working interactively with it by writing your own numbers, and seeing the results. When this workshop is used in a cloud-based environment (such as Cloud Labs), you will not be able to use a camera or webcam thus the Paint software program will be used to create and test handwriting samples instead.

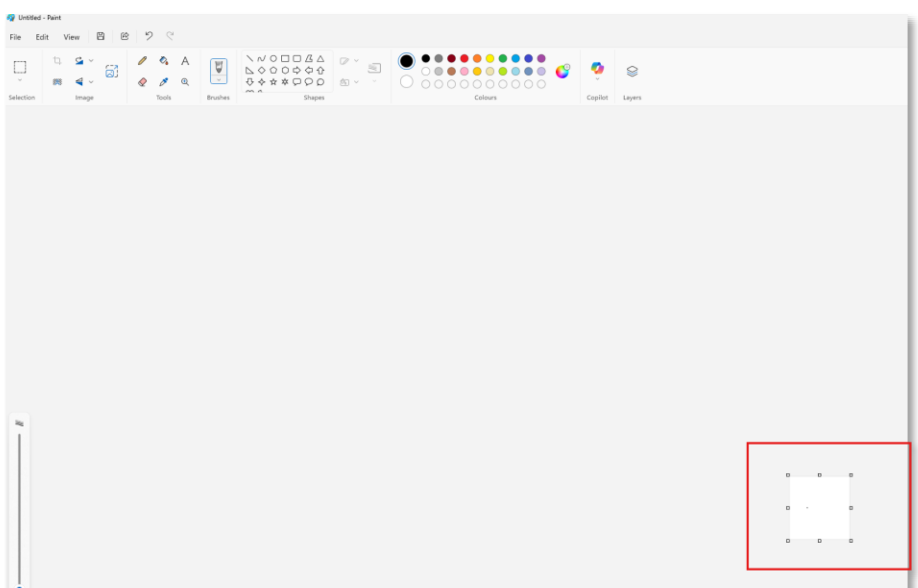
13.1 Create a Handwritten Image

13.1.1 Go to the Windows Start bar. Enter the word Paint

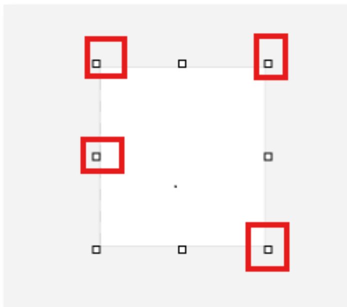
The Paint software program will now open as



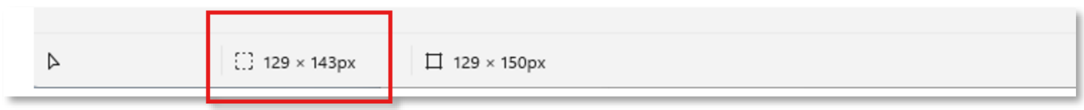
In the middle of the window, you could have a selection.



Select any corners of the window as follows:

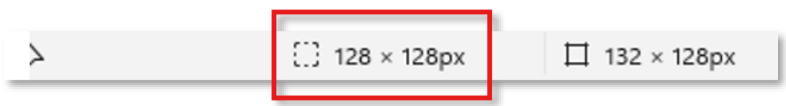


Notice in the bottom left corner that there are numbers for width and length that change



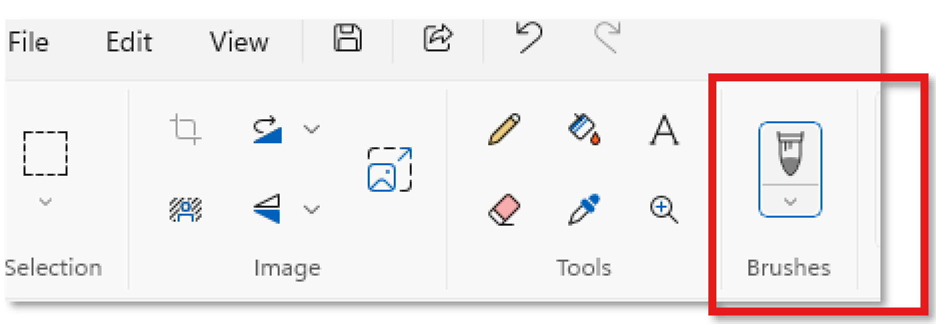
13.1.2 Select the corners until the numbers are **128** × **128** or close to this value.

If you get close, for example around 130, it will still work.

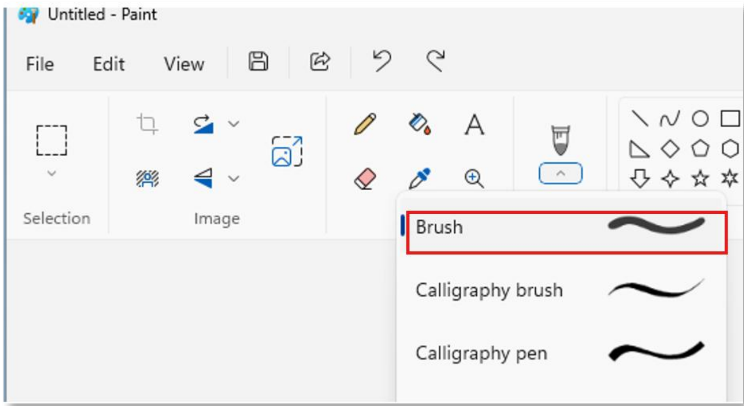


The reason for 128×128 is that the numbers in the dataset, as well as the model were trained for 128×128 pixels dataset.

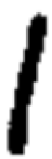
13.1.3 Select Brushes in the Paint App



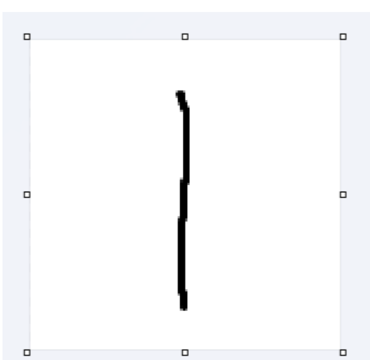
You can use Brush as follows:



You should use a “thick” brush because the model was trained on thick brush strokes. These thicker strokes can be observed in the dataset, which features bold, well-defined lines such as the number 1 in the train dataset as

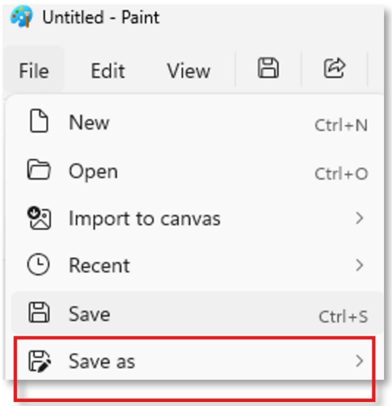


13.1.4 Write the number 1 in the Paint application.



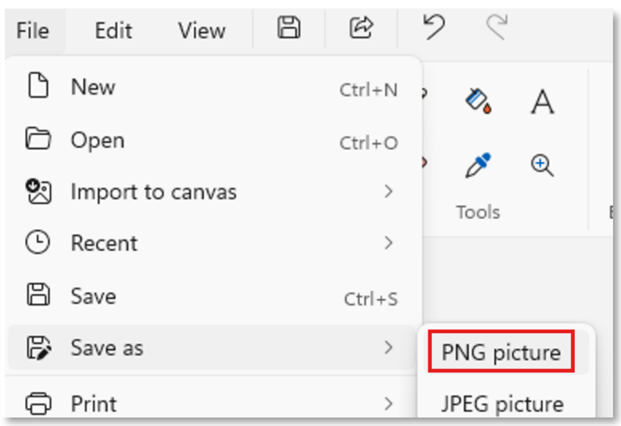
13.2 Save the Image

13.2.1 Save the file by selecting Save As.

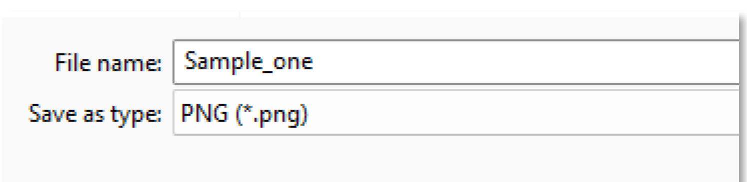


13.2.2 Select the .png file format.

OneWare uses the .png file format for the data in the datasets.



13.2.3 Name the file as sample_one (it could be another name if you wish) and click on the Save button.



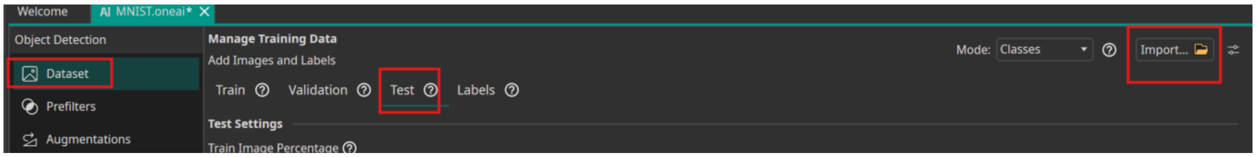
13.2.4 Select the OneWare directory:

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\MNIST\MNIST

This is a higher level than the dataset files, but this will be changed in the next step.

13.3 Import the Image into OneWare Studio

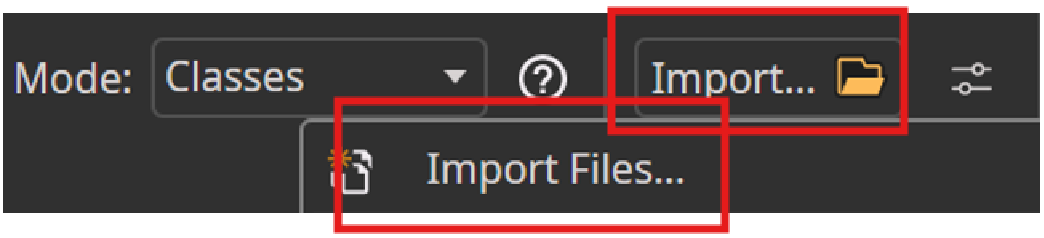
13.3.1 Within the OneWare software, select Dataset, and Test Data and then select Import as



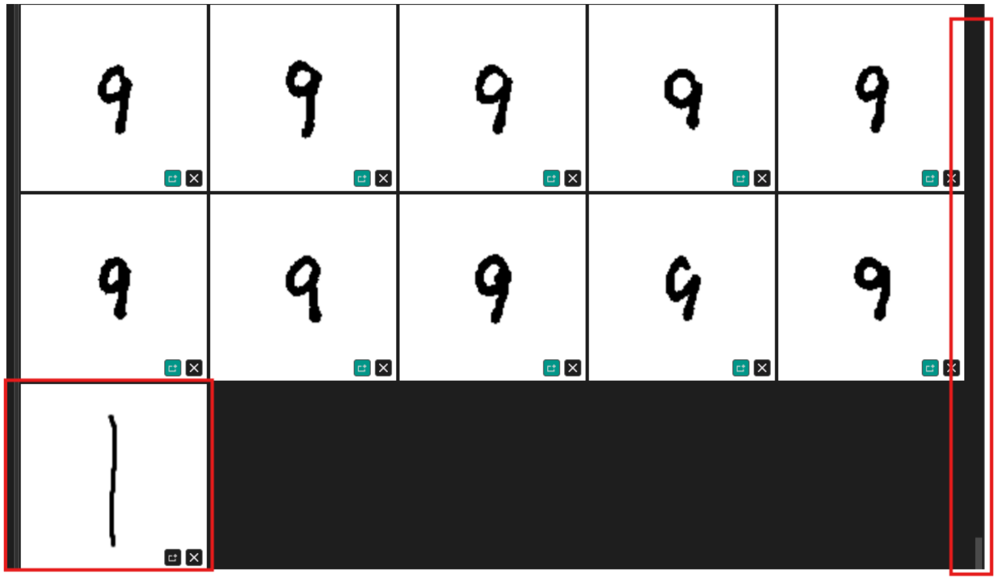
It is important to use the Test directory, because you will be testing your handwriting not training with it.

13.3.2 Select Import Files and then select your sample.png from

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\MNIST\MNIST

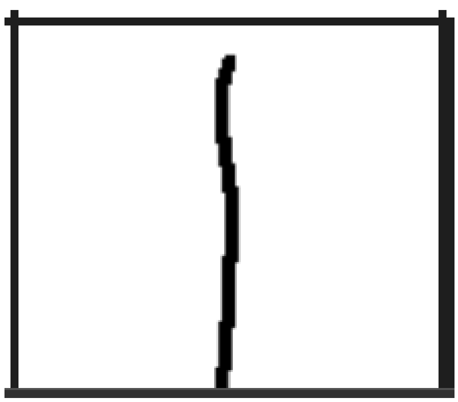


13.3.3 Use the slider on the side (as seen in the red box below) to under the numbers 9 to see your new number.

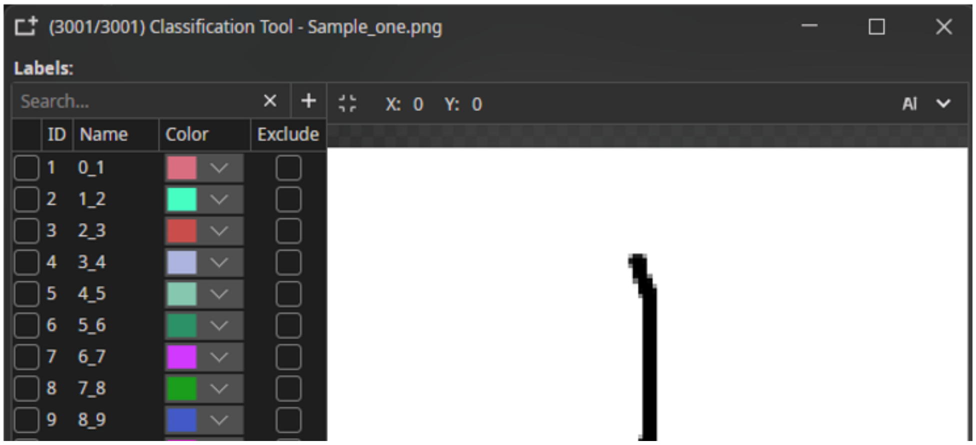


13.4 Auto Label the Image

13.4.1 Double click on this new number.



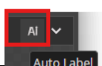
The Classification tool will now appear as follows:

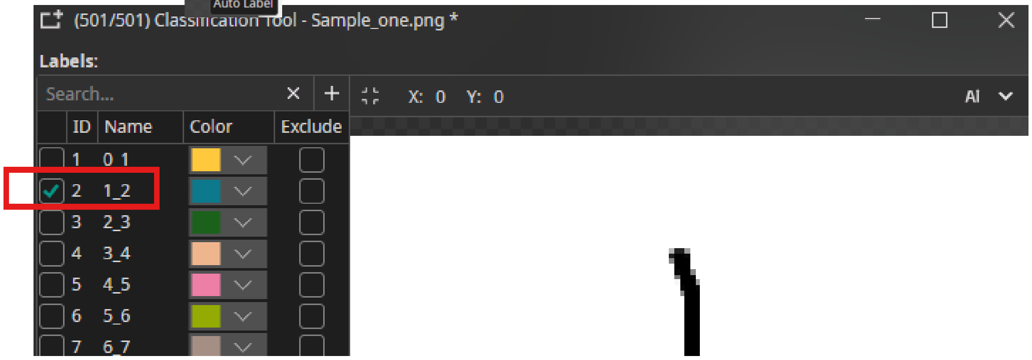


13.4.2 Select the pull down menu next to AI in the top right hand corner.

13.4.3 Select the pull down menu next to Select a model to add.

13.4.4 Click on the MNISTmodel and click the  sign to add the model.

13.4.5 Hover over  Click on AI and the tool will auto label the image



The auto label occurs when the AI tool runs the newly created character through the new model that you have created.

What do you see in the corner with the checkmark? Do you see a 2?

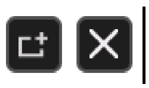
Why is the checkmark by the number 2 instead of 1 (which is expected)?

Note: This is not an error in this case, because the ID and associated Name in this case are all incremented by 1 digit (i.e. there is no number 0). You can see this by clicking the other number ones in the Test Dataset.

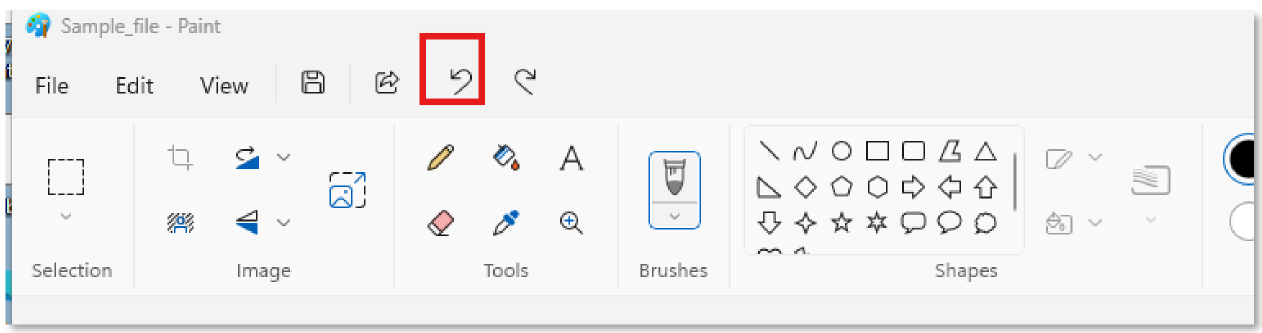
Note: The data was pre-sorted into folders by OneWare and sharedtheirn the github site. The One AI tools expects a specific numbering format for each folder. The first number represents the ID of the number and the second represents the name. OneWare reserves the ID ‘0’ so all

13.4.6 Select X in the top window to close the digit.

13.4.7 Delete this number in the Test set by selecting



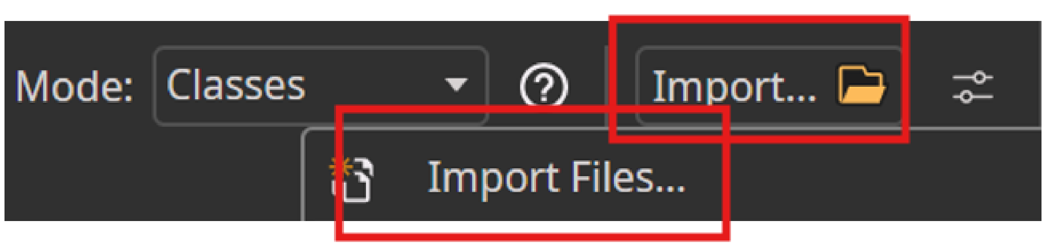
Now, for another experiment, open the same Paint file and select the back button.



Use Brushes and write any other number from 0 to 9.

Save the file.

In OneWare, use the Import Files from c:\OneWare\MNISt\MNIST and re-import the file



Scroll down to the bottom of the window. This new number will be seen at the bottom of the window.

Double click on this new number and select AI

Is this number correct i.e. does the model's prediction correspond to what you expect?

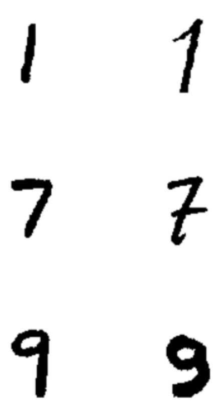
You can repeat the above steps with other numbers as desired.

It is expected (and has been observed) that various numbers that you write will not match the predicted results.

The reasons for these errors include:

- 1) Your handwriting style differs from the samples in the training dataset, especially if you are writing in a European vs a North American style.

  Note: There is a difference between European and North American style. The North American hand-writing version is on the left-hand column, while the European numbers (more rounded) style is on the right-hand column.



Due to this difference, there is another dataset variation, nist_sd_19_subset_random, on the OneWare website. This dataset can be found at [OneAI_demo_datasets/datasets/nist_sd19_subset_random.zip at main · one-ware/OneAI_demo_datasets · GitHub]. This set has more random SD19_dataset values, thus it will catch more of these numbers. If you download the .zip, import the dataset (same steps as done earlier), set configuration settings and re-train it, you should see much higher accuracy results because it takes into account more European style handwriting.

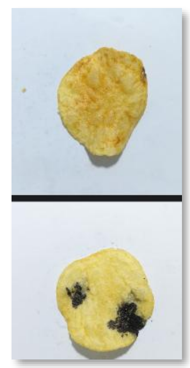
- 2) The dataset is relatively small, with only 3,000 digits. If it were much larger, such as 60,000 numbers in the NMIST dataset, the model would have a better chance of producing correct predictions.

  One thing to note is that handwriting recognition tends to be more variable, which makes it more challenging for smaller datasets to achieve high accuracy unless you are aware of and account for the differences.

You might ask: *Why even use a small dataset?* There are several reasons:

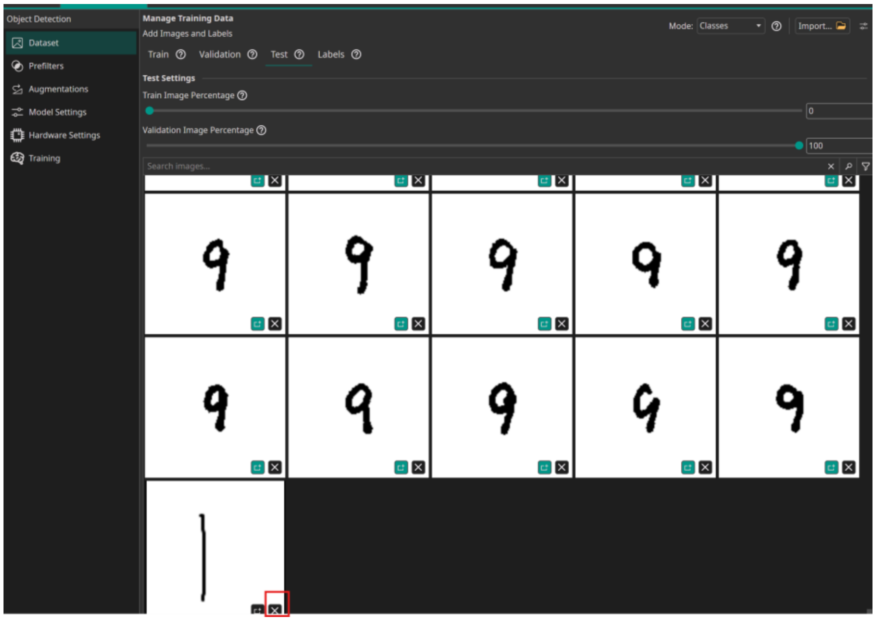
- The time for this workshop can only be so long.
- Small datasets are easier to train and test compared to large datasets. For example, training a dataset of 3,000 images in OneWare Studio, as was done in this case, is much quicker to train than importing and training 30,000 images.

  (Just for reference, some datasets are in the hundreds of millions and even more, thus for large datasets, it takes a very long time to train those datasets)



- Depending on the application, small datasets can perform very well. For instance, in assembly lines inspections, such as checking printed circuit boards where variation is minimal or even detecting defective (burnt) potato chips in a food processing line, a small dataset can work very well.

After you have finished experimenting, you can select in the dataset in your handwritten number(s) and select X (in the red box) to delete your custom numbers.


Object Detection
Dataset
Prefilters
Augmentations
Model Settings
Hardware Settings
Training
Manage Training Data
Add Images and Labels
Mode: Classes
Import...
Train
Validation
Test
Labels
Test Settings
Train Image Percentage
0
Validation Image Percentage
100
Search images...

14 Visualization using the ONNX file

After you set various settings in the OneWare software, trained the model, and experimented with the model, you might have the following questions:

1) What does the model look like?
2) How does the model compare to the LeNet-5 mode covered in the background material?

There are several ways to visualize the model. One method is to view the results using the ONNX file, while another is to perform a simulation. This section will focus on visualization using the previously exported ONNX file.

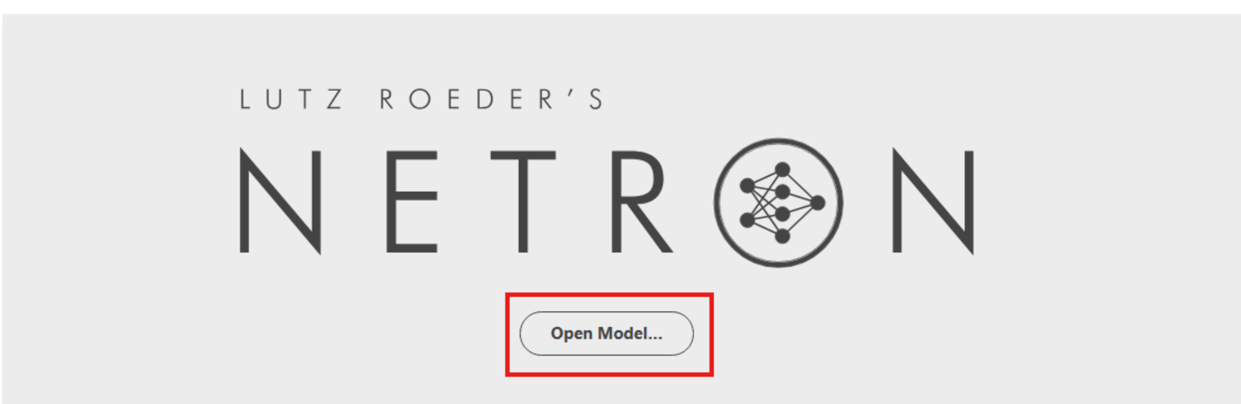
ONNX, which stands for the Open Neural Network eXchange, is an open standard used for machine learning. This file allows interoperability and portability between different frameworks (software). Different software such as OneWare, Tensor Flow (provided by Google), etc. can generate ONNX files, and then the file can be imported into another software.

When you entered settings into the OneWare and AI extension, there was no manual code entry nor knowledge needed to create a non-standard AI model. No manual code entry reduces typing errors, but from the GUI it is not clear on how the model looks and how it correlates to CNN basic building blocks.

As such, a visualization website, Netron will be used to input the ONNX file and output the model in a flow chart/graph approach.

🧑‍💻 Go to the website Netron using the following link and click on Open Model... button:

https://netron.app



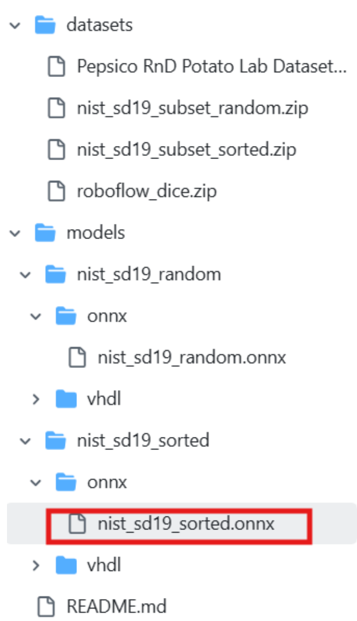
🧑‍💻 Select the

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\MNIST\MNIST\Models directory

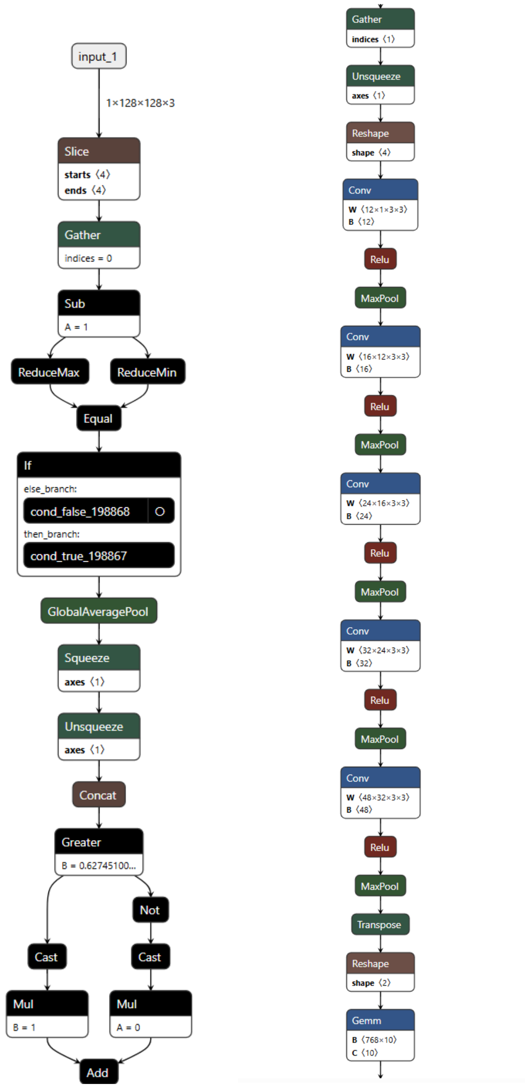
where there is the file called MNISTmodel<date>.onnx.

If you could not train the model, for any reason, there is an ONNX file on the OneWare website at

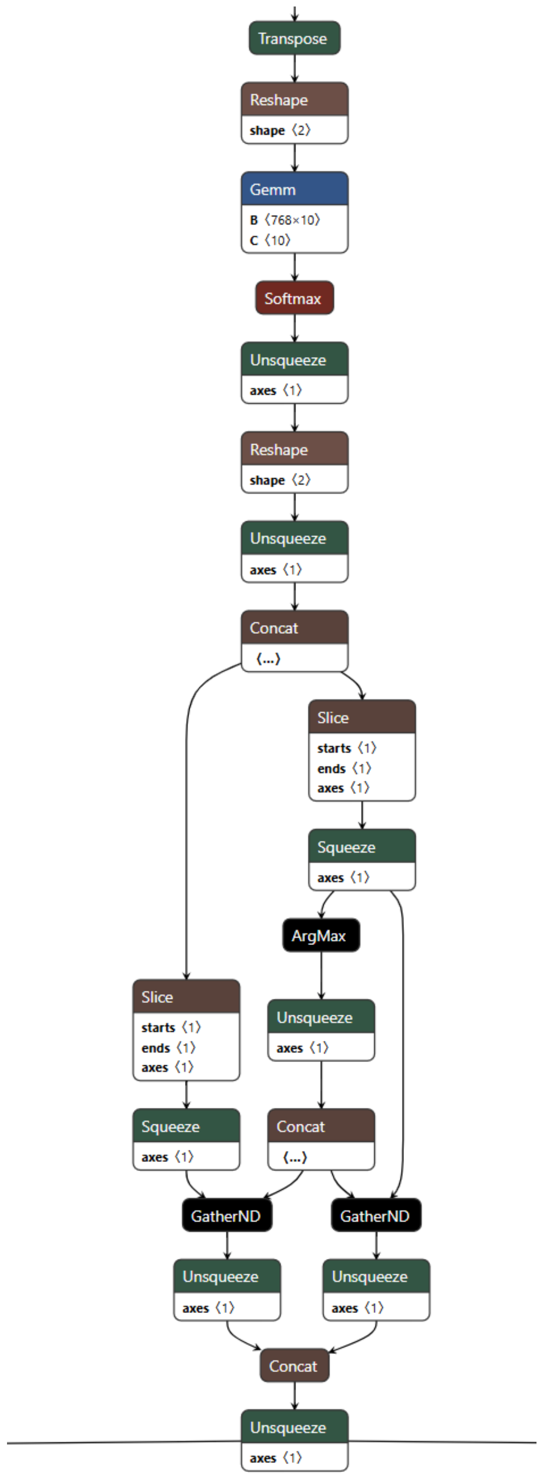
https://github.com/one-ware/OneAI_demo_datasets/blob/main/datasets/nist_sd19_subset_random.zip that you could use



The graph will show the following:

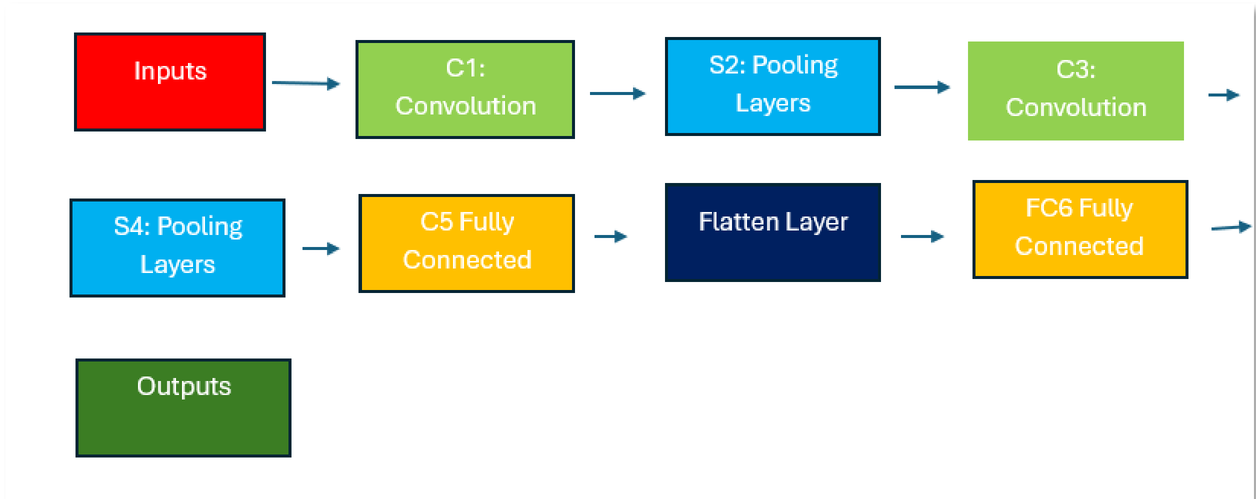


and



Even if you have a slightly different variation, this will be OK. This flow illustrates the custom model created based on the selected settings. While different settings in the OneWare software may produce variations in the resulting model, the same basic concept applies.

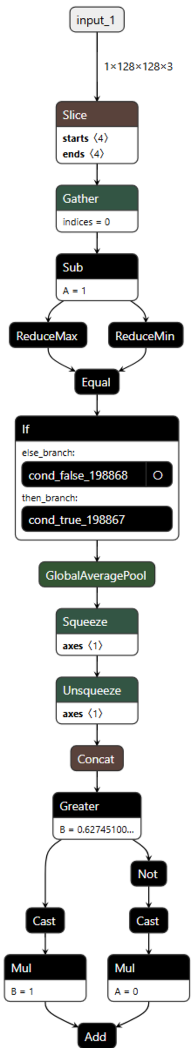
As a refresher, a simple CNN model LeNet-5, which can be used for recognizing handwritten digits from 0 to 9, has the following structure:



The model contains two convolution layers, two pooling layers and two fully connected layers.

Now, let's examine how the flowchart of a custom AI model corresponds to or differs from the LeNet-5 model describe above.

The first section of the model (shown below) is equivalent to the input section. It receives the 128×128 input image as *input_1*. Note that the input also shows a "3", representing three channels. This is because, by default, CNNs for color images use three channels: Red, Green and Blue. In this case, the image is grayscale (black and white). In the OneWare AI Extension, the two unused color channels were removed using the Channel Filter (in the filter section).



You can even select any item in the ONNX model and see more options.

GlobalAveragePool is a method of down-sampling that can be applied at the beginning of the CNN flow.

Squeeze and *Unsqueeze* are operations used to add or remove dimensions to the matrix, respectively.

Concat, which stands for concatenation, combines multiple matrices to form a larger matrix containing various ML features.

Cast changes the data type from one format to another. Converting from floating point to fixed point is an example of cast.

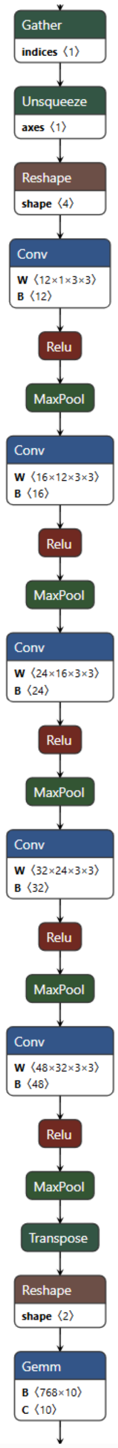
The next section (shown below) resembles the LeNet-5 model. This custom model contains five convolution blocks, whereas LeNet-5 has only two.

Having more convolution blocks enables the model to extract a greater number of features from the data.

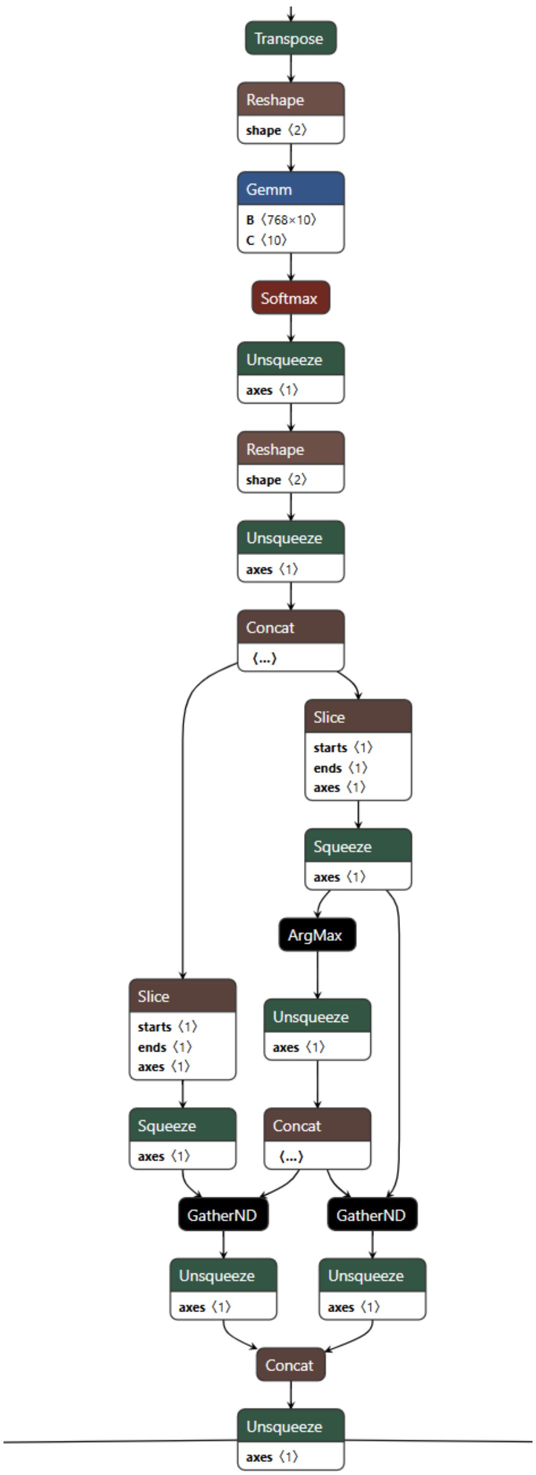
The activation function *ReLu*, which introduces non-linearity into the model, is used here just as in LeNet-5 model.

Softmax function is also the same in both models.

GEMM (General Matrix Multiplication) is employed for convolution operations and fully connected layers.



The final stage performs output processing, which includes functions such as reshaping, concatenation, squeeze, and unsqueeze. Squeeze reduces the vector (tensor) dimension by 1 to reduce the vector size and thus makes the size compatible for later steps. Unsqueeze means that the vector(tensor) dimension is increased by 1 which reshapes the vector. For example, if the vector is (2, 3) and then there is an unsqueeze of 1, the result would be (2, 1, 3). Unsqueeze makes the vector compatible with further steps.



15 Simulation

Simulation is a method used to verify the functionality of a design and ensure the design operates as expected. It offers several advantages over debugging hardware, including:

- Ability to verify all signals, including internal ones.
- Overcoming the difficulty of probing FPGA internal signals with external oscilloscopes due to pinout limitations.
- Testing corner cases for input conditions.
- Flexibility to test small or large portions of the design to ensure correctness.
- Capability to perform design testing even when hardware is not available.

15.1 Launch Questa-Sim

Questa-Intel FPGA Starter Edition Includes many precompiled libraries, which can be used directly without recompilation, simplifying the steps required to run a simulation.

The Questa-Intel FPGA Starter Edition comes with a free one-year license. This is the version used in this workshop.

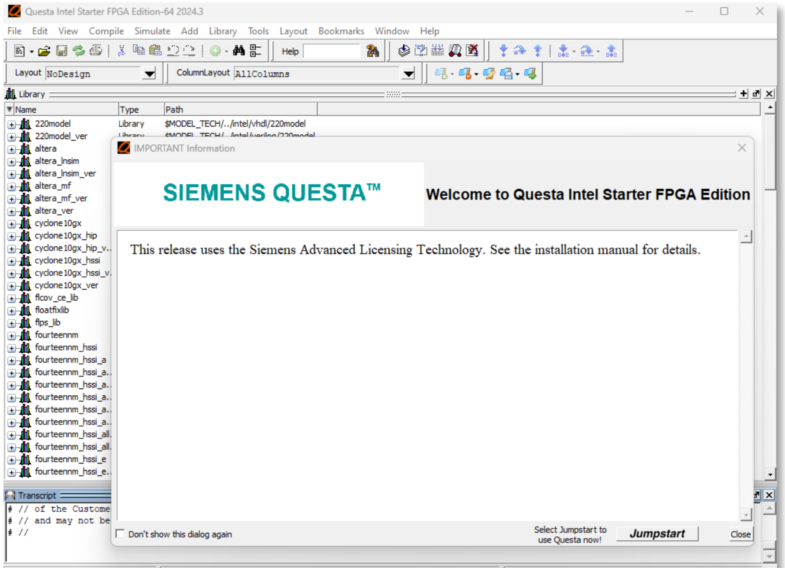
15.1.1 Open the QuestaSim simulation tool.

15.1.1.1 If you are working **in CloudLabs** right-click on the **Questa FSE icon**. Then select **Open**.

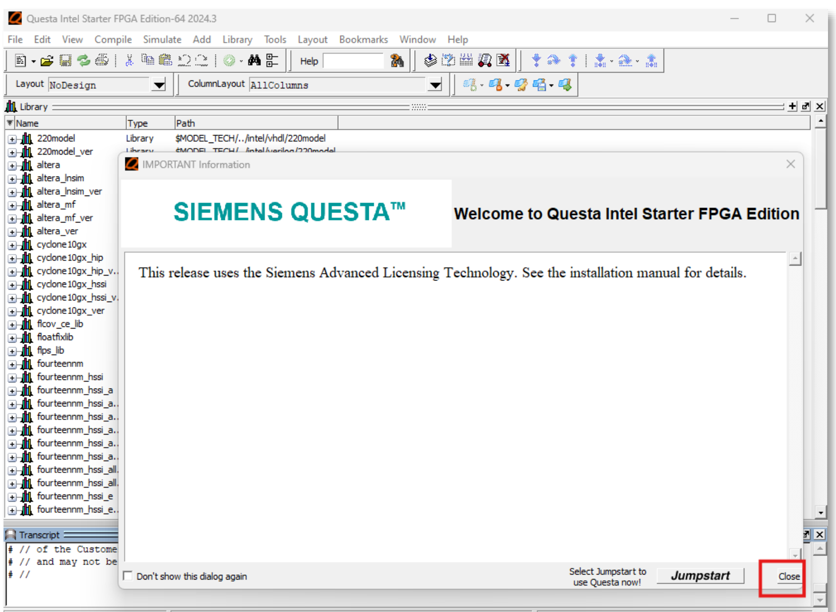
15.1.1.2 If you are working **locally** on a laptop launch from the from the **Windows Start** button.

If QuestaSim does not appear in the Windows Start menu, but it is installed and licensed on your computer, navigate to the ***C:\altera_pro\25.1.1\questa_fae\win64*** and select **questasim** to launch it.

Once opened, the software will appear as follows:



15.1.2 Click Close to remove the Welcome screen.



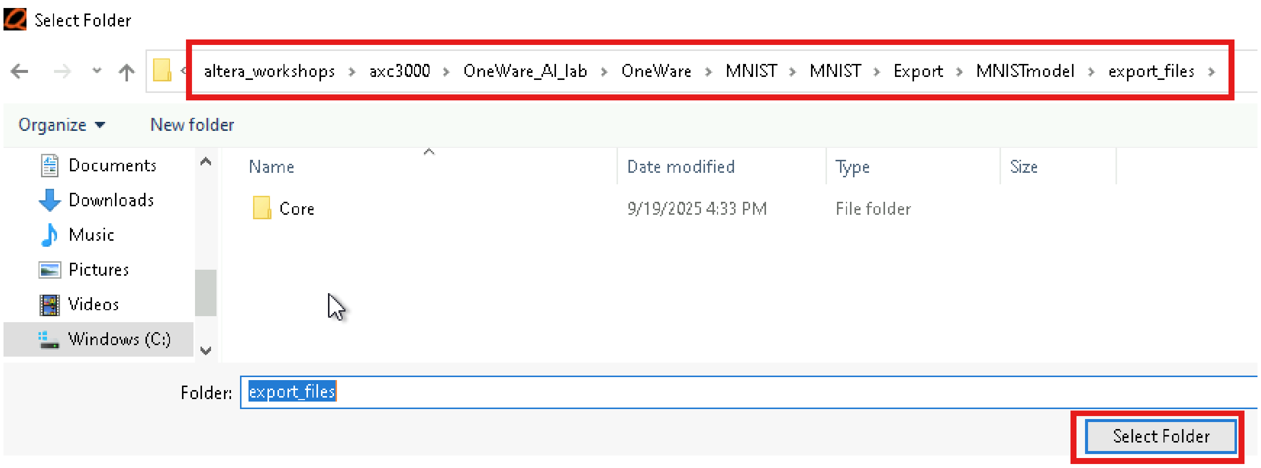
15.2 Navigate to the exported VHDL model files.

15.2.1 Select File →Change Directory and navigate to the /Export folder, which was created automatically when you downloaded the VHDL file.

The Export folder contains the VHDL files that were generated as an output of the OneWare parameter selections. The path is listed below

C:\altera_workshops\axc3000\OneWare_AI_lab\OneWare\MNIST\MNIST\Export\MNISTmodel\export_files

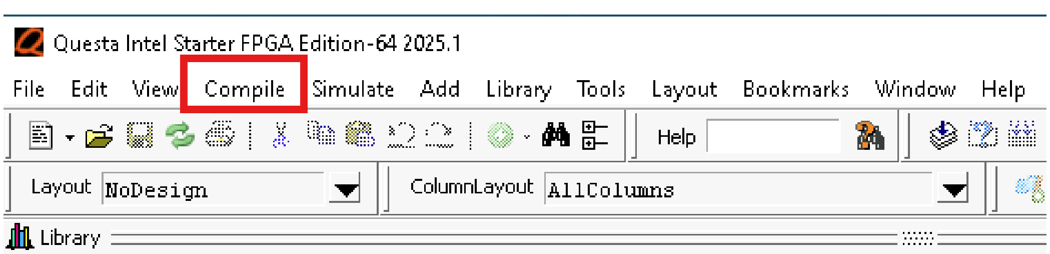
15.2.2 Select the folder.



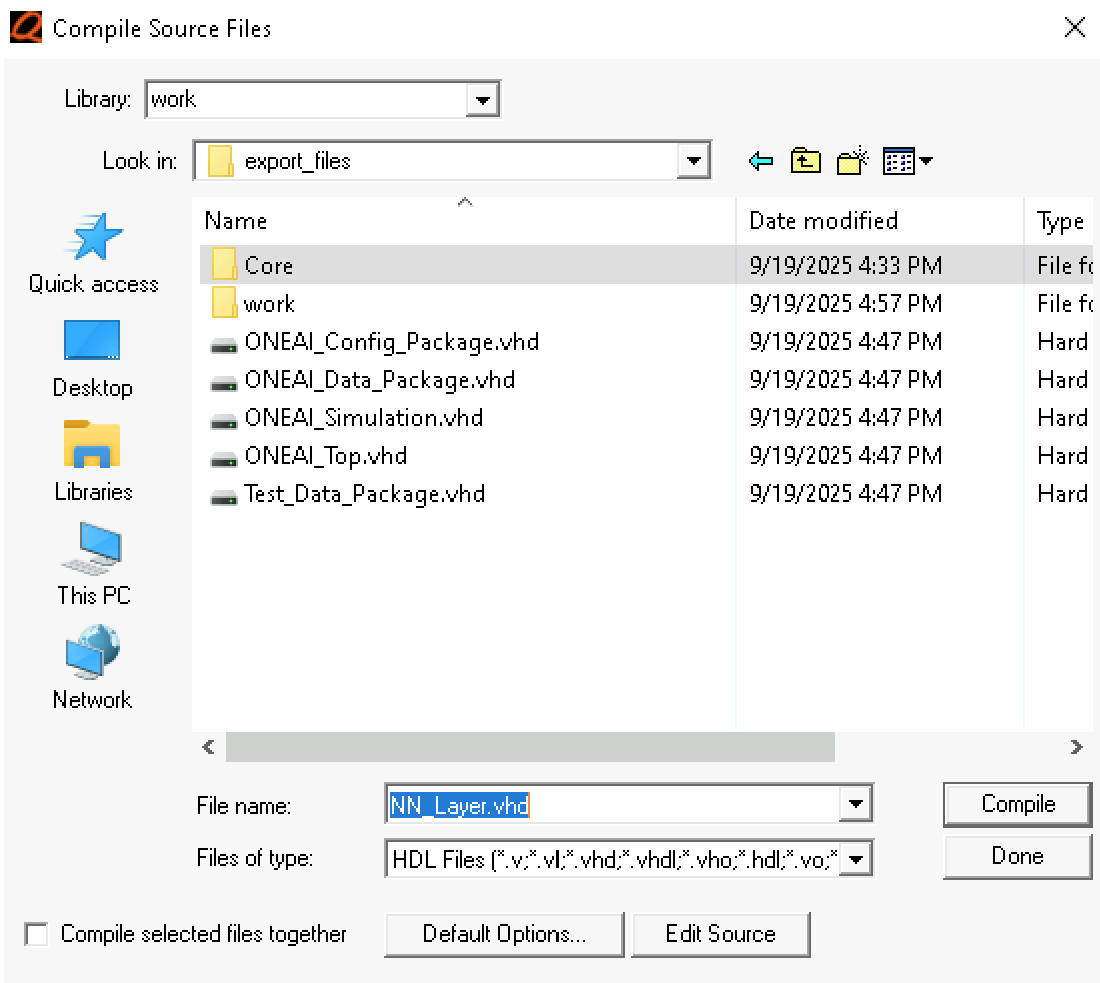
15.3 Compile the Model source files

To use the files for simulation, they need to be compiled. Questa will check the files for syntax issues and create a database that it will use to run simulation.

15.3.1 Click Compile → Compile



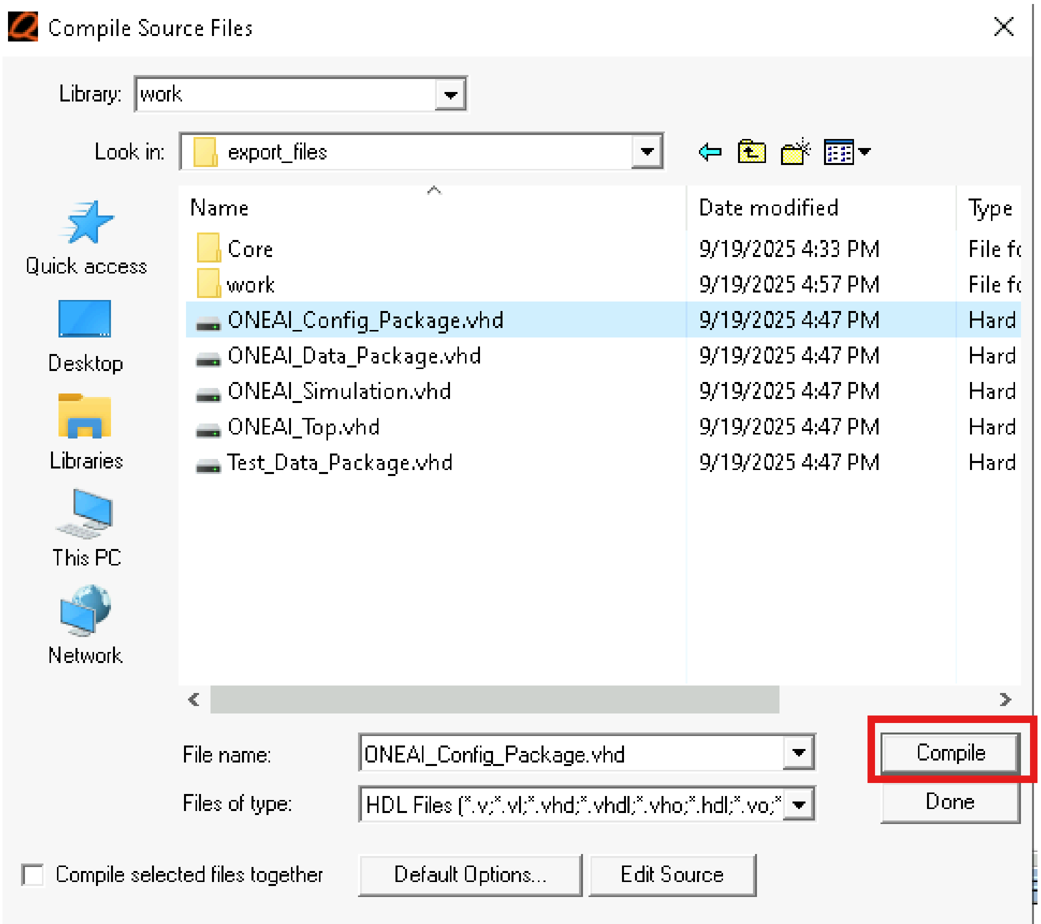
A compile source window will appear as:



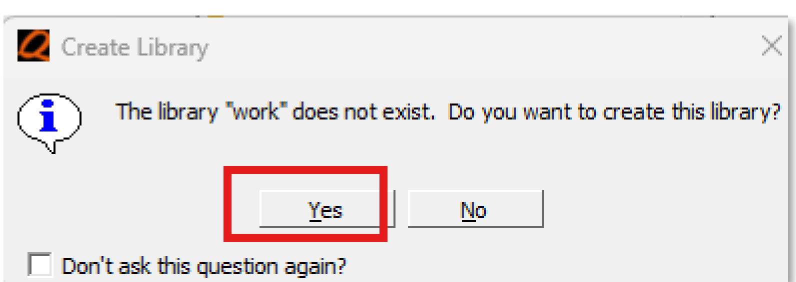
Since the files are written in VHDL, they must be compiled in order from the lowest to the highest hierarchy. (Verilog files, on the other hand, do not have hierarchy dependent). For reference, to determine the file hierarchy, open the VHDL files and review the libraries listed at the top. These libraries can help identify which file is at the lowest level.

After looking in the library headers, it is determined the files to compile (from the lowest level) to the highest level are ONEAI_config_package.vhd, OneAi_Data_package.vhd, CNN_Convolution.vhd, CNN_Normalization.vhd, CNN_Pooling_Efficient.vhd, CNN_Row_Buffer.vhd, CNN_Row_Expander.vhd, NN_Layer_vhd, Test_data_package.vhd, OneAI_Top.vhd and OneAI_Simulation.vhd. The next steps will describe how to compile these files.

15.3.2 Select ONEAI_Config_Package and click Compile.



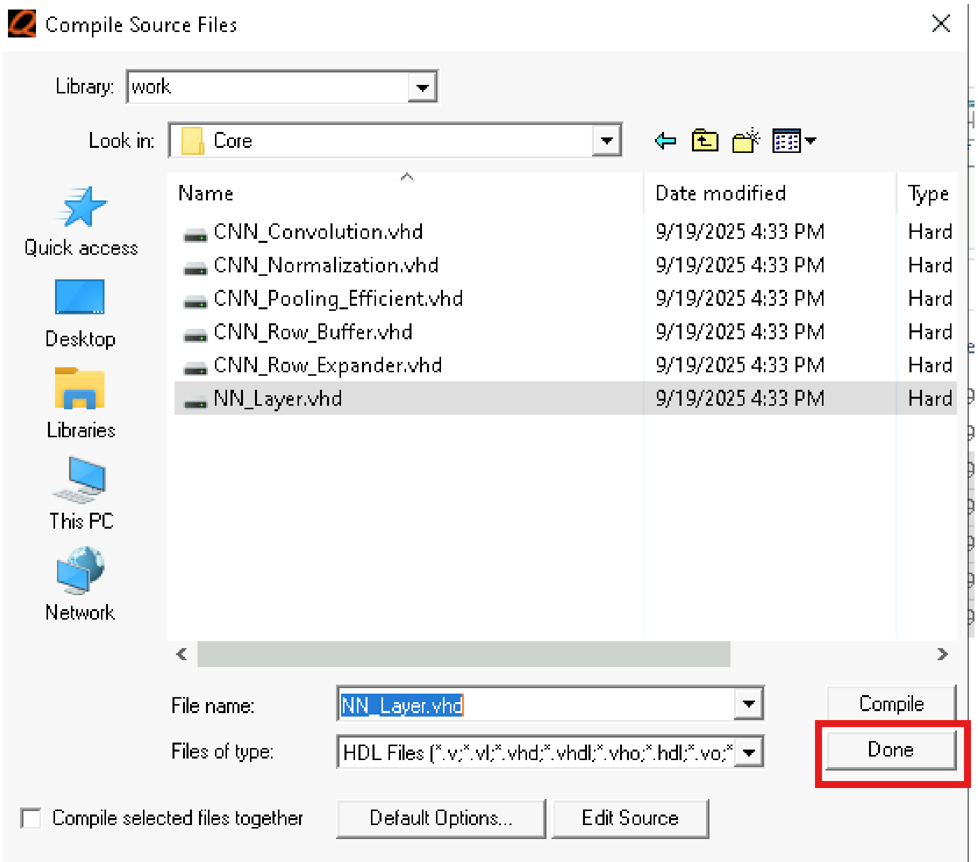
15.3.3 A window will appear asking if you want to create the work library. Select Yes.



There should be no errors at the bottom transcript window

Repeat the previous step to compile the OneAI_Data_Package.vhd file.

Go to the Core directory, select the files in the order indicated below, and click the Compile button.

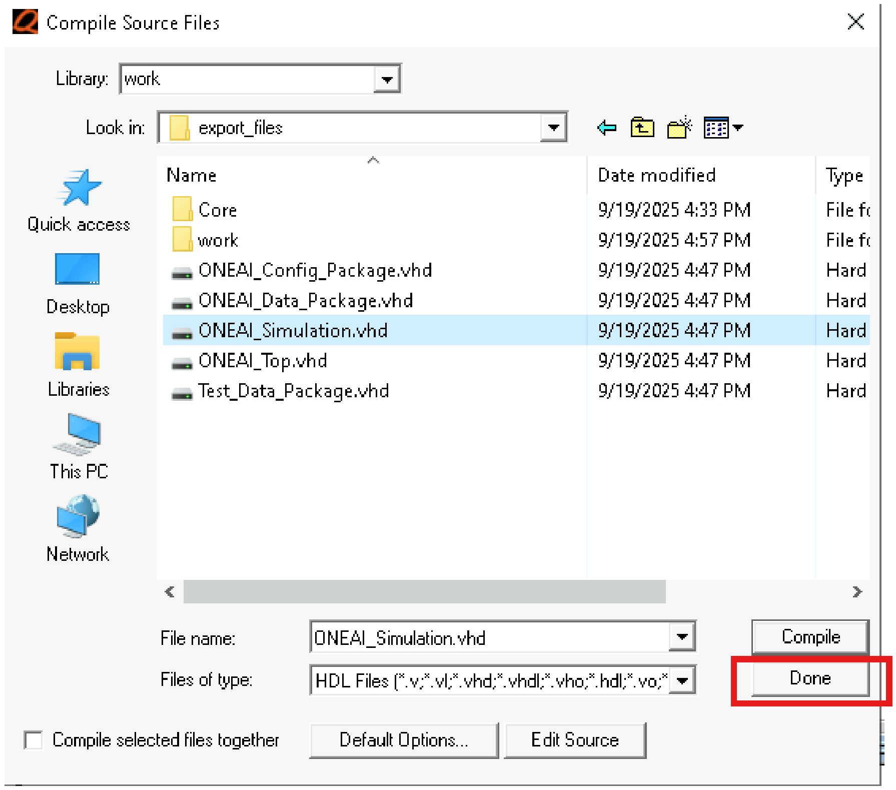


NOTE: When compiling the CNN_Normalization.vhd file, the compiler will display a warning for this file in the transcript window; this warning can be safely ignored.

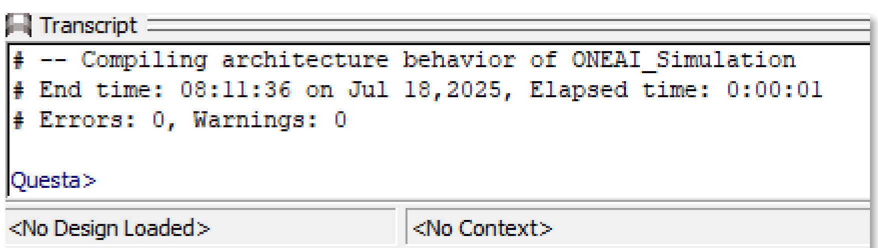
15.3.4 Go back to the top level, to the *OneAi* folder, then select the following files, in order, one by one, compiling each file after selection.

- Test_data_Package.vhd
- OneAI_Top.vhd
- ONEAI_Simulation (top level)

15.3.5 Select **Done**.

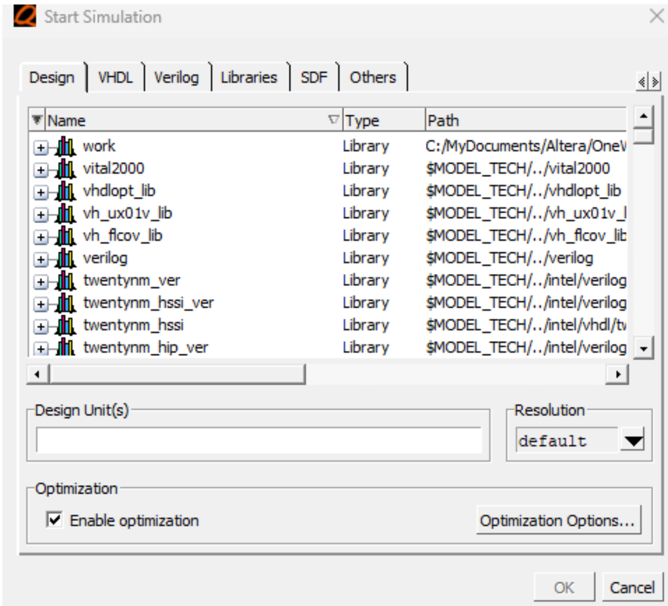


15.3.6 Type ***vsim*** in the transcript window and press Enter



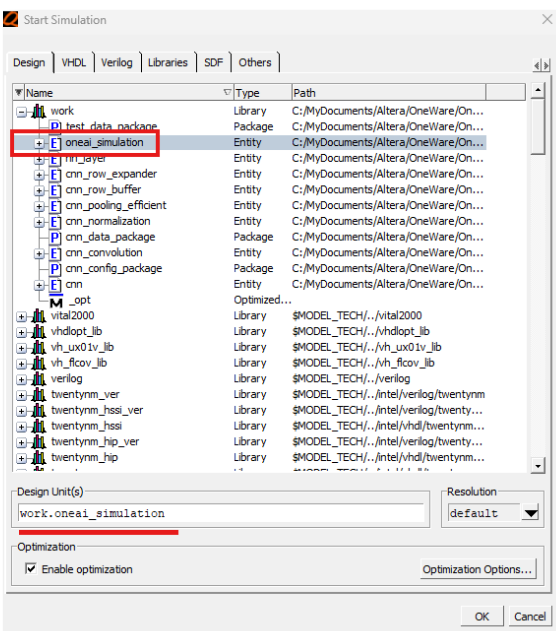
The ***vsim*** command opens the Starts Simulation window in Questa-Intel Starter Editon. This step, after a delay, will open the simulation window. The simulation window allows the selection of the files that need to be simulated and simulation options to be selected.

The window will look like:



15.3.7 Expand the **work** directory and select ***oneai_simulation***. This file is the top-level of the design.

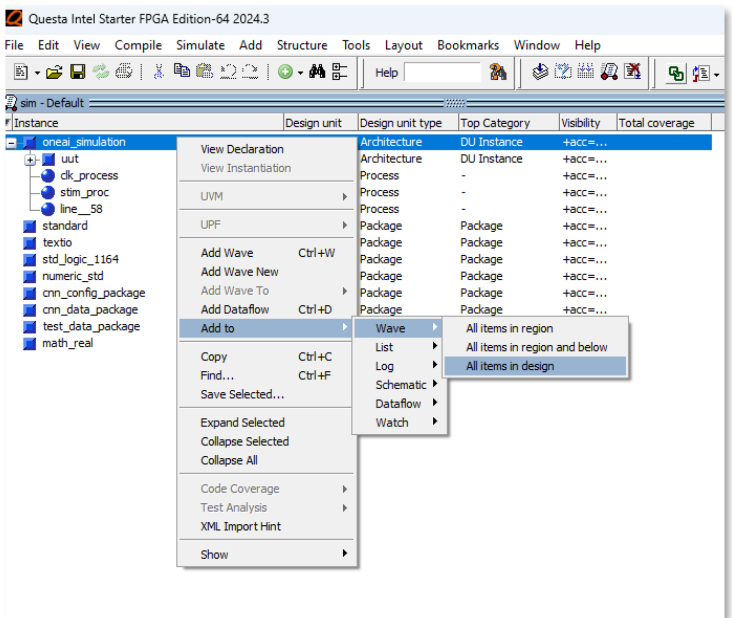
15.3.8 Click OK



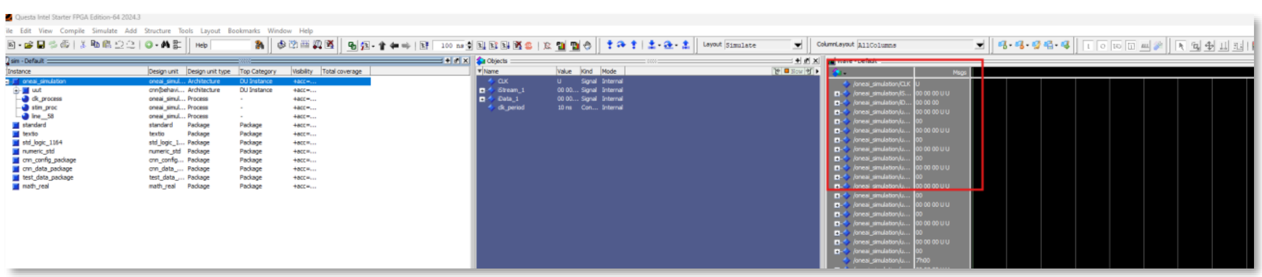
15.4 Simulate the model

15.4.1 Right click on the oneai_simulation. Select → **Add to** → **Wave** → **All items in Design** .

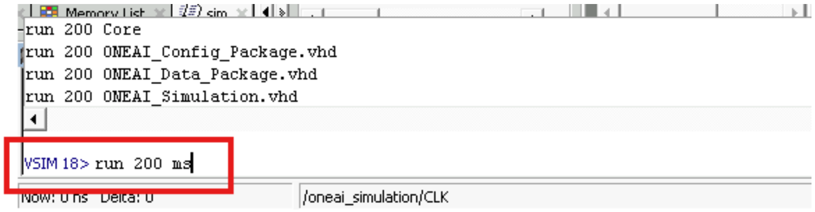
This step adds various signals from the testbench file and below (what the testbench calls) to the Wave window**.**



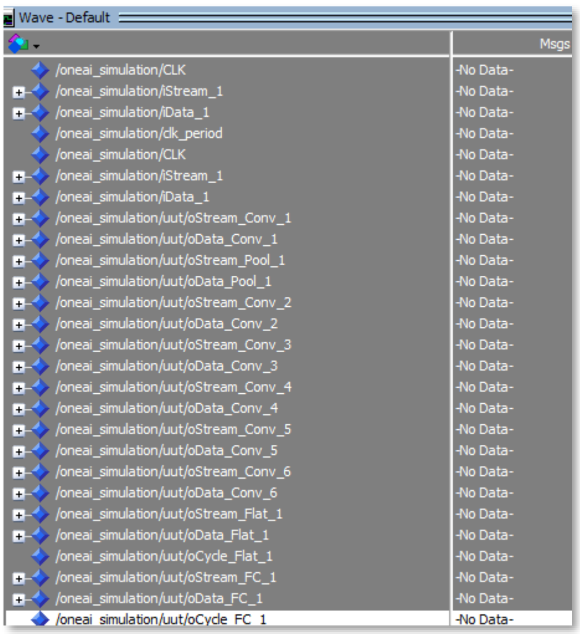
The window will show all the signals on the simulation window:

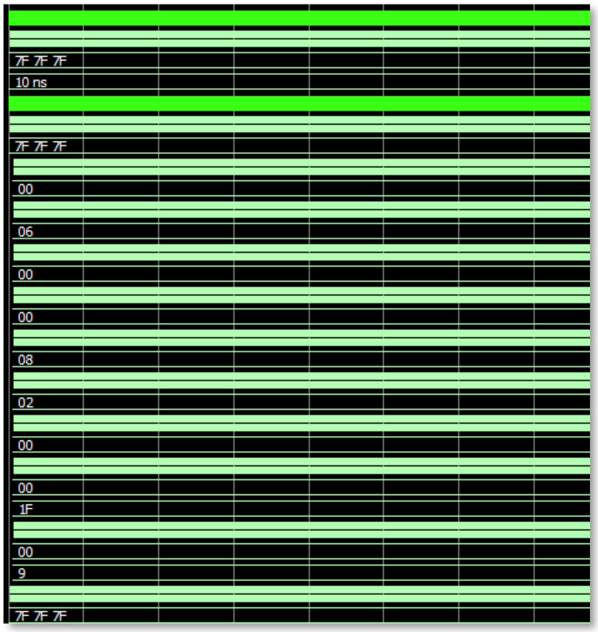


15.4.2 In the transcript window, type the command ***run 200 ms*** and press the **Enter** key.



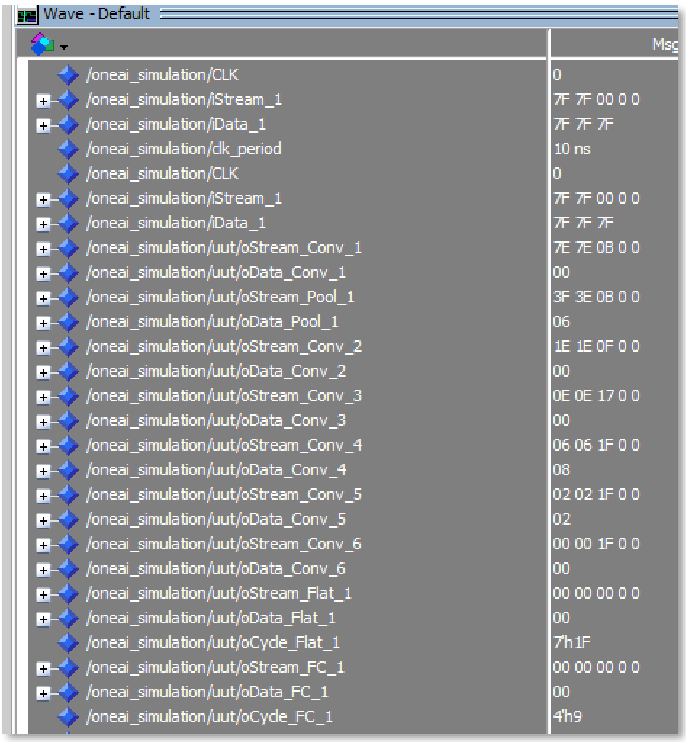
The results are captured below.





Here are some important points to note:

- This workshop will not run through the entire simulation, as it will take much too long.
- Simulation time is affected by the FPS (frames per second) setting, simulator, pixel size, and the *fmax* value configured in the OneWare software. In this case, the model was set to a minimum of 60 FPS, meaning there is one frame processed per second. The simulator is the free version with Quartus, which is the slower simulation version. The fmax was set to very low. The pixel size is large (128×128).
- By examining the signals, you can observe the differences between the LeNet-5 model and this custom model in terms of the number of settings.



iData_1 signal represents the input values. In this case, the values **7F 7F 7F** indicates an all-black input.

This model contains:

- 6 convolution stages (Conv_1 to Conv_6),
- 1 pooling stage (for downsampling),
- 1 flatten stage (which converts the data into a 1D array required for the fully connected layer), and
- 1 fully connected layer (FC_1).

A common question is: Why does this model have a different number of stages compared to the LeNet 5 model?

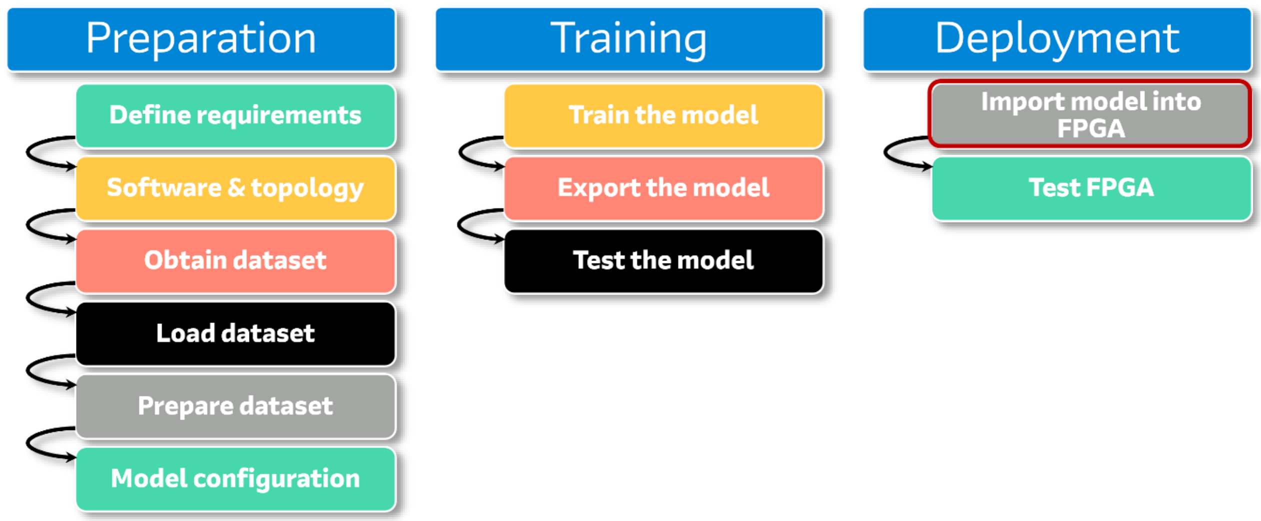
When you use OneWare software, it generates a custom AI model based on the parameter you select. In this case, the specific settings chosen for filters and model configuration produced the resulting architecture. The input size is also much larger (128×128) compared to LeNet-5 ‘s common (28×28) (and sometimes 32×32) input size.

The associative files represent the model's output. If different parameters are entered, the model is structure will change accordingly. Using OneWare allows you to build AI models without manually writing the code, enabling you to experiment directly with your data and test different model configurations.

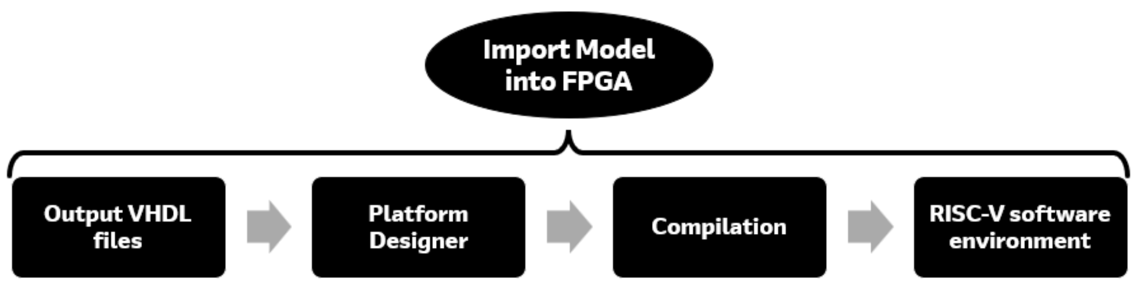
15.4.3 Close the simulator.

16 Import Model into FPGA

After the model is verified via software, the next step is to implement the AI model in hardware as shown in the red box.

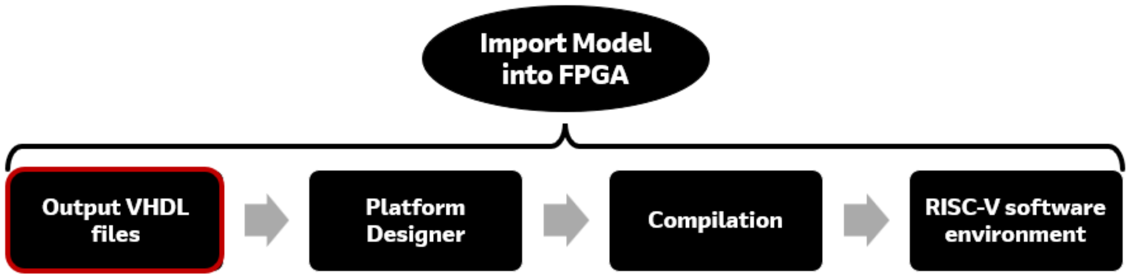


The *Import Model into FPGA* stage consists of several sub-steps, as outlined below.

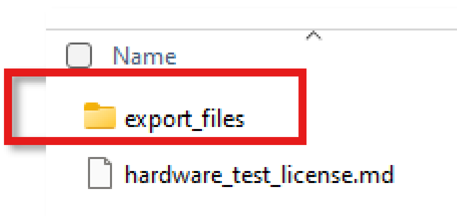


17 Output VHDL Files

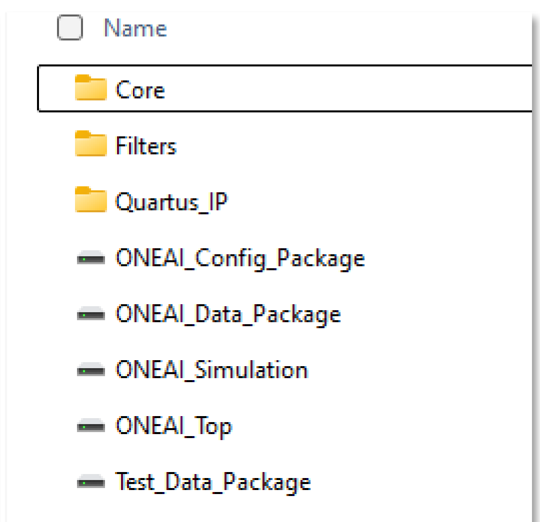
The first step of this stage is generating the VHDL output files.

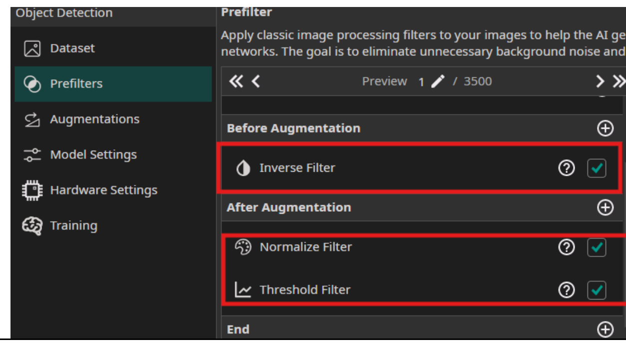


These files were exported after the training stage. In the Windows Explorer, go to the directory …\ ***OneWare\MNIST\MNIST\Export\MNISTmodel*** there are several VHDL files under the ***export_files*** folder.

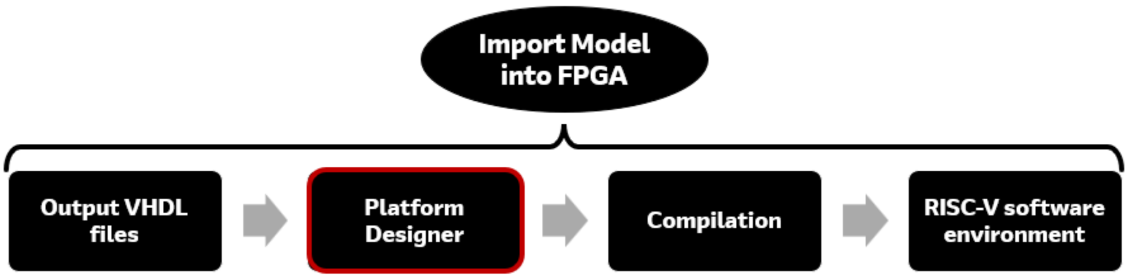


There are files under each sub-directory of Core, Filters, Quartus IP and some stand-alone files.



Directory or File	Details
Core	CNN associated files as Convolution, Normalization, and Pooling (down sampling)
Filters	Associated with the Prefilters that you set earlier such as inverse, normalize and threshold i.e. 
Quartus IP	Files that are used with the FPGA software, Quartus, more specifically the Platform Designer, so that they can be easily connected to other components
OneAI_Config_Package	Used for the model configuration
OneAI_Data_packages	Defines the values of constants including the model weights
OneAI_simulation	Can be used for simulation
OneAI_Top	Top level that “calls” the files in the Core and Filters sub-directories
Test_Data_Package	Contains a test where there are numbers between 0 or 127. Generally, 0 is for the color blackcolor black and 127 is for the color white.

18 Platform Designer



Platform Designer, a GUI based system integration tool within Altera's Quartus software, enables quick integration of various IPs, IP functions and systems. It supports "plug-and-play" IPs, multiple industry standard interfaces and automatic HDL generation.

A lab project folder has been pre-populated with several files and directories. These represent the complete design minus the AI IP block that was generated in the previous section by OneWare Studio. Use will be made of the platform Designer tool to integrate the AI IP into the system.

The contents of the project are shown below

- Qpro
- scripts
- software
- sources

Directories	Description
Qpro	FPGA software files
Scripts	Various scripts including details for the camera sensor
Software	Setup files for the soft core (Nios V) processor
Sources	Top level file, pin files and timing constraint file

18.1 Move OneWare AI IP

Before using Platform designer, the AI IP generated by OneWare Studio must be placed in the correct location so that Quartus can properly load them.

18.1.1 Using Windows Explorer, create a folder

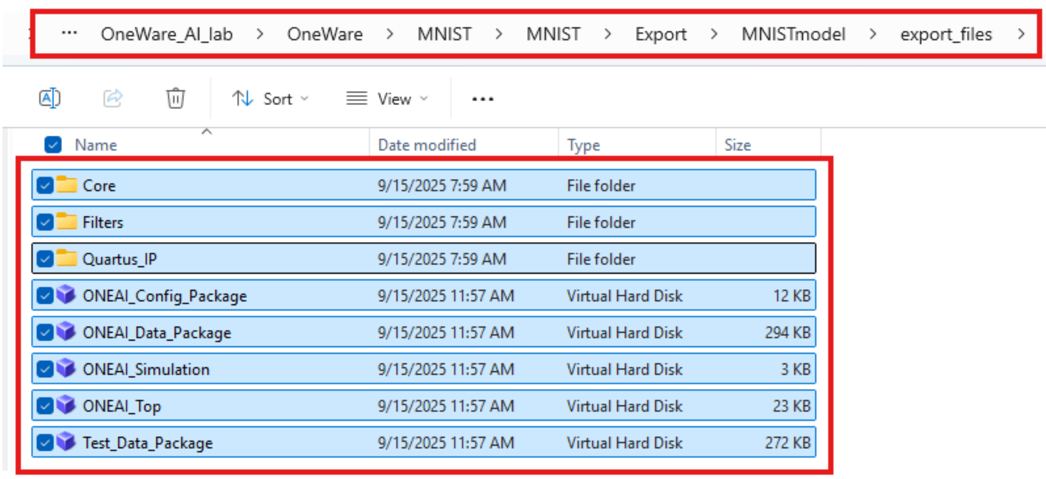
c:\altera_workshops\axc3000\OneWare_AI_lab\Qpro\one_ai

18.1.2 Select all the files in the export files subdirectory and copy them into the

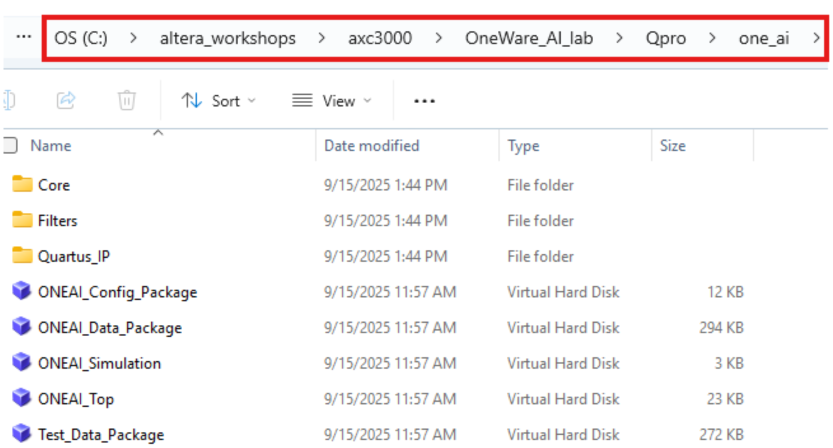
c:\altera_workshops\axc3000\OneWare_AI_lab\Qpro\one_ai

subdirectory.

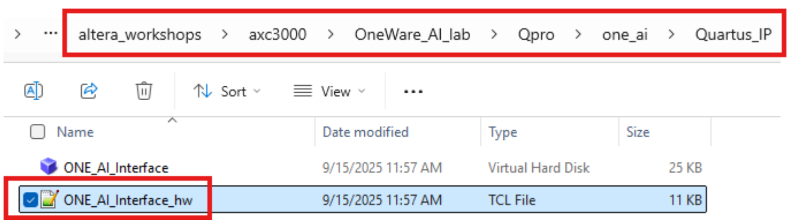
Source directory



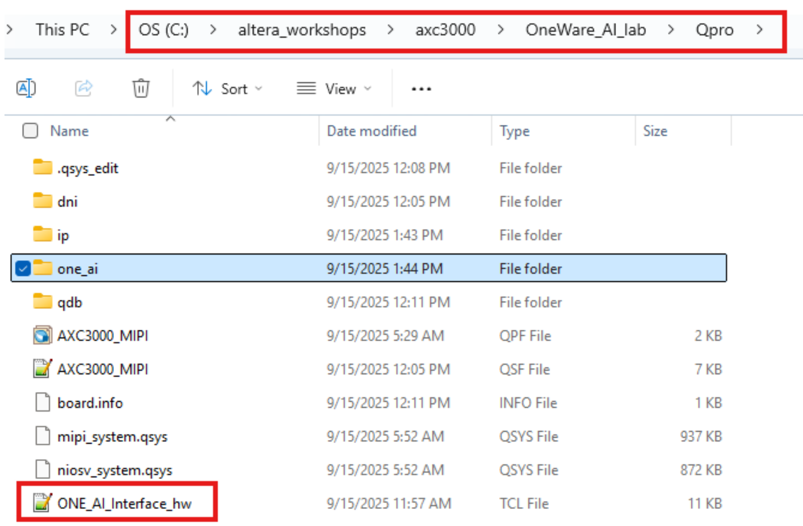
Destination directory



18.1.3 Move the file ONE_AI_Interface_hw.tcl from the one_ai\Quartus_IP subdirectory



to the **QPro** directory.



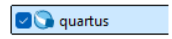
This file is read by the library manager in Platform Designer. It describes the hardware features of the ONE_AI IP core.

18.2 Open Quartus project

The next step is to use Altera's Quartus software to connect, via the Platform Designer, the generated AI files with other pre-created files to create a complete design containing an AI model in it.

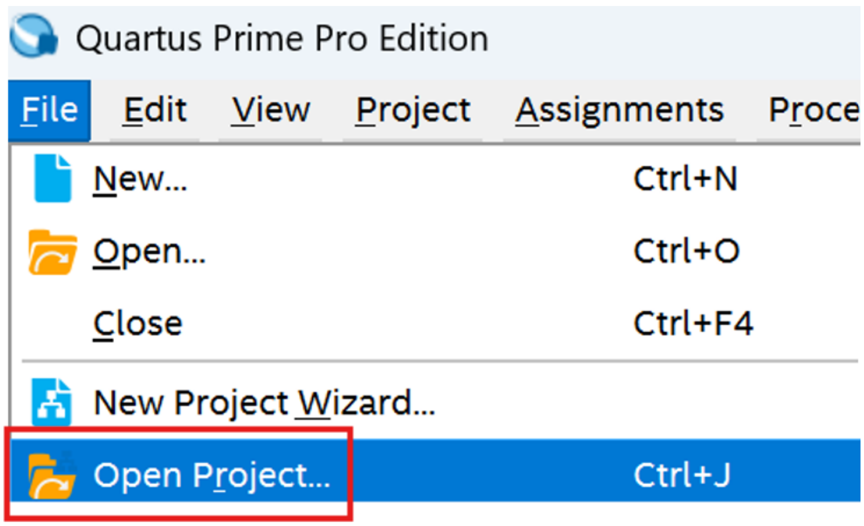
Quartus Prime software is a powerful, design environment that supports all the steps of FPGA development including design entry, synthesis, optimization, verification and programing

18.2.1 Launch from the Windows Search → 25.1.1

If by chance, you do not see the Quartus in the Search Window, go to c:\altera_pro\25.1.1\quartus\bin64\ and select the 

application.

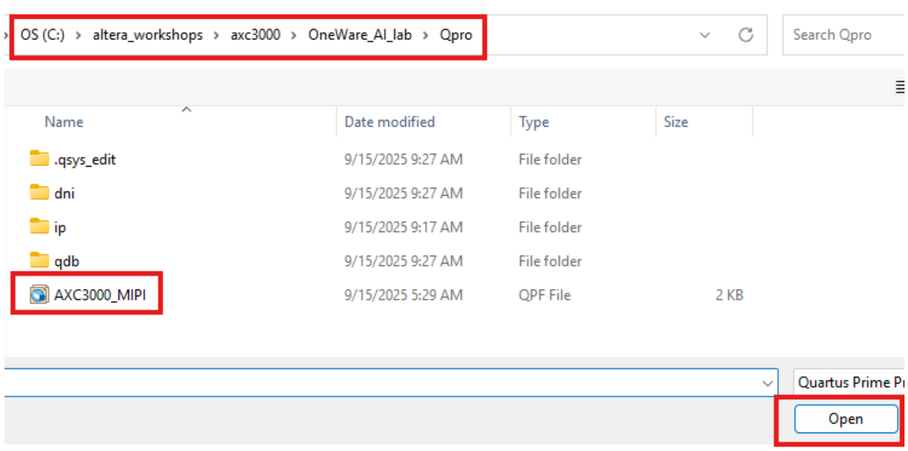
18.2.2 Select in Quartus: **File** → **Open Project**



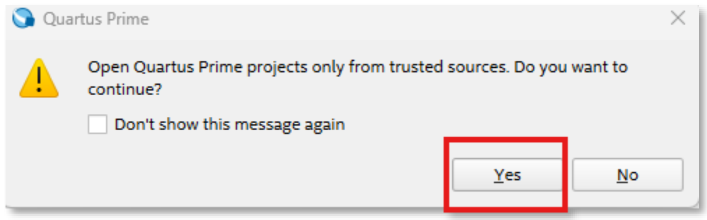
18.2.3 Navigate to

C:\altera_workshops\axc3000\OneWare_AI_lab\Qpro

18.2.4 Select **AXC3000_MIPI** and click Open.

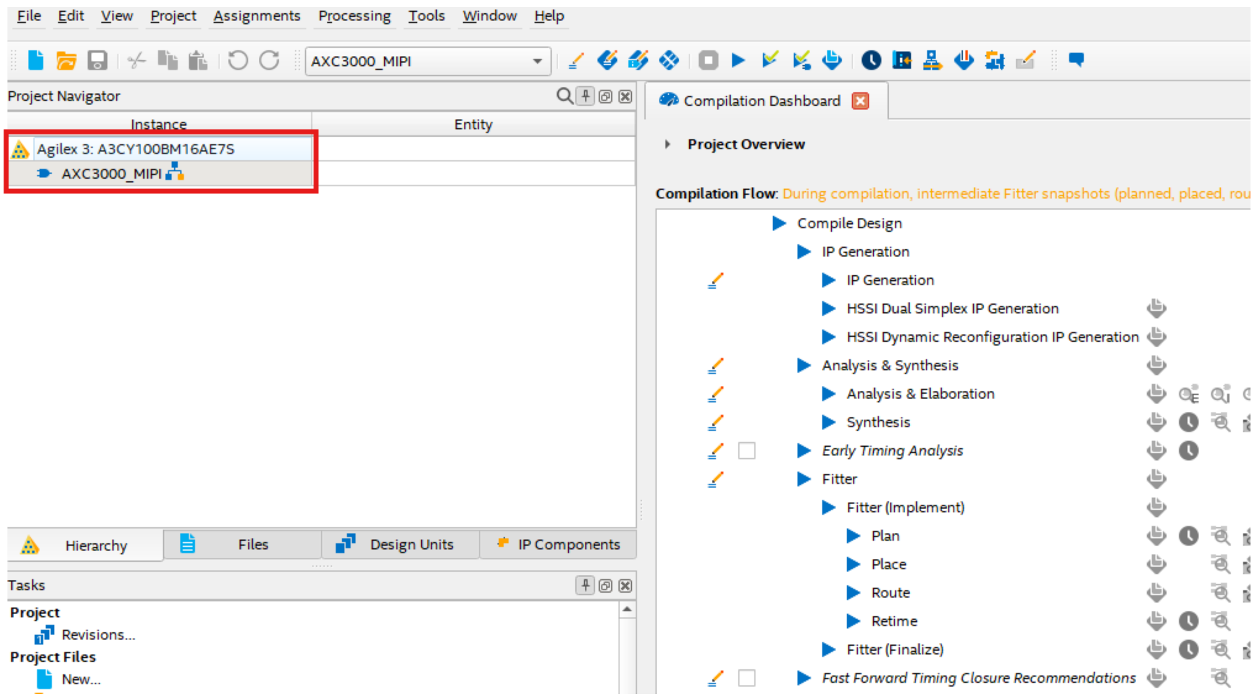


18.2.5 When the pop up window appears for the trusted source, select Yes



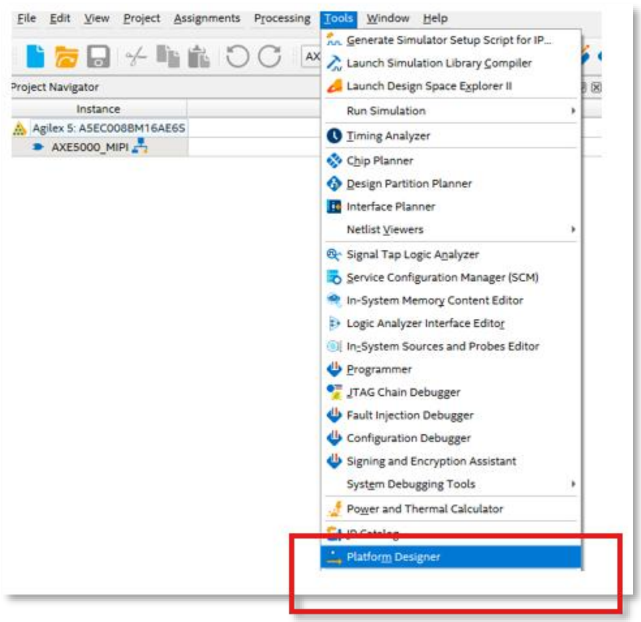
The Quartus window will now be launched.

Note the Agilex 3 device, the device on the dev board that will be used, is already selected.



18.3 Open Platform Designer project

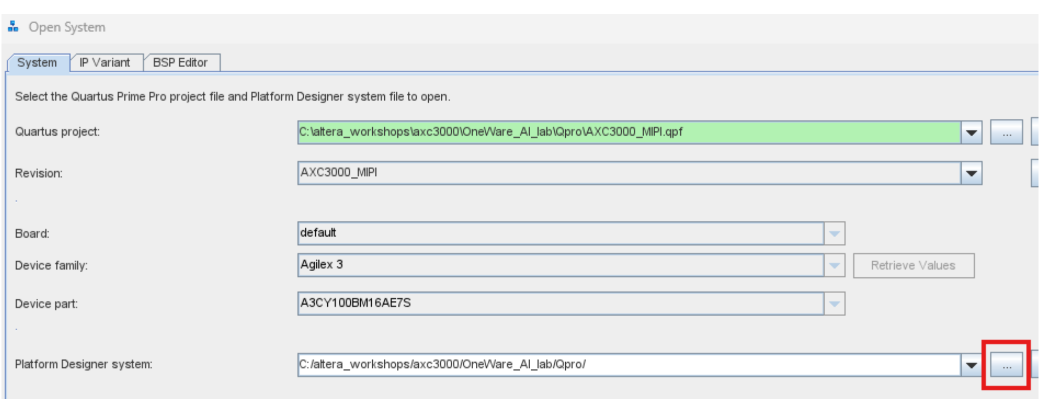
18.3.1 In Quartus select: Tools → Platform Designer



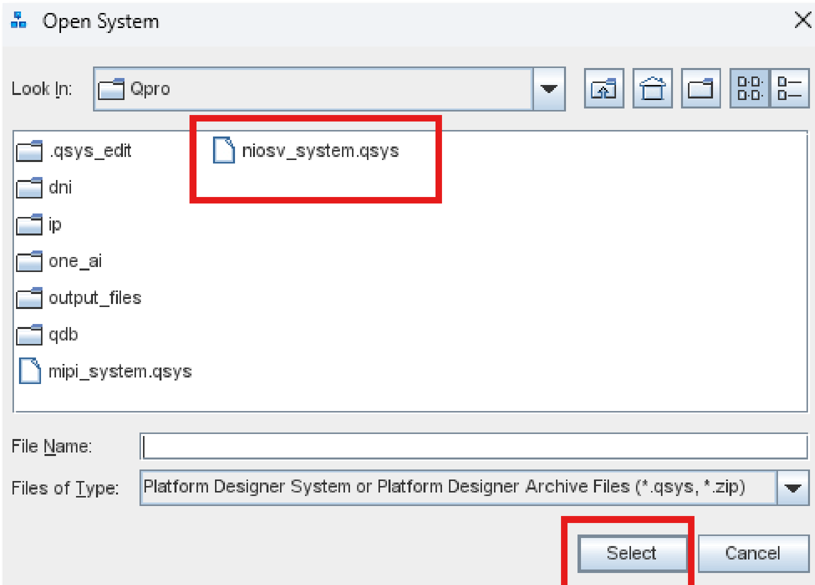
18.3.2 Close the **What's New Platform Designer** window by selecting the X as this window is not needed.

Platform Designer is a system integration tool. It is required that a system file is selected.

18.3.3 Select Search button



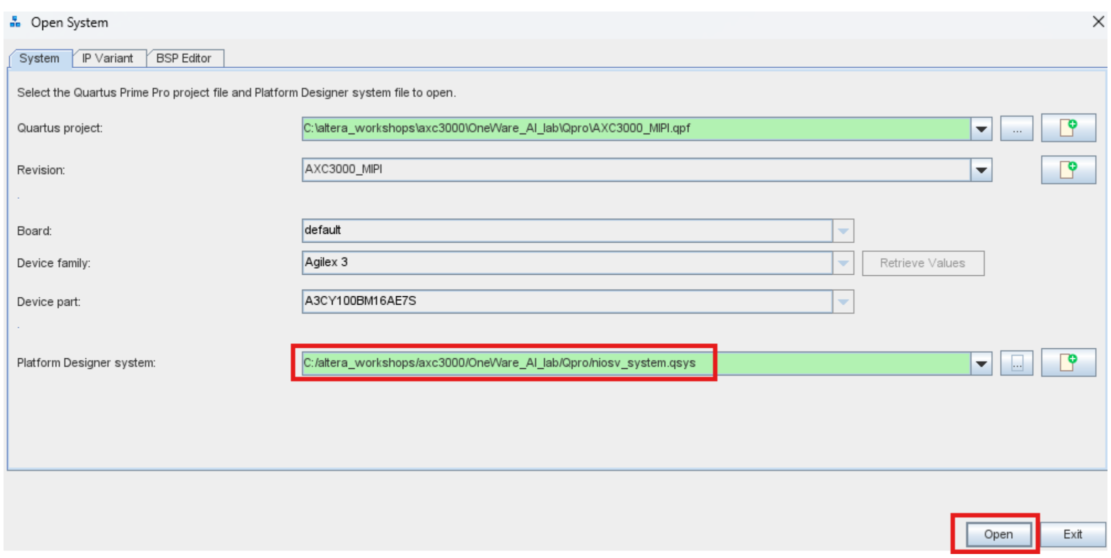
18.3.4 Select the **Niosv_system.qsys** file and then choose the Select button.



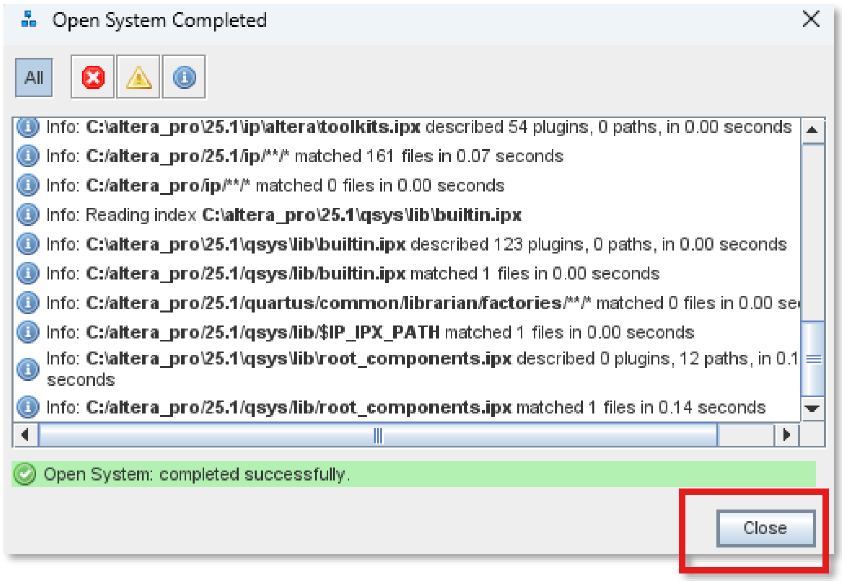
The Niosv_system.qsys file is a file that contains a list of various **parametrized components** including a soft-core processor that is used.

The window will now appear as shown below (highlighted in green).

18.3.5 Click **Open**.

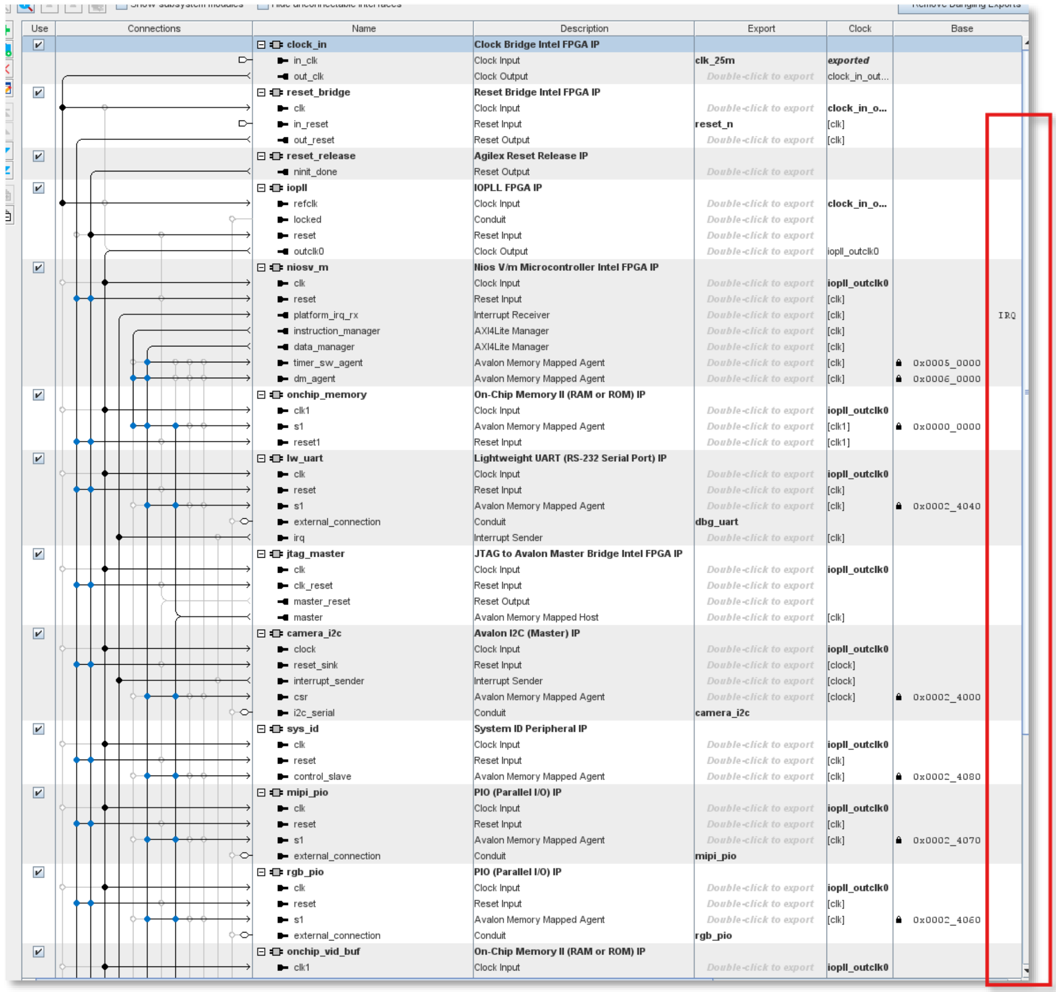


18.3.6 A window will appear as follows. Select **Close** once it shows completed successfully (in green).

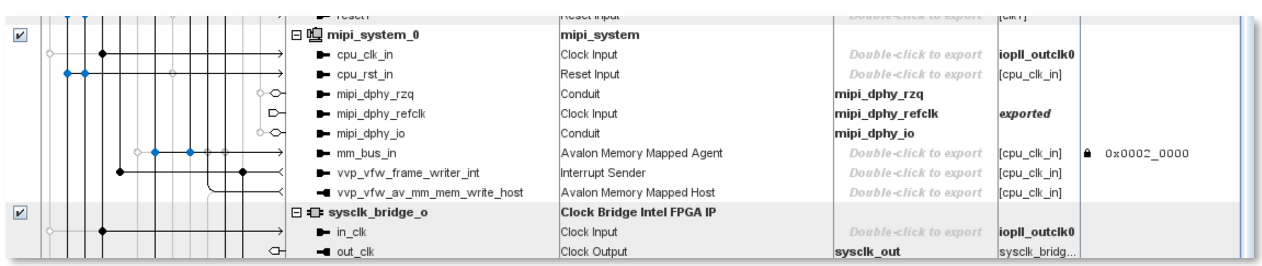


In this step, it is loading the various components.

The Platform Designer offers many pre-built yet customizable IP functions for use in a design. The main window will look like:

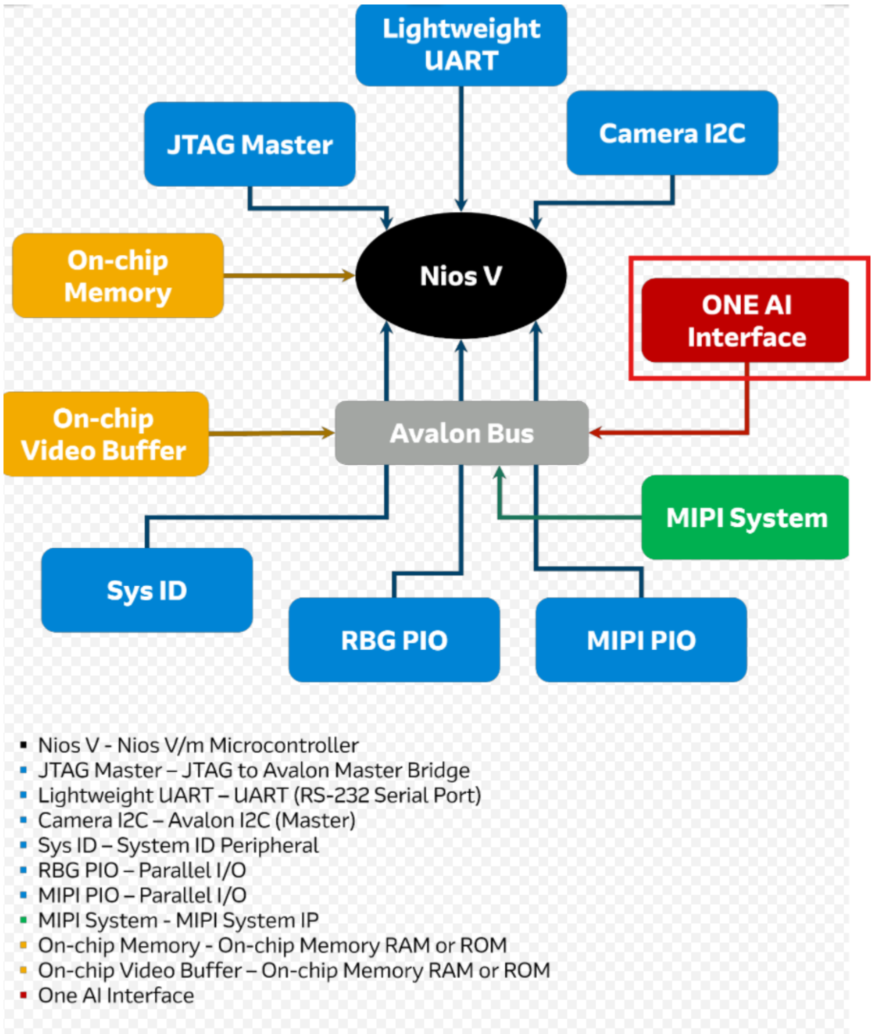


If you select the scroll bar on the right-hand side, there will be more IPs. The order of the IPs in the Platform Designer does not matter.



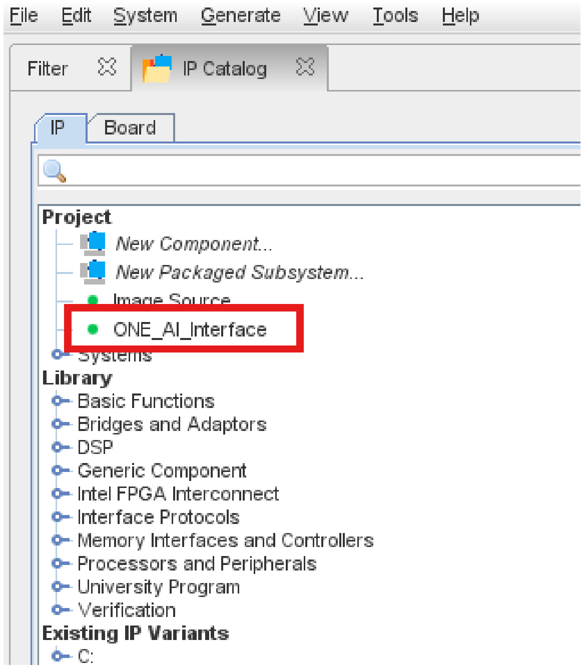
Since this workshop focuses on AI rather than adding additional components, many of the necessary connections have already been made for you.

The following block diagram lists the IP cores and shows how they are connected. The design includes components such as the JTAG Master, PIO, MIPI system, Nios V Microcontroller (soft-core processor based on RISC-V), and more. A brief description of the IPs can be seen after the diagram. The block in red, (OneWare AI model), is the only component that you will be adding and connecting.



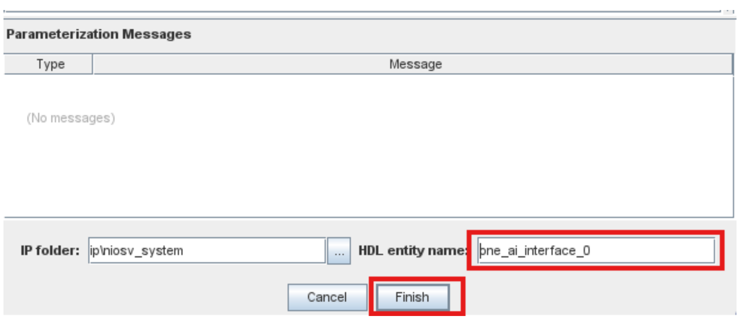
18.4 Add OneWare AI IP block to the System

18.4.1 Go to the left-hand side of the window and double-click on the **ONE_AI** Interface as shown below.



The ONE_AI_Interface is entered as a reusable, custom HDL component that you can use in your design. The next steps are to connect the AI model to the rest of the components.

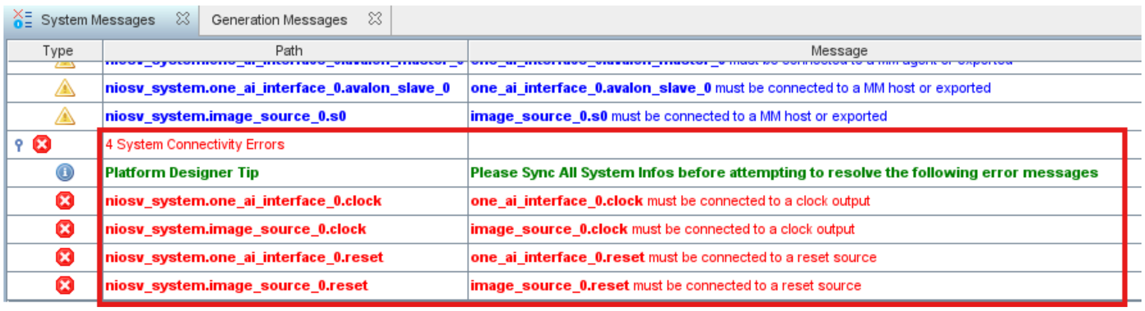
18.4.2 A window will open. Edit the HDL entity name to be **one_ai_interface_0**.



The **BUFFER_BASE_ADDRESS** defaults to **0x0008 0000**. This is the address of the onchip_vid_buf **video buffer memory**. The camera will be writing frames of video into this buffer, and the ONE_AI IP needs to read those frames from the same location in order to process them.

18.4.3 Click Finish.

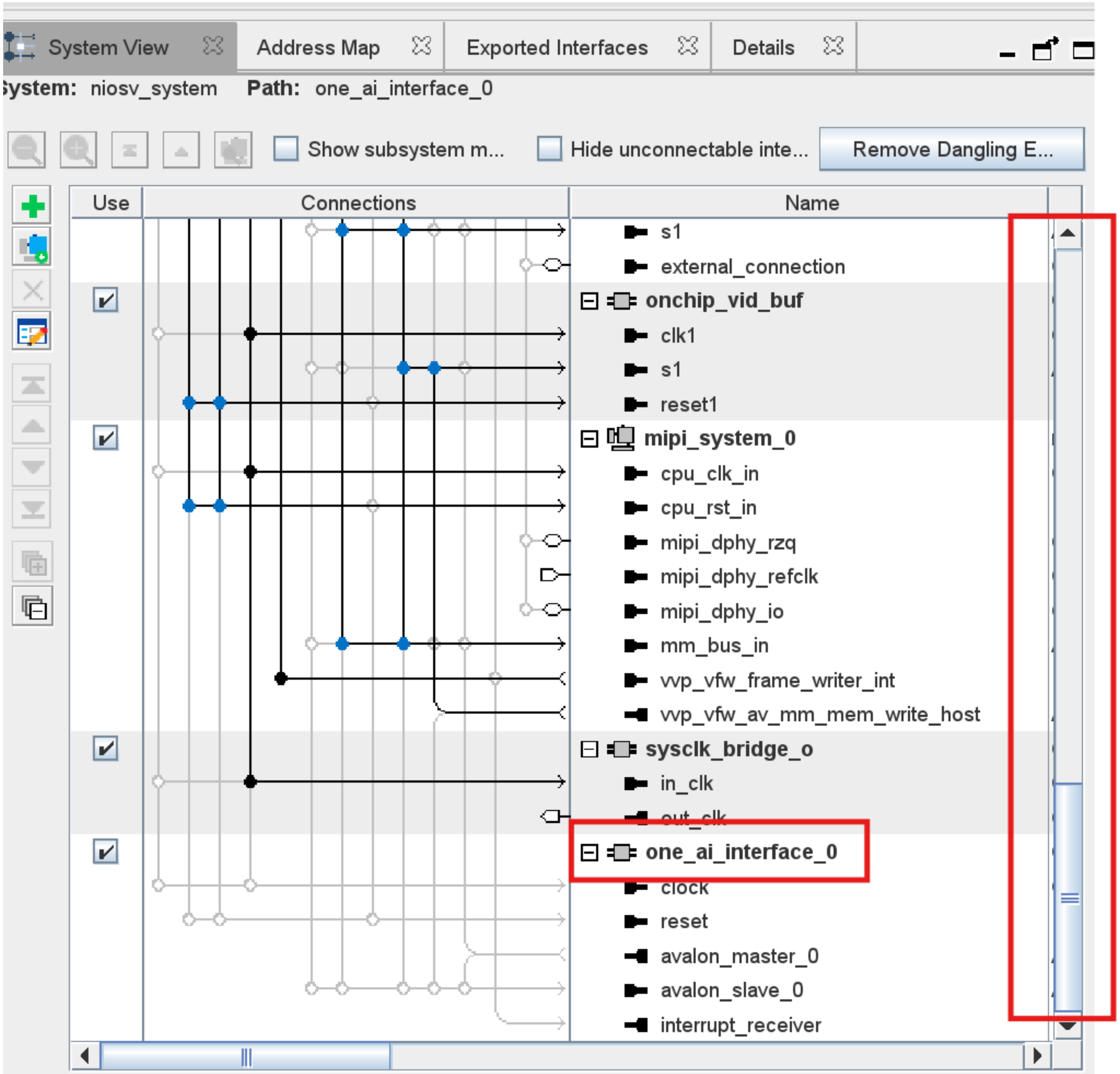
You will see warnings and errors at the bottom of the window as:



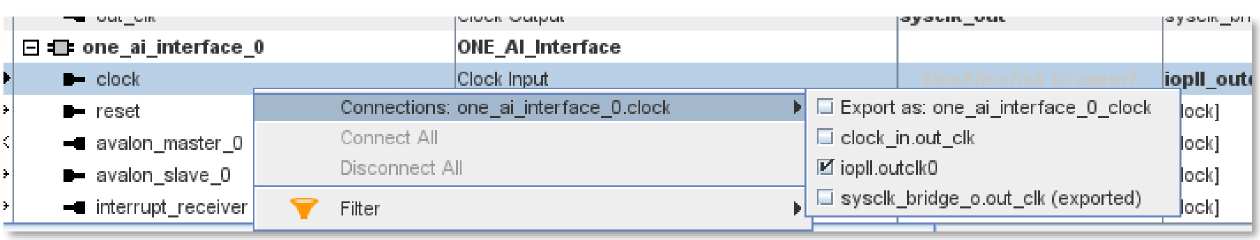
These warnings and errors, which are expected, are because this component is not yet connected to the other components. The error messages and warning messages will be removed in the following steps where the connections are made.

The next step is to connect the One_AI component to the other components.

18.4.4 Scroll down the Platform Designer window by using the **slider** window to get to the one_ai_interface.



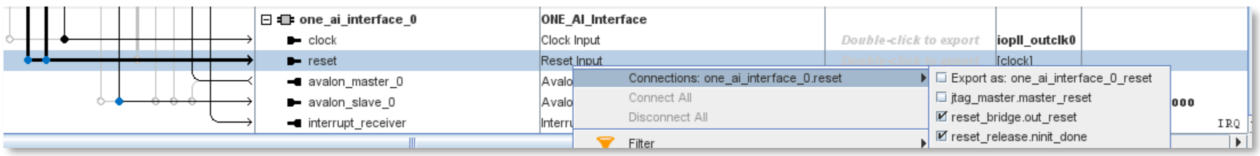
18.4.5 Right click on the ***Clock Input*** signal, select connections and select the ***iopll.outclk0*** to connect it.



You can just move your mouse away to see the connections.

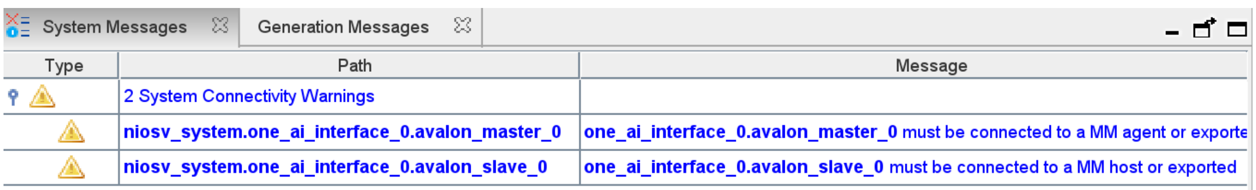
There will now be a connection for the clock of the *one_ai_interface* to the *top-level clock*. One error will now be removed.

18.4.6 Right click on the *Reset Input* signal, select Connections and select ***reset_bridge_out*** and ***reset_release_init_done***.



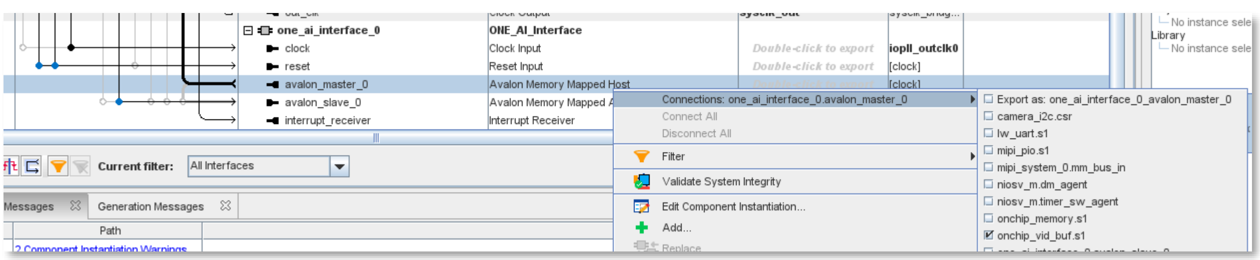
Another error message will be now removed, as the reset is now connected to other reset signals.

There will be two system connectivity warnings as



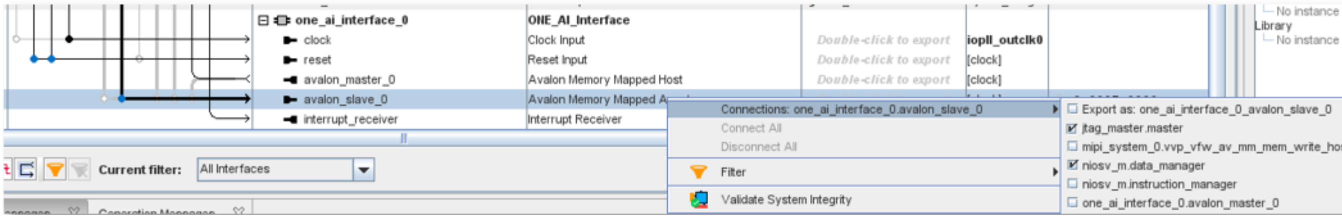
Which will now be removed.

18.4.7 Right click on the ***avalon_master*** signal, select Connections, and select the ***on-chip_vid_buf_s1*** where this is the video buffer.



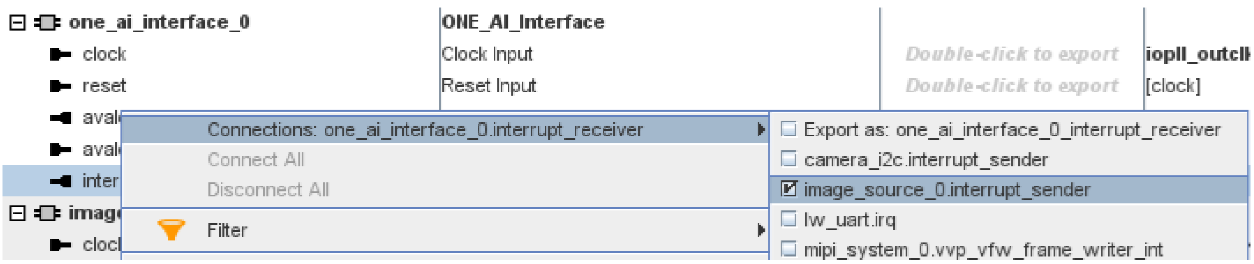
This means that the OneAI will be connected to the video buffer.

18.4.8 Right click on the **avalon_slave** signal, select Connections and connect to **nios_v_data_manager** and the **jtag_master.master** signal.

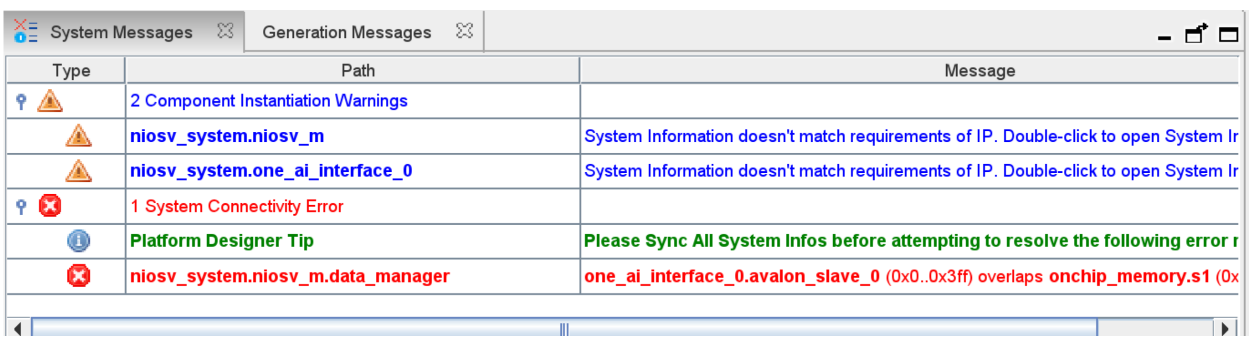


An error will appear that there are overlapping memories, but these will be removed shortly.

18.4.9 Right click on the ***interrupt receiver*** signal, select connections and select the ***image_source_0.interrupt_sender*** to connect it.

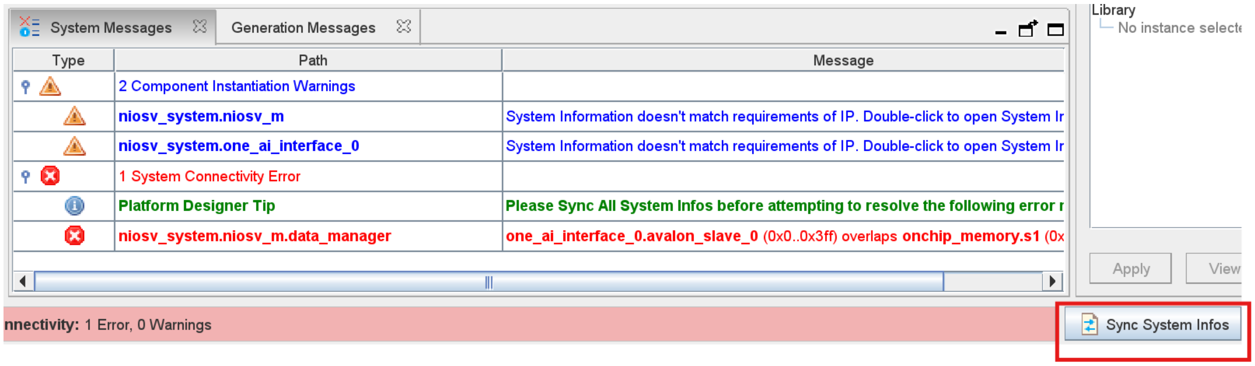


You should see the following messages



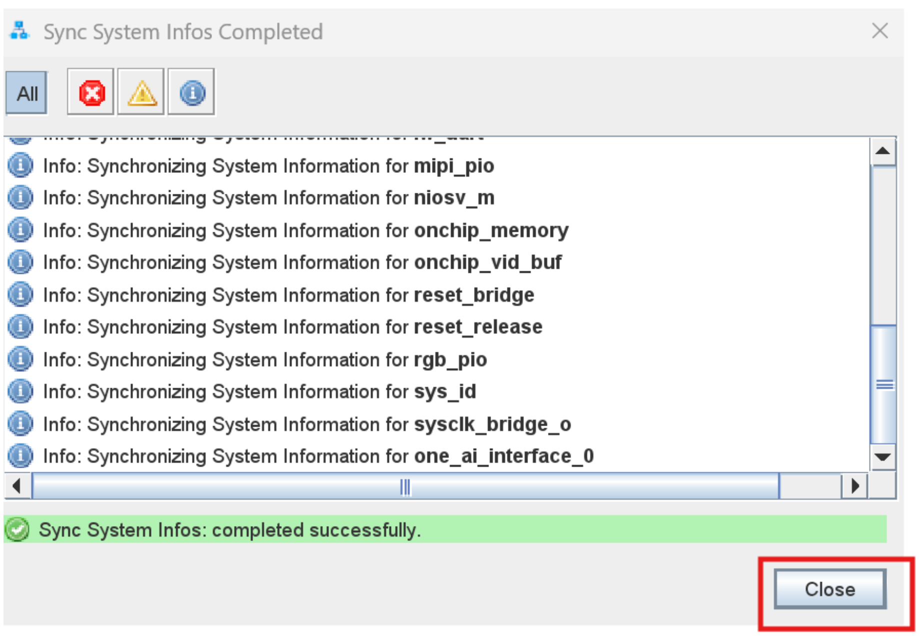
There are 2 main items. One is that the system info does not match, and the base addresses of different components overlap each other.

18.4.10 Select **Sync System Infos** button

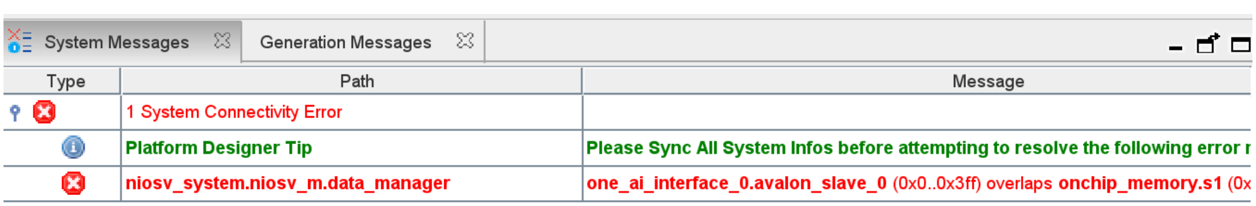


The step resolves the mismatches between the IP and the actual parameters.

18.4.11 Select Close

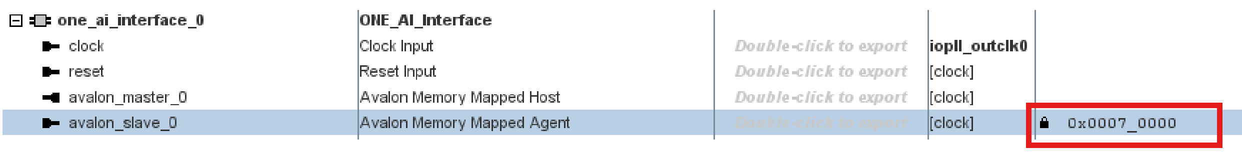


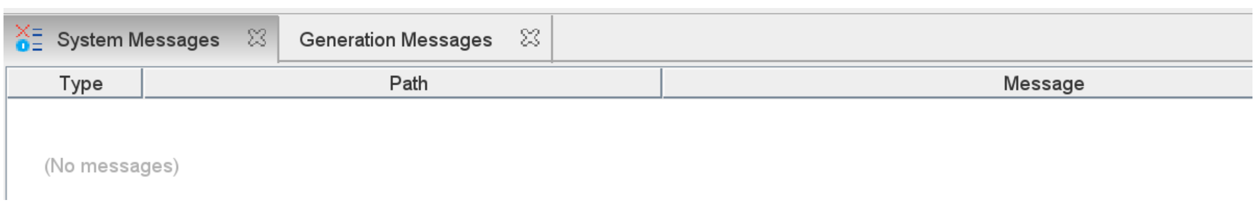
Notice how the number of errors and warnings are reduced to



The next step is to modify the **overlapping** memory address.

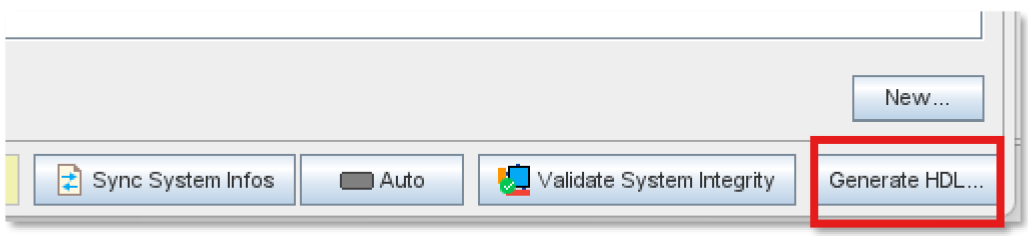
18.4.12 Double click in the base Address field for the one_ai_interface_0 component. Modify the value to be 0x0007 0000. Click the lock symbol to ensure the address cannot change if other components are added to the system.





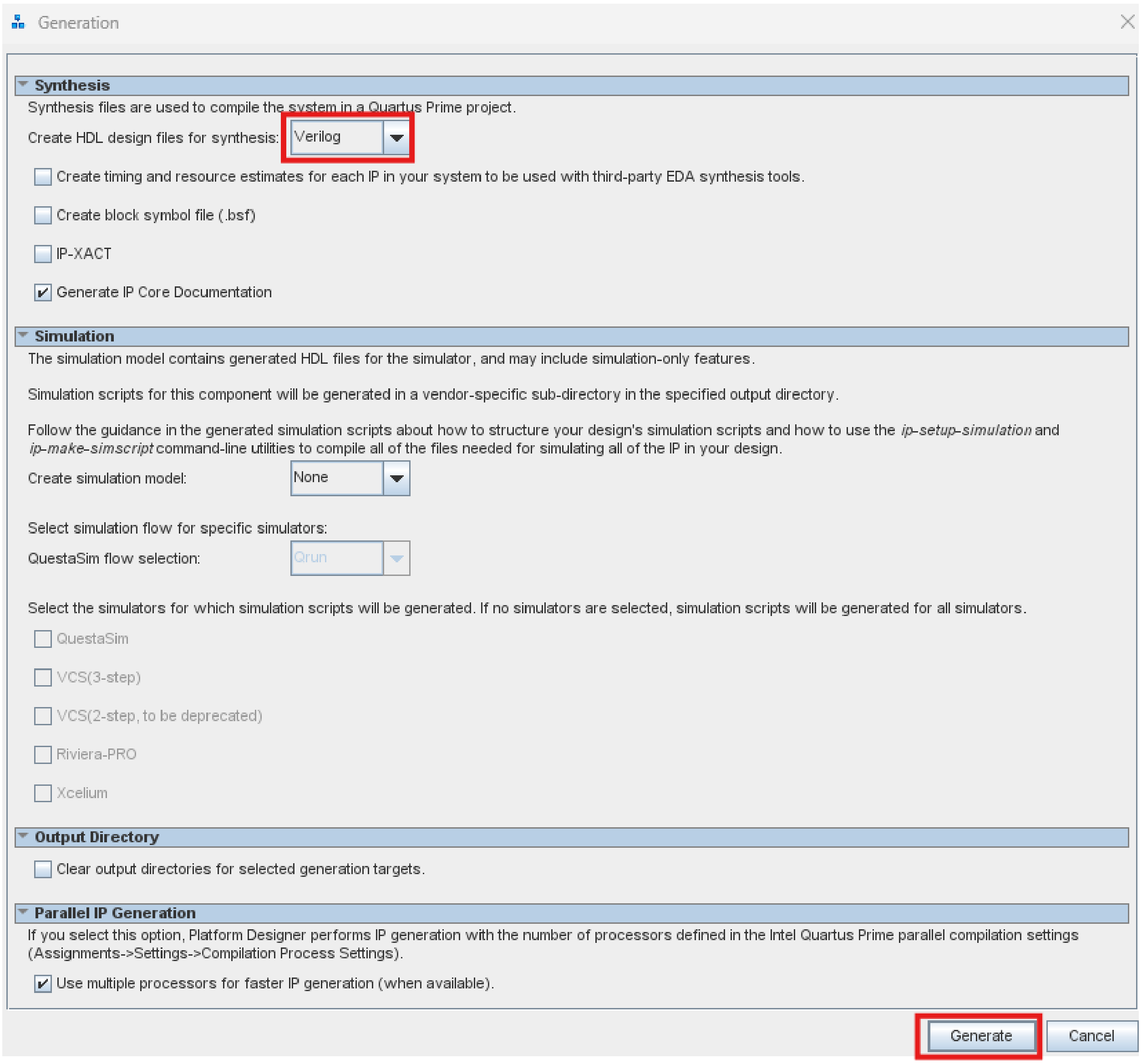
18.5 Generate the Platform Designer system

18.5.1 Click **Generate HDL**.



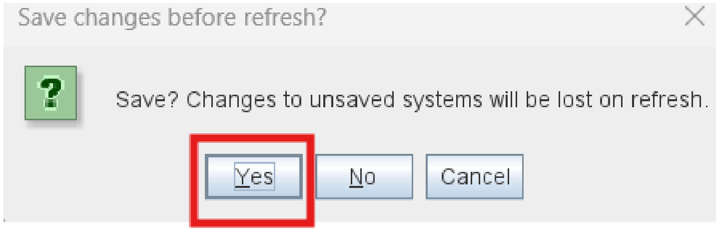
Platform Designer will generate all the system source files that will be compiled by Quartus into an FPGA bitstream image.

18.5.2 A new **Generation** window will appear. Select Verilog and click on Generate.

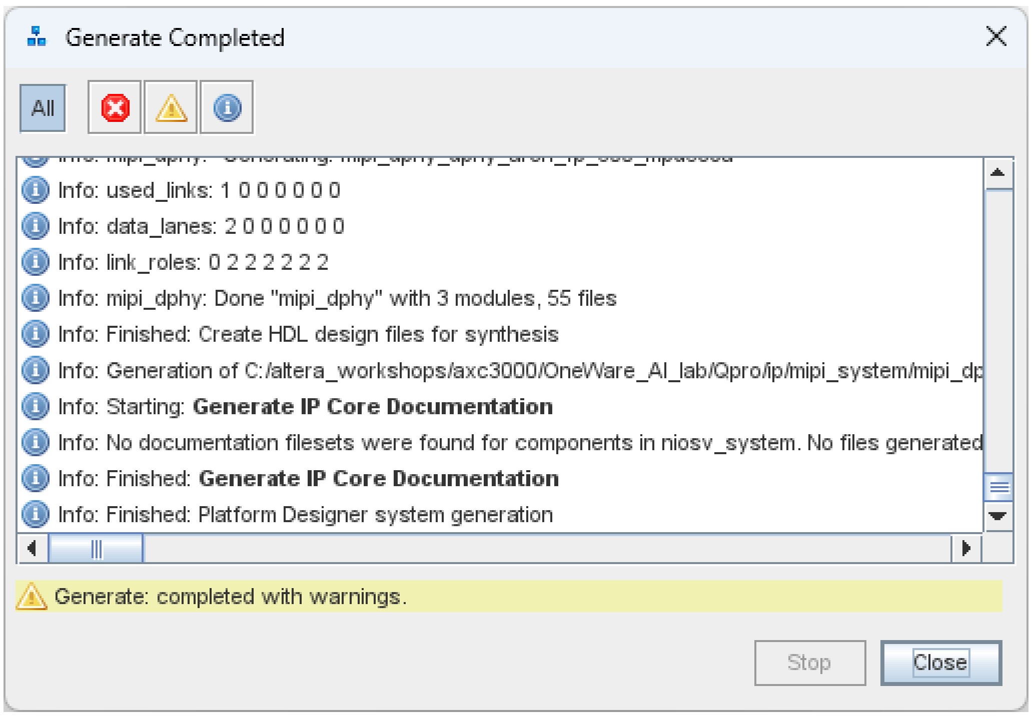


This step generates the HDL (hardware description language) files, as needed, and the associated interconnect so that Quartus will have synthesizable code that it can use for the next steps. As it goes through the steps, you will see it generating code.

18.5.3 It will ask you to save changes as seen below. Select **Yes**



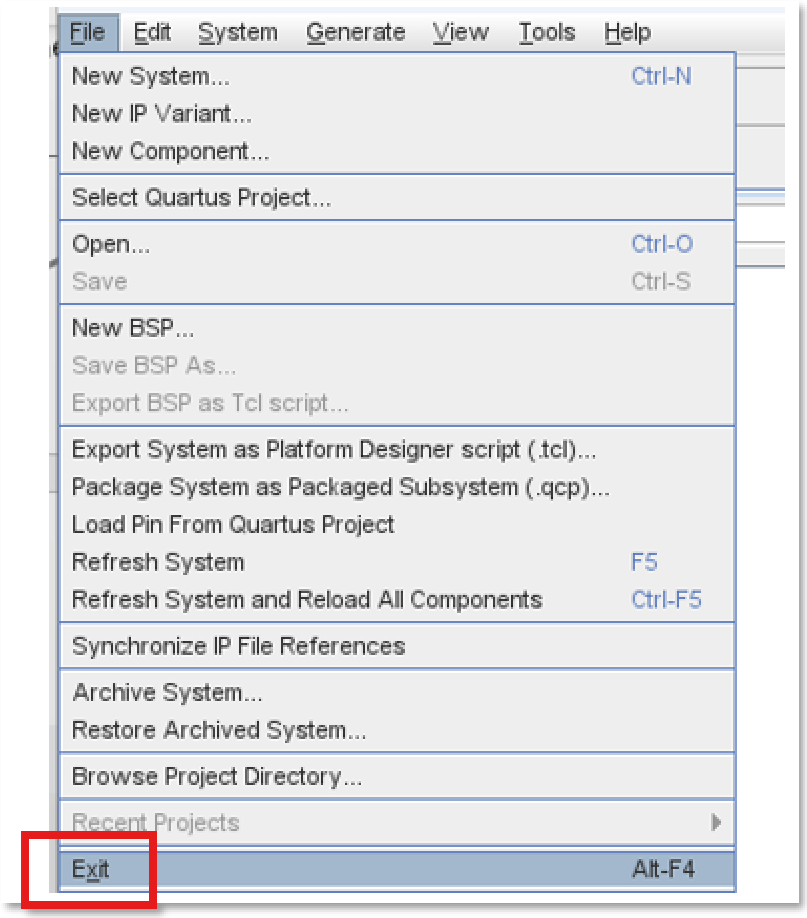
Once it is finished generating, it will show the following:



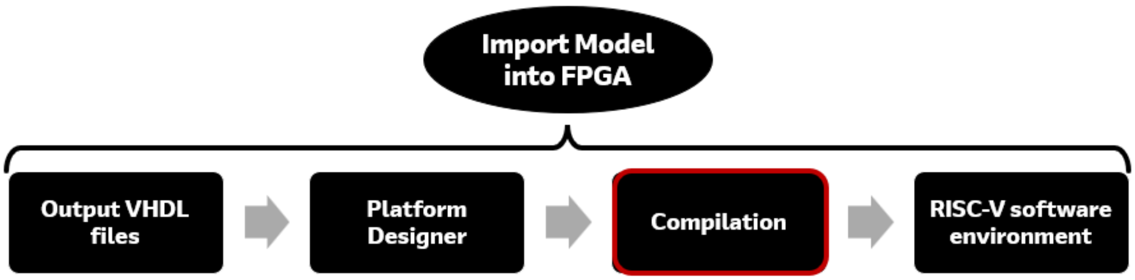
The warnings can be ignored.

18.5.4 Click **Close**.

18.5.5 In the Platform Designer Window, select **File** → **Exit** to close this window.



19 Compilation



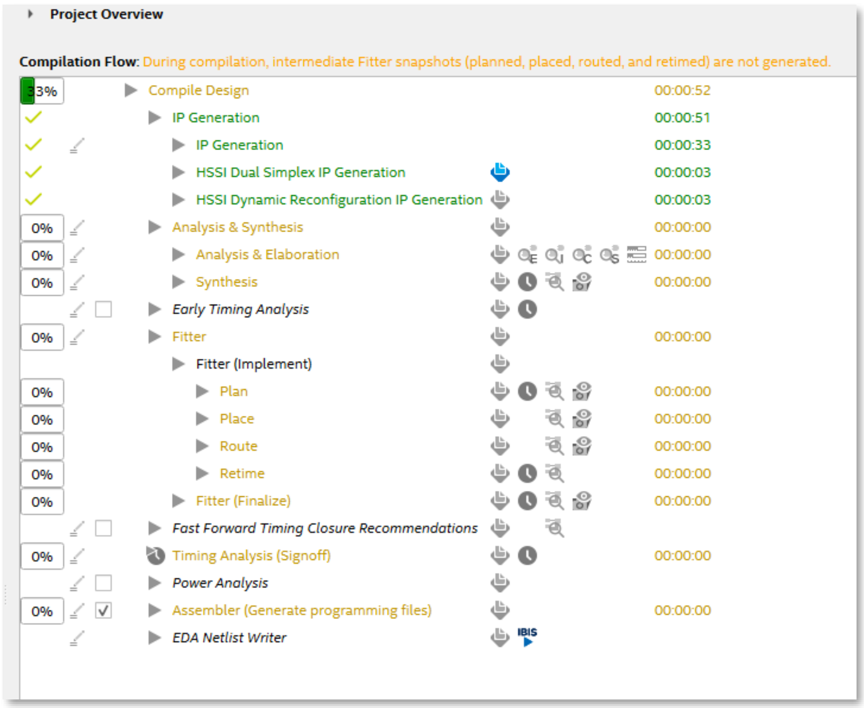
Compilation is the process in which the software checks syntax errors, creates an internal database, applies pinouts and timing constraints, and proceeds with placement and routing to connect the design within the device. During the *Assembler* stage of compilation, a programming file is generated, which can then be used to program the device.

In this design, the pin out and timing constraints have been preconfigured to reduce the time required for the workshop. If needed, details on how to add timing constraints and pin assignments can be found in the *Introductory FPGA workshop*.

19.1 Compile the design

19.1.1 Click on the blue arrow or Processing → Start Compilation to begin the compilation.

As the process runs, messages will appear at the bottom of the window and in the status bar, such as:

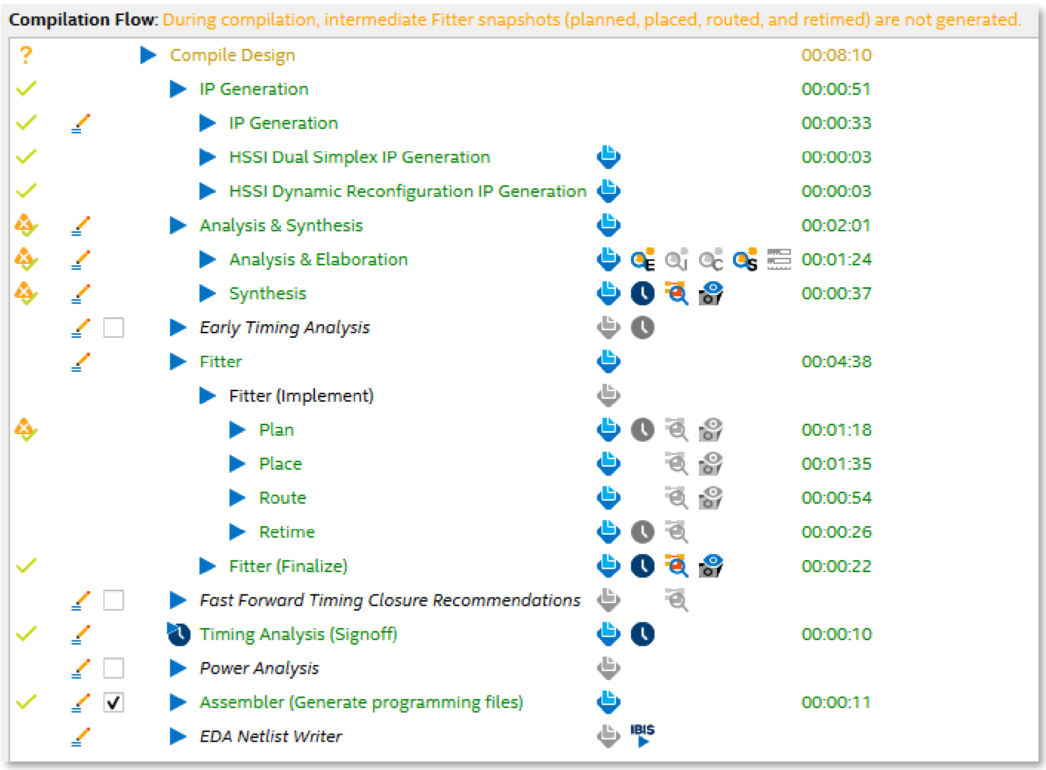


The compilation time will vary depending on the computer used. On one system, it took approximately 7 ½ minutes. The text of each step will turn green as it completes the step. When the compilation is complete, the status text will appear in green. The *Assembler* step produces the programming file.

There should be no errors during compilation. If there are errors, they should be fixed before re-compilating. A green checkmark next to the Compile steps in the **Compilation Flow** window indicates that the compilation was successful.

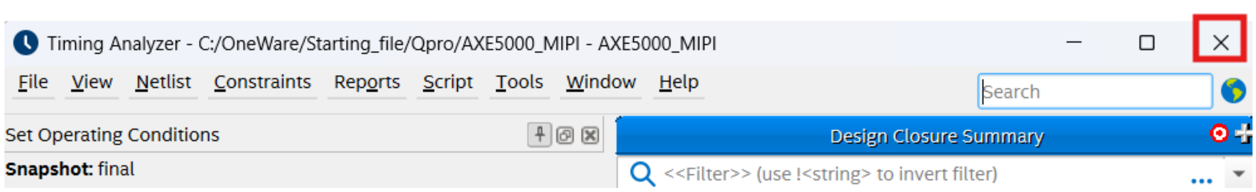
As Quartus compiles the design, there will be a progress bar that shows the steps that it does and the checkmarks as the steps are completed. Notice the checkmarks as steps are completed. The time used for each step depends on the computer used such as the number of processors, amount of memory, and processor clock speed.

If you did a re-compile from the beginning, there will be a **Timing Analysis** Window that appears at the end, but the focus is on the compilation.



19.1.2 You can close the Timing Analyzer Window by selecting X.

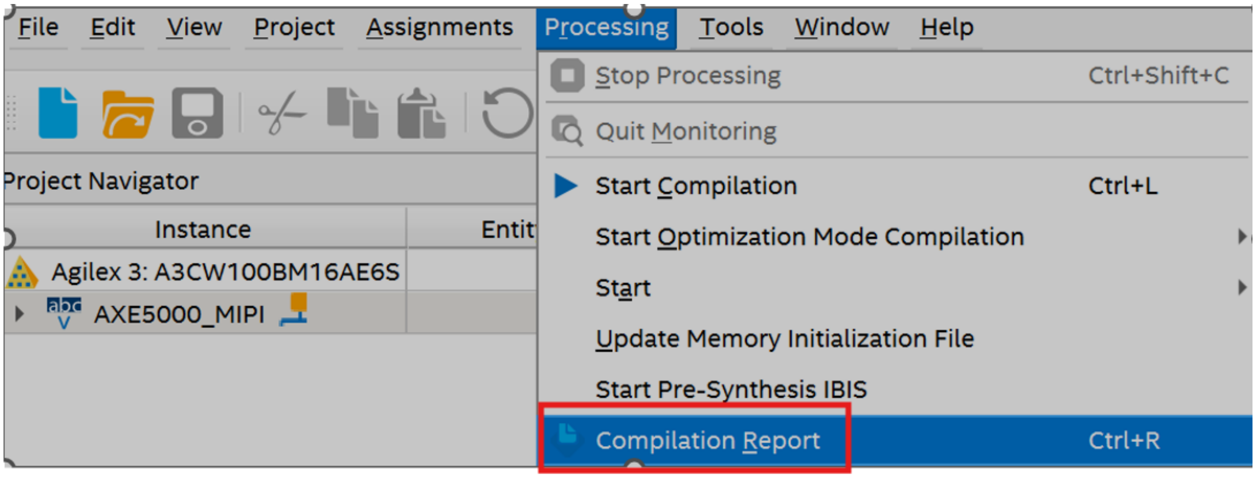
This window will not be used.



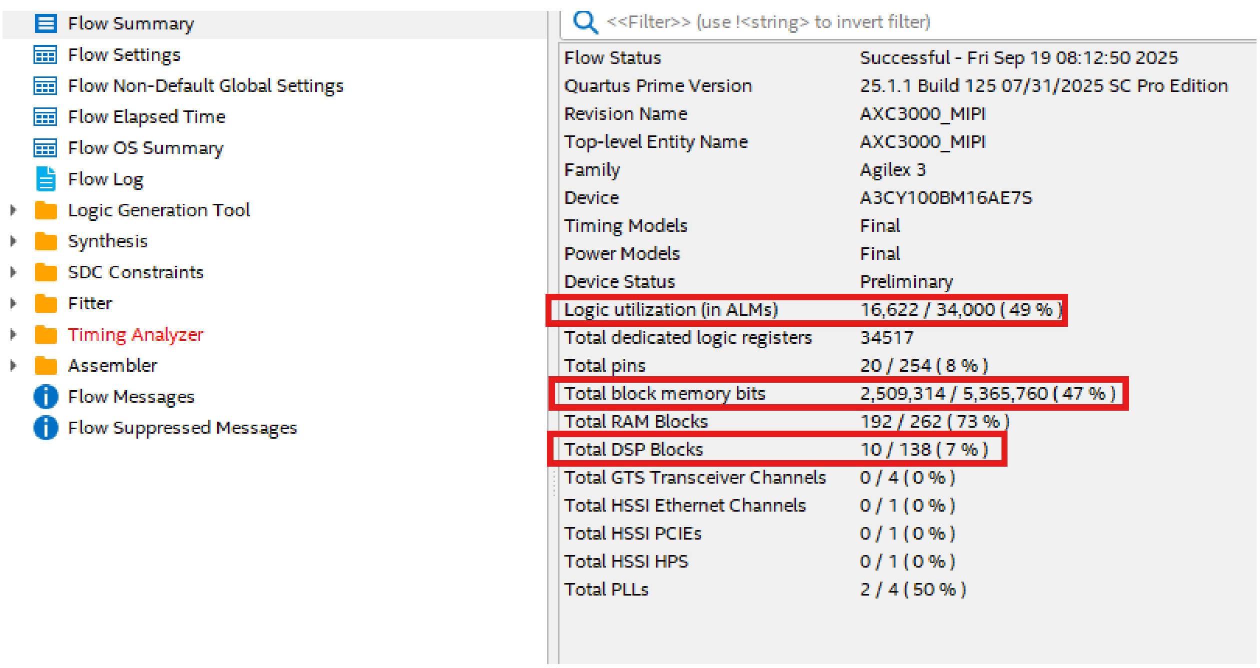
19.2 AI IP Resource usage

How many resources has the AI IP block consumed?

19.2.1 These resources can be found by selecting Processing → Compilation Report



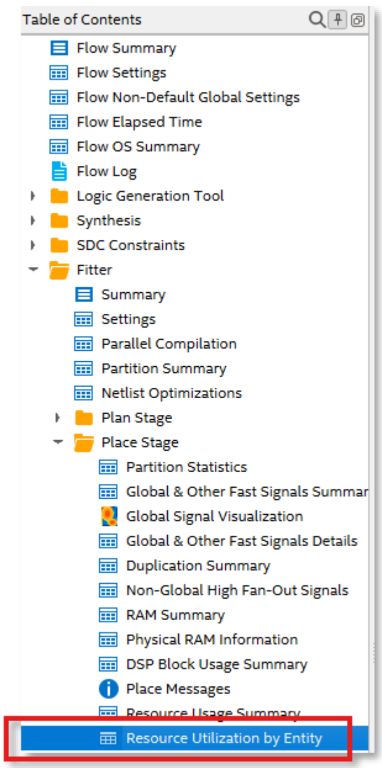
The first item is to see the resources for the entire design. The entire design uses ***49%*** of logic, ***47%*** of memory, ***64%*** of RAM blocks and 7% of the DSP blocks. However, these resources are for all the features including MIPI camera interface, Nios V soft core processor and many more.



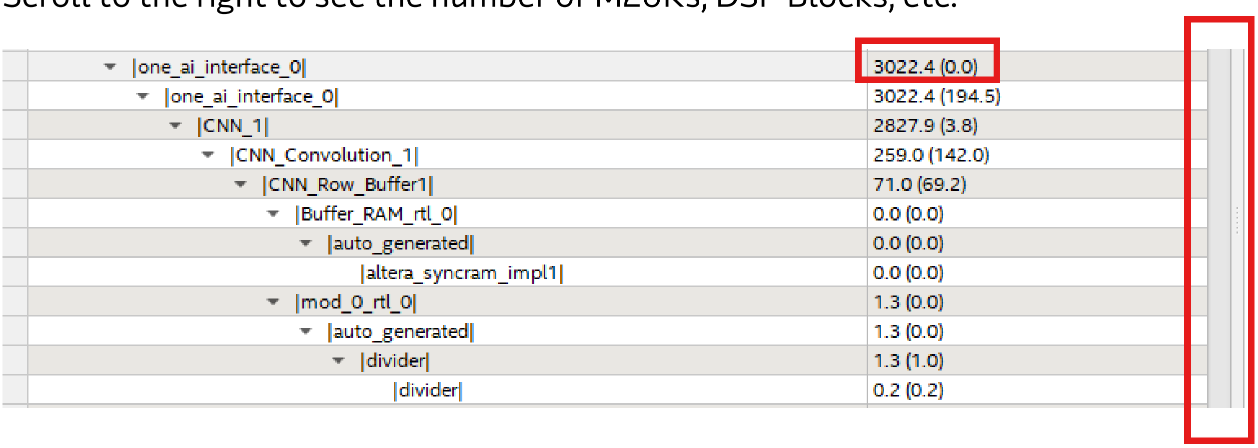
The Timing Analyzer section show some timing issues, but they can be ignored at this time.

19.2.2 To view the section that shows AI model's resources usage, expand the **Fitter** section → **Place Stage**, and select **Resource Utilization by Entity**.

This will show the number of resources for every entity.



19.2.3 Use the scroll bar on the right hand side until you see the one_ai_interface. Scroll to the right to see the number of M20Ks, DSP Blocks, etc.



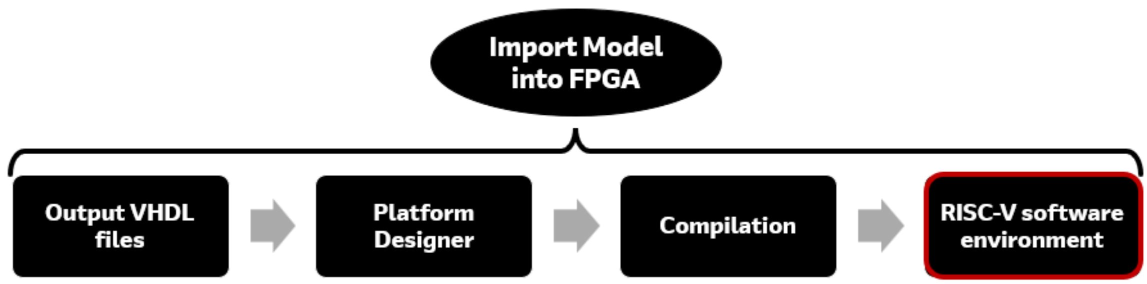
|one_ai_interface_0|	3022.4 (0.0)
|one_ai_interface_0|	3022.4 (194.5)
|CNN_1|	2827.9 (3.8)
|CNN_Convolution_1|	259.0 (142.0)
|CNN_Row_Buffer1|	71.0 (69.2)
|Buffer_RAM_rtl_0|	0.0 (0.0)
|auto_generated|	0.0 (0.0)
|altera_syncram_impl1|	0.0 (0.0)
|mod_0_rtl_0|	1.3 (0.0)
|auto_generated|	1.3 (0.0)
|divider|	1.3 (1.0)
|divider|	0.2 (0.2)

The One AI block uses relatively small resources. It uses 3,022.4 ALMs out of the 34,000 ALMs available on the device, or about 6% of the total. In terms of the memory, it consumes 639,450 memory bits out of the 5.12 Mbits available, which is also a modest amount. This means that the focus can be on smaller devices.

20 RISC-V software environment

Once the processor hardware is ready, you can start building the software design using the the **Ashling RiscFree™ IDE software for Intel FPGAs**.

This step can be seen in the red box below.



The RiscFree IDE is an Eclipse-based IDE that supports C and C++ software development. With the RiscFree environment, there are three main steps consisting of:

a) Creating a C project environment with the hardware project imported,

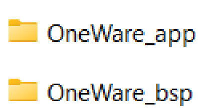
b) building an application C code

c) compiling the code

Before the RiscFree IDE can be used, there are some set up items that need to be done first.

To ensure an optimized compilation flow, it is recommended that you create a directory tree within your working directory.

This is already set up for you in the project. There is a folder called software, and there are two subdirectories called OneWare_app and OneWare_bsp as



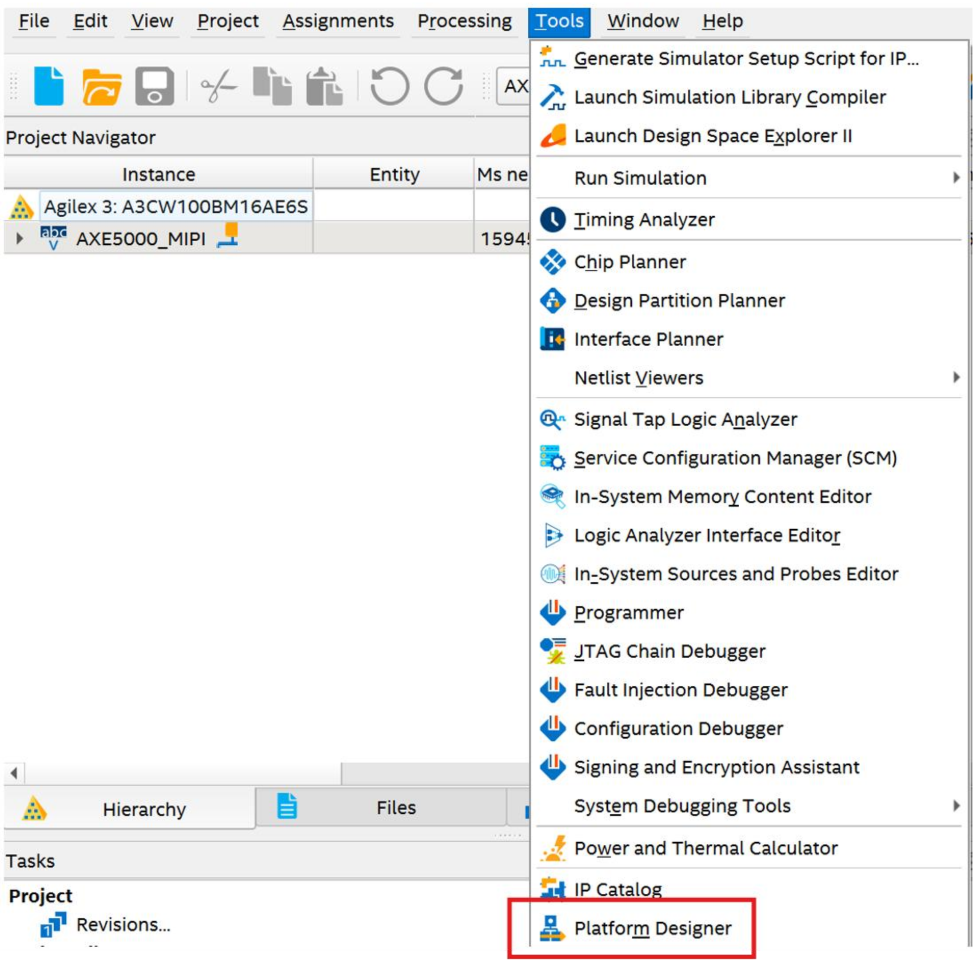
The app folder will have the C program and the bsp folder is where the Board Support Package will be created.

A board support package (BSP) is an integral part of software development. A BSP is a collection of software code that acts as a bridge between hardware and software by abstracting low-level hardware details. This allows higher-level software to be developed without needing to know specific hardware details, enabling faster application development. The BSP takes in the memory-mapped addresses assigned to memory and peripherals attached to the CPU bus and creates a Hardware Abstraction Layer (HAL) which contains the system definitions in a file called **system.h** and collects the available API routines for the peripherals being used.

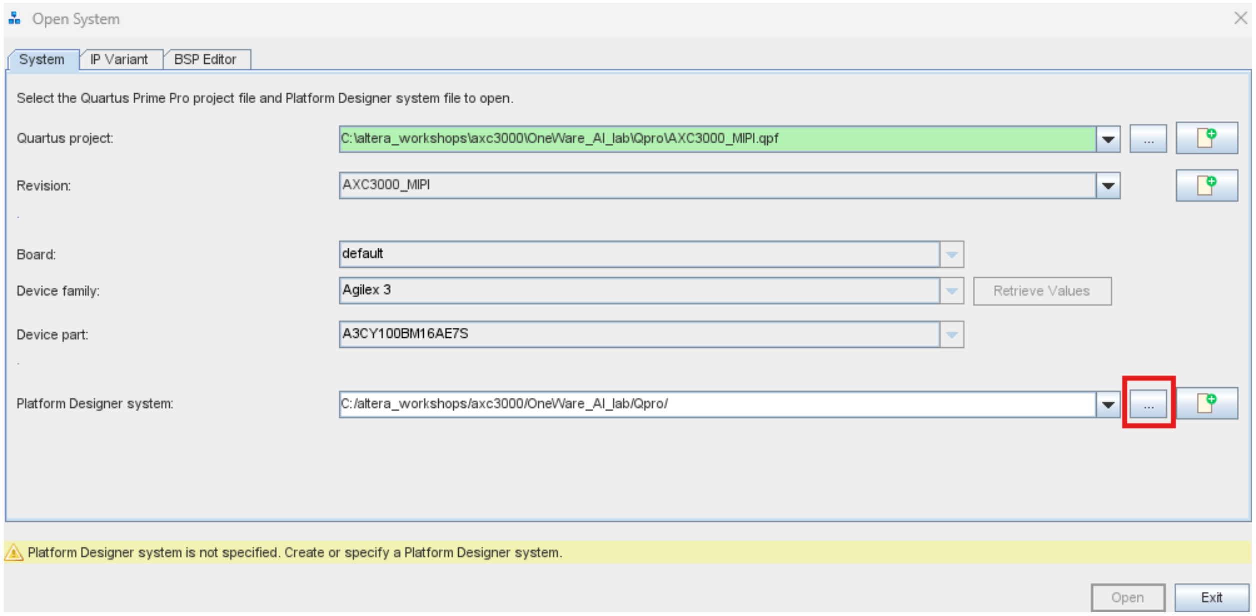
20.1 Create the Board Support Package

The BSP files are generated in Quartus within the Platform Designer.

20.1.1 Go to the Tools menu → Platform Designer

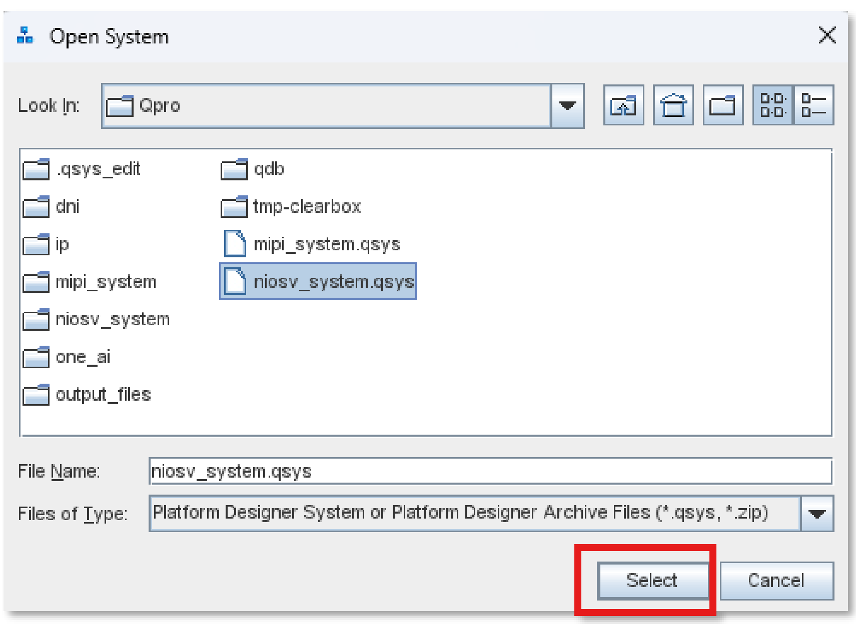


20.1.2 Select the **browse** button.

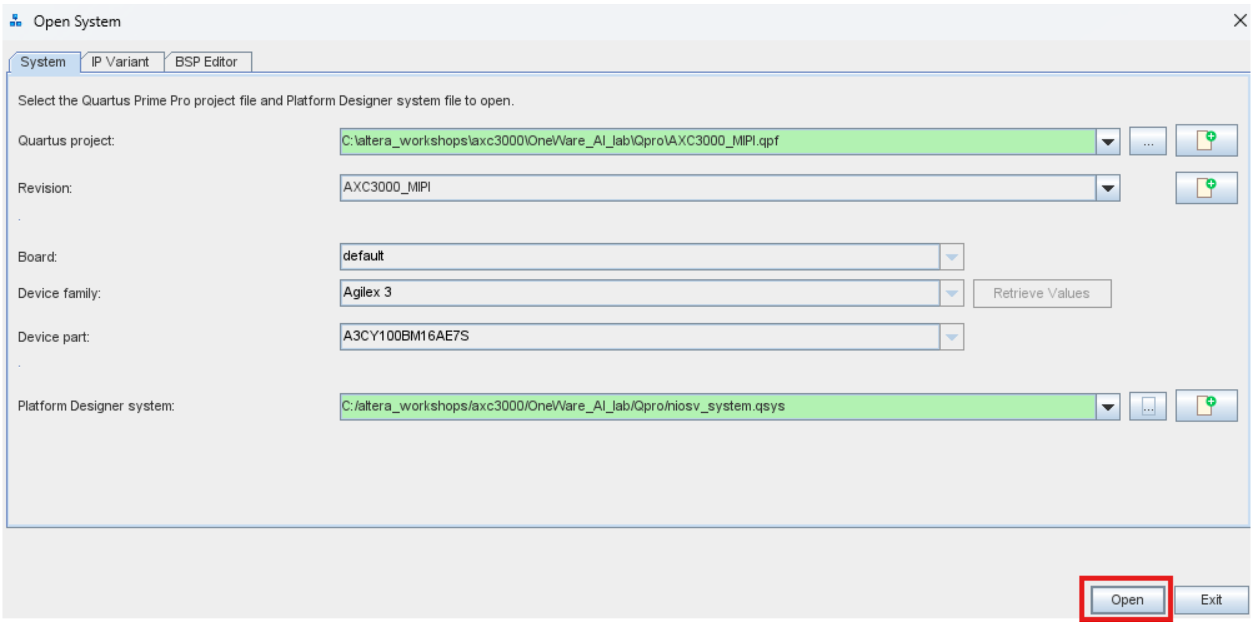


20.1.3 Select **niosv_system.qsys** and click **Select**.

The Nios V System is the Platform Designer System that is pre-built with various IP components that are connected (including the AI model)

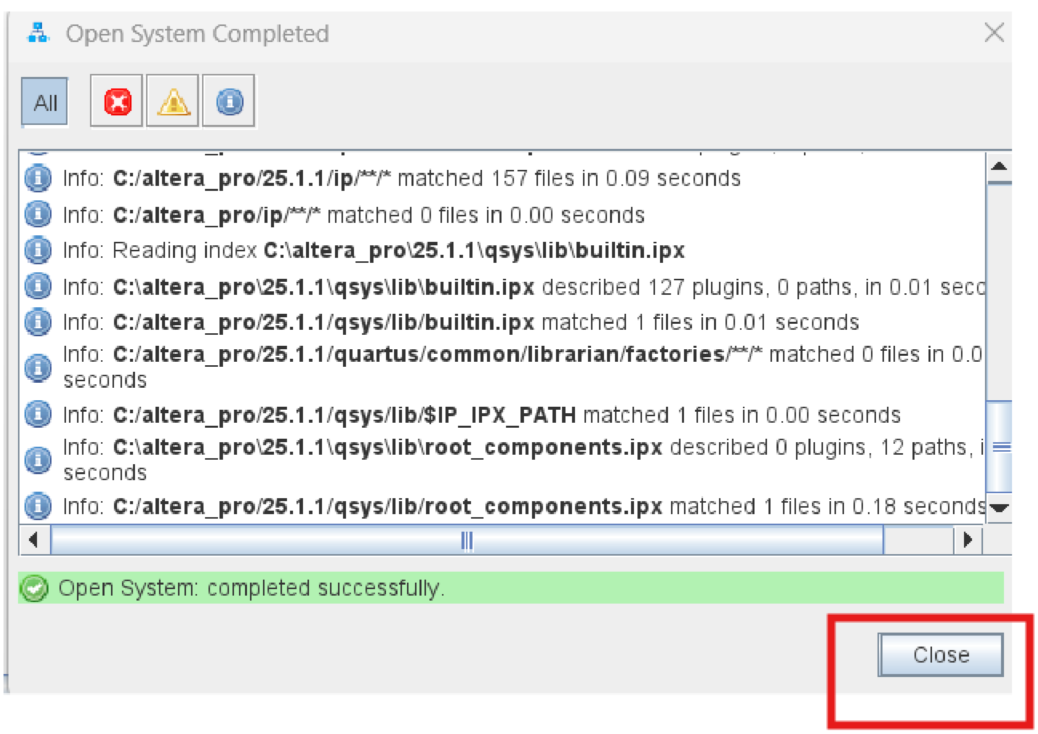


20.1.4 Click **Open**.

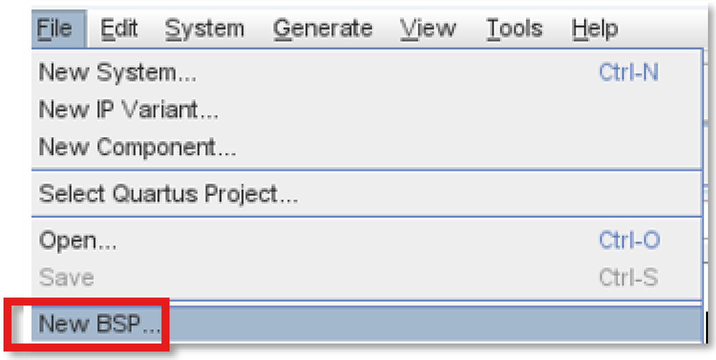


It will show the Open System Completed window

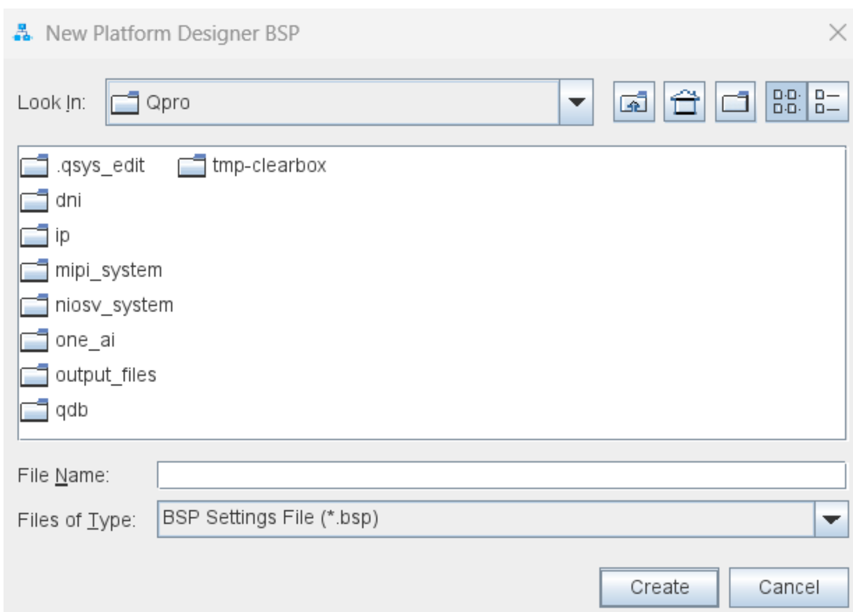
20.1.5 Click **Close** after it completed successfully.



20.1.6 Within Platform Designer select **File** → **New BSP**.



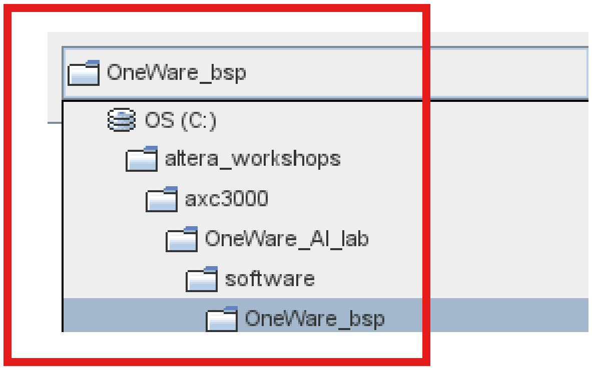
There will be New Platform Designer window that appears as



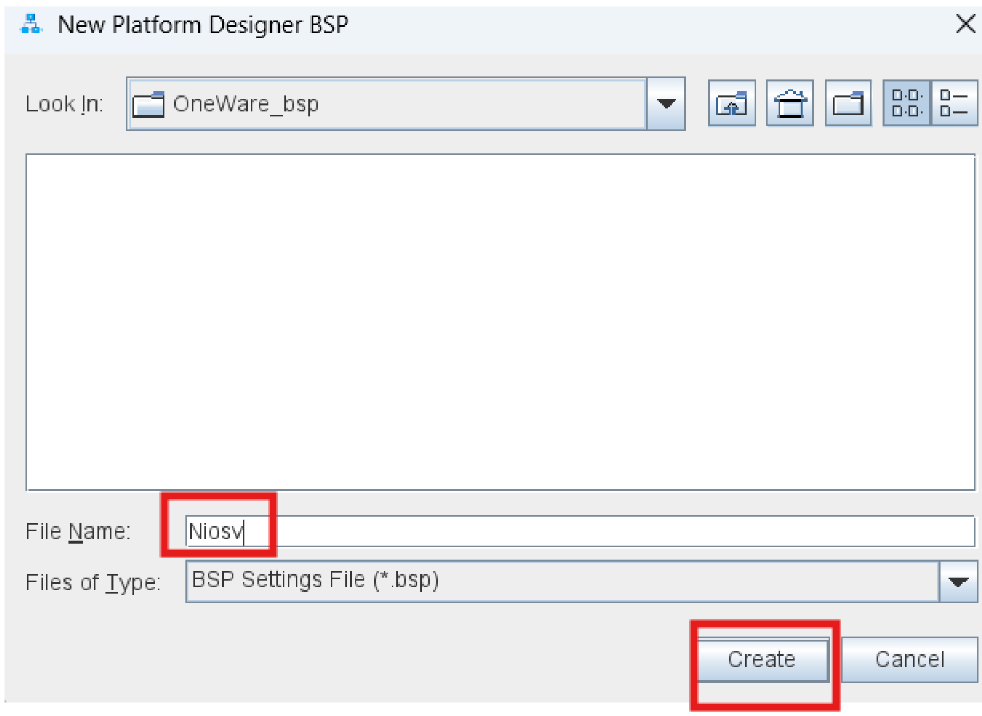
The BSP Editor requires a new file name and the location of where the file will be located.

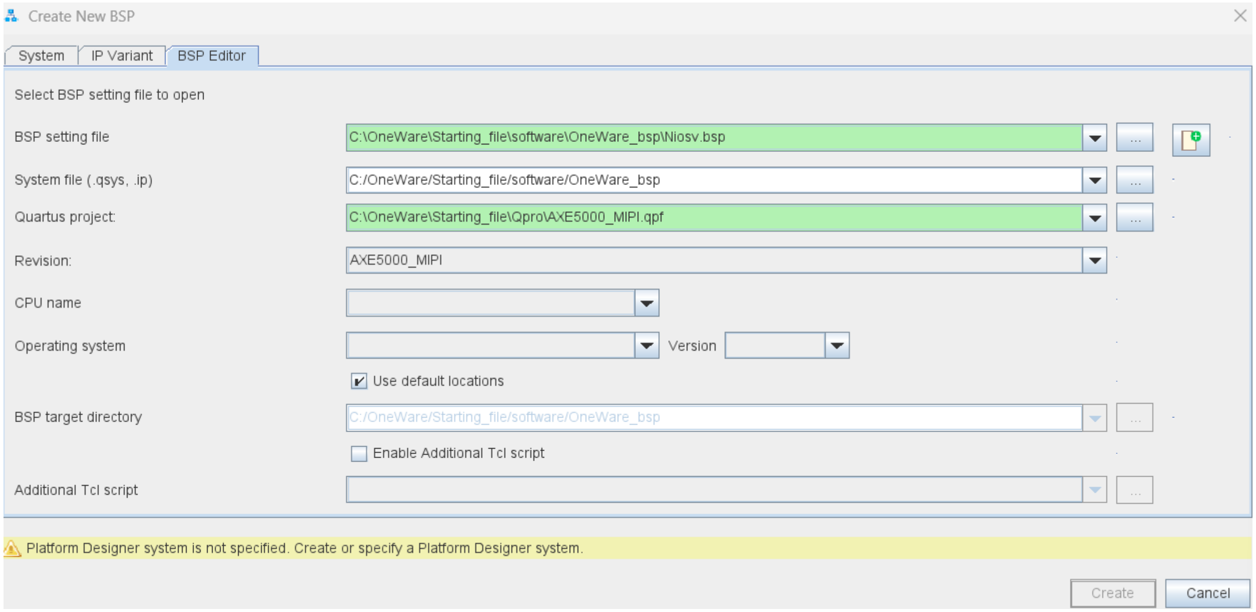
20.1.7 Navigate to the OneWare_bsp folder at

C:\altera_workshops\axc3000\OneWare_AI_lab \software\OneWare_bsp



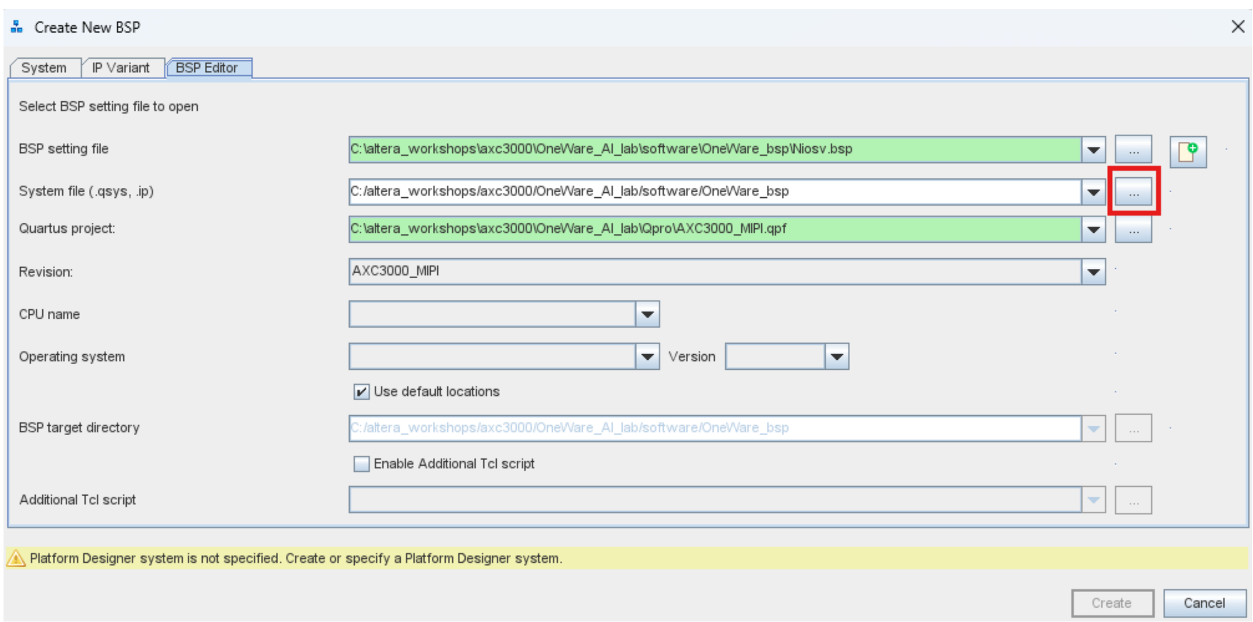
20.1.8 Enter the name **Niosv** and click **Create**.



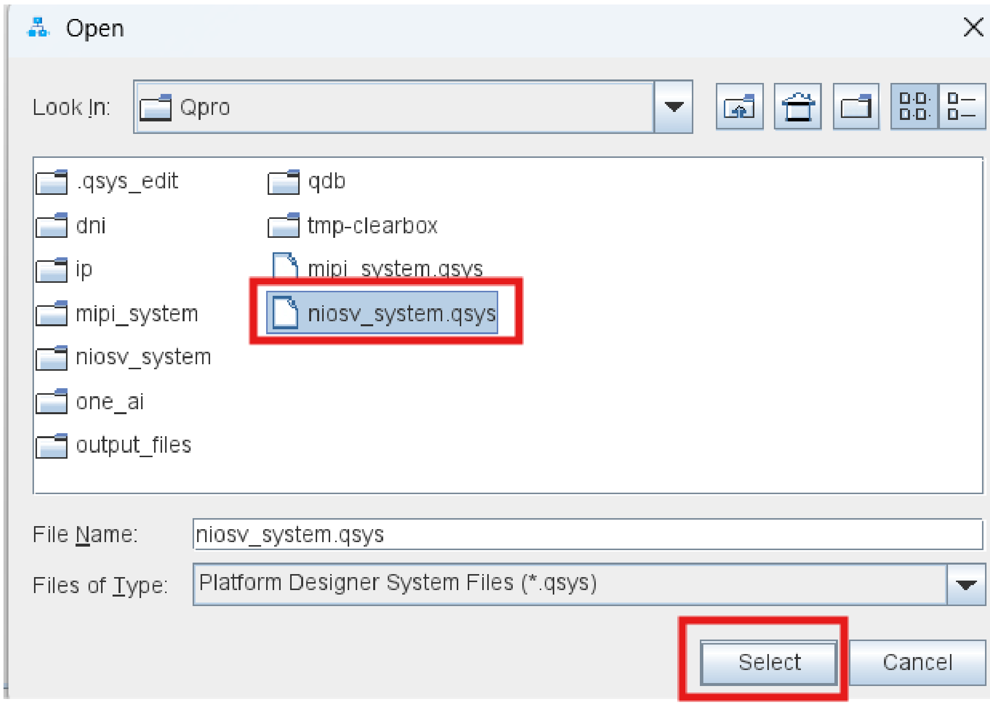


Since the Platform Designer is not assigned (and thus there is a yellow line), the Platform Designer files need to be assigned.

20.1.9 Click **browse** and navigate to the **Qpro** directory.

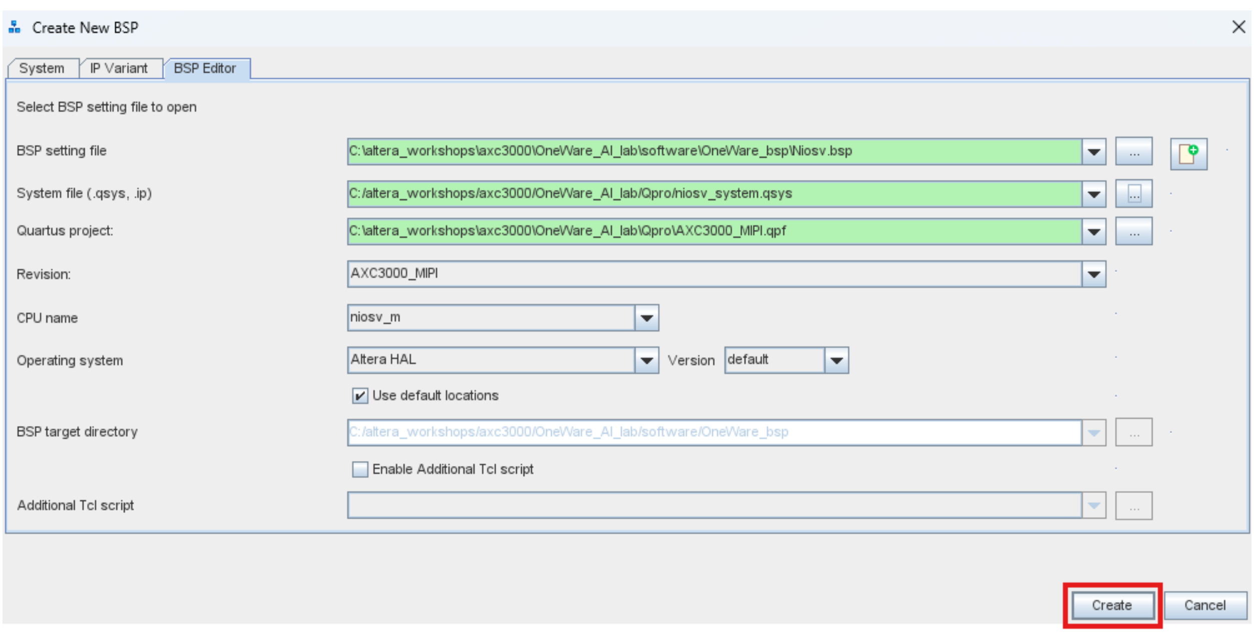


Click to select **niosv_system.qsys**. Click Select.



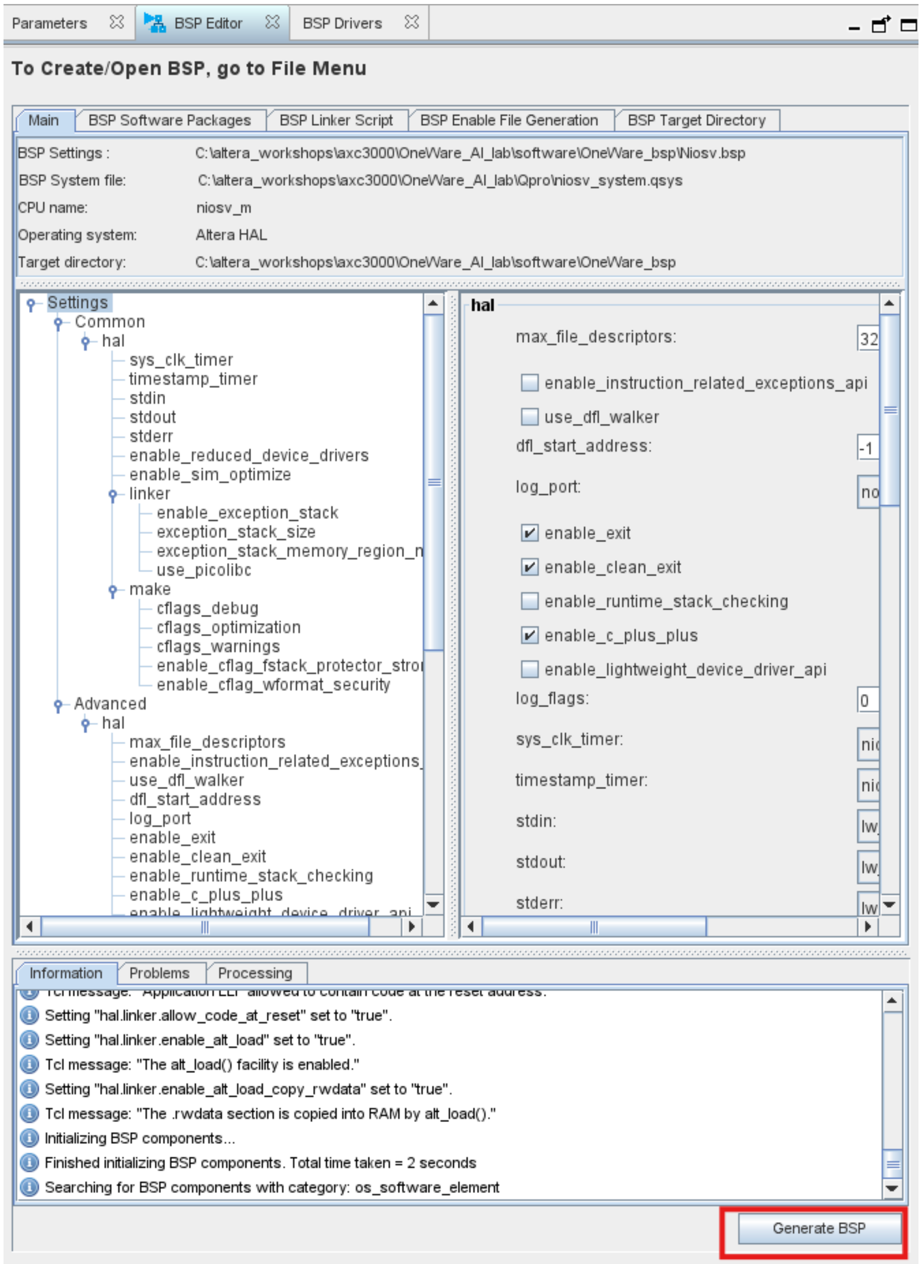
The window will now show, after a slight delay, green lines.

20.1.10 Click **Create**.



The green lines means that software has the items needed to create the BSP files

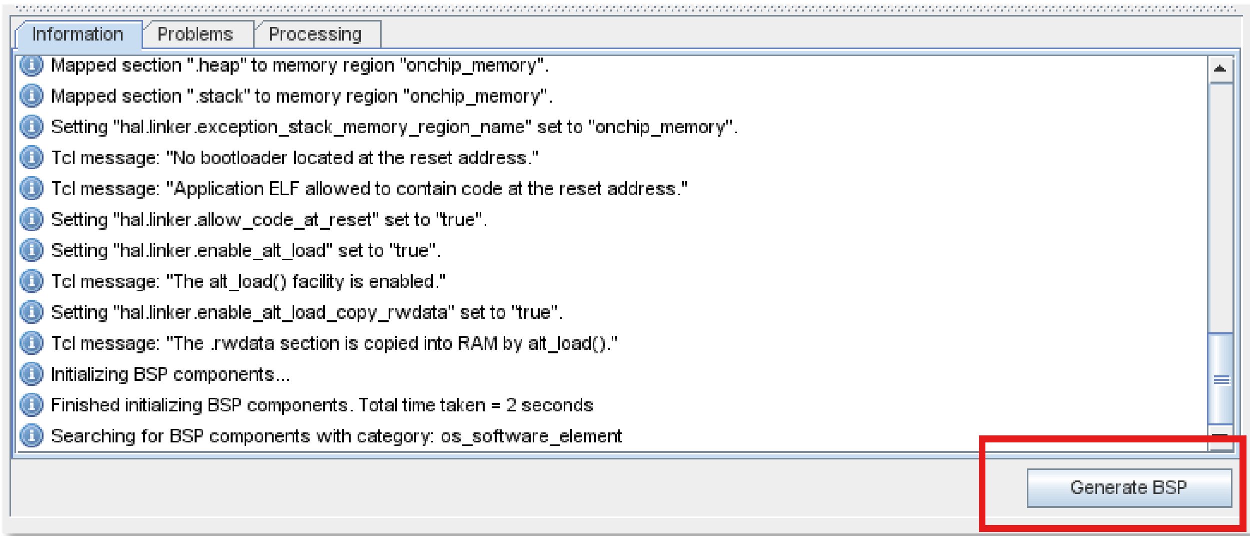
A New window for the BSP will appear as:



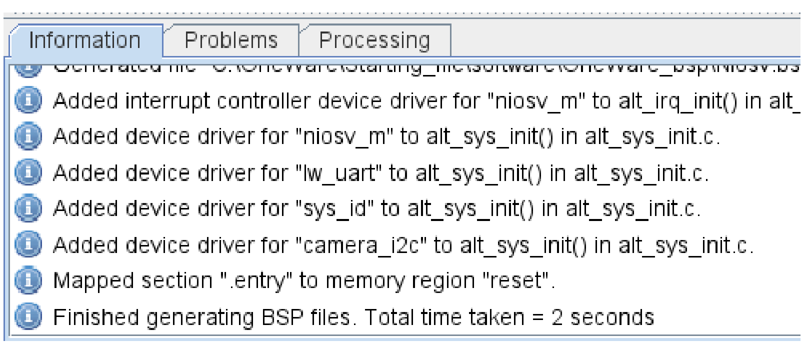
Various settings can be configured to customize the BSP, including optimization options, linker settings, and controls for how the code will reset, among others.. In this case, all default settings are used.

20.1.11 At the bottom of the page, select **Generate BSP**.

You may need to widen the left side of the window to view the details, **if you do not see it**.



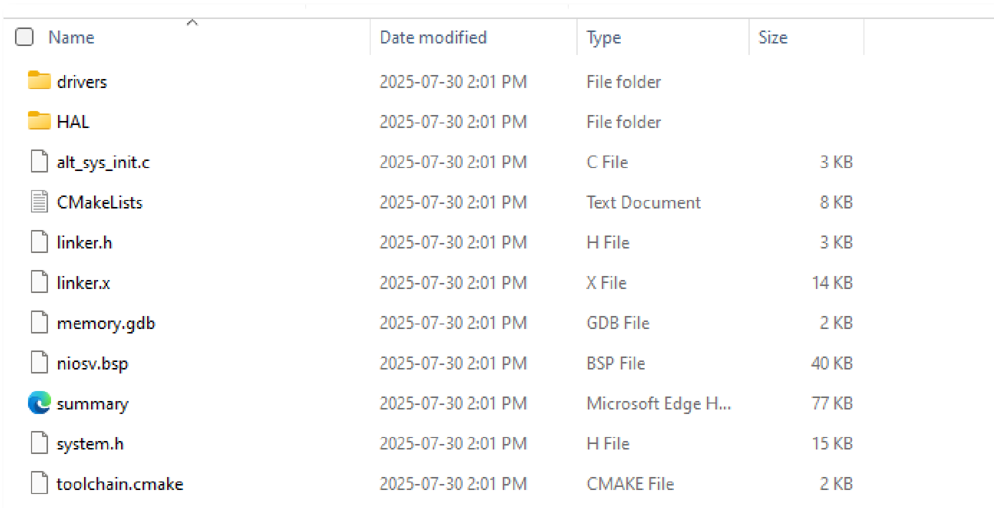
The window will show that it generates the BSP file very quickly.



20.1.12 In Windows, navigate to

c:\altera_workshops\axc3000\OneWare_AI_lab\software\OneWare_bsp.

You will now see that the following files have been generated:



Name	Date modified	Type	Size
drivers	2025-07-30 2:01 PM	File folder	
HAL	2025-07-30 2:01 PM	File folder	
alt_sys_init.c	2025-07-30 2:01 PM	C File	3 KB
CMakeLists	2025-07-30 2:01 PM	Text Document	8 KB
linker.h	2025-07-30 2:01 PM	H File	3 KB
linker.x	2025-07-30 2:01 PM	X File	14 KB
memory.gdb	2025-07-30 2:01 PM	GDB File	2 KB
niosv.bsp	2025-07-30 2:01 PM	BSP File	40 KB
summary	2025-07-30 2:01 PM	Microsoft Edge H...	77 KB
system.h	2025-07-30 2:01 PM	H File	15 KB
toolchain.cmake	2025-07-30 2:01 PM	CMAKE File	2 KB

Several files are generated in the BSP directory.

- **Drivers** directory contains various device drivers such as the uart and I2C
- **HAL** refers to the hardware abstraction layer, which means that low level details do not need to known by the software.
- **alt_sys_init.c** contains the initialization drivers.
- **CmakeLists** is used to build the BSP by specifying the files, directories, and other build settings to be targeted. Note you may have a **CMakeLists** file in the BSP directory and the Application (App) directory.
- **linker.h** provides the memory layout.
- **linker.x** is a linker script (which defines how portions of the program are linked to memory)
- **system.h** defines the Platform Designer peripherals. It has declarations describing the BSP memory map and other system information needed by software applications.

The next step is to have a C Code in the software/OneWare_app folder. The C code is already created for you.

20.2 Review the AI IP CNN Output function

20.2.1 Using Windows, navigate to OneWare_app directory and by using NotePad, open the file called niosv_mria220.c

20.2.2 Scroll down the file until you get to the section display_cnn_outputs.

This code will read the 10 classes (0 to 9).

```
/******************************************************************************/
// display_cnn_outputs : Display CNN output values using sequential auto-increment
/******************************************************************************/
void display_cnn_outputs(void)
{
    alt_u32 cnn_value;
    int i;

    alt_putstr("\r\n=== CNN Output Values ===\r\n");

    // Reset position to start from class 0
    IOWR_32DIRECT(ONE_AI_BASE, ONE_AI_USER_OBJECTS_CLASSES * 4, 0);  // Reset to class 0
    usleep(1000);

    // Read 10 classes by reading address 0x04 multiple times
    for(i = 0; i < 10; i++)
    {
        // Each read from address 0x04 automatically increments to next class
        cnn_value = read_ai_register(ONE_AI_OUTPUT_READ_AUTO);  // Address 0x04
        alt_putstr("Data ");
        dword2decimal(i);
        alt_putstr(": ");
        dword2decimal(cnn_value);
        alt_putstr("\r\n");
    }
    alt_putstr("==========================\r\n");
}
```

The c code reads from the ONE AI interface. The data is read from the interface according to this table:

Address Range	Description	ACCESS	FORMAT
0x00	New Image Ready Flag	R	[0] = ready flag
0x01	Number of Outputs/Output to Read From	R/W	32-bit value
0x02	FPS (Frames Per Second)	R	32-bit value
0x03	Single Output Read	R	32-bit AI output value
0x04	Auto-Increment Output Read	R	32-bit AI output value
0x10-0x1F	Width & Height Control	R/W	[31:16]=H, [15:0]=W
0x20-0x2F	Objects & Classes Control	R/W	[31:16]=C, [15:0]=O
0x30-0x3F	Output Dimensions (Width/Height)	R	[31:16]=H, [15:0]=W
0x40-0x4F	Output Dimensions (Obj/Classes)	R	[31:16]=C, [15:0]=O

Reading with address 0x04 automatically increments the position to read from to speed up the reading process.

No changes are needed in this file but rather it shows how the values are output.

20.2.3 Close the niosv_mria220.c file.

20.3 Create the Application project

In addition to the BSP, the software project needs an application project which is created using a CMake flow. It uses the C program code, the BSP, and the HAL to generate an application environment used by the Ashling RiscFree IDE.

To create the application, this is done with executing commands in the Nios V command shell.

20.3.1 In the Windows search bar enter **Nios**. Select **Nios V Command Shell (Quartus 25.1.1.)**.

If it is not found, navigate to *\altera-pro\25.1.1\niosv\bin\niosv-shell* and select it manually.

This will open the command shell.

20.3.2 **Click** in the niosv-shell.

```
Entering Nios V shell
Microsoft Windows [Version 10.0.22631.5624]
(c) Microsoft Corporation. All rights reserved.
```

If you see any warning that the path to RISC-V needs to be manually added, this could mean that the RISC-V free is not downloaded nor installed. Download and install RISC-V for free.

20.3.3 Change the path using the command

cd C:\altera_workshops\axc3000\OneWare_AI_lab\software

20.3.4 Use the **dir** command to list the two directories: *OneWare_app* and *OneWare_bsp*

```
<DIR>          .
<DIR>          ..
<DIR>          OneWare_app
<DIR>          OneWare_bsp
```

The OneWare_app directory will have niosv_mira220.c file in it.

20.3.5 Enter following command

niosv-app --bsp-dir=OneWare_bsp --app-dir=OneWare_app -s=OneWare_app/niosv_mira220.c

```
[niosv-shell] c:\MyDocuments\Altera\OneWare\QuartusDemo\nist_demo\nist_demo\software> niosv-app --bsp-dir=OneWare_bsp --app-dir=OneWare_app -s=OneWare_app/niosv_mira220.c
```

The breakdown of the command is:

- **--bsp-dir** links to the BSP directory and all the files.
- **--app-dir** links to the application directory.
- **-s** links to the source files, i.e. the .c files.

20.3.6 Press **Enter**. The file CMakeLists.txt has been generated

```
2025.07.30.14:16:59 Info: "OneWare_app\CMakeLists.txt" was generated.
```

A CMakeLists.txt file is a configuration file (or recipe) that defines how to generate build files. One important point to note is that this file is in the application directory, while another CMakeLists.txt file resides in the BSP directory.

When generated using the niosv-app command, the CMakeLists.txt file specifies how to generate build files for a user application or library using user-provided source files, and links them with the specific BSP.

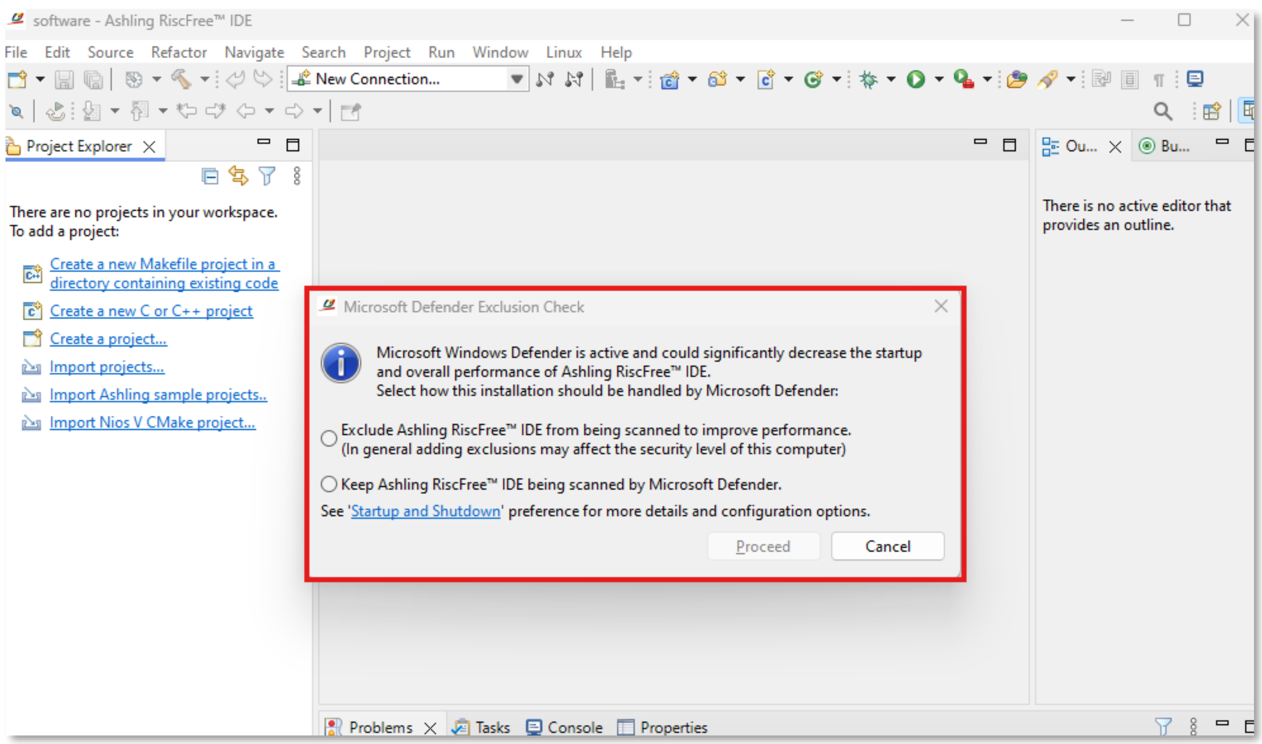
20.4 Build the Application with RiscFree

20.4.1 In the Nios V shell, type **cd ..** to move up one directory level.

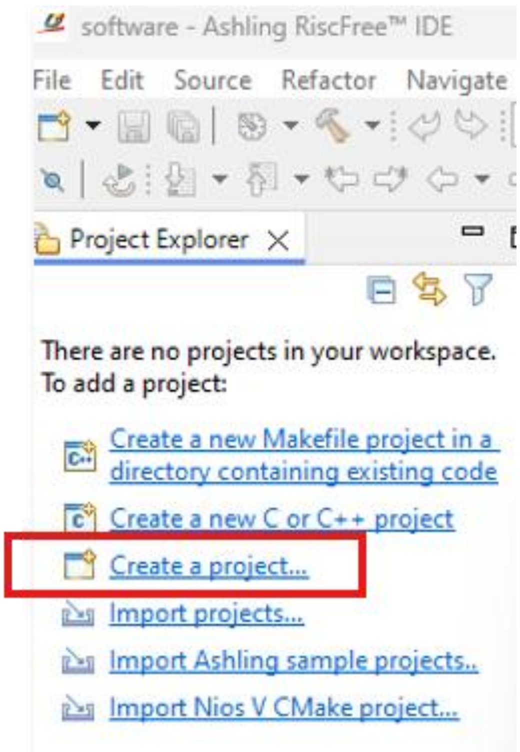
20.4.2 Using the Nios V shell, type the following command to open the Ashling RiscFree IDE.

```
RiscFree.exe -data software
```

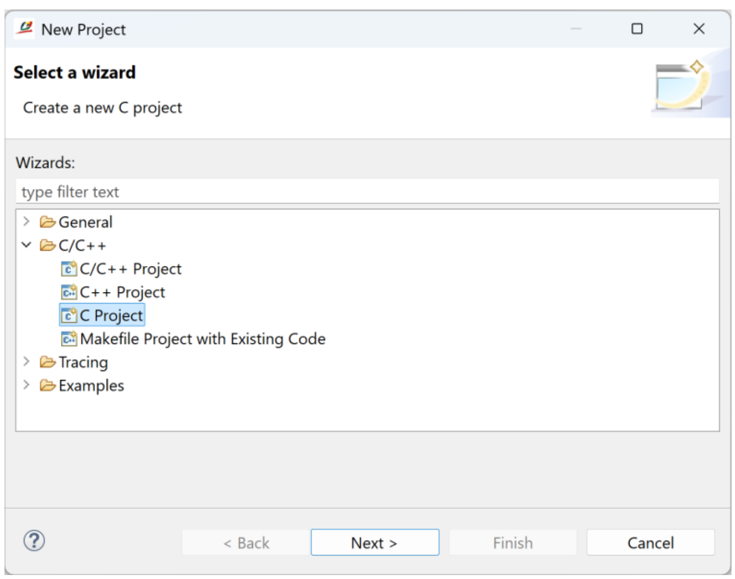
20.4.3 If you see the window below, select **Cancel**.



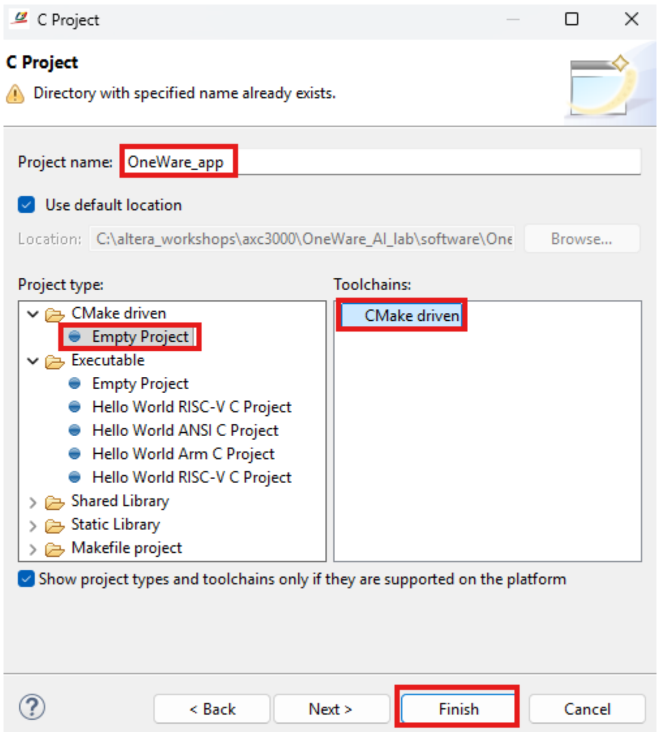
20.4.4 Select **Create a Project** in the **Project Explorer** tab.



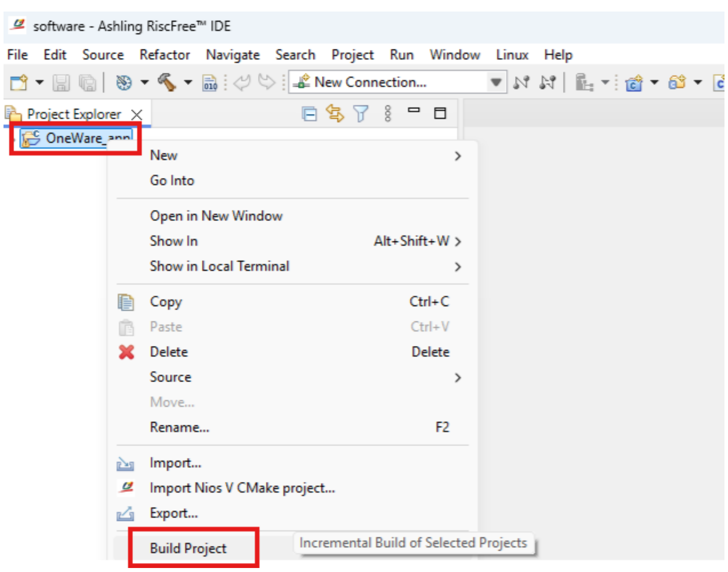
20.4.5 Expand the **C/C++** category, and select C project item. Select **Next**.



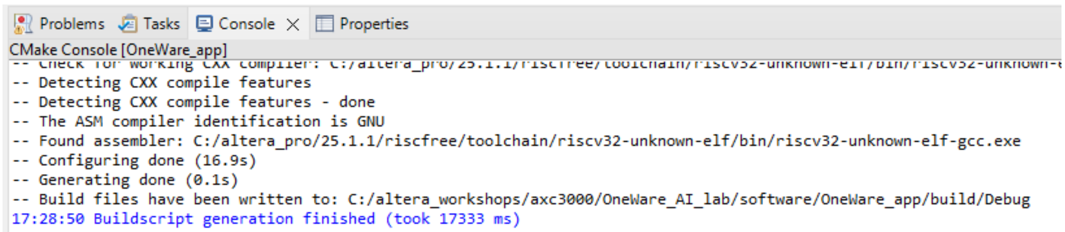
20.4.6 Expand the **CMake driven** Project type and select **Empty Project**. In the Toolchains pane, select **CMake driven**. Enter **OneWare_app** as the **Project Name**.



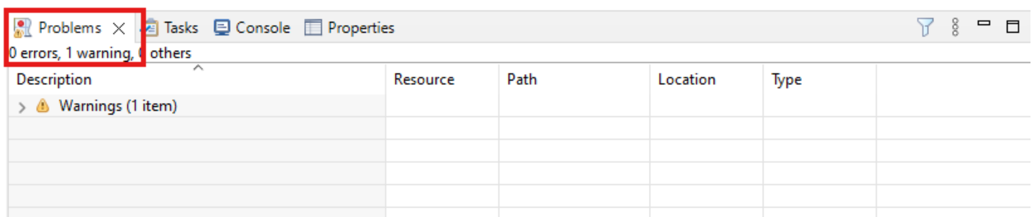
20.4.7 Right click on the **OneWare_app** project in the **Project Explorer Tab**. This will start the software build.



20.4.8 Wait for the compile to complete.



20.4.9 Review the ***Problems*** tab for any error messages.



20.5 Update the FPGA Bitstream file

The hardware project needs to be compiled with no errors in order to get a **.sof** configuration file (FPGA bitstream). The design has been compiled in Quartus.

The contents of the Nios V processor memory have been created, but were not available when the previous Quartus compile occurred.

Quartus provides an abbreviated compile that only updates the contents of the on chip memory.

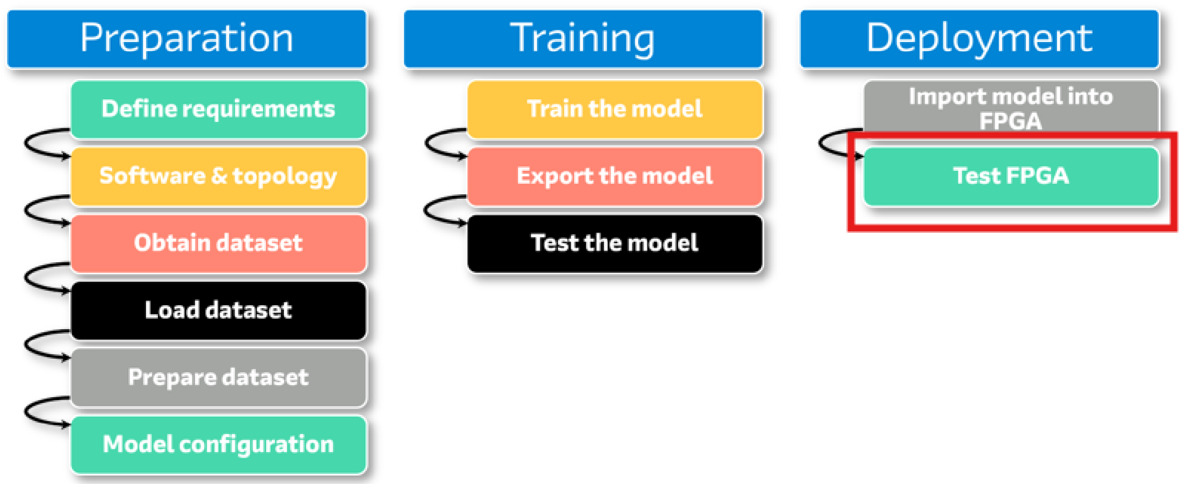
20.5.1 Update the contents of the on chip memory. Processing → Update Memory Initialization File

20.5.2 Update the bitstream file (sof). Processing → Start → Start Assembler

The on-chip memory is initialized with the software project compiled code in a HEX format. This will be included in the FPGA bitstream when the device is programmed and will be run when the Nios V processor is released from reset.

21 Test the Design

The final stage is to test the FPGA that contains the AI mode.



21.1 Test Structure

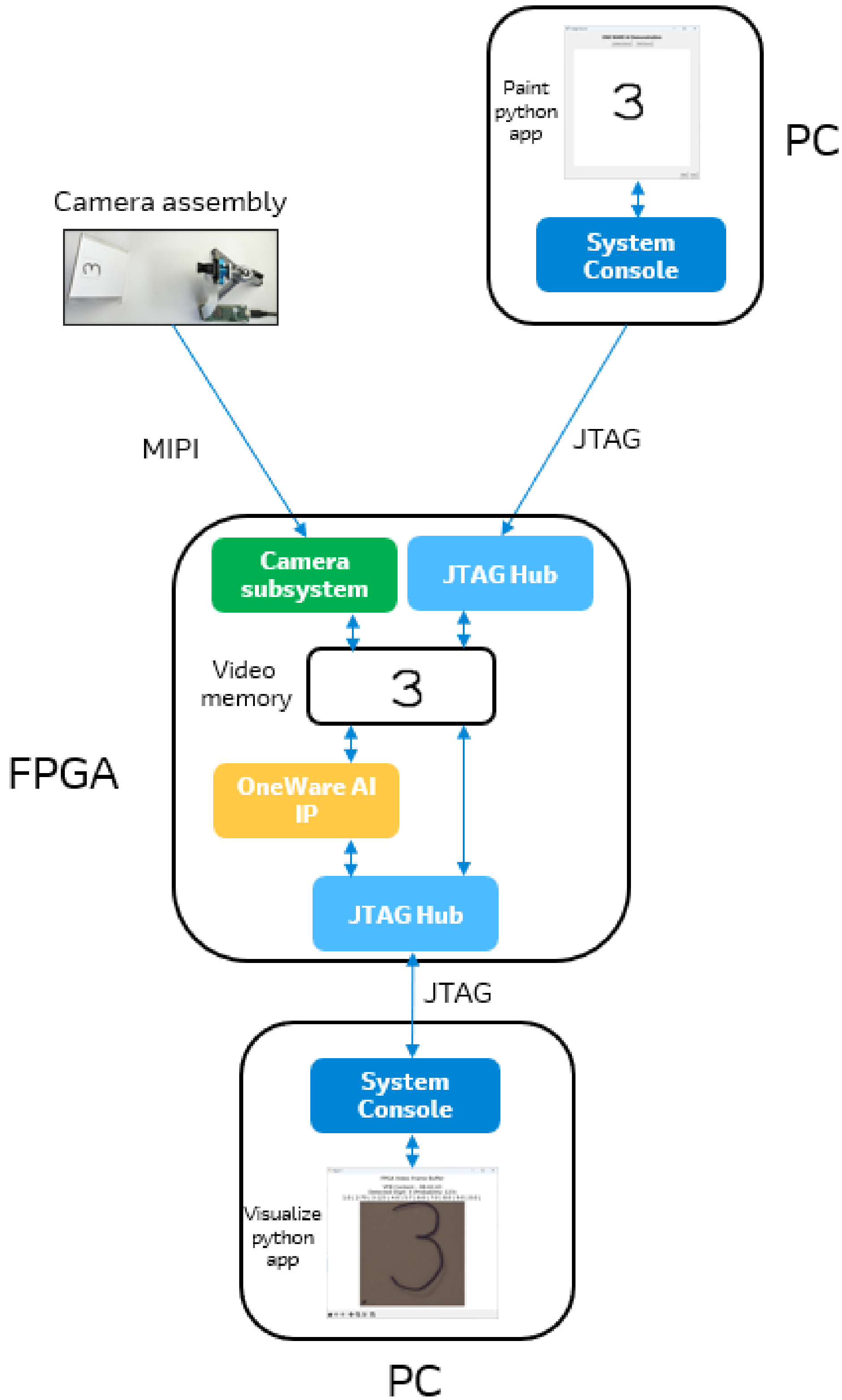
The test environment is shown in the diagram below.

The AI IP block receives images from one of two sources: the camera subsystem or from the one_ai python paint like application. The **one_ai.py** application uses windows system calls to communicate with System Console.

System Console is a system level debug tool that allows access to the design in the FPGA. It can read or write memory mapped registers and memory blocks. It communicates with the FPGA via the JTAG cable and a JTAG hub in the FPGA.

The **one**_ai application has a dual function. It selects which image source writes images into the video frame buffer. It also provides a built-in paint-like function to enable the user to draw digits.

Detected digits are extracted from the OneWare AI IP block by the **visualize_vfb** python application. In addition it displays the probability of the detected digit. It also extracts the video frame buffer contents and displays them. **Please note** that the python application has been written to extract and display the video frame buffer every 2 or 3 seconds. **Agilex 3** and the **AI IP** is capable of processing hundreds to thousands of **frames per second**.


Paint
python
app
3
PC
System
Console
Camera assembly
MIPI
JTAG
Camera
subsystem
JTAG Hub
Video
memory
3
FPGA
OneWare AI
IP
JTAG Hub
JTAG
System
Console
Visualize
python
app
PC

21.2 Design verification

Use the following guide to determine which verification section to complete

- If the design was compiled with CloudLabs, and will be verified using the Arrow board farm, then complete Section 22.
- If the design was compiled locally or with CloudLabs, but will be verified locally, then complete Section 23.

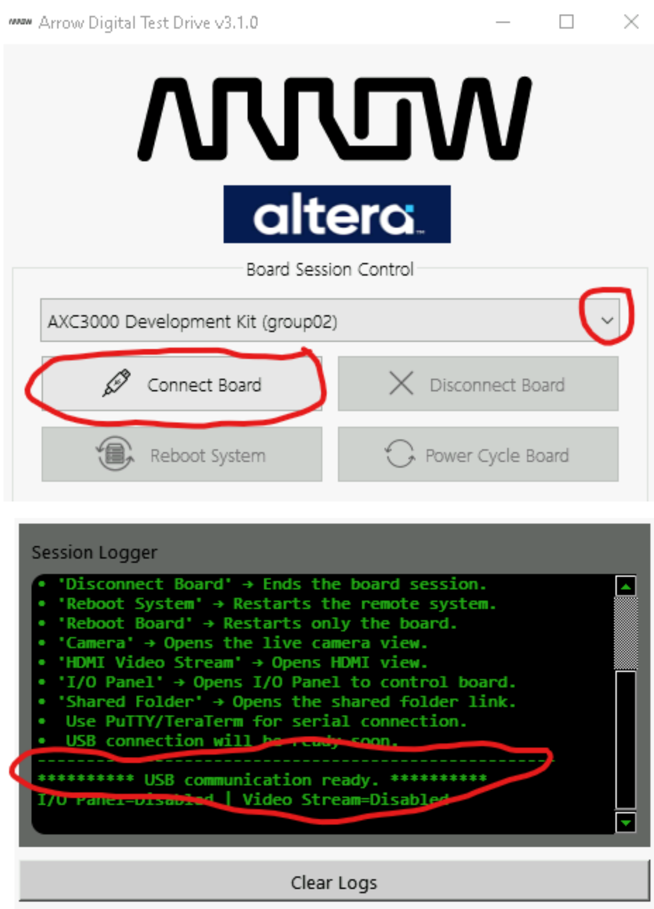
22 Verify the Design Remotely

If the design was built using CloudLabs and will be verified using the board farm, then complete this section.

22.1 Connect to the Board Farm

22.1.1 Connect to the remote AXC3000 Board using the ArrowDTD tool. Click on the ArrowDTD icon to launch the tool.

22.1.2 Select an available board from the dropdown menu and click connect board.

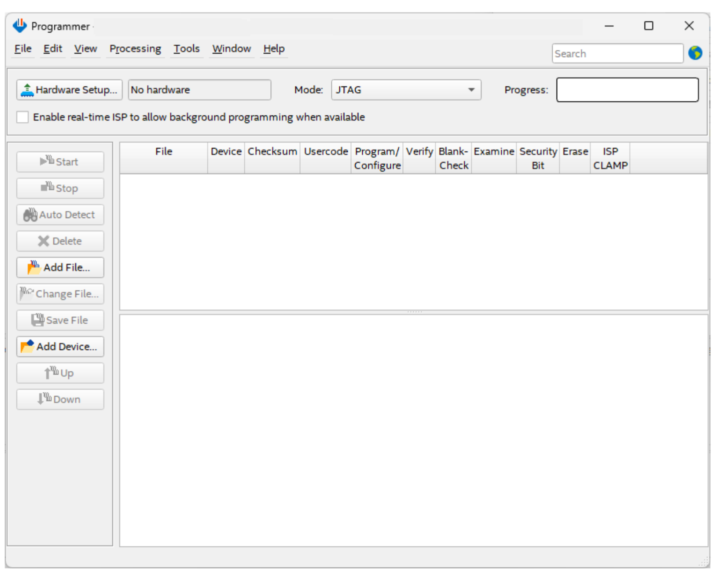


22.1.3 Open the Camera View of the board.

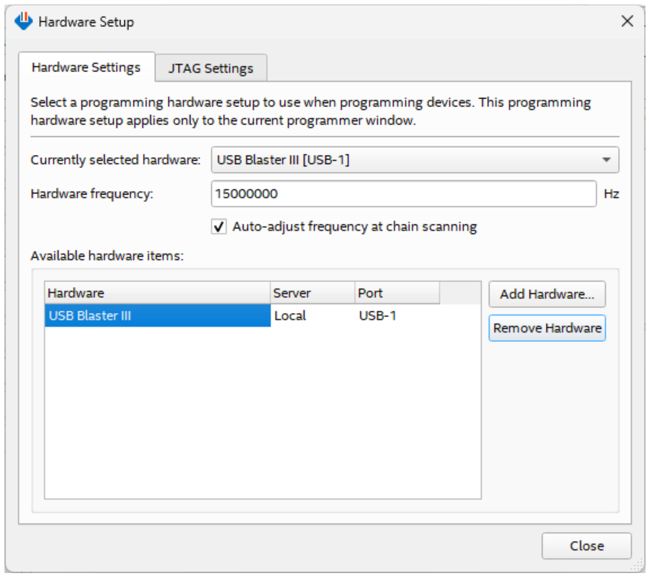
22.2 Program the FPGA

The next step is to program the design into the FPGA on the board. During the Assembler stage of the compilation, a programming file was generated.

22.2.1 Open the Programmer. Use the pull-down menu in Quartus to select Tools → Programmer.



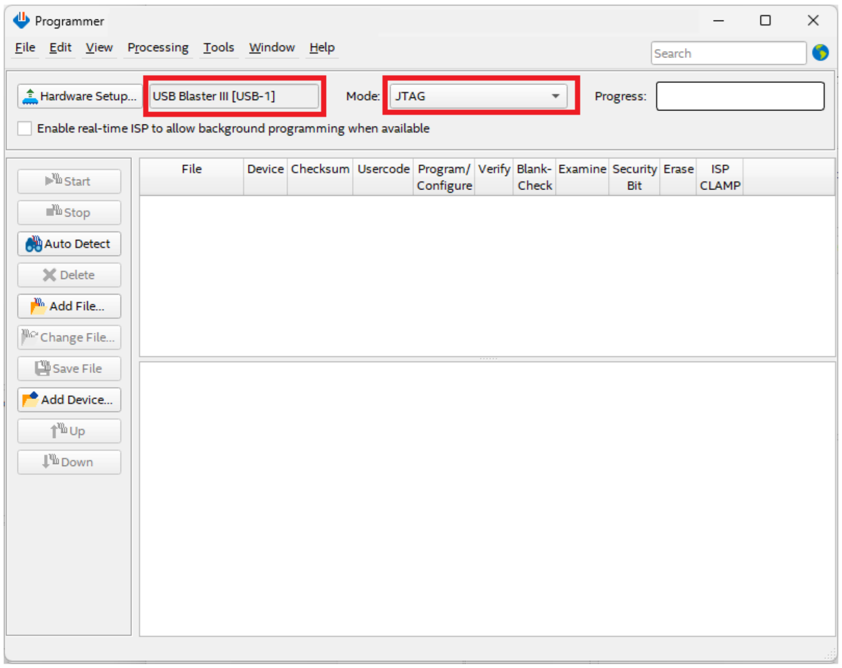
22.2.2 Click **Hardware Setup...** and double click **USB Blaster III** entry in the Hardware Setup window. **Currently selected hardware** should now show USB Blaster III [USB0] (the USB port number may be different depending on your PC).



22.2.3 Click **Close**.

22.2.4 If the device and programming file have been **automatically populated** then, **skip to section 22.2.9**

22.2.5 Make sure the hardware setup is Altera-USB-Blaster-III [USB0] and the mode is JTAG. Click **Auto Detect**.



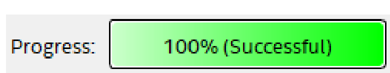
22.2.6 **Double click** the first <**none**> line item to choose the programming file

22.2.7 The sof file will be populated as a line item. If not, navigate to the **sub folder** output_files, and select the .sof file.

22.2.8 Click **Open**.

22.2.9 Make sure the Programmer shows the correct file and the correct part in the JTAG chain and check the **Program/Configure** checkbox.

22.2.10 Click **Start** to program the board. When the configuration is complete, the Progress bar should show 100% (Successful).

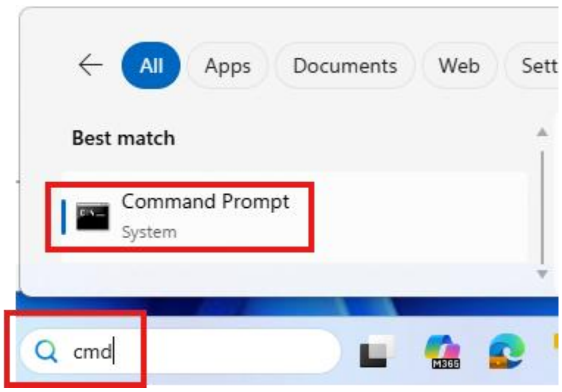


22.3 Open the One AI Image Source utility

This is a python based application It allows the user to select a paint like tablet or a live camera to provide an image source for the One AI IP block.

22.3.1 Open a Command Prompt.

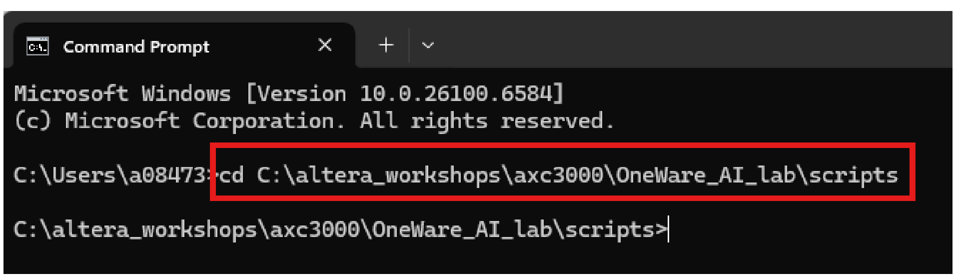
22.3.1.1 Type '**cmd'** in the Windows search bar. Click on **Command Prompt**.



22.3.2 Navigate to the script directory

22.3.2.1 cd to the path shown below in the Command Prompt shell

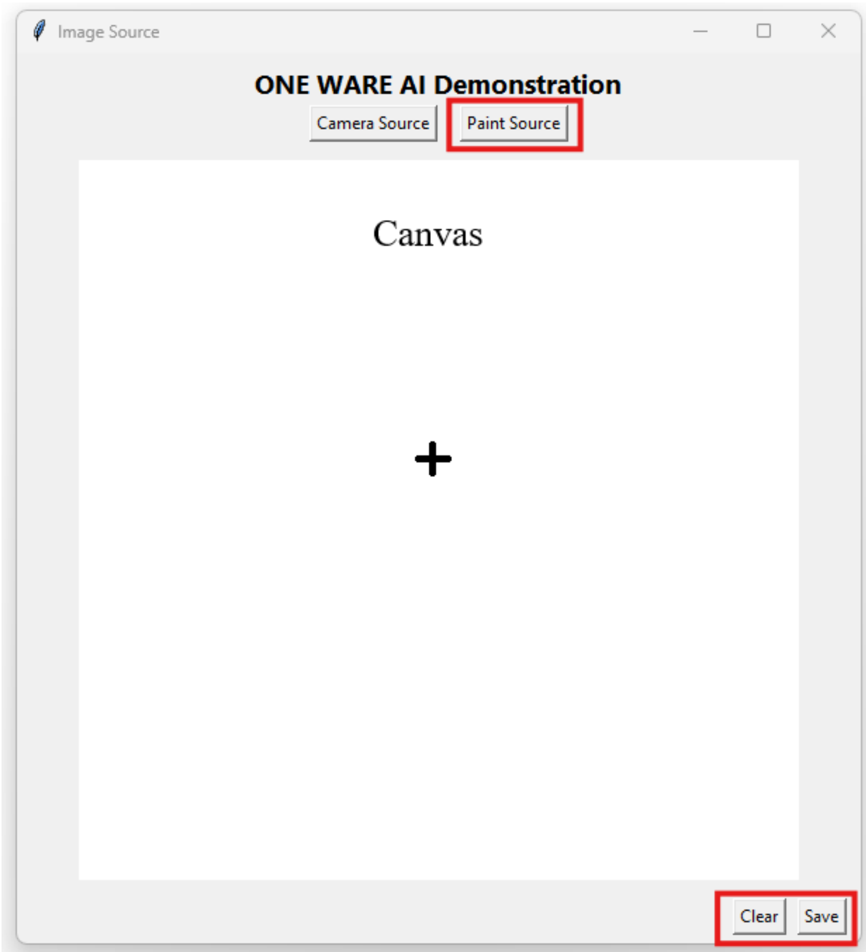
C:\altera_workshops\axc3000\OneWare_AI_lab\scripts



22.3.2.2 Type '**python one_ai.py**' at the prompt in the shell. Minimize the shell.

The application will launch after a few seconds.The GUI is shown below

There are four buttons in the GUI. The top two buttons select between a digit from the paint tablet or one sourced from a live camera. Since this is being deployed with hardware in a board farm, the camera option will not be available. Pressing the Paint button sends a message to AXC3000 board to source images into the frame buffer memory from the application.



Hovering the mouse over the canvas will change the cursor to a **cross**. Moving the mouse while the left button is clicked will allow the user to draw a digit. Click the **Save** button to transmit the digit to the **AI IP block** in the AXC3000 board. Click the **Clear** button to erase the digit from the canvas.

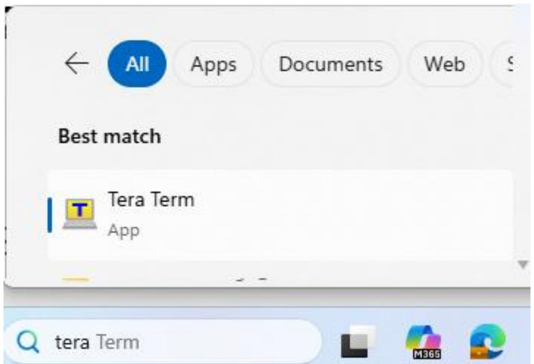
Note: The paint function performs better when the mouse is moved slower while drawing.

22.4 Open Tera Term

The inference results are output using the Nios V processor and a serial port.

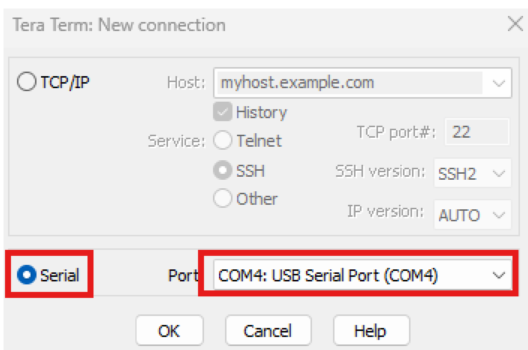
The probability of the Detected Digit is displayed numerically.

22.4.1 Type '**Tera'** in the Windows search bar. Click on **Tera Term**.



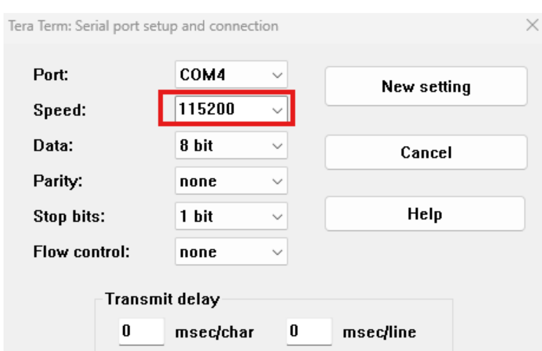
22.4.2 Setup the New connection

22.4.2.1 Select the 'Serial' connection option

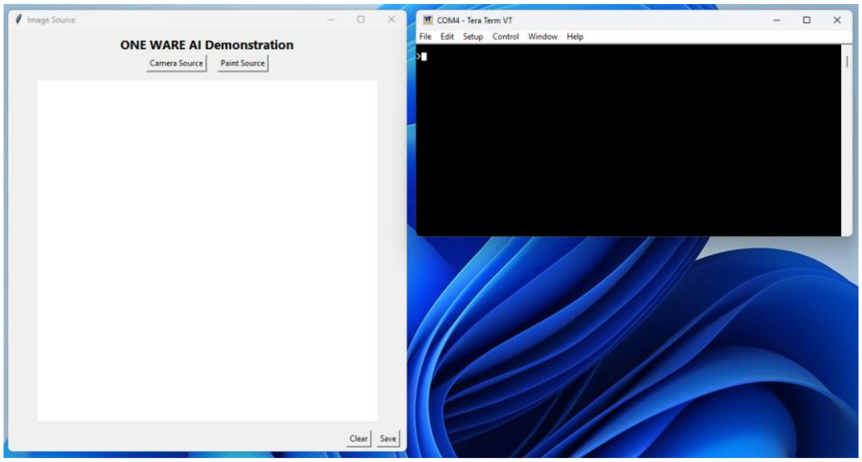


22.4.2.2 More than one serial port will be listed in the Port list. Select the one listed as '**USB Serial Port**'

22.4.2.3 Select Setup → Serial port. Select the serial port speed to be 115,200. Press enter.



22.4.3 **Move** the two applications so they fit comfortably side by side in the CloudLabs desktop.



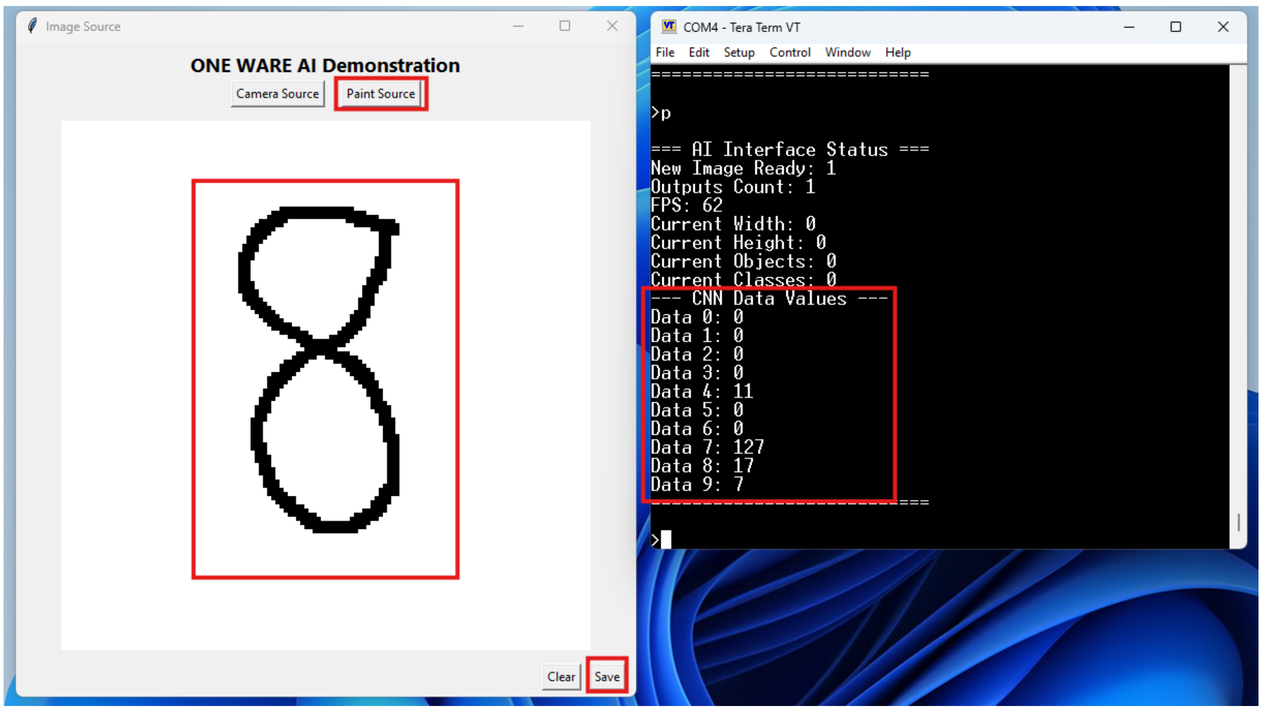
22.5 Classify the Image

22.5.1 Draw a digit.

22.5.1.1 Press the **Paint Source** button.

22.5.1.2 **Drag the mouse** over the canvas, press the left mouse button and create a digit

22.5.1.3 Press the **Save** button to transmit the digit to the AXC3000 board.



22.5.2 Review the inference results. Click the Tera Term window. Enter a lower case letter '**p**'. Press **enter**.

22.5.2.1 Take note of the **Detected Digit and its associated probability**.

The inference results are listed in the serial terminal output. Data 0 through Data 9 represent the 10 inference containers that are the output of the model. Data 0 represents the digit '1', Data 1, the digit '2' and so forth. Data 9 represents the digit '0'.

The maximum inference value is 255 and is a sum of all the inference containers. The container that has the highest value represent the inferred result.

A well written digit will have a higher probability. A digit that mimics any of the images in the Training folder will also have a higher probability. To convert from a probability number to a percentage, divide the inferred container value by 255.

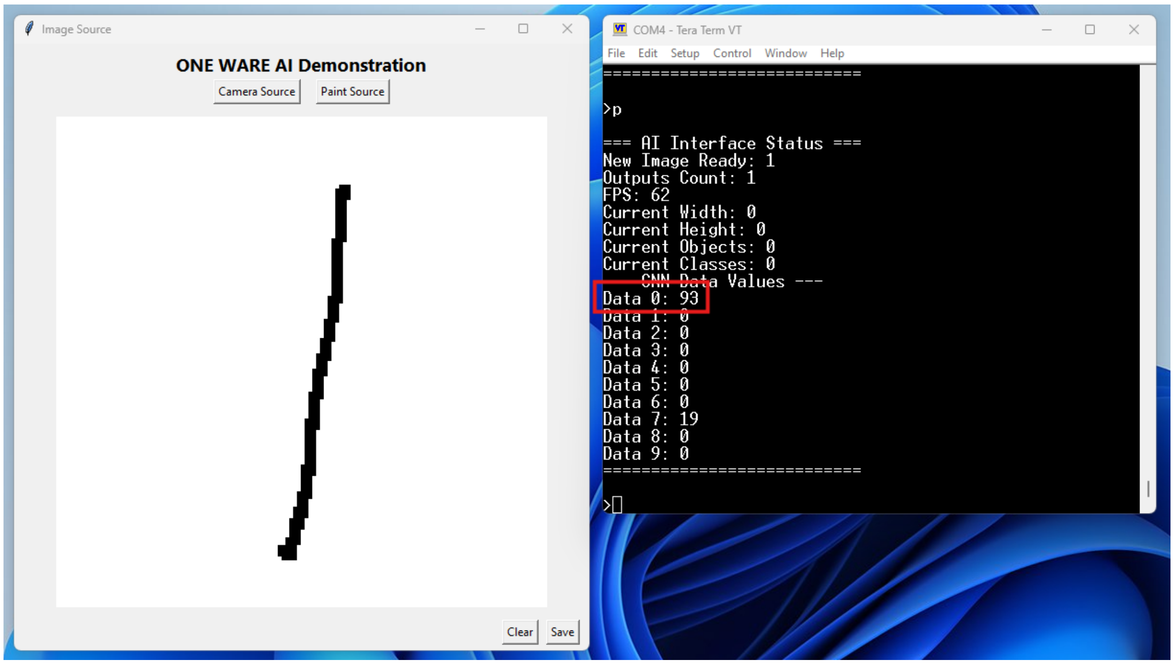
In the example shown above, there is an almost 50% chance that the digit is an '8', followed by a 6.7% chance it is a '9' and a 3.9% chance it is a '5'.

22.6 Classify a different digit

22.6.1 Press the Clear button to erase the digit.

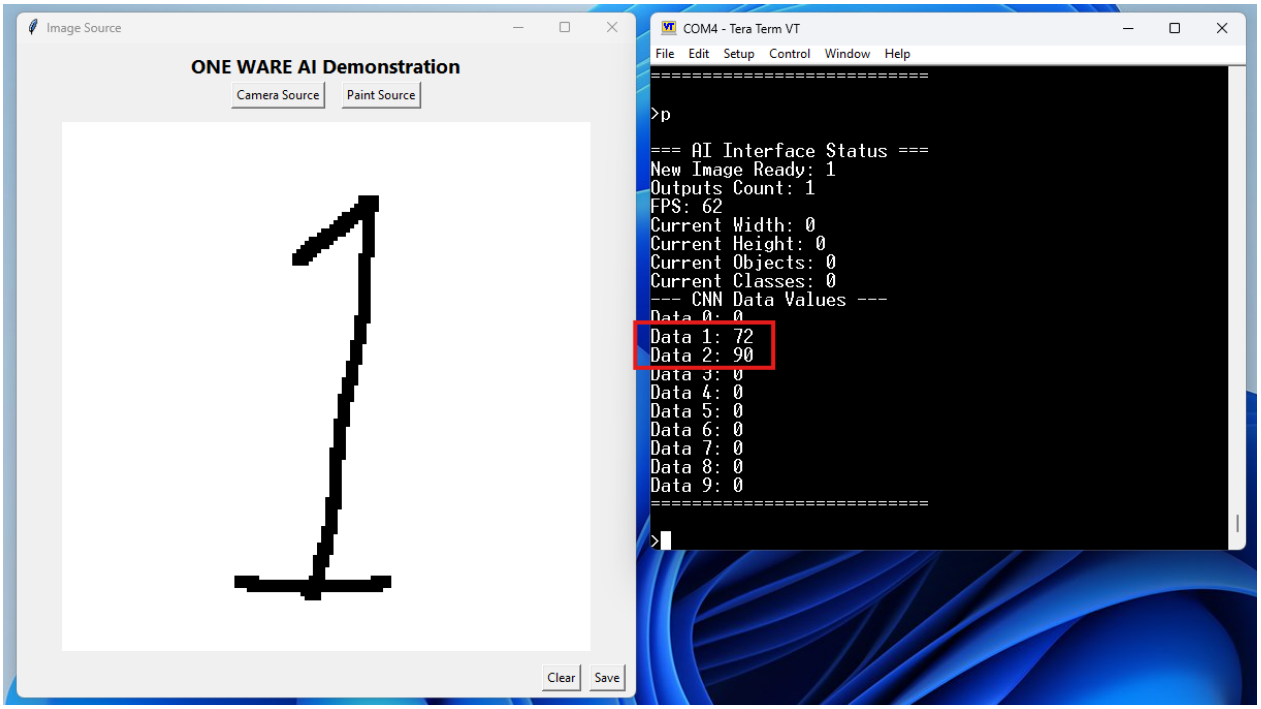
22.6.2 Write a ‘1’. Note the Detected Digit and its probability.

22.6.3 Select the Pencil tool and draw a new digit.   Press Save



22.6.4 Modify the digit to be more traditionally European. Add a slant at the top and a line across the bottom.

Note the results. It has misidentified the digit. It thinks it could be a two or a three. Why? There were no European style ‘1’s in the training or testing database.



Experiment with other digits and styles.

22.7 Close the applications

22.7.1 Close the One AI Image Source application

22.7.1.1 Click on the GUI to select it.

22.7.1.2 Press Alt F$ to close it

22.7.2 Close Tera Term.

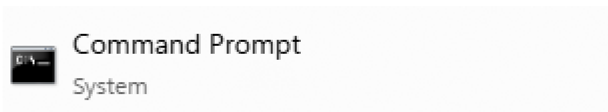
23 Verify the Design Locally

If the design was built using **CloudLabs** but is being verified locally then **complete section 23.1**. If the design was **built locally then skip to section 23.3**

23.1 Clone the workshop repository

Download Python and tcl scripts necessary for the verification process.

23.1.1 Open a Command prompt



23.1.2 Change directory and clone the repository

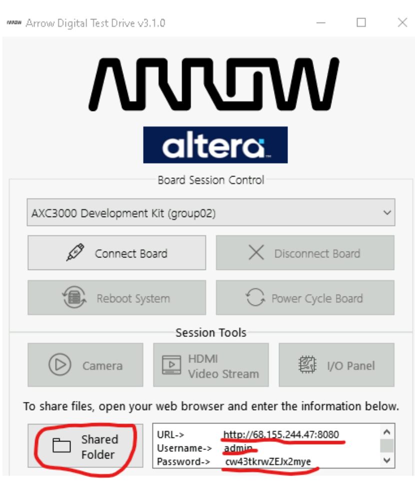
$ cd c:\

$ git clone https://github.com/ArrowElectronics/altera_workshops.git

23.2 File Download

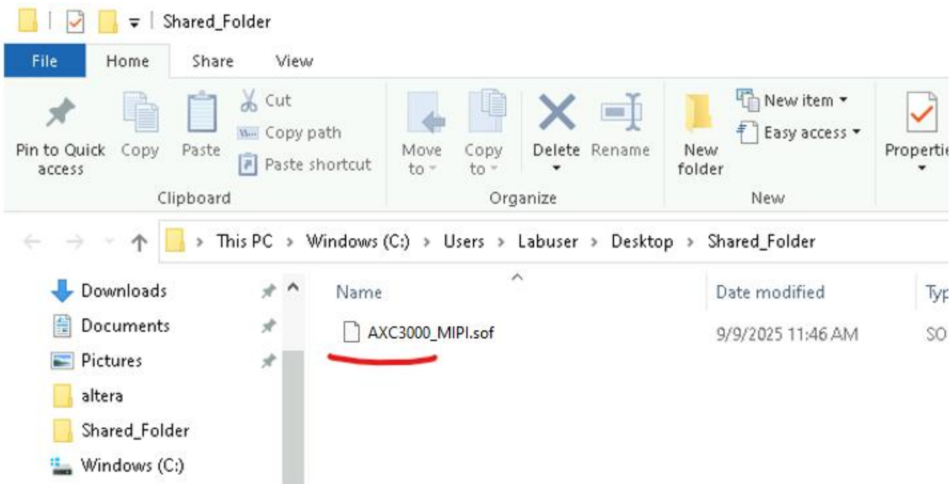
23.2.1 Click on the ArrowDTD icon in CloudLabs to launch the tool

23.2.2 Open the **Shared Folder** utility in the CloudLab VM. Open a browser in the local PC. Navigate to the **URL** indicated. Use the provided username and password.



23.2.3 Drag and drop the following file into the **Shared** folder.

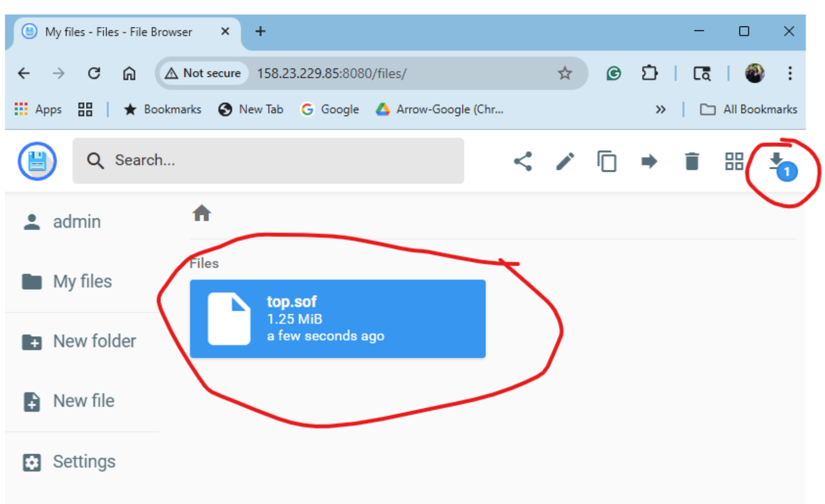
c:\altera_workshops\axc3000\OneWare_AI_lab\Qpro\output_files\AXC3000_MIPI.sof



23.2.4 This will be reflected in the **browser** on the local PC.

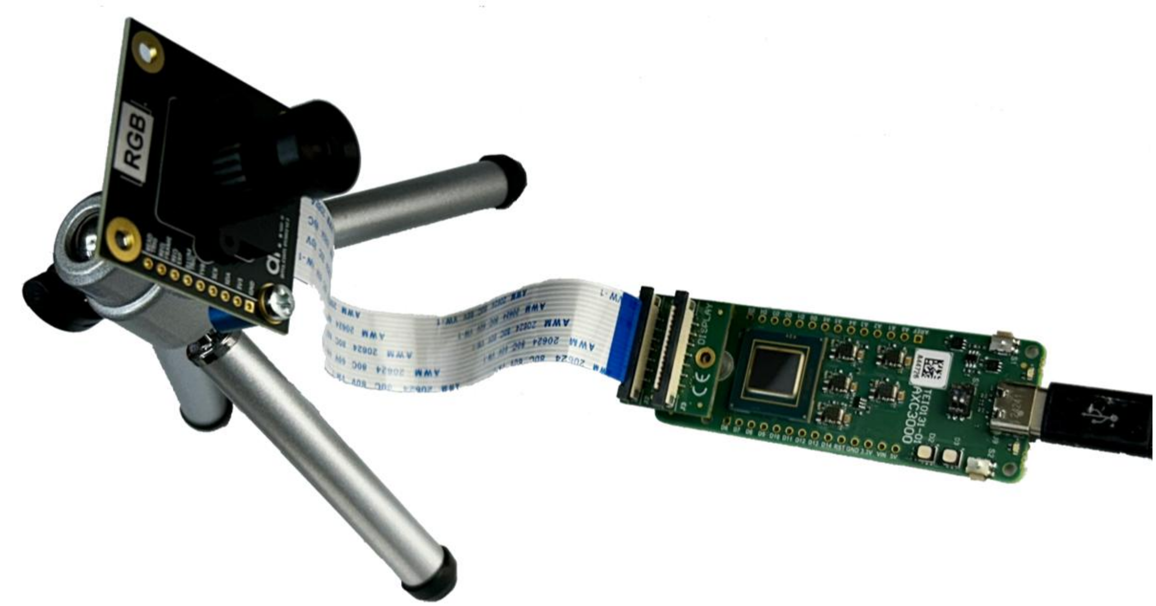
23.2.5 **Download** the SOF file to

c:\altera_workshops\axc3000\OneWare_AI_lab\Qpro\output_files\AXC3000_MIPI.sof



23.3 Connect the camera to the AXC3000 board

The ams OSRAM Mira220 camera comes with a 15-pin flat ribbon cable. The connection to the AXC3000 board is done via the CRUVI adapter (TEC0278) J2 connector. The figure below shows the assembled camera demo.



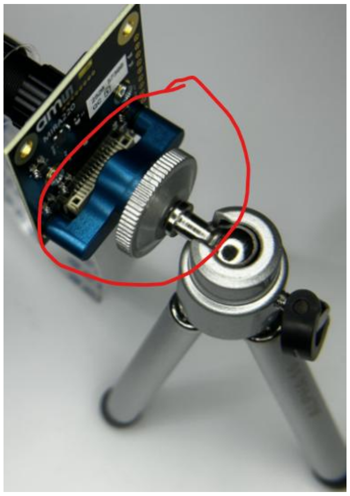
Note: If you want to acquire your own camera, you will need these 2 part numbers:

1. Imager board Mira220_RGB_mini_SB, which comes with the flat ribbon cable
2. Lens assembly AB04620MG from https://www.alaudoptics.com/AB04620MG.html

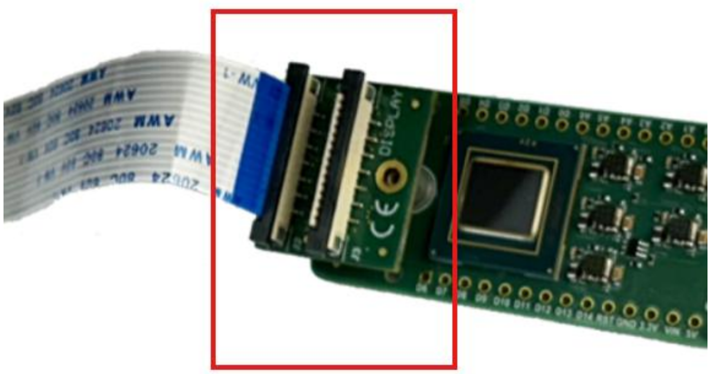
The camera assembly is shipped as two main pieces. The first is the tripod. The second is the camera body assembly (camera attached via a flat ribbon cable to the CRUVI adapter).

Use the following steps to assemble these pieces.

23.3.1 Attach the Camera body assembly to the tripod using the visible thread.



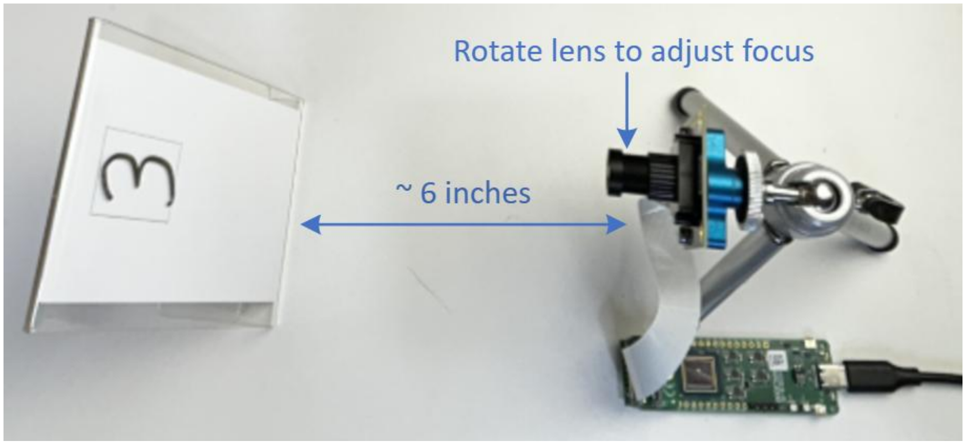
23.3.2 Insert the CRUVI adapter in the AXC3000 board. Orient the AXC300 and the Camera assembly as shown in the picture below.



23.4 Orient the Camera assembly and the Picture Frame

A plastic picture frame holder is also provided for the workshop. To obtain the best workshop experience, position the camera about 6 inches from the picture frame. The camera might need to have its focus adjusted, although it probably does not. You will view the camera image in a later segment and will have the opportunity to adjust it then.

It will be useful to mark the position of the picture frame on a piece of paper underneath the frame. When you are writing digits on its surface it might shift and you can use the marker to reposition it.

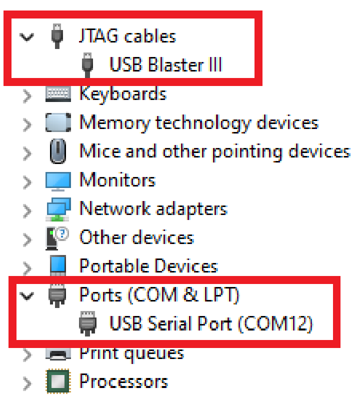


23.5 Program the FPGA

The next step is to program the design into the FPGA on the board. During the Assembler stage of the compilation, a programming file was generated.

23.5.1 Connect your AXC3000 Board to your PC using a USB C cable.

Since the USB Blaster III should be already installed, the Window's Device Manager should display the following entries, highlighted in red (COM port number may differ depending on your PC):



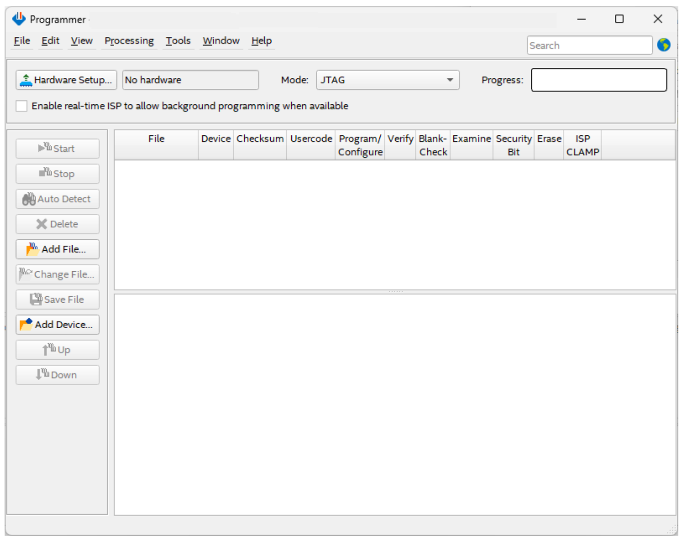
23.5.2 Open the Programmer

If you built remotely using Cloudlabs then complete step 23.5.3

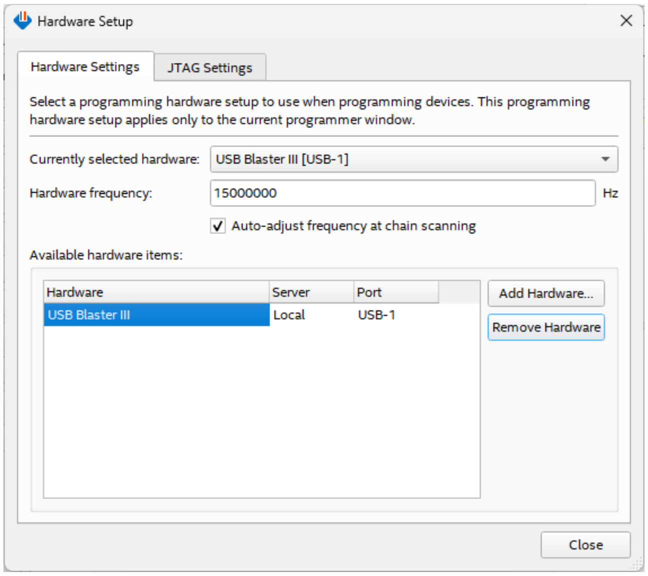
If you built locally then complete step 23.5.4

23.5.3 From the Windows Start menu **open Altera 25.1.1.125 Pro Edition --> Programmer.**

23.5.4 Open the Programmer. Use the pull down menu in **Quartus** to select **Tools → Programmer**.

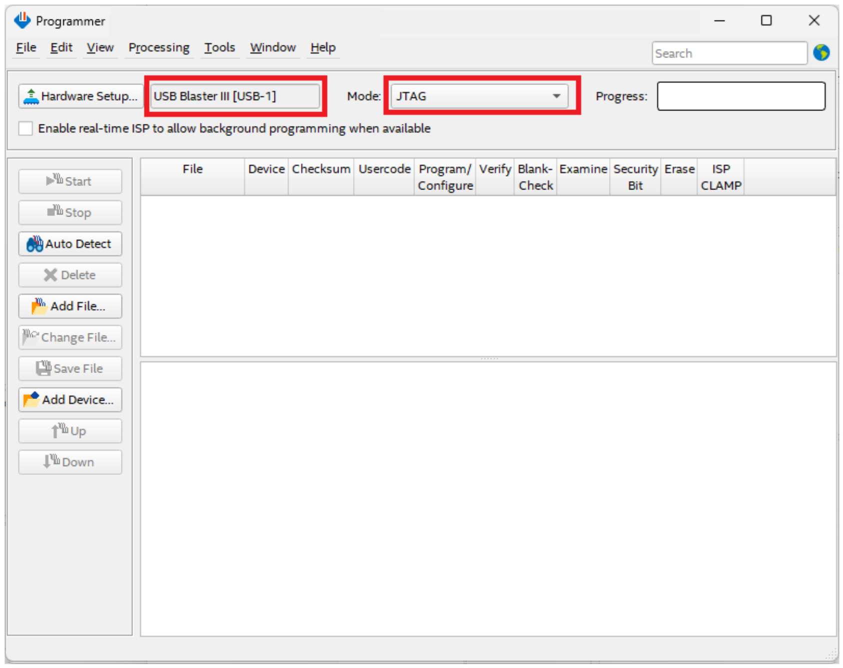


23.5.5 Click **Hardware Setup...** and double click **USB Blaster III** entry in the Hardware Setup window. **Currently selected hardware** should now show USB Blaster III [USB0] (the USB port number may be different depending on your PC).



23.5.6 Click **Close**.

23.5.7 Make sure the hardware setup is Altera-USB-Blaster-III [USB0] and the mode is JTAG. Click **Auto Detect**.



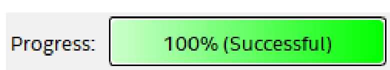
23.5.8 **Double click** the first <**none**> line item to choose the programming file. If it is already populated then skip to section 23.5.11

23.5.9 The sof file will be populated as a line item. If not, navigate to the **sub folder** output_files, and select the .sof file.

23.5.10 Click **Open**.

23.5.11 Make sure the Programmer shows the correct file and the correct part in the JTAG chain and check the **Program/Configure** checkbox.

23.5.12 Click **Start** to program the board. When the configuration is complete, the Progress bar should show 100% (Successful).

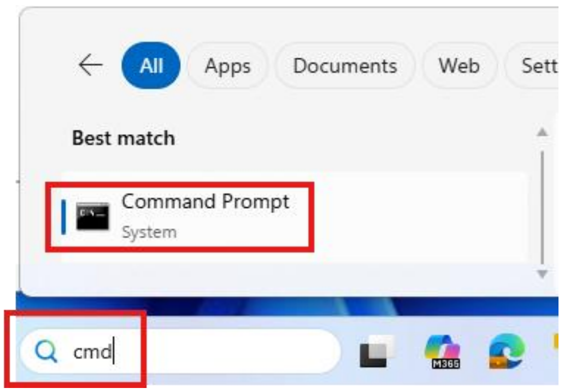


23.6 Open the One AI Image Source utility

This is a python based application It allows the user to select a paint like tablet or a live camera to provide an image source for the One AI IP block.

23.6.1 Open a Command Prompt.

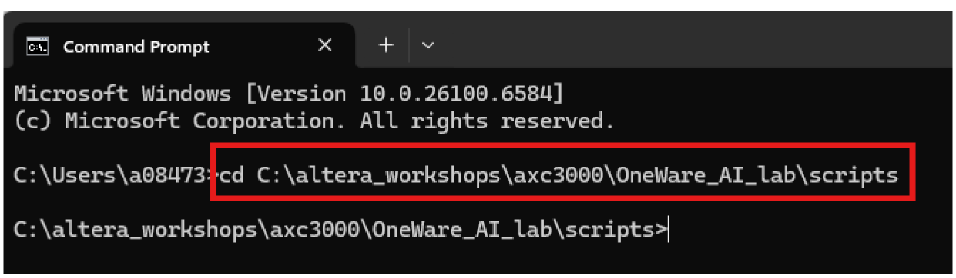
23.6.1.1 Type ‘**cmd’** in the Windows search bar. Click on **Command Prompt**.



23.6.2 Navigate to the script directory

23.6.2.1 cd to the path shown below in the Command Prompt shell

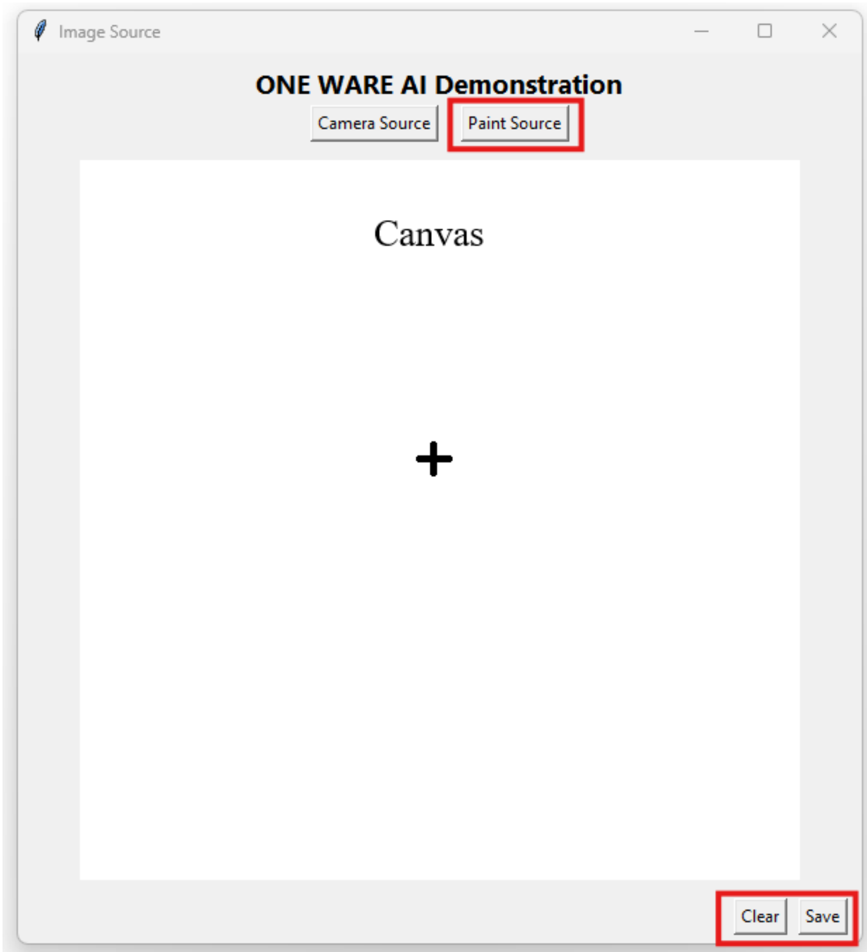
C:\altera_workshops\axc3000\OneWare_AI_lab\scripts



23.6.2.2 Type ‘**python one_ai.py**’ at the prompt in the shell. Minimize the shell.

The application will launch after a few seconds.The GUI is shown below

There are four buttons in the GUI. The top two buttons select between a digit from the paint tablet or one sourced from a live camera. Since this is being deployed with hardware in a board farm, the camera option will not be available. Pressing the Paint button sends a message to AXC3000 board to source images into the frame buffer memory from the application.



Hovering the mouse over the canvas will change the cursor to a **cross**. Moving the mouse while the left button is clicked will allow the user to draw a digit. Click the **Save** button to transmit the digit to the **AI IP block** in the AXC3000 board. Click the **Clear** button to erase the digit from the canvas.

Note: The paint function performs better when the mouse is moved slower while drawing.

23.7 Open the One AI Visualize utility

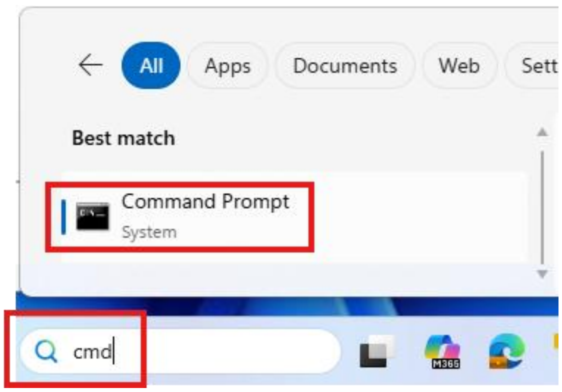
This is a python-based application It performs two functions.

It extracts the image in the frame buffer in the AXC3000 and displays it. It also displays the probability of the Detected Digit.

Please note that the python application has been written to extract and display the video frame buffer every 2 or 3 seconds. **Agilex 3** and the **AI IP** is capable of processing hundreds to thousands of **frames per second**.

23.7.1 Open a second Command Prompt.

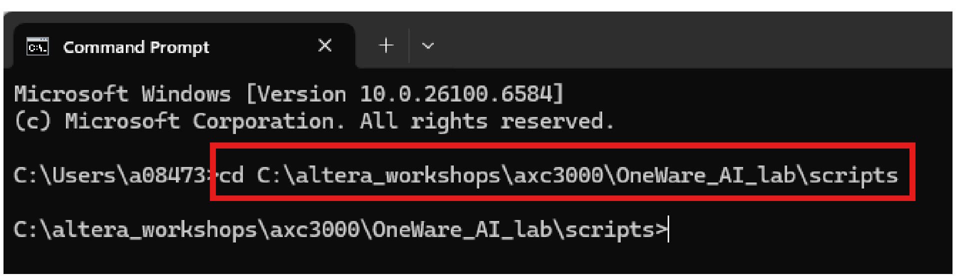
23.7.1.1 Type **'cmd'** in the Windows search bar. Click on **Command Prompt**.



23.7.2 Navigate to the script directory

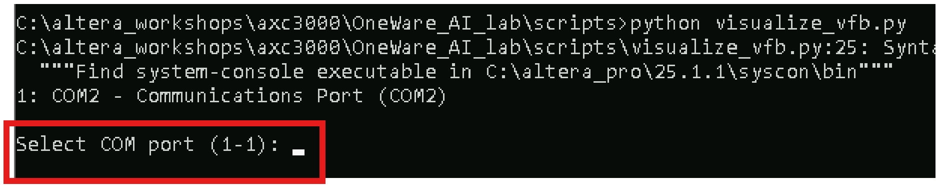
23.7.2.1 cd to the path shown below in the Command Prompt shell

C:\altera_workshops\axc3000\OneWare_AI_lab\scripts



23.7.2.2 Type '**python visualize_vfb.py**' at the prompt in the shell.

23.7.2.3 The command shell will prompt for a **COM port** selection. If there is only one item in the list, **select 1 and press enter**. Minimize the shell.



The application will launch after a few seconds.The GUI is shown below

The GUI will display the detected digit. The probability shown is simply the highest number allocated to one of the ten classification outputs from the One AI IP block. All ten output probabilities add up to a maximum of 255. It is not unusual to see values in at least two locations. For example, a **'3'** could be detected as a three and also partially as an **'8'**. This is more likely for a handwritten digit captured by the camera rather than one that is painted.

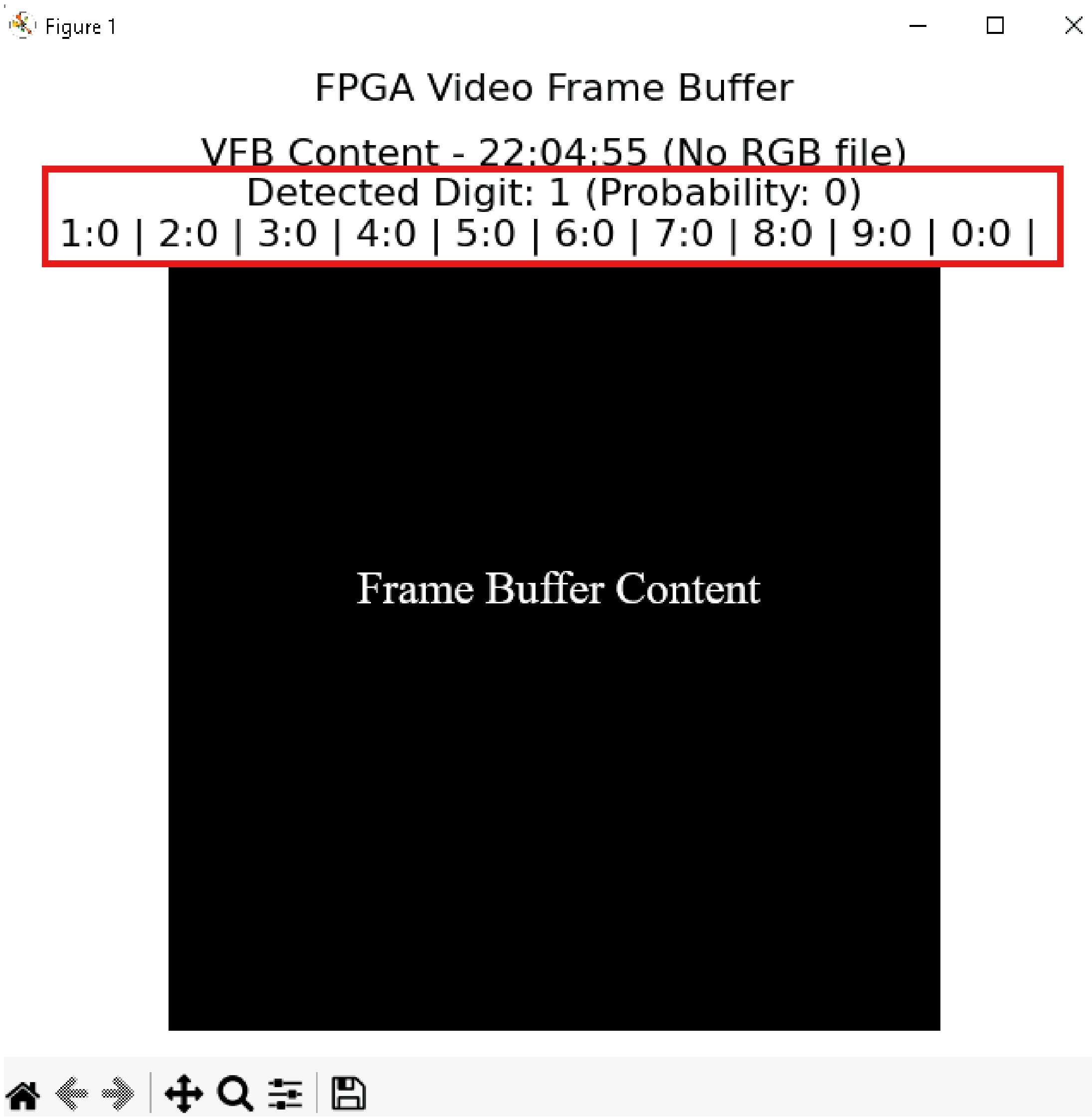
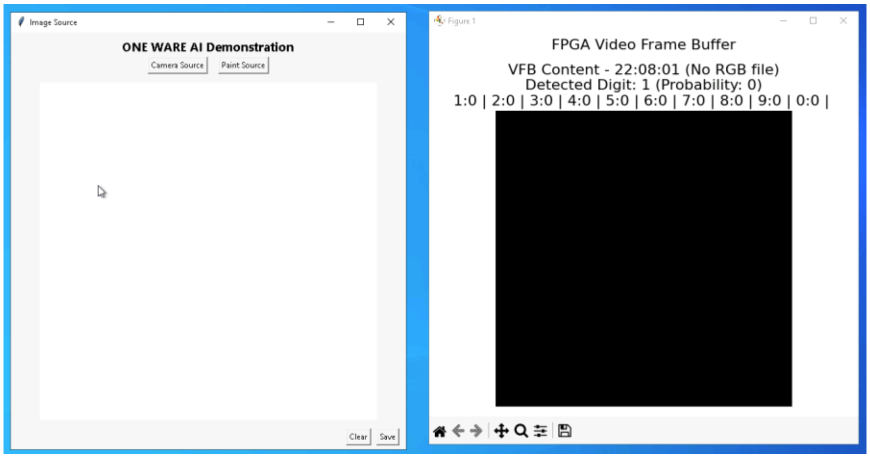


Figure 1
FPGA Video Frame Buffer
VFB Content - 22:04:55 (No RGB file)
Detected Digit: 1 (Probability: 0)
1:0 | 2:0 | 3:0 | 4:0 | 5:0 | 6:0 | 7:0 | 8:0 | 9:0 | 0:0 |
Frame Buffer Content

23.7.3 **Resize** the two applications so they fit comfortably side by side in the CloudLabs desktop.



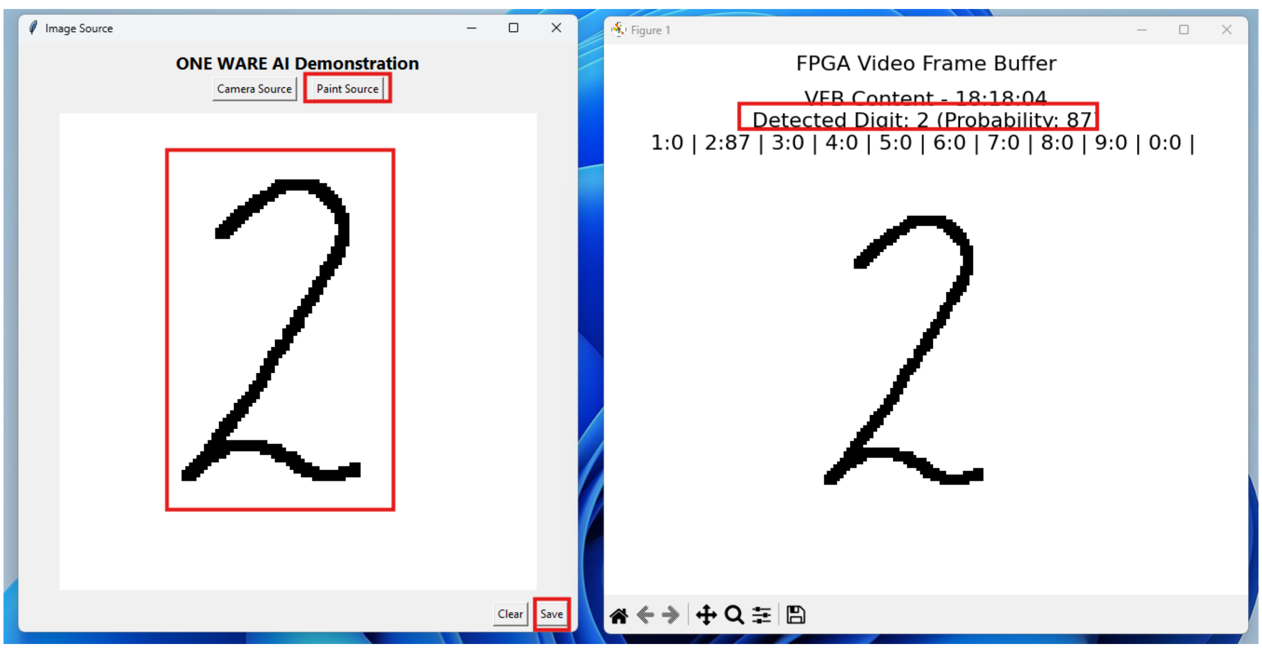
23.8 Classify the Image with Paint

23.8.1 Draw a digit.

23.8.1.1 Press the **Paint Source** button.

23.8.1.2 **Drag the mouse** over the canvas, press the left mouse button and create a digit

23.8.1.3 Press the **Save** button to transmit the digit to the AXC3000 board.



23.8.2 Review the inference results

23.8.2.1 Take note of the **Detected Digit and its associated probability**.

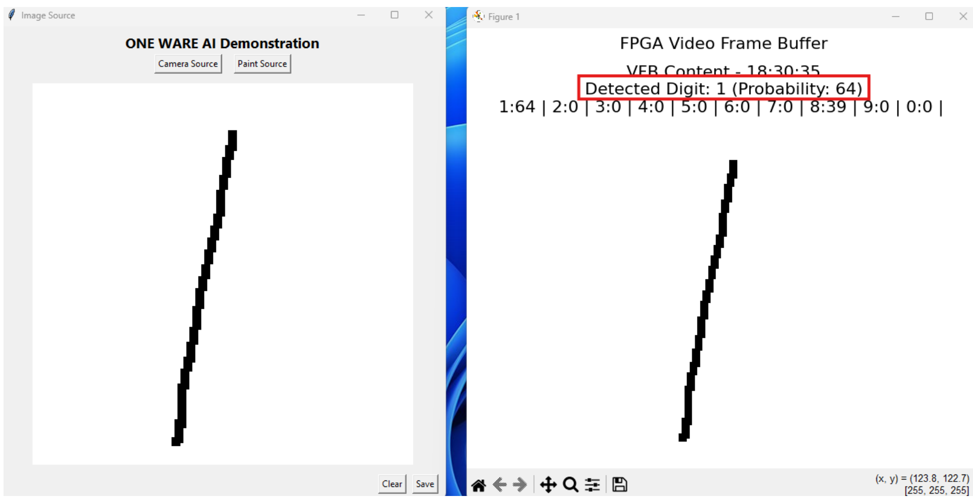
A well written digit will have a higher probability. A digit that mimics any of the images in the Training folder will also have a higher probability.

23.9 Classify a different digit

23.9.1 Press the Clear button to erase the digit.

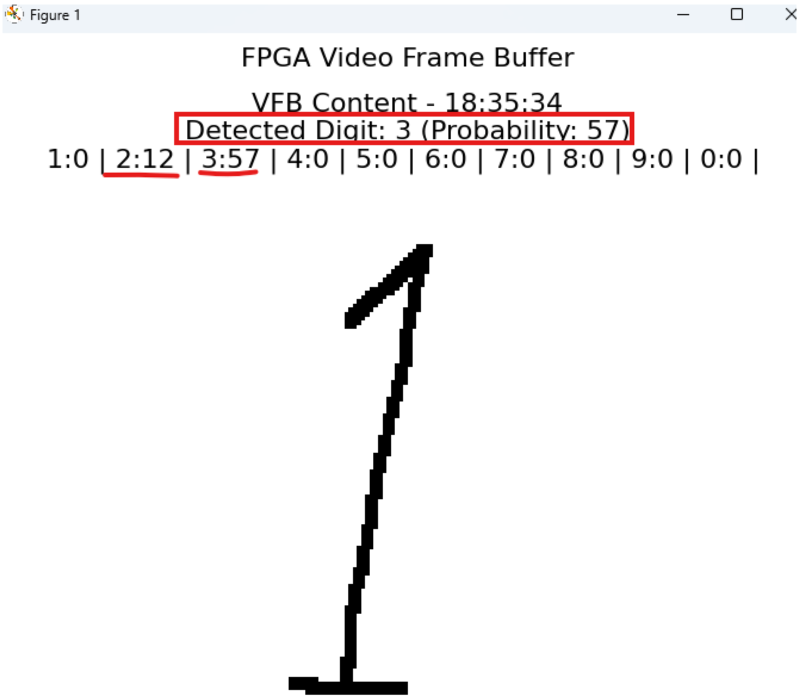
23.9.2 Write a '1'. Note the Detected Digit and its probability.

23.9.3 Select the Pencil tool and draw a new digit.   Press Save



23.9.4 Modify the digit to be more traditionally European. Add a slant at the top and a line across the bottom.

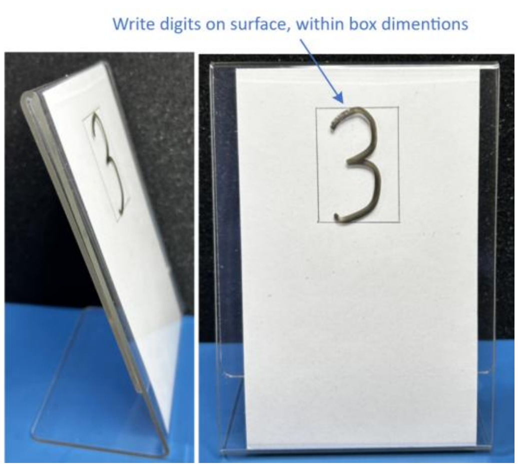
Note the results. It has misidentified the digit. It thinks it could be a two or a three. Why? There were no European style '1's in the training or testing database.



Experiment with other digits and styles.

23.10 Classify the Image with the Camera

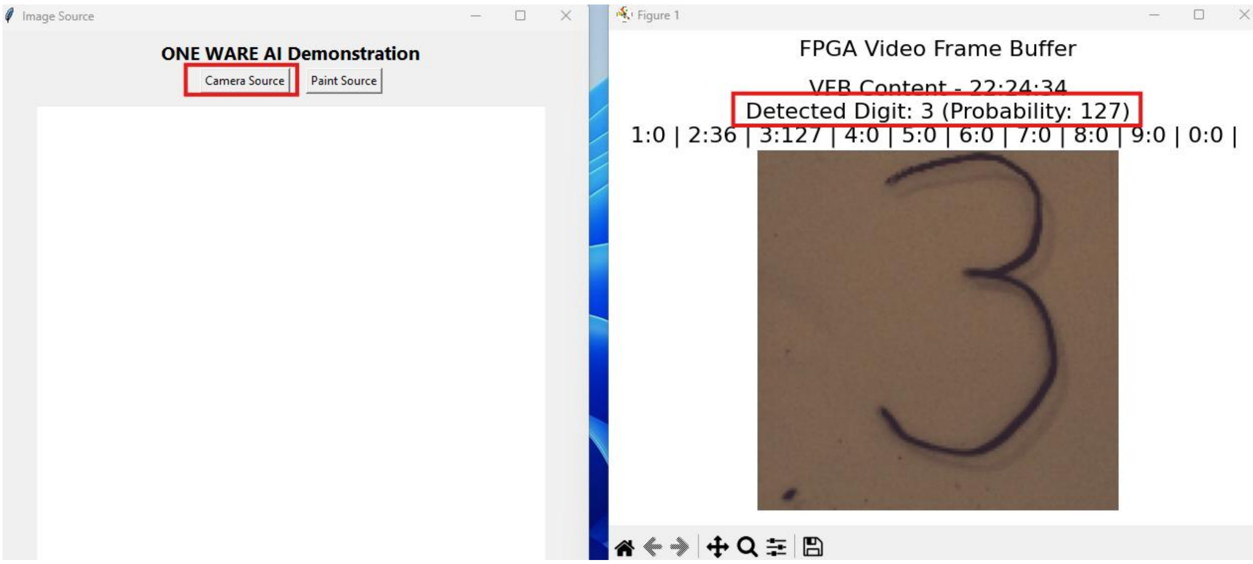
23.10.1 Write a digit on the surface of the picture frame holder with a dry erase marker.



23.10.2 Press the Camera Source button

The Detected Digit should update almost immediately. The snapshot of the written image can take up to a few seconds to appear.

23.10.3 Focus the camera lens, if the picture is blurry. Turn the lens a quarter turn one way and wait a few seconds for the image to update. Continue until the image is in focus.



23.10.4 Review the results. Is the probability correct?

23.10.5 Erase the digit with a cloth. Try other digits.

23.11 Close the applications

23.11.1 Close the One AI Image Source application

23.11.1.1 Click on the GUI to select it.

23.11.1.2 Press Alt F$ to close it

23.11.2 Close the One AI Visualize utility.

23.11.2.1 Maximize the Command Prompt shell.

23.11.2.2 Press Control C.

24 Congratulations on working through this workshop!

You worked through this workshop where you were exposed to considerable information on the AI flow including

- Design requirements
- Software (OneWare) and network topology
- Obtaining a dataset
- Loading (and modifying) a dataset
- Changing a model
- Training a model
- Testing the model to evaluate what works and what does not with your own handwriting
- Visualization of the model with the output file, ONNX
- Simulation
- Inference of the Model by importing a model in FPGA
- Testing the FPGA (for those who have actual boards)

25 Legal Disclaimer

ARROW ELECTRONICS

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The purpose of this evaluation board is solely intended for evaluation purposes. Any use of the Board beyond these purposes is on your own risk. Furthermore, according the applicable law, the offering Arrow entity explicitly does not warrant, guarantee or provide any remedies to you with regard to the board.

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Before You handle or use the Evaluation Board, You shall comply with all such warnings and other instructions and employ reasonable safety precautions in using the Evaluation Board. Failure to do so may result in death, personal injury, or property damage.

You shall not use the Evaluation Board in any safety critical or functional safety testing, including but not limited to testing of life supporting, military or nuclear applications. Arrow expressly disclaims any responsibility for such usage which shall be made at Your sole risk.

WARRANTY

Arrow warrants that it has the right to provide the evaluation board to you. This warranty is provided by Arrow in lieu of all other warranties, written or oral, statutory, express or implied, including any warranty as to merchantability, non-infringement, fitness for any particular purpose, or uninterrupted or error-free operation, all of which are expressly disclaimed. The evaluation board is provided “as is” without any other rights or warranties, directly or indirectly.

You warrant to Arrow that the evaluation board is used only by electronics experts who understand the dangers of handling and using such items, you assume all responsibility and liability for any improper or unsafe handling or use of the evaluation board by you, your employees, affiliates, contractors, and designees.

LIMITATION OF LIABILITIES

In no event shall Arrow be liable to you, whether in contract, tort (including negligence), strict liability, or any other legal theory, for any direct, indirect, special, consequential, incidental, punitive, or exemplary damages with respect to any matters relating to this agreement. In no event shall arrow’s liability arising out of this agreement in the aggregate exceed the amount paid by you under this agreement for the purchase of the evaluation board.

IDENTIFICATION

You shall, at Your expense, defend Arrow and its Affiliates and Licensors against a claim or action brought by a third party for infringement or misappropriation of any patent, copyright, trade secret or other intellectual property right of a third party to the extent resulting from (1) Your combination of the Evaluation Board with any other component, system, software, or firmware, (2) Your modification of the Evaluation Board, or (3) Your use of the Evaluation Board in a manner not permitted under this Agreement. You shall indemnify Arrow and its Affiliates and Licensors against and pay any resulting costs and damages finally awarded against Arrow and its Affiliates and Licensors or agreed to in any settlement, provided that You have sole control of the defense and settlement of the claim or action, and Arrow cooperates in the defense and furnishes all related evidence under its control at Your expense. Arrow will be entitled to participate in the defense of such claim or action and to employ counsel at its own expense.

RECYCLING

The Evaluation Board is not to be disposed as an urban waste. At the end of its life cycle, differentiated waste collection must be followed, as stated in the directive 2002/96/EC. In all the countries belonging to the European Union (EU Dir.

2002/96/EC) and those following differentiated recycling, the Evaluation Board is subject to differentiated recycling at the end of its life cycle, therefore: It is forbidden to dispose the Evaluation Board as an undifferentiated waste or with other domestic wastes. Consult the local authorities for more information on the proper disposal channels. An incorrect Evaluation Board disposal may cause damage to the environment and is punishable by the law.